User/Operator Manual

English

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Tempus Pro

Patient monitor

PHILIPS

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18.1 Symbols used 268

1 About this manual

This manual is for the Tempus Pro patient monitor.

All Tempus Pro monitors have a standard configuration that provides ECG monitoring, NIBP, pulse oximetry (SpO_2) and impedance respiration. Additional functions are available depending on Tempus Pro part number:

- 00-1004-R: standard configuration with invasive pressure (2 channels), $ETCO_2$ and contact temperature (1 or 2 channels).
- 00-1007-R: standard configuration with invasive pressure (2 channels), $ETCO_2$, contact temperature (1 or 2 channels) and Bluetooth headset.
- 00-1024-R: standard configuration with printer, $ETCO_2$ and contact temperature (1 or 2 channels).
- 00-1026-R: standard configuration with printer, invasive pressure (2 channels), $ETCO_2$ and contact temperature (1 or 2 channels).

For a full list of Tempus Pro features, see “1.6 Features list”.

Tempus Pro pulse oximetry provides the standard SpO_2, pulse rate and perfusion index (PI) measurements but also has the capability to monitor carboxyhemoglobin saturation (SpCO), methemoglobin saturation (SpMet), total hemoglobin concentration (SpHb), total oxygen content (SpOC) and pleth variability index (PVI).

The Tempus Pro is supplied in different configurations depending on the customer’s requirements. This manual is written to cover all Tempus Pro features and therefore details regarding optional features will not be applicable to all units.

Note	If any serious incident has occurred in relation to the device it should be reported to Remote Diagnostic Technologies Limited and the competent authority of the Member State in which the user and/or patient is established.

1.1 Document library

Consult the instructions for use. These are provided in electronic format in the Philips Document Library.

www.philips.com/IFU

Note	Please check the Philips Document Library regularly for new versions of this document

A paper copy of this document may be requested from the Philips Document Library or by sending a request to RDT_Customerservice@philips.com.

1.2 CE statement

0413

Marking by the above symbol indicates compliance of this medical device to the Medical Devices Directive (MDD) 93/42/EEC (as amended) and the Radio Equipment Directive (RED) 2014/53/EU (as amended). The CE mark is accompanied by the number 0413 which is the reference number for the MDD 93/42/EEC Notified Body who certify RDT's quality system.

The Tempus Pro is a class IIb device under the MDD and is a class I device (harmonized frequencies) under the Radio Equipment Directive (RED) 2014/53/EU.

A Declaration of Conformity in accordance with the above regulations has been made and is on file with RDT, see "RDT contact details".

The wireless portion of this equipment may be operated in GB, France, Italy, Switzerland, Germany, Holland, Portugal, Spain, Sweden, Norway, Denmark and Finland.

1.3 FDA prescription statement

Federal law (USA) restricts the use or sale of this device by, or on the order of, a physician.

1.4 Proprietary notice

The purpose of this document is to provide the user with adequately detailed information to efficiently install, operate, maintain and order spare parts for the Tempus Pro. Every effort has been made to keep the information contained in this document current and accurate as of the date of publication or revision. However, no guarantee is given or implied that the document is error free or that it is accurate with regard to any specification. RDT reserves the right to change specifications without notice.

Tempus Pro, Corsium, Corsium Crew, ReachBak, i2i and TempusNET are all trademarks of RDT.

The Bluetooth name and logo are owned by the Bluetooth SIG Inc. and any use of this name or mark is under license.

The following trademarks are the property of Oridion Medical Ltd.: Oridion, Microstream, FilterLine, (FilterLine Set; FilterLine H Set, Nasal FilterLine), CapnoLine (Smart CapnoLine, Smart CapnoLine O2, Smart CapnoLine Plus, Smart CapnoLine Plus O2, Smart CapnoLine H Plus, Smart CapnoLine H Plus O) and Integrated Pulmonary Index (IPI).

NO IMPLIED LICENSE

Possession or purchase of this device does not convey any expressed or implied license to use the device with unauthorized sensors or cables that would, alone, or in combination with this device, fall within the scope of one or more of the patents relating to this device.

1.5 Use of this manual

The instructions and safety precautions provided in this manual must be observed during all phases of the operation, usage, service or repair of the Tempus or its accessories. Failure to comply with the information contained in this manual e.g. warnings, precautions, instructions etc. will violate the safety standards of design, manufacture and intended use of the products. Remote Diagnostic Technologies Ltd. assumes no liability for customer failure to comply with the information contained in this manual.

WARNING	**The Tempus is intended to be operated by clinically qualified personnel only. This manual, as well as accessory directions for use and all precautionary information and specifications should be read before use.**
CAUTION	Separate manuals are provided for the setup and maintenance of the device. These manuals are intended for the communications engineers and Bio-medical engineers responsible for such activities and are available from RDT upon request.

1.6 Intended users

RDT assumes that users of the product are clinically trained in how to take and interpret a patient's vital signs.

The Tempus Pro and this manual are intended to be used by such clinically-trained personnel. While the Tempus Pro has been designed as a good quality monitor; RDT reminds users that no monitor can replace good clinical judgment or the care and attention of a clinician. Before attempting to work with this equipment, the user must read, understand note and strictly observe all warnings, cautions and safety markings in this manual, other associated labelling and on the equipment.

It is the user's responsibility to ensure they are properly prepared to use the product. No formal training should be required so long as the user reads this manual thoroughly and familiarizes themselves with the product before use. RDT can provide direct training courses if preferred.

1.7 Features list

Note	Not all features are supported in all territories.

All Tempus Pro monitors come with the following items:

- Lithium-ion battery
- Non-invasive blood pressure hose
- Non-invasive blood pressure cuff
- Masimo patient cable
- Tempus Pro ECG cable
- Tempus mains power supply and mains cable
- Tempus Pro user manual set (CD-ROM)

The following additional items may be included:

- Configurations with a built in capnometer (optional) are provided with an Oridion Military and Air Medical Sample Pack.

Configurations can include an optional 2 Channel Invasive Pressure Module (part number 01-2017).

1.8 Patent and warranty

1.8.1 Patent claims

RDT has applied for patents covering the Tempus Pro and its communications technology in the following jurisdictions: Patents Pending (PCT/GB2018/051798, PCT/EP2019/079312 & other areas).

The capnography component of this product is covered by one or more of the following US patents: 6,428,483; 6,997,880; 6,437,316; 7,488,229; 7,726,954 and their foreign equivalents. Additional patent applications pending.

PATENT MARKING: This device is covered under one or more of the following U.S.A. patents: 6,157,850, 6,263,222, 6,501,975, 7,469,157and other applicable patents listed at http://www.masimo.com/patents.htm.

1.8.2 Limited warranty

Remote Diagnostic Technologies Limited ('RDT') warrants each new Tempus to be free from defects in workmanship and materials under normal conditions of use and service. For details please refer to the Terms and Conditions of Sale. Consumable items are expressly excluded from this Warranty. RDT's sole obligation under this warranty will be to repair or (at RDT's option) replace products that prove to be defective during the warranty period. The foregoing shall be the sole warranty remedy. Except as set forth herein, RDT makes no warranties, either expressed or implied, including the implied warranties of merchantability and fitness for a particular purpose. The warranty shall be void if the Tempus is in any way modified or used with non-approved consumables, unless specifically authorized in writing by RDT, and RDT shall not be liable in any event for incidental or consequential damage. This warranty is not assignable.

Full terms and conditions of sale are available from RDT and are provided with your order confirmation.

All specifications quoted in this manual are nominal unless detailed otherwise.

1.8.3 Service support and returns

Repairs made under warranty to any Tempus must be made by the manufacturer. If the device requires repair or return for any reason, please contact your local distributor or Remote Diagnostic Technologies in order to first obtain a returns reference (RMA) number, see “RDT contact details”.

RDT reserves the right not to accept returns which have not first been provided with an RMA number. When calling, please be ready to quote the serial number of the device.

The Tempus Pro is designed to be as maintenance free as possible. The only user replaceable and user serviceable parts are those listed in section 11 of this manual.

In the event that the Tempus Pro fails to operate correctly or in a way that is not described in this manual, stop using the device immediately and switch the device off immediately. Contact the manufacturer or distributor at once. Do not attempt any kind of corrective action and do not connect the device to a patient. If the device malfunctions and may have caused or contributed to a serious injury of a patient or user, RDT must be notified immediately by telephone, fax or written correspondence.

2 Warnings and cautions

2.1 Safety and note icons

This manual uses the following icons are used to indicate safety, caution and warning messages
(per ISO 3864-2):

DANGER	**Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.**

WARNING	**Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.**

CAUTION	Indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury.

The note icon is used for important and helpful information:

Note	A point of particular interest or emphasis intended to provide more effective or convenient operation.

2.2 EMC information

The Tempus Pro Patient Monitor has been tested and approved to IEC/EN60601-1-2:2014. This means that the Tempus meets or exceeds the requirements for electrical medical equipment in terms of its levels of emitted electromagnetic (EM) radiation and its susceptibility to electromagnetic radiation from other devices.

In addition, the Tempus Pro has been tested according to the requirements of RTCA DO-160-G section 21 category M.

It should be noted that the Tempus Pro may be affected by high levels of stray EM radiation from other electronic devices (even those which comply with relevant CISPR emission standards) that are being used in close proximity to it.

As required by international medical device standards, the Tempus Pro is intended for use in electromagnetic environments of ±8kV static contact (±15kV air discharge) and magnetic fields of 30 A/m (50/60Hz). The Tempus Pro is proof against radiated RF emissions from 80MHz to 2.7GHz to a level of at least 3V/m. In the event that the Tempus Pro will be used in environments with RF levels exceeding this, please contact RDT for further information.

WARNING	**Operating high frequency electrosurgical equipment in the vicinity of the monitor can produce interference in the monitor and cause incorrect measurements.**
WARNING	**Do not use the monitor with nuclear spin tomography (MRT, NMR, NMT) as the function of the monitor may be disturbed.**
WARNING	**This system is intended for use by healthcare professionals only. This system may cause radio interference or may disrupt the operation of nearby equipment. It may be necessary to take mitigation measures, such as re-orienting or relocating the Tempus or shielding the location.**

CAUTION	Tempus has been tested in a “heavy” wireless environments consisting of different wireless technologies (Bluetooth, WiFi 802.11 b and cellular communications) with multiple transmitters used simultaneously. Users deploying the Tempus in environments where other wireless technologies are being used should evaluate the potential risk of interference and aggregate spectrum usage.

2.3 Indications for use

The Tempus Pro is a portable vital signs monitor intended to be used by clinicians and medically qualified personnel, for the attended or unattended monitoring of single or multiple vital signs in clinical and pre-hospital care applications. The device is indicated for: 3-, 4- and 5-lead ECG monitoring; 12-lead ECG recording with interpretation; real-time arrhythmia detection/alarming; QT measurement/alarming and ST measurements/alarming; impedance pneumography; non-invasive blood pressure (NIBP); end-tidal CO2 (ETCO2) and respiration rate; pulse oximetry (SpO2); contact temperature; and invasive pressure and extended pulse oximetry capability including; carboxyhemoglobin (SpCO), methemoglobin (SpMet), total hemoglobin (SpHb) and total oxygen content (SpOC) measurements.

The monitor is intended to be used as a stand-alone monitor or as a telemedicine system (transmitting patient data to other medical professionals located elsewhere).

The device is indicated for adults, pediatrics and neonates.

The monitor can be used to display images from Interson 3.5 MHz General Purpose (GP) and 7.5 MHz Small Parts/Vascular (SR) USB ultrasound probes or a Karl Storz C-MAC S USB video laryngoscope. The monitor can also be used to display the following readings from a Masimo ISA OR+ gas module*: end-tidal and fractionally inspired CO2, O2, N2O, Halothane, Isoflurane, Enflurane, Sevoflurane, and Desflurane. These optional accessories are to be used in accordance with their indications for use.

* Not cleared for use with the Tempus Pro in Canada, the USA or Singapore.

2.3.1 Contraindications

The Tempus Pro does not replace a physician’s care. The device is not an apnoea monitor.

The Tempus Pro is not intended to be used in strong magnetic or electro-magnetic fields which are generated for medical purposes e.g. MRI.

2.4 General warnings, cautions and notes

DANGER	**ELECTRICAL SHOCK HAZARD when covers are removed. Do not remove covers. Refer servicing to qualified personnel authorized by RDT.**
DANGER	**Explosion Hazard: DO NOT use the Tempus Pro in the presence of flammable gasses such as fuels. Use of the Tempus Pro in such environment may present an explosion hazard.**
WARNING	**The FilterLine may ignite in the presence of O2 when directly exposed to laser, ESU devices, or high heat. When performing head and neck procedures involving laser, electrosurgical devices or high heat, use with caution to prevent flammability of the FilterLine or surrounding surgical drapes.**
WARNING	**The use of the or the or the symbols indicates that the user must read the user manual before using the product.**
WARNING	**When used with the mains power supply, to avoid the risk of electric shock the power supply must only be connected to a supply mains with a protective earth.**
WARNING	**Do not connect the Tempus Pro to an electrical outlet controlled by a wall switch or dimmer.**
WARNING	**To ensure patient electrical isolation, only connect Tempus Pro to other systems which are compliant with the relevant IEC standard e.g. IEC60950, and which employ suitably electronically isolated circuits. Signal input and output connectors are only for connection to equipment complying with relevant IEC safety standards and must be configured to comply with IEC60601-1-1 or IEC60601-1:2005.**
WARNING	**Attaching different medical systems together can cause increases in leakage currents. If the Tempus is connected to other medical devices, or if the patient has multiple devices attached to them at the same time, the cumulative effect of leakage currents must be considered.**
WARNING	**Do not touch the patient at the same time as touching the body or contacts of any of the medical or communications or power connectors.**
WARNING	**Many environmental variables, including patient physiology and clinical application, can affect the accuracy and performance of the monitor. The clinician must verify all vital-signs information prior to patient intervention.**
WARNING	**Do not autoclave, ethylene oxide sterilize, or immerse in liquid the Tempus or any of its cables or accessories as this may cause sensor damage which may result in inaccurate readings.**
WARNING	**Attention should be paid to the following EMC information prior to installing or using the device.**
WARNING	**The Tempus should not be stacked next to other equipment. If this is necessary verify normal operation if utilizing device adjacent to or stacked with other electrical equipment.**

WARNING ⚠	**Portable and mobile Radio Frequency (RF) communication equipment may interfere with the operation of the device.**
WARNING ⚠	**Computers, cables and accessories not tested to IEC/EN60601-1-2 or equivalent IEC standards may result in increased emissions or decreased immunity of the device.**
WARNING ⚠	**Follow precautions for electrostatic discharge (ESD) and electromagnetic interference (EMI) to and from other equipment.**
WARNING ⚠	**Only use Tempus Pro with the relevant cables and peripherals provided by RDT. Use of any other accessories could result in inaccurate measurements. Use of any other cables or accessories could affect defibrillator protection.**
WARNING ⚠	**Exposure of the wireless communication features of the Tempus Pro or its accessories may be interfered with by other devices which operate at the same frequencies.**
WARNING ⚠	**The sensors of the Tempus Pro are only for contact with intact and undamaged skin.**
WARNING ⚠	**Any device or accessory that has been dropped, damaged or subjected to harsh usage or extreme environmental conditions should be inspected by qualified service personnel prior to use to ensure proper operation.**
WARNING ⚠	**The ECG device is not intended for use in a sterile environment. Do not use for direct cardiac application.**
WARNING ⚠	**Do not attempt to insert the ECG device (including patient cables) into an electrical outlet.**
WARNING ⚠	**The ECG recorder and monitoring functions are for resting ECG and should not be used in stress testing environments.**
WARNING ⚠	**Though false positive errors will intentionally outnumber false negative errors, both will occur, thus the necessity for over reading by a qualified physician of any computer-interpreted ECG. The computer interpretation does not produce a definitive diagnosis.**
WARNING ⚠	**Ensure electrodes are connected only to patient.**
WARNING ⚠	**Conductive parts of electrodes and connectors, including neutral electrode, should not contact other conductive parts including earth.**
WARNING ⚠	**Always keep motion to a minimum. Motion artefact can potentially affect the accuracy of patient readings.**
WARNING ⚠	**Do not connect more than one patient to a monitor. Do not connect more than one monitor to a patient.**

WARNING	**Do not use the Tempus in a Magnetic Resonance Imaging (MRI) suite or a hyperbaric chamber.**
WARNING	**Do not place the monitor or its accessories in locations where they could fall onto the patient. When moving the product take care that any connected leads and sensors are not jerked so that they become detached or damaged.**
WARNING	**The Tempus Pro is protected against defibrillator discharge but rate meters and displays may be temporarily affected during defibrillator discharge but will rapidly recover.**
WARNING	**During defibrillation, keep the defibrillator's paddles or electrodes away from the monitor's ECG wires, electrodes and any other sensors or conductive parts of the monitor.**
WARNING	**Pacemaker signals can differ from one pacemaker to the next. The Association for Advancement of Medical Instrumentation (AAMI) cautions that "in some devices, rate meters may continue to count the pacemaker rate during occurrences of cardiac arrest or some arrhythmias. Do not rely entirely upon rate meter alarms. All pacemaker patients should be kept under close or constant observation."**
WARNING	**Device may not operate effectively on patients who are experiencing convulsions or tremors.**
WARNING	**Misuse or improper handling of the device or its sensors or cables can cause damage which may lead to equipment failure or inaccurate readings.**
WARNING	**Do not apply excessive tension to any cable. Using a damaged patient sensor may cause inaccurate readings, possibly resulting in patient injury or death. Inspect each sensor. If a sensor appears damaged, do not use it. Use another sensor or contact your authorized repair center for help.**
WARNING	**The Tempus and all its cables and accessories should be inspected frequently to check for damage. Any worn or damaged items should be replaced. Failing to thoroughly and regularly inspect and maintain the product could result in hazards to patients or equipment failure.**
WARNING	**Using a damaged patient cable may cause inaccurate readings, possibly resulting in injury or death. Inspect the patient cable. If the patient cable appears damaged, do not use it. Contact your authorized repair center for help.**
WARNING	**Ensure that close attention is paid to the alarm configuration settings of the product. If alarm volumes are turned down or if the product is used in noisy environments, then the alarm may not be audible. If alarms are muted this will be indicated on the screen. Always ensure the screen of the Tempus can be seen in case the audible alarms cannot be heard or are turned off.**
WARNING	**Do not attempt to charge a non-rechargeable battery. Never over charge, crush, heat or incinerate, short-circuit, deform, puncture, dismantle or immerse the batteries in any liquid.**
WARNING	**Only use rechargeable batteries and battery chargers specified by RDT.**

WARNING ⚠	**Ensure patient cabling or tubing is carefully routed on device to reduce the possibility of patient entanglement or strangulation.**
WARNING ⚠	**All numerical, graphical and interpretive data should be evaluated with respect to the patient's clinical and historical picture.**
WARNING ⚠	**Do not attempt to insert any connections from the Tempus Pro (including patient cables) directly into an electrical outlet. (IEC60601-2-34:2011, 2011).**
WARNING ⚠	**Failure of Operation: If the Tempus Pro fails to respond as described in this user guide; DO NOT use it until approved for use by qualified personnel.**
WARNING ⚠	**Reuse, disassembly, cleaning, disinfecting or sterilizing of any single use items (such as the Capnometer cannula) may compromise functionality and system performance leading to a user or patient hazard. Performance is not guaranteed if an item labelled as single patient use is reused.**
WARNING ⚠	**Do not touch the patient at the same time as touching the body or contacts of any of the medical or communications or power connectors.**
WARNING ⚠	**The USB connection must only be connected to non-mains powered peripherals (such as a mouse or keyboard) or to interface accessories provided by RDT (such as the USB-Serial Cable part number 01-1022). Any connections made to the USB port must be to medical or IT peripherals or communications systems which comply with the applicable IEC safety standard (i.e. IEC60601-1 or IEC60950). Any connection arrangements must be made in a manner compliant with IEC60601-1-1 or IEC60601-1:2005.**
WARNING ⚠	**The alarm functions of the Tempus are intended to be used by the attendant user only. If the device is connected to a Response Center this is for the purpose of sharing vital signs data in real time, between two users for the purpose of obtaining additional clinical support. The system is not a distributed alarm system (e.g. nurse monitoring station system) in the terms of IEC60601-1-8. The i2i system at the Response Center is not equipped with alarm silencing or suspending controls.**
WARNING ⚠	**When connected to a PC running the i2i application at a Response Center, streamed data, such as the waveforms displayed on the Tempus Pro, will be transmitted and displayed automatically on that PC's display. Users are advised that streamed data are transmitted using the UDP protocol. The UDP protocol includes error checking but does not retransmit data. Therefore, any data that is dropped, lost or delayed during the streaming process will not be retransmitted. In the event that packets of waveform data are lost, they will appear as gaps in the waveform that is displayed on the i2i interface.**
WARNING ⚠	**Medical data (vital signs data, photos, ECG recordings, patient details, TCCC cards etc.) are transmitted using TCP/IP, this includes error checking and retransmission so missing or dropped packets are therefore retransmitted.**

WARNING	**Do not disassemble the device. The device must only be serviced by trained and authorized biomedical engineers following RDT approved procedures and using parts provided by RDT. Any other changes or processes are unauthorized and should not be performed. No other modifications are allowed. Unauthorized processes, repairs, modifications, reworks or service not expressly approved by RDT could void any guarantees and warrantee and could be hazardous. Do not modify this device without authorization from RDT. If the device is modified, appropriate inspection and testing must be conducted to ensure continued safe use of the device.**
WARNING	**If ECG interpretation is on but Glasgow interpretation is off, Tempus Pro will provide standard ECG interpretations.**

CAUTION	The Tempus may be used on ambulances or on vehicles using radios. As detailed in section 14.5, it has been tested to determine operation in field strengths greater than 3V/m. Users are advised to maintain the best distance possible between radio antennas and the Tempus as best practice.
CAUTION	This device is intended for use by persons trained in professional health care. The operator must be thoroughly familiar with the information in this manual before using the device.
CAUTION	The Tempus Pro may not operate correctly if used or stored outside the relevant temperature or humidity ranges described in the performance specifications of this manual.
CAUTION	Only use only approved accessories supplied by RDT. Conventional 4mm snap-on, pre-gelled ECG electrodes may be used so long as they comply with AAMI EC12. Users are reminded to read and adhere to the instructions for such electrodes including noting single use only status, maximum usage times and patient indications.
CAUTION	Do not clean the Tempus or its accessories except as directed in this guide.
CAUTION	In the event that the device displays an error that is not described within this manual e.g. applications errors, turn the device off and then on again. This should clear the error and allow normal operation to resume. Do not continue to use the device if such an error is displayed. If symptoms persist, please contact RDT.
CAUTION	The Tempus records all data that it measures. In order to ensure that data from different patients is not confused, the device must be switched off between taking readings from different patients or the patient discharge function used.
CAUTION	Should the device become wet, wipe off all moisture and allow sufficient time for drying before operating. Take care to ensure that water or liquids are not spilt over the device or into its ventilation holes in the side corners.
CAUTION	If the accuracy of any measurement is in question, verify the patient's vital sign(s) by an alternative method and then check the monitor for proper functioning.
CAUTION	Follow local government regulations and recycling instructions regarding disposal and recycling of device and device components.

CAUTION ⚠	The Tempus Pro and its accessories use different types of batteries which includes rechargeable and non-rechargeable types. If any battery fails to hold a charge or otherwise becomes inoperable, the battery should be replaced and the old battery should be disposed of properly.
CAUTION ⚠	Dispose of batteries in accordance with applicable regulations which vary from country to country. In most countries, the disposal of used batteries is forbidden in general domestic and commercial waste and the end-users are invited to dispose of them properly, typically through not-for-profit profit organizations, mandated by local governments or organized on a voluntary basis by professionals.
CAUTION ⚠	Pressing buttons or the touch screen with sharp or pointed instruments may cause permanent damage. Only fingers should be used to press these keys.
CAUTION ⚠	Do not reconnect the headset to its docking pin when the main battery is very low or flat (less than 10% charge – as represented by a single flashing LED on the battery charge indicator). Doing this could reduce the battery charge into a “deep discharge” state (where no battery lights come on).
CAUTION ⚠	Use of monitoring during continuous nebulized medication delivery will result in damage to the device which is not covered by the warranty. Disconnect the Capnometer sample line from the device, or switch off the device, during medication delivery.
CAUTION ⚠	Observe proper battery polarity (direction) when replacing batteries. The batteries slide easily into place when correctly oriented and should not be forced.
CAUTION ⚠	The mobile RF communications equipment contained within the device and its accessories can affect other medical devices that are in close proximity to the device.
CAUTION ⚠	Use of the RF communications equipment contained in the device and its accessories may be prohibited in a number of areas. These include: on aircraft in-flight (including during take-off and landing), near defibrillators (that are in use), near other electronic medical devices and in hospitals.
CAUTION ⚠	In addition, the use of the RF communications equipment contained in the device and its accessories may be prohibited in explosive atmospheres e.g. in fuelling areas, near fuel or chemical transfer or storage areas and in areas containing chemicals or particles such as grain, dust or metal powders.

CAUTION

The use of the RF communications equipment contained in the device and its accessories may cause interference with implanted pacemakers and other medically implanted equipment.

A minimum distance of 2.3 m (7.5 ft) must be maintained between the device and its accessories (containing RF communications equipment) and other medical equipment (including implantable medical devices such as defibrillators and pacemakers). If such medical equipment has an electromagnetic interference immunity level of less than 3V/m (or 10V/m for implantable devices), this distance should be increased in line with the requirements of IEC60601-1-2:2014.

If the intended patient has an implantable device (e.g. pacemaker), do not use any of the Tempus Pro's RF communications equipment (e.g. Bluetooth® or WiFi) before using the device to record the patient's physiological data. After the data recording session is completed, move the device at least 2.3 m (7.5 ft) away from the patient, and then use it normally to communicate with the base station. Otherwise, radiofrequency radiation from the device (up to 63mW) may adversely impact the implantable pacemaker in the patient. If the patient's implantable device has an immunity level less than 10 V/m, the separation has to be greater than 2.3 m (7.5 ft).

If you suspect interference is being caused, disconnect from the Response Center by pressing . Interference could include visible interference on equipment displays, audible interference e.g. buzzing, from speakers of other equipment, or equipment unexpectedly changing state e.g. functions starting or stopping.

Example of a PC display without interference:

Example of a PC display with interference:

CAUTION ⚠	When using the device with GAN terminals, in order to avoid the risk of interference from the output beam from the antenna of the terminal with the operation of the device, ALWAYS ensure that the device is situated at least 6m behind the face of the antenna. Since the power of the GAN terminal's beam is high (25W approx.), care should be taken to ensure that the antenna remains fixed and to maintain the device away from the face (and therefore the beam) of the antenna.
CAUTION ⚠	RF energy may affect some electronic systems in motor vehicles, such as car stereo, safety equipment, etc. Check with your vehicle manufacturer's representative to be sure that your product will not affect the electronic system in your vehicle.
CAUTION ⚠	Do not use the Tempus Pro's Bluetooth® or WiFi communications on-board any aircraft where its use is prohibited.
CAUTION ⚠	Take care to minimize the risk that trailing cables will be caught by passers by.
CAUTION ⚠	Prior to deployment of Tempus Pro there are settings that user organizations need to configure. Carefully check all the relevant settings on each Tempus Pro before it is deployed. For more information, see "4.1 Before deployment".
CAUTION ⚠	If you use mounting systems that are not supplied or approved by RDT, we cannot guarantee the performance of the Tempus Pro.
CAUTION ⚠	Do not attempt to calibrate the Tempus Pro or its accessories. If there is any indication that calibration is necessary, for example a warning message, contact RDT support.
CAUTION ⚠	Tempus Pro must only be used with the devices listed in this manual. Do not use it with any other devices or general purpose equipment.
CAUTION ⚠	The Tempus Pro software must always be updated when RDT issues a security update.

Note	The Tempus has been tested and found to comply with IEC/EN 60601-1-2.
Note	If all the battery lights remain off when the battery button is pressed, the battery may be in a "deep discharge" state. The battery is not damaged when in this state but will require an extended period on an external charger (additional 24 hours) in order to restore normal operation.
Note	Important! The Tempus Pro is intended for use in the electromagnetic environment(s) specified in this manual. Users of this equipment should ensure that it is used in such environment(s).
Note	After the life cycle of the Tempus Pro and its accessories have been met, disposal should be accomplished following national and/or local requirements.
Note	Operation of the device may be adversely affected in the presence of conducted electrical transients or strong electromagnetic or radio frequency sources such as electrosurgery and electrocautery equipment, HF radio transmission antenna, x-ray machines and high intensity infrared radiation.
Note	All user and patient accessible materials are non-toxic.

Note	Hazards arising from software errors have been minimized. Hazard analysis was performed to meet the requirements of EN ISO 14971 and IEC 62304.
Note	Each external connection and part of the device is electrically isolated.
Note	Performance and safety test data are available on request from RDT, see "RDT contact details".
Note	GSM usage (if enabled) is restricted by the network availability, roaming agreements and local provision of circuit mode connections.
Note	IP sealing is not guaranteed if the device is subject to rough handling, impact, improper use or rapid decompression.
Note	Device should be returned for service if it is subject to rough handling and IP sealing is needed to be relied upon.
Note	The device specifications are subject to change without notice.
Note	It is recommended that the device is connected to the Response Center every month for a test patch.
Note	The iAssist help processes on your Tempus Pro may differ from the example iAssist help process used in this manual; however, the process always follows the same key elements.
Note	Always ensure that you read the complete iAssist help process in order and do exactly what it requires.
Note	For optimum performance of the wireless communications, please make sure that there is no metal surrounding the Tempus Pro.
Note	Over bending the folding foot or RapidPak clip could cause them to be damaged. Do not over-bend these items.
Note	Take care when repacking cables to ensure they cannot be snagged or damaged in the RapidPak clip and the folding foot.
Note	The Tempus Pro should be repacked following the relevant instructions. Lost or damaged cables and accessories should be replaced with spares ordered from RDT.

3 Introduction to the Tempus Pro

The Tempus Pro is a multi-parameter vital signs monitor designed for use in pre-hospital and remote locations. It provides the user with conventional patient monitoring parameters for use in measuring and monitoring patient's vital signs.

The device also provides the user with the ability to capture patient incident data in electronic format. This data can be passed from one Tempus to another to enable the patient's record to remain with them as they move from one caregiver to another.

In addition, the device also gives the user the ability to transmit all data in real-time to a Response Center using a variety of integrated communications interfaces. If data is being transmitted, the device also allows the user to transmit still or moving pictures via an integrated digital camera and also to speak to the Response Center using a wired or wireless headset.

The Tempus Pro is intended to be used by trained clinical staff.

The Tempus Pro can provide the following information about the patient from its sensors:

- ECG Monitoring;
- Real-time arrhythmia detection and alarming;
- Real-time ST and QT measurement and alarming;
- 12 Lead Diagnostic ECG Recording;
- 12 lead ECG recording interpretation;
- Impedance pneumography (respiration rate measured through ECG leads);
- Heart rate (from ECG) or pulse rate (from pulse oximetry);
- Non-invasive blood pressure;
- End tidal CO2 (ETCO2);
- Respiration rate;
- Pulse Oximetry including Oxygen saturation (SpO2), Total Hemoglobin (SpHb), Methemoglobin saturation (SpMet), Carboxyhemoglobin (SpCO), Oxygen Content (SpOC), Pleth Variability Index (PVI);
- Temperature;
- Invasive pressure.

Tempus Pro is also able to display ultrasound and video laryngoscopy images on its large color display utilizing third party ultrasound probes and video laryngoscopy accessories. For more information, please refer to *Tempus Pro Ultrasound Supplement Guide* (Part number 42-2003) and *Tempus Pro Laryngoscope Supplement Manual* (Part number 42-2004).

To display anesthetic agent gas vital signs on the Tempus Pro, you can use the Masimo ISA OR+ Anesthetic Gas module (available from RDT). For more information, see the *Tempus Pro AA Gas Supplement Guide*.

3.1 Overview of the Tempus Pro

The Tempus Pro consists of a PC-ABS plastic enclosure which is over-moulded with TPU material to make it resistant to shock. The enclosure also includes the RapidPak™ clip which provides storage for the SpO_2 sensor, the adult NIBP cuff and hose and the ECG cable.

The unit has a clear window on the top which contains the alarm bar. This lights yellow or red for clinical alarms. The alarm bar can be configured by the user to light green in all non-alarm conditions. The alarm bar has a small aperture at the rear of the case which allows the alarm status to be seen from the rear. In addition, light from the alarm bar is distributed by the translucent handle which sits across the top of the device.

3.2 Key features of the unit

3.2.1 Tempus Pro front

On the front of the unit is a large, color display which is fitted with a touchscreen. The touchscreen may be used with a gloved hand or plastic stylus. Around the touchscreen are two panels of membrane buttons (left/below and right).

Also on the front are grilles covering the alarm speaker. Care should be taken not to probe through the grilles with any sharp or pointed object.

The Tempus Pro

3.2.2 Tempus Pro base

The base of the Tempus Pro houses the battery. The battery can be removed by pushing the latches at each end of the battery together. This will release the battery allowing it to be pulled out.

The battery includes a charge level button and four indicator lights.

3.2.3 Tempus Pro rear

Non-printer models

If the Tempus Pro is not fitted with an internal printer (part number 00-1004-R or 00-1007-R):

- The rear houses the RapidPak clip (discussed above).
- Also on the rear is the aperture for the camera and backlight.
- The clip carries a general product label for regulatory purposes and also two labels which help guide the user to repack the SpO_2 sensor and the ECG cable.

If the Tempus Pro is a non-printer model with Bluetooth headset (part number 00-1007-R):

- The Bluetooth headset is docked onto a connector which enables the Tempus Pro to top its charge up automatically on a regular basis, thus ensuring the headset is always ready to use.
- The Bluetooth headset is a Sennheiser Presence or Sennheiser VMX200.

The Rear of the Tempus Pro (non-printer model)

Printer models

CAUTION	**Printer models with 4G dongle only:** Ingress protection of the 4G dongle might be compromised if the dongle kit or the dongle sleeve is damaged.
Note	Always store the Tempus Pro with a paper roll fitted in the printer.

If the Tempus Pro is a printer model (part number 00-1022-R, 00-1024-R or 00-1026-R):

- The rear houses the internal thermal printer.
- Also on the rear are the camera, backlight and moveable foot.
- The RapidPak clip is absent.
- The ECG cables, SpO2 sensor and NIBP cuff are stored in the bag.

The Rear of the Tempus Pro (printer model)

3.2.4 Tempus Pro left side

The left side of the device contains seven connectors for:

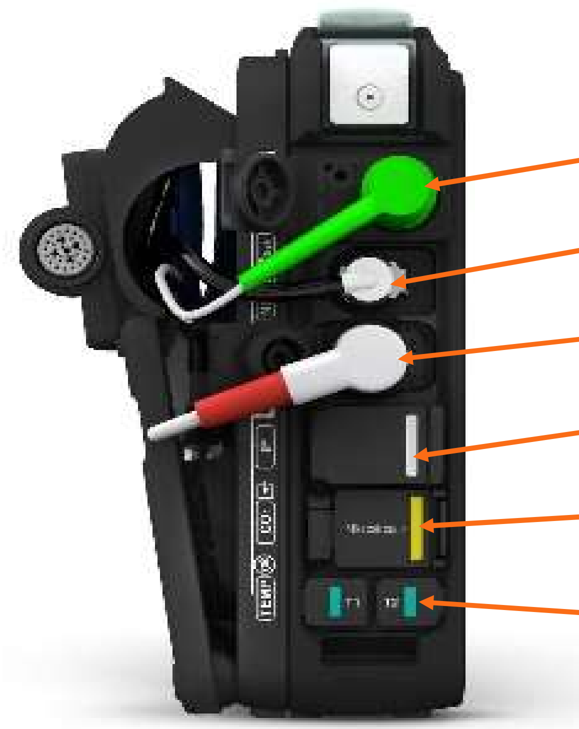
- ECG monitoring and Diagnostic ECG - green;
- NIBP - white latching connector;
- Pulse oximeter (SpO2 and pulse rate) – light grey
- Invasive pressure – white;
- Capnometer (ETCO2 and respiration rate) - yellow;
- Contact temperature (2 sockets marked T1 & T2) - turquoise.

WARNING	**Do not touch the patient at the same time as touching the body or contacts of any of the medical or communications or power connectors.**

CAUTION ⚠	Consideration needs to be made to the potential for leakage currents to be created if the Tempus or its connectors are made wet or dusty. In the event that any connectors (but particularly patient leads i.e. ECG, Invasive Pressure, SpO_2, contact temperature) or their respective sockets become wet or contaminated with sand or dust, they should be cleaned and dried before use – see "11.2.4 Cleaning the NIBP cuffs and hose". Connectors and their immediate surrounding areas should be clean and dry at all times.

Normally the ECG, NIBP and SpO_2 connectors will have their sensor cables attached at all times.

The Tempus' connectors are rated IP66 in their own right and are therefore water proof and sand proof against ingress of water or particulate into the device's enclosure without the covers. The covers provide additional protection to prevent nuisance ingress of sand into the connectors in the event that a cable or sensor is not in place.



Left side of the Tempus – medical connectors

3.2.5 Tempus Pro right side

The right side of the Tempus houses the data communication and power connections. These comprise:

- Two USB 1.0 & 2.0 sockets – these are reserved ONLY for non-mains powered radios (or other communications devices), non-mains powered USB peripherals (such as keyboards) or non-mains powered medical accessories. Any accessory or peripheral attached to the Tempus MUST be approved for use with the Tempus Pro by RDT.
- Mains Power – only use the Tempus Pro approved mains power supply part number 01-2049 provided by RDT.
- Audio – this is only for use with the Wired Headset (part number 01-1019) supplied by RDT or for use with the Tactical Headset Adaptor Cable (part number 01-2041) supplied by RDT (this cable adapts to Peltor® or similar military-style headsets).
- Tactical Switch – this switch allows the user to switch the Tempus' visual and audible outputs to a minimal setting and is described in section 4.3.2.

WARNING	**The USB connection must only be connected to non-mains powered peripherals approved by RDT and documented in this manual, or to interface accessories provided by RDT. Any connections made to the USB port must be to medical or IT peripherals or communications systems which comply with the applicable IEC safety standard (i.e. IEC60601-1 or IEC60950). Any connection arrangements must be made in a manner compliant with IEC60601-1. Users must ensure that they prevent the patient from coming into contact with the peripheral and also must ensure the user does not touch the peripheral while touching the patient. Alternatively, a suitable isolation device can be used. RDT recommends the use of appropriate opto-isolation devices such as the** Ulinx™ Model UH401 or the Baaske USB 1.1 isolator**. Further details can be obtained from RDT upon request.**
WARNING	**Always use a Baaske MI 1005 isolation device when connecting from the Ethernet socket to any mains-powered communications device (e.g. router, access point, hub, communications terminal etc.).**
WARNING	**The Ethernet connection must only be connected to battery powered (non-mains-powered) communications devices such as laptop computers or satellite communications terminals such as BGAN or VSAT terminals. Any connections made to the Ethernet port must be to medical or IT peripherals or communications systems which comply with the applicable IEC safety standard (i.e. IEC60601-1 or IEC60950). Any connection arrangements must be made in a manner compliant with IEC60601-1:2005 or IEC60601-1-1:2001. Users must ensure that they maintain the communications device 1.5 m away from the patient and also must ensure the user does not touch the communications device while touching the patient. If such devices require mains power then the device must be powered using a power supply compliant with the electrical isolation requirements IEC60601-1.**
WARNING	**The audio connector must only be connected to unpowered microphone headsets or battery powered microphone headsets (such as Peltor® or Bose®). Further details can be obtained from RDT upon request.**
WARNING	**Do not touch the patient at the same time as touching the body or contacts of any of the medical or communications or power connectors.**

CAUTION	Do not use USB hubs (powered or unpowered) with the Tempus. This could cause the USB sockets to cease operating.
CAUTION	Consideration needs to be made to the potential for leakage currents to be created if the Tempus or its connectors are made wet or dusty. In the event that any or their respective sockets become wet or contaminated with sand or dust, they should be cleaned and dried before use – see "10.2.3 Cleaning connectors". Connectors and their immediate surrounding areas should be clean and dry at all times. Ensure the communications connectors are securely covered with their dust covers (if fitted) at all times.
Note	All Tempus Pro functions will work when the device is operating on battery power. Interruption of the mains power supply will not affect any functions, even if the mains power interruption exceeds 30 seconds.

Right Side of the Tempus – Communications Connectors

3.3 Theory of operation

All of the measurements made by the Tempus Pro are displayed on the screen. Attaching a sensor to a patient will initiate measurements by that particular parameter. Monitoring will continue until the sensor is removed from the patient or the unit (and the consequent alarm is silenced). In the case of non-invasive blood pressure, measurements start once the user presses the 'start' button and will continue on a regular basis when in automatic mode until either the cycle is stopped by the user or the unit is unable to take a reading e.g. cuff is removed.

3.3.1 Pulse rate and oxygen saturation (SpO_2)

Pulse oximetry measures functional oxygen saturation.

Pulse oximetry is based on the following:

- The difference in the absorption of red and infrared light (spectrophotometry) by oxyhemoglobin and deoxyhemoglobin
- Changes in the volume of arterial blood in tissue during the pulse cycle (plethysmography), and hence, light absorption by that blood.

A pulse oximeter determines Spot Oxygen Saturation (SpO_2) by passing red and infrared light into an arteriolar bed and measures changes in light absorption during the pulsatile cycle. Red and infrared low power light emitting diodes (LEDs) in the oximetry sensor serve as light sources; a photodiode serves as the photo detector.

Since oxyhemoglobin and deoxyhemoglobin differ in light absorption, the amount of red and infrared light absorbed by blood is related to hemoglobin oxygen saturation. To identify the oxygen saturation of *arterial* hemoglobin, the monitor uses the pulsatile nature of arterial flow.

During systole, a new pulse of arterial blood enters the vascular bed and blood volume and light absorption increase. During diastole, blood volume and light absorption reach their lowest point.

The monitor bases its SpO_2 measurements on the difference between maximum and minimum absorption (measurements at systole and diastole). The focus of light absorption by pulsatile arterial blood eliminates the effects of nonpulsatile absorbers such as tissue, bone, and venous blood.

Pulse rate and oxygen saturation are detected by a soft reusable finger probe packed on the rear of the device (disposable sensors may also be used). In particular it is also important that the sensor is not used on the same arm as the blood pressure cuff, because false readings may occur when the cuff is inflated. Readings will not be obtainable or may be inaccurate from patients with some colors of nail varnish or polish.

Masimo rainbow SET measurements use a multi-wavelength sensor to distinguish between oxygenated blood, deoxygenated blood, blood with carbon monoxide content and blood with oxidized hemoglobin. The sensors have various light emitting diodes (LEDs) that pass light through the site to a photodiode (photodetector). The maximum radiant power of the strongest light is rated at 22 mW. The photodetector receives the light and converts it into an electronic signal. Masimo rainbow SET signal extraction technology then calculates the measurement values.

3.3.2 Non-invasive blood pressure

The Tempus Pro uses oscillometric technology to measure the patient's blood pressure non-invasively. A pump within Tempus Pro inflates the reusable blood pressure cuff around the patient's arm. Circulating blood within the arm causes slight changes (oscillations) in the cuff pressure, which can be detected and measured. As the inflation pressure changes, the systolic, diastolic and mean arterial pressure can be measured.

This method of blood pressure measurement provides accurate readings provided that the correct size of cuff is used, it is attached to the patient correctly and the specified operating precautions are observed.

3.3.3 Electrocardiograph (ECG)

The ECG is intended to monitor patients of all ages in hospital and pre-hospital applications (including transport) for general cardiac monitoring. Electrical currents influenced by the cardiac impulse flow through the body tissue around the heart. The Tempus Pro can be used with:

- a 3-lead cable to monitor ECG leads I-III;
- a 4-lead cable to monitor ECG leads I-III, AvR, AvL and AvF;
- a 5-lead cable to monitor ECG leads I-III, AvR, AvL, AvF and a user-placeable V lead.

Twelve lead diagnostic recording functionality is available as an option. To support this, use one of the standard or modular 12-lead cables (including precordial chest electrodes) available from RDT.

ECG cables can be used with conventional disposable pre-gelled electrodes (not supplied) for ECG monitoring and alarming.

3.3.4 Impedance pneumography

Impedance pneumography operates by continuously measuring changes in the impedance of the patient’s body. This is done using the two or more ECG electrodes and thus allows the device to measure the patient’s impedance through the chest. Impedance is measured using a low-current, high frequency alternating current signal. As the patient breathes, the chest expands and contracts giving a rising and falling impedance reading. The Tempus measures the time between successive peaks and troughs of the impedance measurement and translates the time into an interval or rate of respiration.

The impedance method is an indirect measurement of respiration and is also subject to muscle noise and electrode placement and therefore is inferior to the respiration rate measurement provided through capnography.

3.3.5 End tidal CO_2 ($ETCO_2$) and respiration rate

Tempus uses Microstream® non–dispersive infrared (NDIR) spectroscopy to continuously measure the amount of CO_2 present at the end of exhalation ($EtCO_2$) and the Respiratory Rate.

Infrared spectroscopy is used to measure the concentration of molecules that absorb infrared light. Since absorption is proportional to the concentration of the absorbing molecule, CO_2 concentration can be determined by comparing its absorption to that of a known standard.

The Microstream® $EtCO_2$ consumables deliver a sample of the inhaled and exhaled gases from the ET tube adaptor or directly from the patient (via an oral/nasal cannula) into the monitor for CO_2 measurement. Moisture and patient secretions are extracted from the sample, while maintaining the shape of the CO_2 waveform.

The 50 ml/min. sampling flow rate reduces liquid and secretion accumulation, decreasing the risk of obstruction in the sample pathway in humid ICU environments.

Once inside the Microstream® CO_2 sensor, the gas sample goes through a micro-sample cell (15 microliters). This extremely small volume is quickly flushed, allowing for fast rise time and accurate CO_2 readings, even at high respiration rates.

The Micro Beam IR source illuminates the micro-sample cell and the reference cell. This proprietary IR light source generates only the specific wavelengths characteristic of the CO_2 absorption spectrum. Therefore, no compensations are required when different concentrations of N_2O, O_2, anesthetic agents and water vapour are present in the inhaled and exhaled breath. The IR that passes through the micro-sample cell and the IR that passes through the reference cell are measured by the IR detectors. The microprocessor in the monitor calculates the CO_2 concentration by comparing the signals from both detectors.

3.3.6 Temperature

Two channels of contact temperature are measured through two sockets on the Tempus. These sockets are compatible with YSI type 400 probes only. A range of third-party YSI 400 series compatible transducers may be used with the Tempus. These transducers may be re-usable or disposable.

The Tempus measures temperature by measuring the changing resistance in the tip of the transducer. When a transducer is correctly attached to the Tempus and to the patient e.g. in the rectum or axilla, the Tempus obtains a constant (“direct”) reading of resistance that equates to the temperature at that site. There is no offset or reference body site. The device contains a self-checking system which checks the thermometer against a known (internal) reference point (every second). If this cross-check fails at any time the thermometer will be disabled and an error posted on the display.

The thermometer’s reading can be presented in ºC or ºF.

3.3.7 Invasive pressure

Two channels of invasive pressure are measured through a socket on the Tempus. An interface cable is required to be attached to this socket, the interface cable splits into two ends, each of which can be connected to a third-party pressure transducer (with a sensitivity of 5 µV/V/mmHg). A range of third-party transducers (both reusable and disposable) are available. A list of compatible sensors is provided in section 14.8. A different interface cable is required to adapt the Tempus’ socket to the specific connectors of each type of transducer.

The Tempus measures pressure by measuring the changing resistance in the transducer. When a transducer is correctly attached to the Tempus and to the patient’s blood stream e.g. in the pulmonary artery, the Tempus obtains a constant reading of resistance that equates to the systolic and diastolic pressure. From this mean arterial pressure and heart rate are also calculated.

The option to use two additional channels invasive pressure measurement is provided by an external USB IP Module. The 2 Channel IP Module can be located adjacent to the Tempus Pro or clipped onto a new feature on the rear clip. There is no setup required to connect the USB device to the Tempus Pro, other than plug the USB cable in. The Tempus Pro will automatically recognize that the module is attached.

3.3.8 Controls and indicators

The controls and indicators are divided into three types, items on a touchscreen, membrane buttons and LED indicators. The Tempus Pro can be operated with or without gloves.

Note	Touchscreen and membrane buttons should be pressed once. As with all touchscreens if your finger wavers or quivers, the Tempus Pro may register it as more than one press.
Note	To display ultrasound and video laryngoscopy images on the Tempus Pro's large color display, you can use third-party ultrasound probes and video laryngoscopy accessories. For more information, see the *Tempus Pro Ultrasound Supplement Guide* and *Tempus Pro Laryngoscope Supplement Guide*.

3.3.9 Summary Record of Care / Tactical Combat Casualty Care

The Tempus contains an electronic patient record card known as a Summary Record of Care (SRoC). This is completed semi-automatically through the collection of vital signs and the use of the Event button. SRoC allows the user to record patient trauma details and to review historic events and trends.

SRoC is automatically augmented with patient encounter data when:

- An arrhythmia is detected. The Tempus Pro automatically logs an event and captures a snapshot of the waveform.
- A waveform snapshot is manually captured. To capture a waveform at any time, press and release the Camera & Waveform Snapshot button.
- A 12 Lead ECG is recorded.
- A camera or laryngoscope image is recorded. For laryngoscope details refer to *Tempus Pro Laryngoscope Supplement Guide*.
- An ultrasound image is recorded. Images can be taken during general and FAST exams. Refer to *Tempus Pro Ultrasound Supplement Guide*.

Tempus Pro can also be configured to record a Tactical Combat Casualty Care (TCCC) card (this feature only applies to Tempus units configured with the TCCC patient report type); for details see "9.3.2 Updating the TCCC card".

3.3.10 Previous activity button

The Previous Activity button allows the user to return to the last non-monitoring function that they were using i.e.:

- TCCC card
- Summary Record of Care
- Enter patient details
- Discharge patient

3.3.11 Trends

All vital signs are captured on the device. These can be easily viewed as a graphical or tabular trend. Also included in the trends are all alarm states and Events.

3.3.12 Data handover

The Events, Trauma Record and Trend data (as well as photos and 12 Lead ECGs) are automatically collated together to form a Summary Record of Care patient report. This can be exported from the Tempus via USB and handed to the next caregiver within the care path. Should the next caregiver have a Tempus Pro, the data may then be viewed and augmented on that unit.

3.3.13 Email

An encrypted PDF formatted Summary Record of Care can be shared using email. Encryption and data transfer are performed using FIPS- 140-2 compliant technology.

3.3.14 ePCR export

Summary Record of Care data can be exported to an electronic Patient Care Reporting system (ePCR). Encryption and data transfer are performed using FIPS- 140-2 compliant technology. This feature is optional.

3.3.15 Communications

GPS

The Tempus contains a Global Positioning Satellite (GPS) receiver. This can be used to detect signals from multiple geostationary satellites to obtain a location (expressed in terms of longitude and latitude). The GPS receiver needs a clear view of the sky to work and so must be used outside.

Digital camera & waveform snapshot

A miniature digital camera is mounted in the unit. Still photos of the patient can be taken which are then automatically added to the patient record.

Photos can be transmitted to a Response Center at the time (if the device is connected to a Response Center) or transmitted later. In addition, if the Tempus is connected then live moving video can be transmitted (one way from the Tempus to the Response Center). This enables the user to show to colleagues the patient and what is happening to them.

The Camera & Waveform Snapshot button is a two function key. Pressing the button will record the previous ten seconds and the following ten seconds of the waveforms for review. Pressing and holding the button will take you to the Camera & Video Options Menu.

Long range voice and data communications

In addition to being able to operate as a vital signs monitor, the Tempus Pro can optionally transmit all of its measurements via a communications link to a Response Center. The device allows the Response Center to see exactly what is on the Tempus' display in real time. Therefore, the Response Center user can see what the patient's vital signs are, if they are in an alarm condition etc. In addition, the Response Center user can control the device remotely (except for the alarm controls on the membrane keypad which are not provided to the Response Center user).

The Tempus also allows the user to talk to the Response Center user via a wired or wireless headset.

The user at the Response Center is able to receive still photos from the device and is also able to annotate these pictures (with words, symbols and markings) and send back to the Tempus device to better illustrate the verbal instructions being given to the operator at the remote location. The Response Center user can also receive moving video from the Tempus.

The Tempus Pro can connect over either wired or wireless (WiFi) Ethernet networks.

All data, voice and video that is transmitted can be secured using FIPS- 140-2 compliant encryption technology.

If the device is connected to a Response Center, then all the readings are transmitted via a communications link to a computer at a Response Center which enables the user there to see exactly the same information that is being displayed on the device in real time. The only exceptions to this are when a 12 Lead ECG is being monitored or recorded and when a still photo is being taken – in both cases the data (ECG or picture) is first recorded and then transmitted to the Response Center.

ECG data and digital pictures take the following amount of time to send to the Response Center:

- 12 lead ECG recordings – 1 minute
- Digital picture – up to 30 seconds

Note	These times are for guidance only and are based on the use of a 64kbaud or greater connection. Real time data and waveforms are transmitted with up to a 5 second delay.
Note	RDT recommends that users perform a test connection to the Response Center every month in order to verify that their communications remain open for the Tempus to use.

Bluetooth® communications

The Tempus is fitted with a Bluetooth® radio. This enables it to communicate with other Bluetooth® enabled devices. The Tempus can be provided with an optional Bluetooth headset for providing voice communications.

3.3.16 On-screen user instructions

The Tempus provides the user with a conventional vital signs monitoring user interface. It can also be set to provide additional on-screen instructions in the form of text and graphics. This is called iAssist mode. It is intended for use by a trained healthcare professional, who may benefit from additional instructions on how to attach the monitor to a patient. It is assumed that a more skilled or experienced user will turn the iAssist feature on before handing the device to the more junior colleague.

iAssist is intended to make the device easier to use, provide on-screen instructions and provide context sensitive errors in order to help the user apply the device. It does not change the behaviour of the patient or technical alarms except to provide more on-screen and graphical instructions in some cases.

Using iAssist means that the format of the results (Home Screen) changes from a 4 channel (2 ECG, Plethysmogram and Capnogram) layout to a 3 channel (1 ECG, Plethysmogram and Capnogram) layout.

iAssist instructions are available for ECG monitoring, Non-Invasive Blood Pressure, Pulse Oximetry, Contact Temperature and Capnography. iAssist instructions are not available for invasive pressure as this process requires specific training and is surgically invasive. Instructions are not included for use of a 10-wire ECG cable as this requires specific training.

4Setting up

4.1 Before deployment

Prior to deployment of Tempus Pro there are settings that user organizations need to configure. Carefully check all the relevant settings on each Tempus Pro before it is deployed. When deploying multiple units, you may set up a single unit and export the configuration from one unit to the others (cloning the settings).

Three types of configuration are exporting and imported:

- events (summary record of care);
- new patient defaults;
- settings.

Summary record of care events include:

- summary record of care assessment, intervention, drug and fluid events;
- summary record of care injury and burns body map tags;
- summary record of care general screen events selection.

New patient defaults include:

- patient alarm limits for all parameters;
- default patient mode (adult, pediatric or neonate);
- alarm silence and suspend times;
- default NIBP settings (mode, auto-time and inflation pressures);
- default ECG settings (HR/PR source, ST/QT etc.).;
- printer default settings.

Settings include:

- communications settings (e.g. IP address);
- communications menu;
- email settings;
- ePCR settings;
- patient record type, language, passwords.

Note	Configuration needs to be carried out by a biomedical engineer or suitably qualified person following the instructions in the *Tempus Pro Configuration Utility User Guide*.
Note	The unit name is not cloned from one device to another.
Note	Licensed options and enabled features are not cloned from one device to another.

4.2 Unpacking the Tempus Pro

The Tempus Pro is supplied from the factory in protective outer packaging. No special precautions are required when unpacking the Tempus Pro. RDT recommends that you keep the packaging.

RDT recommends that the equipment is inspected and tested on receipt to confirm that the unit has not been damaged and that all expected items and accessories have been received and are in working order. New batteries should be charged up for at least 4 hours on receipt.

Users should ensure that all expected items are received with the Tempus when first opening the delivery case.

The Tempus Pro

As detailed in section "1.6 Features list", the Tempus Pro is supplied in a number of configurations. Confirm that all items ordered have been received.

4.3 Starting and stopping the Tempus Pro and selecting a patient

4.3.1 Switching on

WARNING	**Ensure the latches on both sides of the battery are fully engaged prior to using the Tempus - an incorrectly fitted battery could result in the Tempus losing power during use.**
CAUTION	Do not press any of the controls until the Tempus has started.

To switch on the Tempus Pro, press and hold the [On/Off] button on the front panel for 1 second. The LED on the [On/Off] button will flash green. The device is ready for use when the LED shines green constantly.

When the Tempus starts it will briefly light the alarm bar and sound the patient alarm. This is to demonstrate that these functions are operational and must be checked by the user (with the tactical switch off).

4.3.2 Tactical mode (optional)

WARNING	**To ensure that the Tempus Pro is safe to use when colleagues are using NVGs (night vision goggles), use only the settings in the green shaded NVG areas of the Tactical Menu: LED Indicators set to off, Brightness set to Min, Low or Mid.**
WARNING	**The Tactical Switch is intended for use by military users for scenarios where low emitted light and low emitted sound is required or desired. Users are reminded that while these functions are enabled, the device will not emit visual alarms from the alarm bar, will not emit audible alarms from the speaker and may present a display that may be too dim to see in daylight conditions. Users should therefore ensure that they use these features only when required and recognize that greater levels of patient care and supervision will be required.**
WARNING	**If the maintenance user has configured limited tactical mode, the display will not be dimmed when tactical mode is switched on**

It should be noted that the Tempus may be started in a low emitted light and silent state by using the Tactical switch on the right hand side of the device.

Tempus Pro Tactical Switch

When the switch is set closest to the  symbol the Tempus will start in the following configuration:

- All alarm LEDs off;
- All audible signals off;
- Wireless functions (if enabled by configuration) off (*);
- Display set to mid brightness and standard contrast mode (*).

(*) If the maintenance user has configured **limited tactical mode**, the preconfigured wireless functions and brightness setting will persist in tactical mode.

In this condition, the following menu of display options will be shown:

"NVG/Tactical Mode" menu

Note	To toggle high contrast, press and hold the Display button for two seconds.
Note	Toggling the Tactical Switch will immediately return the display settings to their previous configuration i.e. display brightness, all LEDs active, alarms set to the default level and high contrast off.
Note	The Tactical Switch disables the ac power and charging LEDs; these are only re-enabled using this switch.
Note	The battery charge LEDs are not disabled using the Tactical Switch.

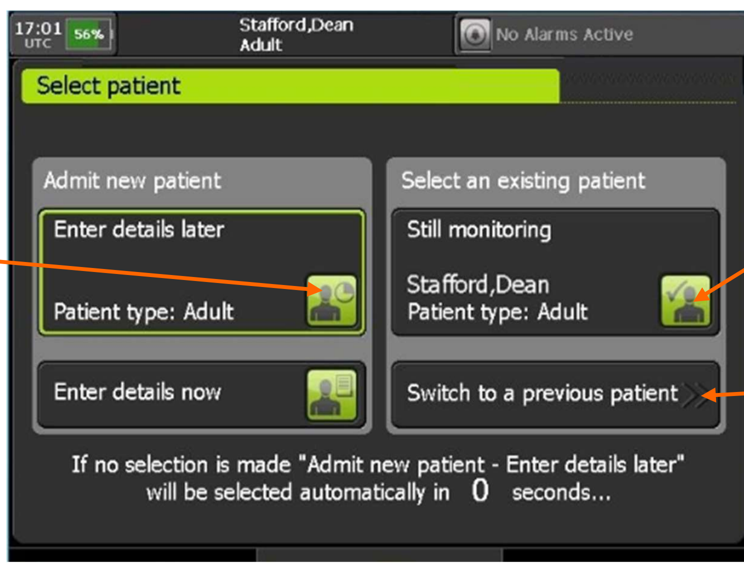
4.3.3 Setting the patient

WARNING	**Ensure that you select the correct patient record. Records can be identified by patient name (last and first), ID number, or incident start time. If you are not sure if the record you wish to select is the correct one, then select new patient. Mixing different patient's records could lead to confusion and misdiagnosis.**

When the Tempus is turned on, it will ask if it is monitoring a new or previous patient. If the Select Previous Patient option is pressed, the Tempus will show all patients monitored in the last 72 hours i.e. any patients whose monitoring started within the last 72 hours.

If any patients have been monitored in the last 72 hours, then the dialog will also include (shown below) an option to select the last patient that was monitored.

Press here to confirm that the Tempus has been put onto a new patient



Press here to confirm that the same patient as before is being monitored

Press here to switch to a previously monitored patient

Patient type is defined as Adult, Child (pediatric) or Neonate. For more information on these groups and how to select them, see "9.3.5 Entering patient details".

4.3.4 Starting and stopping iAssist

The iAssist function gives additional on screen instructions for:

- Using the ECG
- Taking non-invasive blood pressure readings

- Taking blood oxygen readings
- Taking $C0_2$ readings
- Taking temperature readings

To switch it on or off:

1. Press the [Main Menu] button .
2. Slide the iAssist switch to **On** or **Off**. When switched on, iAssist buttons appear at the bottom of the home screen.

4.3.5 Switching off

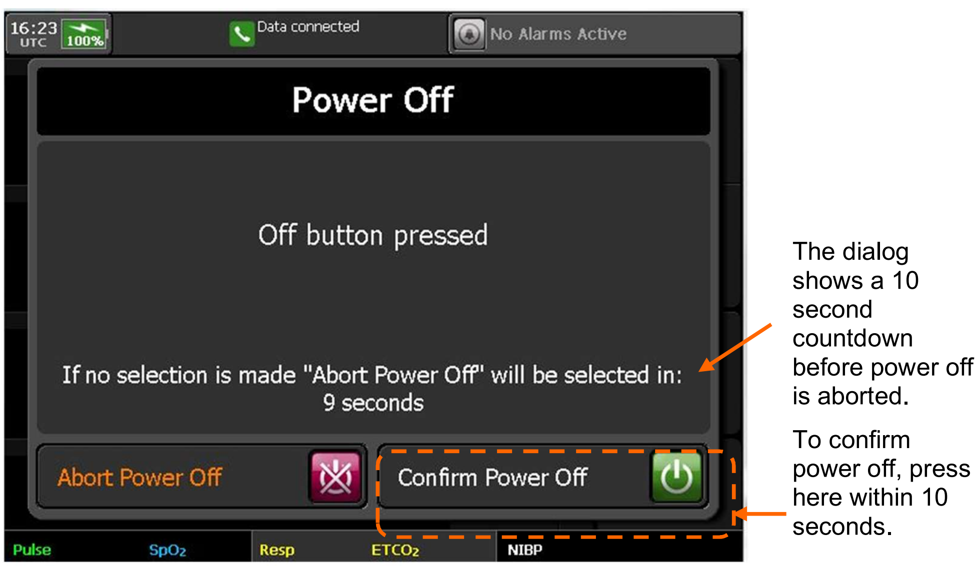
Before switching the Tempus Pro off, you should make sure that it is not in use.

To power off quickly with no countdown or confirmation:

1. Ensure the Tempus Pro is not in use.
2. Press and hold the [On/Off] button  for 2 seconds.

To power off with countdown and confirmation:

1. Ensure the Tempus Pro is not in use.
2. Press and release the [On/Off] button . The LED on the on/off button will start flashing.
3. The Tempus Pro displays a 10 second countdown timer.
   Press the **Confirm Power Off** button.



Power Off Dialog

Note	Before removing the battery, you must switch off the Tempus Pro by pressing the power button. Do not remove the battery when the Tempus is on (unless there is a power supply attached to the device).
Note	If necessary, you can force a hard shut-down by pressing and holding the power button for 10 seconds.

Note	Remember that the battery cannot be removed until the lamp on the front panel has gone out.

5Using the Tempus Pro

The Tempus can be used laying on its back or standing on its foot.

5.1 Controlling the Tempus Pro

Tempus controls

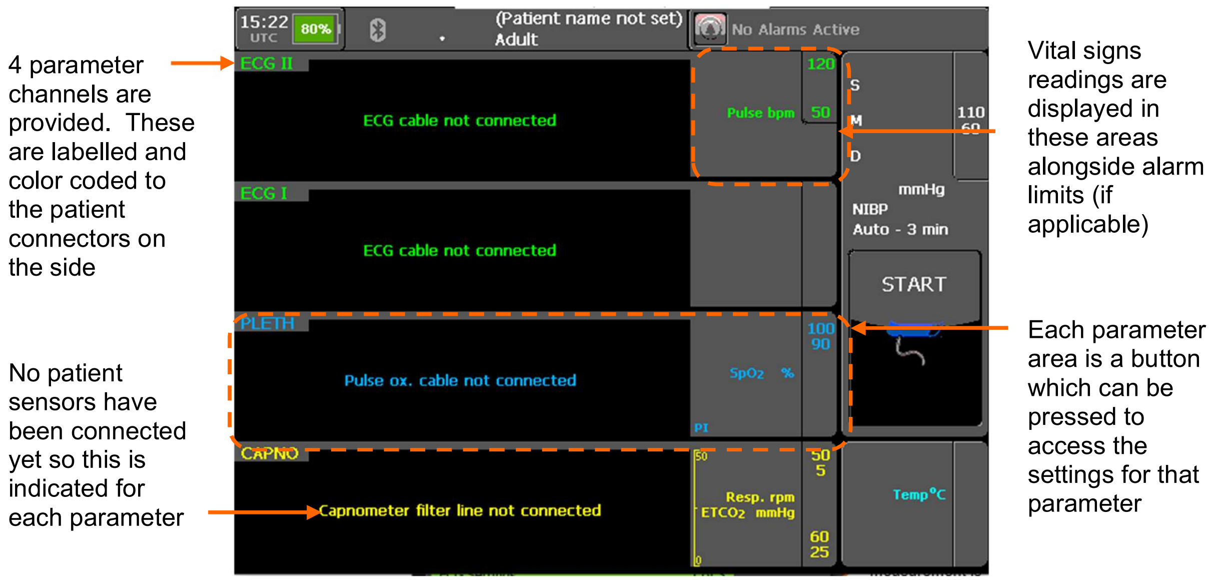
The Tempus can be controlled through two different control interfaces. These are:

- The touchscreen;
- The graphically labelled membrane buttons.

5.1.1 The touchscreen

Note	Take care to press the touchscreen once for any control. Users should press the controls firmly with a finger (gloved or bare skin can be used). Pressing the touchscreen lightly can lead to “double presses” if the user’s finger wavers or quivers. If the touch screen is pressed twice in quick succession and there is a control in the same area where the finger pressed *after* the first control would have disappeared (if the screen changes etc.) then the Tempus will register the second press as activating the second control before the control appears.

When the device is started it will present as follows:



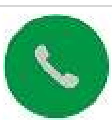
The home screen at start-up

Pressing any of the parameter areas will access the controls and settings menu for that parameter. For example, if the ECG area is touched, the following menu will be shown:

"ECG Settings" menu 1 of 2

5.1.2 Membrane buttons and LED indicators

The Tempus can be controlled through the following membrane buttons:

	The [On/Off] button – pressing and holding down this starts and stops the Tempus – see “4.3.1 Switching on”.
	The [Alarm Suspend] button – pressing this latches all alarms off for 2 minutes (factory default – this is configurable by the operating institution) – see “7.6 Silencing or suspending alarms”.
	The [Alarm Silence] button – pressing this stops any audible alarm signals for 2 minutes (factory default – this is configurable by the operating institution) from monitoring parameters (and clears alarm signals from discrete measurements such as non-invasive blood pressure – see “7.6 Silencing or suspending alarms”.
	The **Home** button - this returns the unit to the results screen. When the home screen is displayed, pressing and holding the **Home** button for 2 seconds will lock the touchscreen.
	The [Data Input/Output] button – pressing this launches a menu which offers options on outputting data to peripherals such as printers or USB memory sticks – see “9.1 Data input and output”. Note that pressing and holding the [Data Input/Output] button for 2 seconds will automatically launch the function that is shown at the top of the “Data Input / Output” menu. This acts as a short cut to make activating this function easier.
	The [Display] button – pressing this launches a menu which offers a range of display options – see “9.2 Display options”. Note that pressing and holding the [Display[button for 2 seconds will automatically switch the display mode into a high contrast mode (black on white) for use in strong daylight conditions. This acts as a short cut to make activating the high contrast display easier.
	The [Patient] button – pressing this allows the user to manage the patient information – see “9.3 Patient information”. Note that pressing and holding the [Patient] Information button for 2 seconds will automatically launch the function that is shown at the top of the “Patient Information” menu. This acts as a short cut to make activating this function easier.
	The [Connect] button – pressing this will bring up instructions on how to connect the device to a Response Center to transmit voice, data and images in real time – see “10 Connecting to an alternate location”.
	The [Disconnect] button – if the device is connected to an alternate location, pressing this will bring up a 10 second countdown after which the voice and data connection will be dropped – see “10 Connecting to an alternate location”.

	The [Camera & Waveform Snapshot] button – pressing and releasing this starts the waveform capture, pressing and holding will start the camera – see “8 Summary Record of Care”. Note that certain events will cause the Tempus to capture a waveform automatically – see “8.1.4 Automatic event capture”. Note for devices with software revisions earlier than v4 this button will only perform the camera function. Please contact RDT for software updates.
	The [Main Menu] button – pressing and releasing this will take you to a full menu to access all features of the product – see “5.2.1 The main menu”.
	The [Event] button – brings up the Summary Record of Care (SRoC) and provides access to trend graphs, trend tables and the TCCC card (if configured).
	The [Previous Activity] button – pressing this will take you back to whatever non-monitoring function you were last performing. For example, this button could be pressed to take you back into the trauma card if you had been editing it and a patient alarm had gone off – see “3.3.10 Previous activity button”.
	The [Battery Gas Gauge button] – pressing this will light the power status LEDs on the battery – see “11.1.1 The battery”.
Also on the membrane are two LEDs, these are indicators and not a button, their function is described below:	
	Mains power LED – lights solid green when the mains charger is attached
	Battery charger LED – flashes green for up to 20 seconds when a battery is first connected and the device is being run from mains power, then lights solid green as long as mains power is attached and the battery is not fully charged – see “11.1.3 Charging the battery”.

5.1.3 Status bar

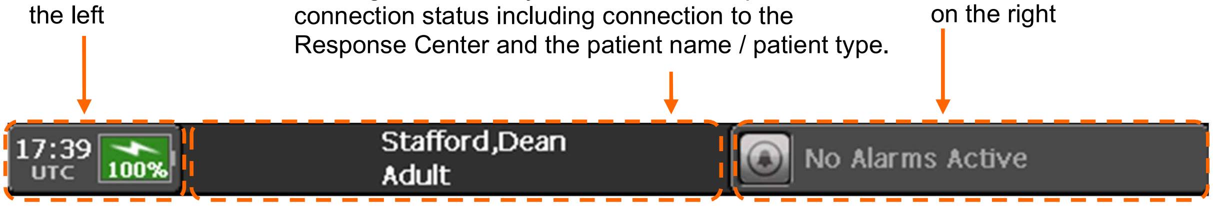
The status bar at the top of the screen shows the time, battery status, communications status and alarms status – see “7 Alarms”.

The current patient name and patient type appears in the center of the status bar until any wired or wireless communications devices are being used then this will be shown in the center of the status bar.

The time and battery status are shown on the left

This area is used to display information related to sending the Summary Record of Care report, connection status including connection to the Response Center and the patient name / patient type.

Alarm status information is shown on the right



The status bar

Battery status indicator

The battery status indicator in the status bar shows the percentage of battery charge remaining:

Icon	Status	Indication/Action
	Green, no flash	Battery charge remaining: 51% - 100%. Tempus Pro battery is not currently charging.
	Green with flash	Battery charge remaining: 51% - 100%. Tempus connected to mains power (charging in progress).
	Orange, no flash	Battery charge remaining: 26-50%. Tempus Pro battery is not currently charging.
	Orange with flash	Battery charge remaining: 26-50%. Tempus connected to mains power (charging in progress)...
	Red, no flash	Battery charge remaining: 0% - 25%. Tempus Pro battery is not currently charging: connect it immediately!
	Red with flash	Battery charge remaining: 0% - 25%. Tempus connected to mains power (charging in progress).
	Unknown battery status	Tempus cannot communicate with battery. Check the following: • Is the battery fitted? • Are the battery clips fully engaged? • Are the battery contacts clean and undamaged? • Is the battery in deep discharge state? For more information, see "11.1.1 The battery".

Note	The battery charge can always be checked by pressing the battery gas gauge button, even when the Tempus is switched off – see "11.1.1 Checking the charge of the battery".

5.1.4 Status bar for communications

The center section of the status bar shows if any communications features are being used. If any Bluetooth® devices are in use, then the Bluetooth® symbol will be shown and along with the number of peripherals that are connected to the device.

Note	It does not identify the specific peripheral connected to the Tempus Pro.

If a WiFi connection is in use, then this will be shown.

If an external Invasive Pressure module is attached, then this will be shown.

The information displayed in the communications area of the status bar is based on the following priorities:

- Email – highest priority
- Response Center communication
- Patient name & type – lowest priority

Email status bar messages

When the Summary Record of Care is being shared using email, the upper part of the status area is used to display the connection progress whilst the lower part of the status area is used to indicate the progress related to the Summary Record of Care report creation and sending.

The connection progress messages (upper part) may be:

- Network Connecting
- <Carrier name> e.g. Vodafone UK (only in GSM mode, if enabled)
- Network Connected (only in Wi-Fi or Ethernet modes)

The SRoC report progress messages (lower part) may be:

- Creating Report
- Sending Report
- Report Sent (displayed for 30 seconds after completion)

In this example, an Ethernet connection has been made and the SRoC report is being created

Status bar when SRoC report is shared via email

Response Center (i2i) communication

If a data connection is in progress this will be shown. The green icon will flash when the connection is in being established and the text will read “Data connecting”. When a link to the data center (gateway) has been made the icon will stop flashing and the text will read “Data connected”. At this time the Tempus will be connected to the data center (gateway) but will not be transmitting any patient data until a Response Center user initiates the connection from their terminal. Once they have done this the text will read “Data Connected”.

Note	To transmit data, you must have a manned Response Center who have the i2i software installed and who have familiarized themselves with its operation. Full installation instructions are provided with the i2i software and support is available from RDT, see “RDT contact details”.

If a voice connection has been initiated, then an additional green icon will appear with the text “Voice” next to it. The icon will flash while the voice connection is being established and will stop flashing once the call has been picked up by the Response Center. If video is being transmitted this will be shown.

This section shows if Bluetooth devices are attached (and how many), if video is being transmitted and if WiFi is being used

This section shows if data is “connecting” (dialling), “connected” (connected to a data center/gateway) or “Connected” (connected to a Response Center)

This section shows if voice is connecting (the icon will flash) or connected (the icon will be solid)

Status bar when response centre communication is established

5.1.5 Instrument readings

When the Tempus is in use each area will become populated with data. For ECG, Pulse Oximetry, Contact Temperature, Invasive Pressure or Capnography the data will begin to appear on the display as soon as the parameter is connected i.e. there is no start control or button for these parameters.

Since the non-invasive blood pressure measurements are single readings this requires the user to both attach the cuff to the patient **and** start the reading. These readings are therefore time-stamped with the time that they were recorded.

5.1.6 Touchscreen lock

The maintenance user can enable, disable and configure the Tempus Pro touchscreen timeout period.

When the home screen is displayed, if you do not press the touchscreen or membrane buttons and the timeout period expires, the touchscreen locks automatically. Tempus Pro continues to perform all monitoring and alarming functions when the touchscreen is locked.

You can control the touchscreen lock:

- To unlock the touchscreen, press any membrane button.
- To lock the touchscreen, press and hold the [Home] button for two seconds.

Note	If NIBP monitoring is in progress when the touchscreen is locked, the NIBP **Stop** button remains active as a safety feature. The timeout period and touchscreen lock only apply when the Tempus is on the home screen.

5.2 Menus

5.2.1 The main menu

If ever you are unsure of what to do, press one of the following membrane buttons:

The [Main Menu] button - this will take you to a full menu to access all features of the product.

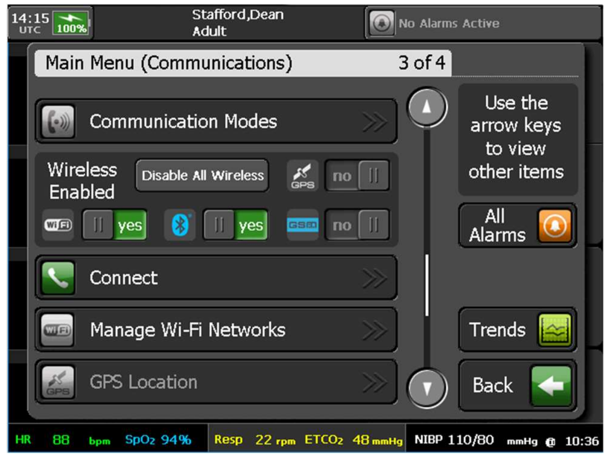
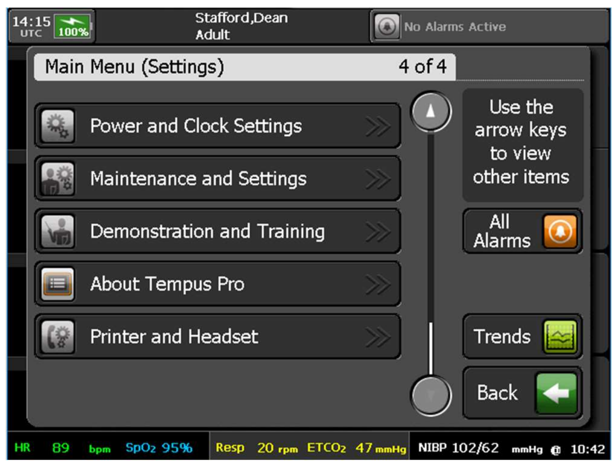
The [Home] button - this returns the unit to the results screen.

Structure of the "Main Menu"

CAUTION ⚠	The Quick Start Guide is intended to provide field-based users with summary information only. It does not replace the need to read this User's Manual or any other device labelling and does not replace proper training on the device.

The “Main Menu” allows access to the following features and controls:

Menu option	Menu screen
TDL Settings – for use with the Tempus ALS system. **Quick Start Guide** – pressing each numbered button will take you through a series of graphics which explain the features of the device. **Medical Parameter Settings** – open a menu from which all medical parameter settings can be accessed, see “6 Taking medical readings”. **Summary Record of Care** - performs the same function as the **Event** membrane button, see “3.3.9 Summary Record of Care / Tactical Combat Casualty Care”. **Waveform Snapshot** - performs the same function as pressing and releasing the membrane button, see “5.1.2 Membrane buttons”. **Display Menu** – performs the same function as the membrane button, see “9.2 Display options”.	 **Note:** The **All Alarms** and **Trends** buttons are present on all pages of the Main Menu, see: “7 Alarms” “8 Summary Record of Care“
Patient Menu – performs the same function as the membrane button, see “9.3 Patient information”. **Data Input / Output Menu** – performs the same function as the membrane button, see “9.1 Data input and output”. **Previous Activity** - performs the same function as the membrane button, see “3.3.10 Previous activity button”. **Camera** – performs the same function as pressing and holding the membrane button, see “9.4 Digital camera”. **Home** – performs the same function as the membrane button. **iAssist** – turn iAssist on or off, see “9.7 iAssist processes”.	 **Note:** Switches are provided to turn functions on and off, e.g. iAssist here.

<table>
<tr><th>Menu option</th><th>Menu screen</th></tr>
<tr><td> Communication Modes – configure the modes, see “9.6.4 Communications modes”.
Wireless Enabled – disable or enable all wireless communications. If any of these features are not enabled (either not purchased or disabled in the device’s maintenance settings) then they will be greyed out.
Connect – performs the same function as the membrane button, see “10 Connecting to an alternate location”.
Manage Wi-Fi Networks – provides access to the Manage Wi-Fi Networks menu. GPS Location – obtain a fix on the device’s location (using the built-in GPS module), see “9.5 GPS location”. </td>
<td>
</td></tr>
<tr><td> Power and Clock Settings – provides access to screen brightness, time, date and power saving settings, see “5.2.3 Power and clock settings”.
Maintenance and Settings – open the Maintenance and Setting Menu. This allows unit name and mode network settings to be modified, see the Tempus Pro Maintenance Manual (available from RDT).
Demonstration and Training – review and update the demonstration settings, see “13.1 Demonstration and training”.
About Tempus Pro – view device identification data and a list of features installed on Tempus Pro, see “9.8 View installed features”.
Printer and Headset – open the settings for the communications interfaces of the product, see “5.2.2 Printer and Headset menu”. </td>
<td>
</td></tr>
</table>

5.2.2 Printer and headset menu

The current unit configuration can be viewed by accessing the configuration screen from the “Printer and Headset” menu – see “5.2.1 The main menu”.

The “Printer and Headset” menu allows access to settings and controls that the user may occasionally need to access. Note that the primary communication settings are designed only to be controlled by maintenance staff so these settings are only available behind the maintenance menu function in the main menu.

The menu allows access to the following functions:

<table>
<tr><th>Menu option</th><th>Menu screen</th></tr>
<tr><td>

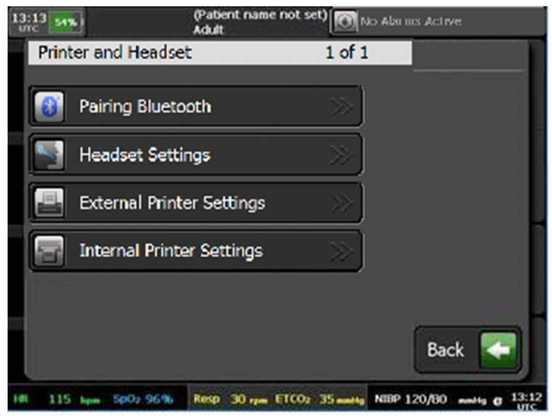
Pairing Bluetooth – provides step-by-step instructions on pairing the device.

Headset Settings – update the headset’s noise filter setting (applicable only to the Bluetooth headset) and change the default headset between wired and wireless.

External Printer Settings – choose the default printer paper - either US Letter (8.5 x 11“) or A4 - and switch between PCL3 & PCL3-GUI printer types.

Internal Printer Settings – only available if the Tempus Pro is fitted with a printer. Change internal printer settings, for example paper type (plain or grid) and auto print snapshots (on or off). See “9.9 Printer maintenance”.

</td><td>



</td></tr>
</table>

5.2.3 Power and clock settings

Using this menu, you can set the power save mode and the screen brightness level. For details relating to the brightness setting, see “9.2.5 Brightness control”.

This menu is accessed using the **Power and Clock Settings** button on the “Main Menu” – see “5.2.1 The main menu”.

<table>
<tr><th>Menu option</th><th>Menu screen</th></tr>
<tr><td>

If the maintenance user has not allowed user control over the date and time, the power and clock settings are:

Power save mode – turn power saving on or off.

Brightness – select the screen brightness level.

Note: The current time and date are also displayed but cannot be updated.

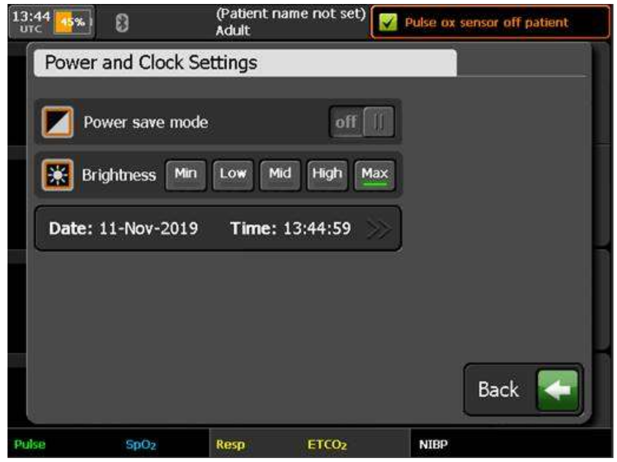
</td><td>

</td></tr>
</table>

<table>
<tr><th>Menu option</th><th>Menu screen</th></tr>
<tr><td>If the maintenance user has allowed user control over the date and time, the power and clock settings are:

Power save mode – as defined above.

Brightness – as defined above.

Date and Time – for update instructions, see “5.2.4 Changing the date and time (optional)”</td><td>
</td></tr>
</table>

5.2.4 Changing the date and time (optional)

You can change the date and time setting of Tempus Pro only if the maintenance user has allowed user control.

CAUTION ⚠	Do not change the date or time during a patient incident, as it will cause a new patient to be admitted.

To change the date and time:

1. Press **Main Menu** button .
2. Press the **Power and Clock** Settings button to display the Power and Clock Settings menu:
   If the Date and Time button contains chevron characters, this indicates that the maintenance user has allowed user control and you can press the button to update the date and time:
   

   If the chevrons are not present and the message panel displays "To change the date or time refer to the maintenance manual", you cannot update the date or time.
3. Press the date and time button (if selectable). The Tempus displays the Date and Time Settings screen.
4. Use the keyboard and arrow controls to set time zone, time, day, month and year.

Note	To avoid confusion, do not adjust timezone and clock at the same time. Always check the clock after making changes.

5. Press the **Save and Back** button.
6. A confirmation dialog is displayed: press **Yes** if you wish to confirm the date/time change and admit a new patient, otherwise press **No**.
7. If you have confirmed the date/time change, the Select Patient screen is displayed.

6 Taking medical readings

Note	When a medical module is first started the word “initializing” may appear for a few seconds in the results area associated to the medical parameter being started.

6.1 Electrocardiography (ECG)

The Tempus Pro allows the user to either monitor the patient’s ECG or take a diagnostic recording of their ECG (this is an optional feature). This section will explain how to use both features. The device defaults to ECG monitoring with either a 3-lead, a 4-lead or a 5-lead cable.

WARNING	**The ECG device is not intended for use in a sterile environment. Do not use for direct cardiac application.**
WARNING	**The ECG device is reusable, other than disposable single-use electrodes.**
WARNING	**Do not attempt to insert the ECG device (including patient cables) into an electrical outlet.**
WARNING	**The ECG is for resting recordings and should not be used in stress testing environments.**
WARNING	**Ensure electrodes are connected only to the patient.**
WARNING	**Conductive parts of electrodes and connectors, including neutral electrode, should not contact other conductive parts including earth.**
WARNING	**The Tempus Pro is rated as being proof against the effects of a defibrillator discharge. Follow these warnings if using an AED or defibrillator with the Tempus Pro:** Follow the instructions of the defibrillator or AED when using it with the Tempus Pro. • Do not touch the patient during defibrillation. • Do not touch the defibrillator’s paddle-electrode surface when discharging the defibrillator. • Keep defibrillation electrodes well clear of other electrodes or metal parts in contact with the patient.

	• Do not touch the patient, bed, or any conductive material in contact with the patient during defibrillation.

WARNING	**The Tempus Pro is protected against defibrillator discharge but rate meters and displays may be temporarily affected during defibrillator discharge but will rapidly recover.**
WARNING	**The Tempus Pro may not operate effectively on patients who are experiencing convulsions or tremors, or who are moving or are in motion (transport) environments. In such cases remember to use best practise for electrode site selection (torso rather than limbs), site preparation (cleaning/scrubbing with an alcohol wipe) and cable securement at the sensor site and as much as possible along the cable's length. In motion environments the Tempus should be secured using a clamp or strap and to obtain the best possible readings the device should be damped to reduce shocks or vibrations if possible.**
WARNING	**The Tempus is not for direct cardiac application.**
WARNING	**Do not use electrodes of dissimilar metals.**
WARNING	**Always set the Tempus Pro to the correct patient age to ensure the QRS detector is set to detect appropriate R wave amplitudes. Per AAMI EC 13, the Tempus uses the definitions of patient age provided as follows:** • **Neonate –** Children 28 days old or less if born at term (based on 37-week gestation), otherwise up to 44 gestational weeks. • **Pediatric (Child) –** Children of 29 days up to and including 12 years old. • **Adult** – Individuals 13 years old or older. **Failure to set the correct age may result in the Tempus mis-detecting R waves and providing incorrect heart rate values.** **Setting the patient age to 8 years or less will mean that the ECG monitor will be set to detect pediatric R waves per AAMI EC13.** **If you do not set the patient age, the Tempus will use the default patient age group (see the *Tempus Pro Maintenance Manual*). Ensure that the patient age/type is always entered for non-adult patients – see "9.3.5 Entering patient details".**

CAUTION	As the following sections will explain, the Tempus Pro will present results in a different layout depending on whether the device is set to ECG monitoring or Diagnostic ECG. The alarm functions will be the same in either case.
CAUTION	Certain line-isolation monitors may cause interference on the ECG display and may inhibit heart rate alarms. Users should ensure electrodes are placed correctly, with the skin prepared in advance as instructed.
CAUTION	The electrodes of the ECG apron must be applied carefully. Follow the electrode manufacturer's instructions in the storage, removal and application of the electrodes to the patient to ensure the electrode is placed optimally.
CAUTION	Care must be taken to ensure that the electrodes do not contact live (electrical) parts or earthed metal parts of local systems or structures.

CAUTION ⚠	If the patient is attached to a cardiac defibrillator, care must be taken not to touch any part of the ECG cables while the patient is being shocked.

Note	The leads and cables of the ECG should be checked for fraying, tears, knots or other signs of damage before and after use.
Note	Ensure that all labelling including instructions for use, printed warnings and use-by dates associated with third-party ECG electrodes are adhered to.

6.1.1 Getting started

1. To measure correct R-wave values, make sure that the right age group is selected in the patient details. For instructions, see “9.3.5 Entering patient details”.
2. Make sure that you have read the monitoring standards in "14.1.1 ECG monitoring".
3. Ensure that the ECG cable is properly inserted into the green connector on the Tempus Pro:

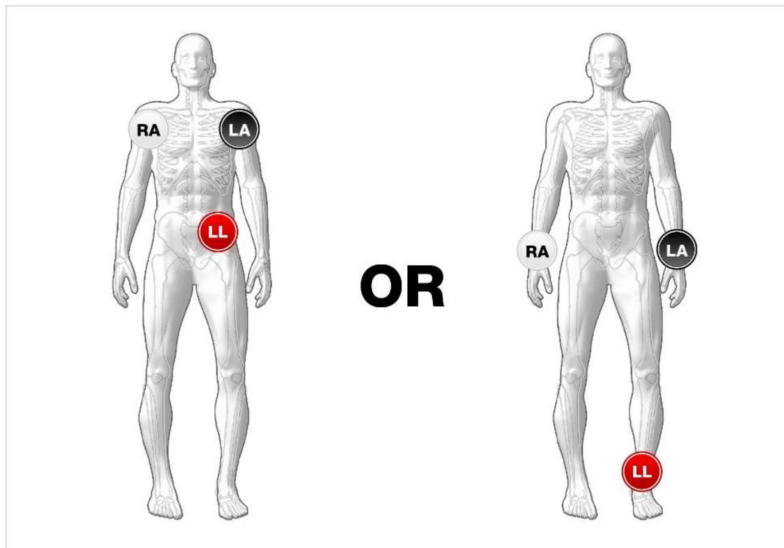
4. Prepare the patient's skin at the electrode sites:
   - Shave or clip excess hair.
   - Clean the skin thoroughly with an alcohol wipe.
   - Rub the skin to dry.
5. If the patient has a pacemaker, turn on pacemaker indication:
   - Press the ECG area of the touchscreen or navigate to the medical parameter setting in the Main Menu.
   - Switch **Pacemaker indication** on.
6. Make sure that the frequency setting matches the frequency of the mains you are using:
   - Press the ECG area of the touchscreen.
   - Check that **ECG power filter** is on.

6.1.2 Monitoring ECG with a 3-lead cable

By default, the Tempus Pro will be set to ECG monitoring when it is supplied.

ECG monitoring with a 3-lead cable allows the user to monitor Leads I, II or III of the patient's ECG. ECG monitoring requires the use of a 3-lead/electrode cable – see "12 Accessories list of the Tempus Pro".

The 3-lead cable may be pre-attached to the Tempus in the green connector. The Tempus Pro will automatically detect which cable type is in use. With a 3 electrode cable one lead waveform (I, II, or III) is available for display at a time.



Electrode positions for 3-lead ECG monitoring

The cable terminates in conventional 4 mm snap connections which can be used with various commonly available disposable ECG electrodes. Two types of cables are provided for compliance with AAMI and IEC guidelines. For details of cable part numbers, see "12 Accessories list of the Tempus Pro".

These are labelled:

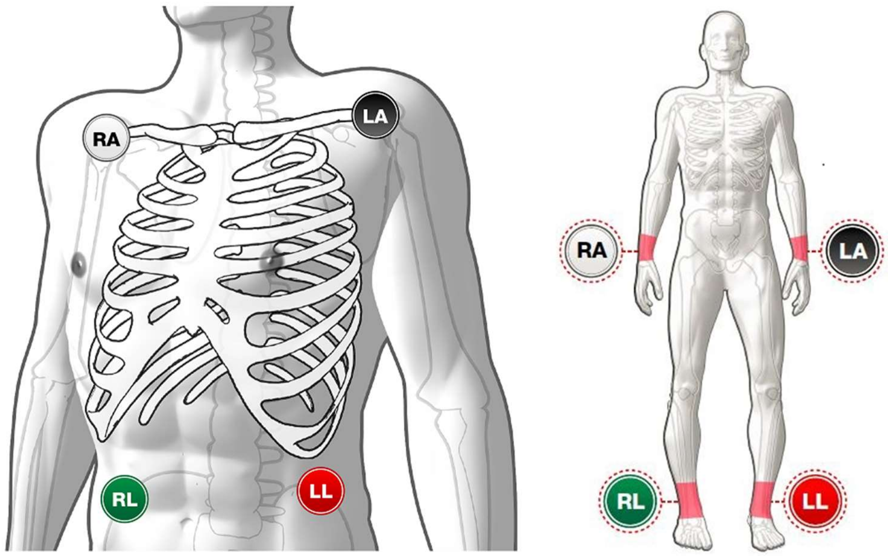
Position	AAMI label/color	IEC label/color
Right mid-clavicular line under clavicle / right wrist	RA White	R Red
Left mid-clavicular line under clavicle / left wrist	LA Black	L Yellow
Left hip / left ankle	LL Red	F Green

Once the electrodes are connected to the patient, the Tempus will begin to display the ECG signals immediately.

6.1.3 Monitoring ECG with a 4-lead cable

ECG monitoring with a 4-lead cable allows the user to monitor Leads I, II, III, AvL, AvR and AvF. Up to two leads can be displayed at any time. Four wire ECG monitoring requires the use of a 4-lead/electrode cable – see “12 Accessories list of the Tempus Pro”.

The 4-lead cable should be pre-attached to the Tempus in the green connector.



Electrode positions for 4-lead ECG monitoring (torso or limbs)

The cable terminates in conventional 4 mm snap connections which can be used with various commonly available disposable ECG electrodes. Two types of cables are provided for compliance with AAMI and IEC guidelines. For details of cable part numbers, see “12 Accessories list of the Tempus Pro”.

These are labelled:

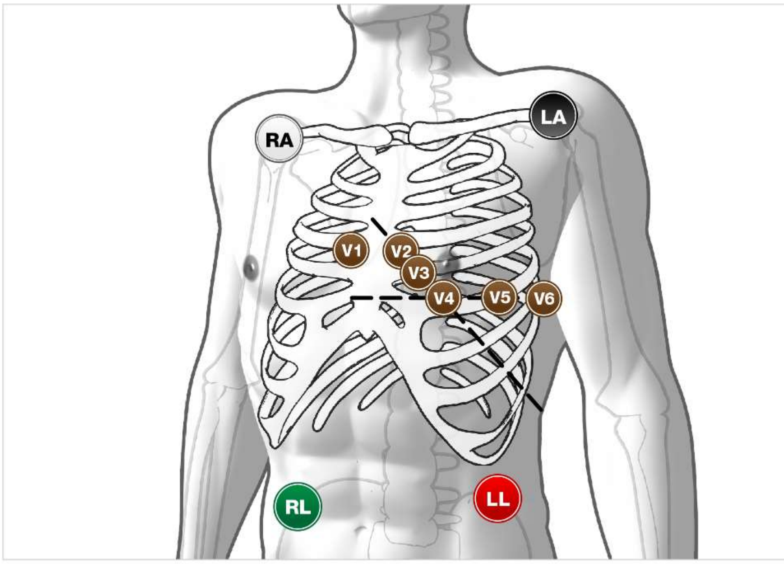
Position	AAMI label/color	IEC label/color
Right mid-clavicular line under clavicle / right wrist	RA White	R Red
Left mid-clavicular line under clavicle / left wrist	LA Black	L Yellow
Left hip / left ankle	LL Red	F Green
Right hip / right ankle	RL Green	N Black

If the ECG is inoperable then the ECG will either show a “Leads off” error for the specific Lead that is unavailable or will show a technical error as described in section “7.4 Technical alarms”.

6.1.4 Monitoring ECG with a 5-lead cable

ECG monitoring with a 5-lead cable allows the user to monitor Leads I, II, III, AvL, AvR, AvF or a user-placeable V Lead. Up to two leads can be displayed at any time. Five wire ECG monitoring requires the use of a 5-lead/electrode cable – see "12 Accessories list of the Tempus Pro".

The 5-lead cable should be pre-attached to the Tempus in the green connector.



Electrode Positions for 5-lead ECG Monitoring

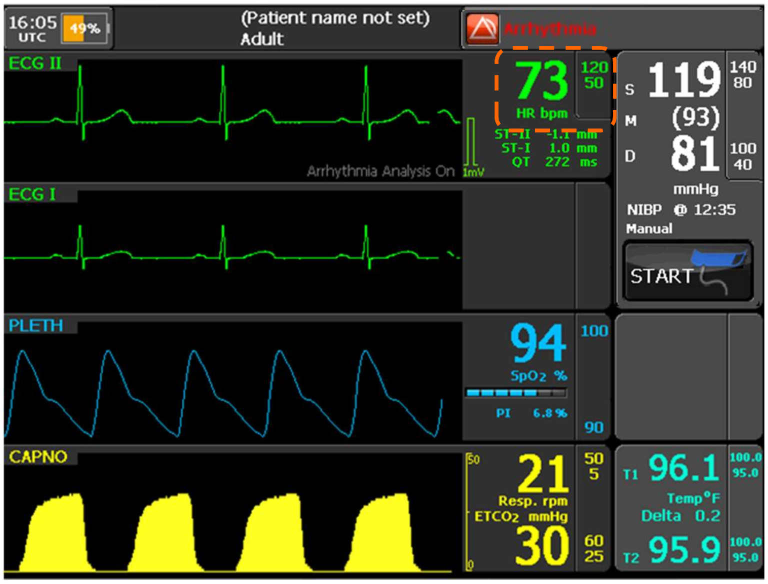
The cable terminates in conventional 4 mm snap connections which can be used with various commonly available disposable ECG electrodes. Two types of cables are provided for compliance with AAMI and IEC guidelines. For details of cable part numbers, see "12 Accessories list of the Tempus Pro".

These are labelled:

Position	AAMI label/color	IEC label/color
Right mid-clavicular line under clavicle / right wrist	RA White	R Red
Left mid-clavicular line under clavicle / left wrist	LA Black	L Yellow
Left hip / left ankle	LL Red	F Green
Right hip / right ankle	RL Green	N Black
This is a moveable pre-cordial electrode. Place it in any position from V1-6 as follows: V1 - 4th intercostal space at right sternal margin. V2 - 4th intercostal space at left sternal margin. V3 - Midway between V2 and V4 leads. V4 - 5th intercostal space at mid-clavicular line. V5 - Same transverse level as V4 at left anterior-axillary line. V6 - Same transverse level as V4 at left mid-axillary line.	V Brown	C White

If the ECG is inoperable then the ECG will either show a "Leads off" error for the specific Lead that is unavailable or will show a technical error as described in section "7.4 Technical alarms".

The picture below shows the Tempus monitoring Lead II and Lead I (using a 5-lead cable). Each ECG trace is labelled in the top left. The heart rate is shown on the right hand side for the uppermost trace.



Tempus monitoring Lead II and Lead I (using a 5-lead cable)

6.1.5 Previewing a 12-lead ECG

The 12 Lead Preview screen is to preview the ECG in advance of acquiring, printing, transmitting and storing a 12-lead ECG. The 12-lead function's preview screen displays real-time 12-lead ECG data to verify signal quality before acquiring a diagnostic 12-lead ECG.

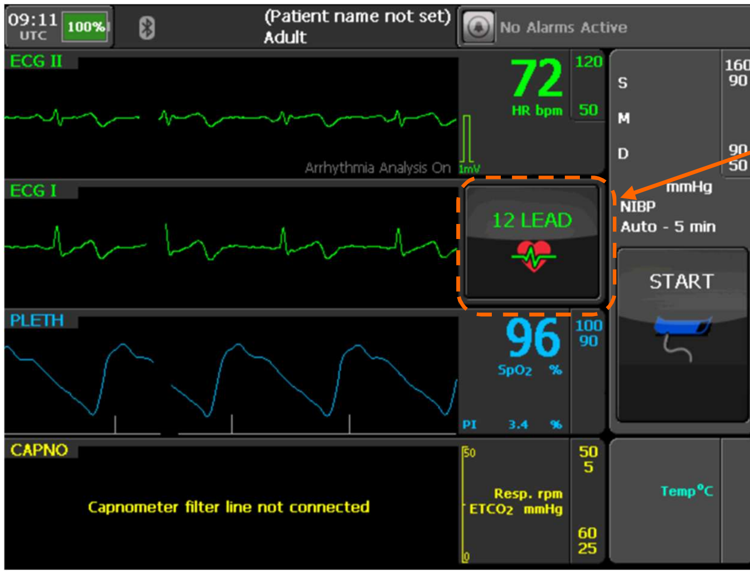
WARNING	**When acquiring a 12-lead ECG, encourage the patient to remain as still as possible.**
WARNING	**Possible misinterpretation of ECG Data: the preview display does not provide the facility for diagnostic and ST segment interpretation. For diagnostic or ST segment interpretation, or to enhance internal pacemaker pulse visibility, a diagnostic quality 12-lead must be recorded and viewed either as a static 12-lead (with background grid) or a 12-lead print out.**
WARNING	**Before previewing a 12-lead ECG, ensure you have selected the appropriate 12 lead filter settings.**
WARNING	**Do not use the Monitoring filter for diagnostic purposes, as it may lead to incorrect diagnosis.**

Diagnostic 12-lead ECG is an option on the Tempus Pro. You can use a standard 12-lead ECG cable or a 12-lead modular cable (a 4-lead cable connected to an additional 6-lead cable). The cable terminates in a 4 mm snap style connector for use with normal disposable electrodes – see "12 Accessories list of the Tempus Pro".

To view a 12-lead ECG, press the **12 LEAD** button on the touch screen.

Note	If the **12 LEAD** button is not available in the selected screen format, there are two alternative ways to view a 12-lead ECG: • Press the ECG area of the home screen to display the **ECG Settings** menu, then press **View 12 lead ECG**.

	• Press the DISPLAY  membrane button, then press the twelve lead diagnostic view icon.
Note	If the Glasgow Interpretation option is used but you have not yet entered the patient's age or sex, the Patient Age/Sex screen is displayed: enter age and sex (if known) and press OK.



With the 12 Lead cable connected, press here to activate the 12 Lead view

12-lead ECG selected

If the 12-lead ECG Cable is attached but the 12-lead View is not selected on the screen, then the Tempus will remain in standard ECG monitoring mode and will continue to show one or two ECG waveforms on screen.

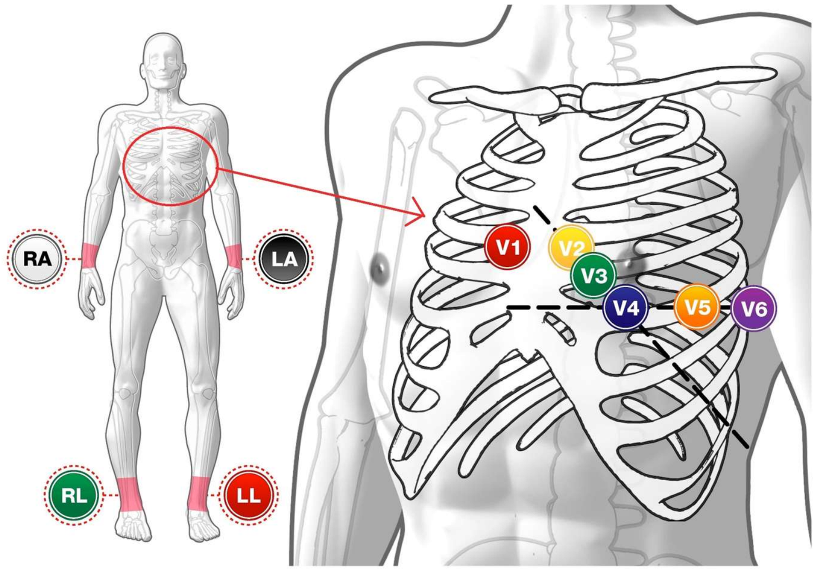
Two types of cables are provided for compliance with AAMI and IEC guidelines. For details of part numbers of cables, see “12 Accessories list of the Tempus Pro”.

These are labelled:

Position	AAMI label/color	IEC label/color
Right mid-clavicular line under clavicle / right wrist	RA White	R Red
Left mid-clavicular line under clavicle / left wrist	LA Black	L Yellow
Left hip / left ankle	LL Red	F Green
Right hip / right ankle	RL Green	N Black
This is a moveable pre-cordial electrode. Place it in any position from V1-6 as follows:		
V1 - 4th intercostal space at right sternal margin.	V1 Red	C1 Red
V2 - 4th intercostal space at left sternal margin.	V2 Yellow	C2 Yellow
V3 - Midway between V2 and V4 leads.	V3 Green	C3 Green
V4 - 5th intercostal space at mid-clavicular line.	V4 Blue	C4 Brown
V5 - Same transverse level as V4 at left anterior-axillary line.	V5 Orange	C5 Black
V6 - Same transverse level as V4 at left mid-axillary line.	V6 Violet	C6 Violet

The electrodes should be attached to the patient as shown below. Before attachment the patient's skin should be prepared according to the instructions of the electrodes e.g. cleaning the skin with alcohol and shaving if necessary.

Remember only to use electrodes which are compliant with AAMI EC12 and to read and follow the instructions for the electrodes.



Electrode placement for 12-lead ECG monitoring

Once in 12-lead ECG Mode, the ECG will monitor 12 leads of ECG.

If the ECG is inoperable then the ECG will either show a "Leads off" error for the specific lead that is unavailable or will show a technical error as described in section 7.4.

In 12-lead mode, the other vital signs data remain displayed across the bottom of the display. All 12 ECG leads are shown in the main display (each lead is labelled).

12-lead ECG preview screen

If no signal or a poor signal is seen on one or more leads, the following steps should be reviewed:

- Has the correct cable been used, is it connected properly?
- Is the patient still?
- Is the equipment setup properly (all electrodes attached)?
- Could there be interference from other electrical equipment?
- Is the mains filter set correctly (see ECG Settings)?
- Are the 12-lead filters set correctly (see ECG Settings)?
- Was skin sufficiently well prepared?
- Could the electrodes be dry or poorly attached?

6.1.6 Performing a diagnostic ECG

The 12-lead function enables users to record a 10 second trace for immediate or subsequent analysis and interpretation.

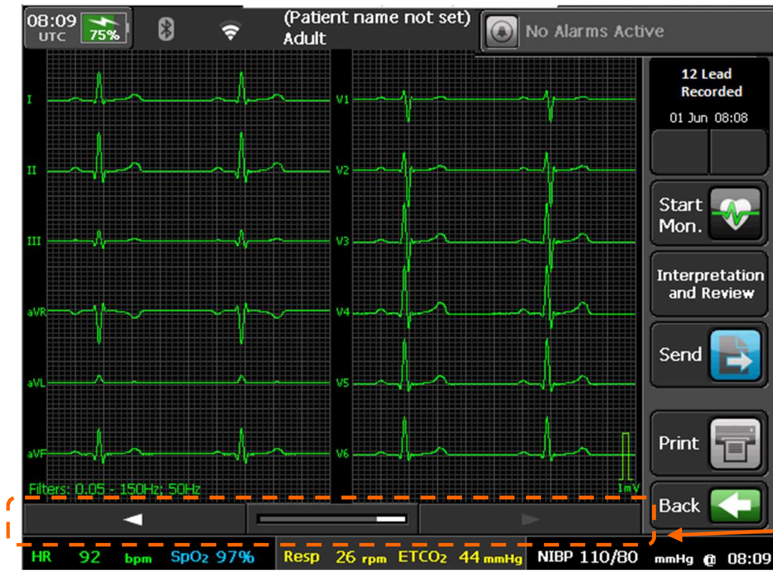
The 10 second recording will be added to the patient record and therefore can be exported by USB stick to another Tempus or to a computer to be read as a .PDF file. The ECG can also be transmitted to a Response Center if this option is in use – see “10 Connecting to an alternate location”.

WARNING	**All numerical, graphical and interpretive data should be evaluated with respect to the patient's clinical and historical picture.**
WARNING	**ECG interpretation performed by software is not a substitute for review and evaluation of ECG recordings by a qualified clinician.**
WARNING	**Motion artefact or noise in the ECG recording could result in erroneous interpretations.**
Note	Interpretations are unconfirmed and need to be reviewed by a clinician.

Making a 12-lead ECG recording

Pressing the **Start Record** button will initiate a 10 second recording.

If the Tempus is connected to a Response Center then this will be transmitted as soon as the recording is complete. The Response Center has tools for measuring the ECG. Otherwise, the recording can be viewed on screen by using the left and right arrow keys.



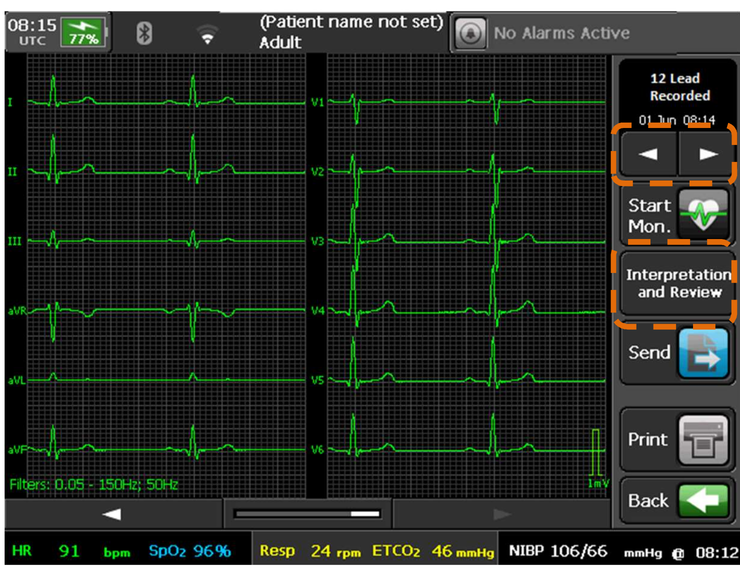
12-lead ECG recorded screen

If an internal printer is fitted to the Tempus, you can print the 10 second recording using the **Print** button.

If the Tempus Pro is fitted with an internal printer and automatic printing is enabled, it will print waveforms as soon as they are captured.

6.1.7 Viewing an ECG recording interpretation

Once a 12-lead ECG has been recorded you can view the ECG interpretation, summary and review by pressing I**nterpretation and Review**.



Selecting ECG interpretation

Your Tempus Pro may be configured with one of two 12-lead ECG interpretation options: either standard or Glasgow.

Note	The Glasgow ECG algorithm (part number 05-2075) may only be installed if it is approved for commercial distribution in your territory.

The key differences between standard and Glasgow ECG interpretation are:

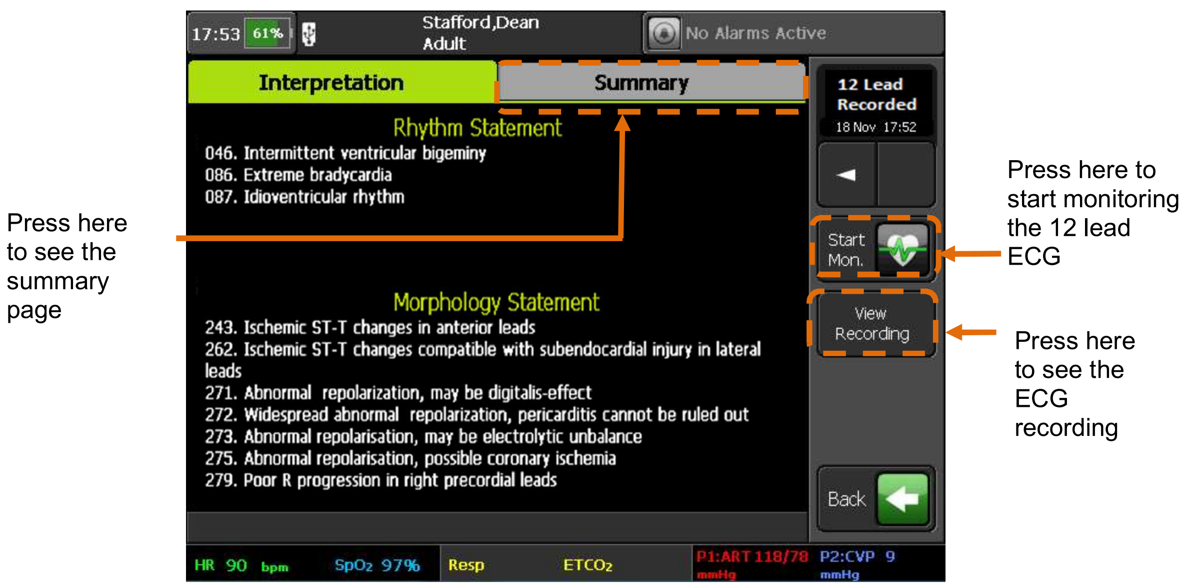
- Standard interpretation is only valid for adults of 18 years or older.
  Glasgow interpretation is valid for any age
- If patient age is not entered, standard interpretation assumes 55 year old male.
  If patient age is not entered, Glasgow interpretation depends on selected age group: adult 55 year old male, pediatric 10 year old male, neonate 0 year old male.

Refer to the *Tempus Pro Maintenance Manual* for full configuration instructions.

Standard 12-lead ECG interpretation (optional)

The Tempus Pro provides standard ECG interpretation of the 12-lead ECG recordings (if configured). This interpretation includes:

- **Rhythm statements** – there are 69 different statements this includes the reporting of detected Arrhythmias such as Ventricular Tachycardia, Atrial Fibrillation, Extreme Bradycardia, and AV Block (please see *QRS Diagnostic Physician's Guide ECG Analysis for Resting 12-lead ECG* for further information, Part number 630000-00).
- **Morphology statements** – there are 121 different statements including ST analysis which includes over 20 Ischemic ST change statements such as: “Ischemic ST-T changes in posterolateral leads” and “Ischemic ST-T changes in lateral leads” (please see *QRS Diagnostic Physician's Guide ECG Analysis for Resting 12-lead ECG* for further information, Part number 630000-00).



ECG interpretation screen (not Glasgow)

Glasgow 12-lead ECG interpretation (optional)

The Tempus Pro provides Glasgow ECG interpretation of the 12-lead ECG recordings (if configured).

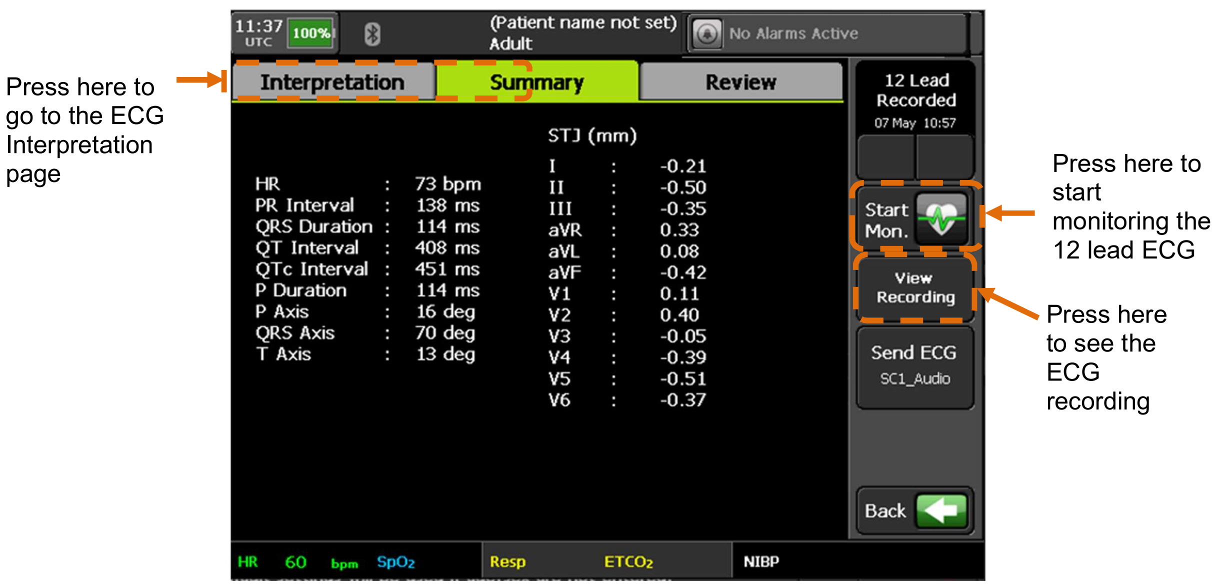
Tempus Pro will display Glasgow interpretation: up to 10 interpretation statements with a summary statement:

ECG interpretation screen (Glasgow)

For more information, please see the *Glasgow 12-lead ECG Analysis Program Physician's Guide*.

Summary

The waveform summary measurements include: HR, PR Interval, QRS Duration, QT Interval, QTc Interval, P Duration, P Axis, QRS Axis, T Axis and STJ (mm) for all leads.



ECG summary screen

6.1.8 ECG specifications

ECG monitoring specifications

The following disclosures are made in accordance with the requirements of AAMI EC13 and IEC60601-2-27:

- The ECG's display includes the isoelectric segments of the QRS.
- For leads off detection, the ECG device applies a 1 kHz triangular wave of approximately 6 mV p-p between the R/RL and L/LL electrodes
- The Heart Rate Detection of the ECG monitor is accurate to +/-10% or +/-5 bpm for a T Wave of amplitude: up to 1.0 mV.
- The heart rate is calculated from the average of the 8 most recent RR intervals. Intervals are logged as "invalid" if the detections are unreliable, or if detections are absent over the maximum RR interval of 2.4 s. Invalid intervals are counted but not included in the average. If there are no valid intervals in the most recent 8, then no heart rate is reported. Heart rate is updated with every new valid or invalid interval.
- Further to clause 4.1.2.1e) of AAMI EC13:
  - The system's response to Fig 3a is 40 bpm ±10% (inverted QRS are counted); an invalid rate will be displayed if **EMS Heart Rate Mode** and **Arrhythmia analysis** are active.
  - The system's response to Fig 3b is 30-100 bpm (small QRS are intermittently counted).
  - The system's response to Fig 3c is 120 bpm ± 10% (all QRS are counted).
  - The system's response to Fig 3d is 60-100 bpm (bipolar QRS are intermittently counted).
- Electrical noise such as power line transients can cause false heart beat detection which may inhibit alarms. The problem may be minimized by ensuring good electrode contact. Power system noise may be reduced by running the monitor from its battery.
- The maximum response time of the heart rate meter to a change in heart rate from 80 to 120 bpm is 10 seconds, and for a change from 80 to 40 bpm is 14 seconds (according to the method in IEC 60601-2-27:2011 section 201.7.9.2.9.101 b).
- The system heart rate alarm will respond to the specified tachycardia signals of Figures 4a and 4b of AAMI EC13 within 5-7 s. In some cases, an out of range alarm may be given, with an indication of no valid heart rate. Invalid ECG rates do not trigger an alarm if the organisational **EMS Heart Rate Mode** is selected and **Arrhythmia monitoring** is active. Note: that tachycardia waveform 4B at half amplitude is not detected when using a single ECG channel (eg. when using a 3 wire ECG cable).
- The heart rate limit alarm will respond to a high or low rate condition 0-10 seconds after being activated.
- Per EC13, 4.1.4.1, heart rate is accurately indicated in the presence of effective pacemaker single pulses without overshoot, having width of up to 2 ms and amplitude up to 700 mV, and such pulses alone do not cause heart rate detection. In the case of ineffectively paced QRS, erratic readings may result for single pulses above 2 mV. Overshoot is specified according to EC13 4.1.4.1 Method B.
- Per EC13 4.2.9.12, the muscle filter will make pacemaker pulses appear smaller than 2mm. Switch muscle filters off (Diagnostic filter setting) to see pacemaker pulses more clearly.
- Measurements of the ECG waveform should be normalized against the scale provided. The scale presents 1 mV as 8 mm with a gain setting of 10mm/mV.
- Note that in the 4 waveform view the channel height of the ECG is 24 mm and thus permits a maximum input signal of ±2.2 mV when using a gain of 5 mm/mV. In the large ECG view a maximum input signal of ±5 mV is achieved using a gain of 5 mm/mV. Large ECG view is described in "9.2 Display options".
- Compliance with EC13 clause 4.2.9.8 b) (and relevant clause of IEC60601-2-27:2011) requirements for frequency response are achieved only with the muscle filters switched off (i.e. using the Diagnostic filter setting).
- The slew rate of the ECG as per AAMI EC13 4.1.4.3 is 3.7 V/s (all filters off).

ECG recording specifications

The following disclosures are made in accordance with the requirements of AAMI EC11 and IEC60601-2-25:

- Frequency and impulse response has been evaluated in accordance with IEC 60601-2-25:2011 section 201.12.4.107.1.1. As per section 201.12.4.107.1.1.1, Methods A and E were used to demonstrate high frequency response compliance. All 60601-2-25 tests have been performed using SCP files recorded on the Tempus Pro and compliance checked using the Corsium ECG viewer. Compliance requires that all optional filters be switched off.
- ECG data in this system are contained in computer files, using a proprietary format based on the SCP-ECG standard. The ECG data are digitized using a sample rate of 500sps and a resolution of 2554nV/bit with 12 bit (4096 level) dynamic range. Per clause 201.12.4.107.3 the skew rate is 0.

The following disclosures apply only when standard interpretation is enabled (i.e. not when Glasgow interpretation is enabled):

- Automated measurements of amplitude and duration of QRS complexes generated from the ECG recording are only valid for adult patients. For younger patients the values should be measured using the electronic calipers within i2i.
- Per clause 201.12.1.101.3 automated measurements of the S segment duration of CAL waveforms are ±15 ms. The automated value can be confirmed using the electronic calipers within i2i
- Automated measurements are only able to distinguish T-waves of amplitudes greater than >50 μV

6.1.9 ECG settings

Pressing either of the ECG trace areas will bring up the “ECG Settings” menu:

<table>
<tr><th>Menu option</th><th>Menu screen</th></tr>
<tr><td>

Heart rate limits (bpm) – open the the alarm limits editor (described below).

Heart Rate Lead – (4-, 5- and 12-lead ECG only) change which lead the heart rate is taken from.

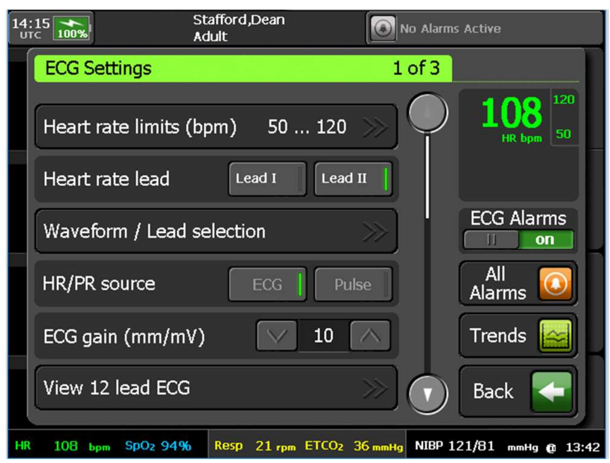
Waveform / Lead selection - when using 4-, 5- or 12-lead ECG, press here to customize the selection of waveforms on the four waveform display, see “9.2.4 Waveform and lead selection”.

HR/PR source – select **ECG** or **Pulse** oximeter as the heart/pulse rate source (see also **HR/PR beat volume** and **EMS Heart Rate Mode** below).

ECG Gain (mm/mv) –select the desired gain (or size) of the waveform.

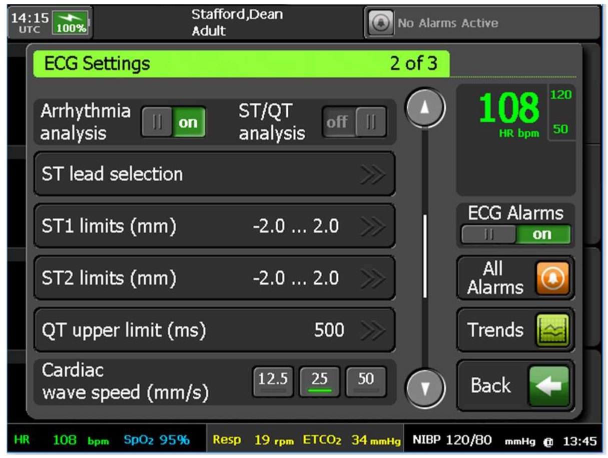
View 12 lead ECG – if the 10 wire (12 Lead) ECG cable is selected, this option will be available. Pressing it will bring up a different display interface that shows all 12 ECG leads.

</td><td>

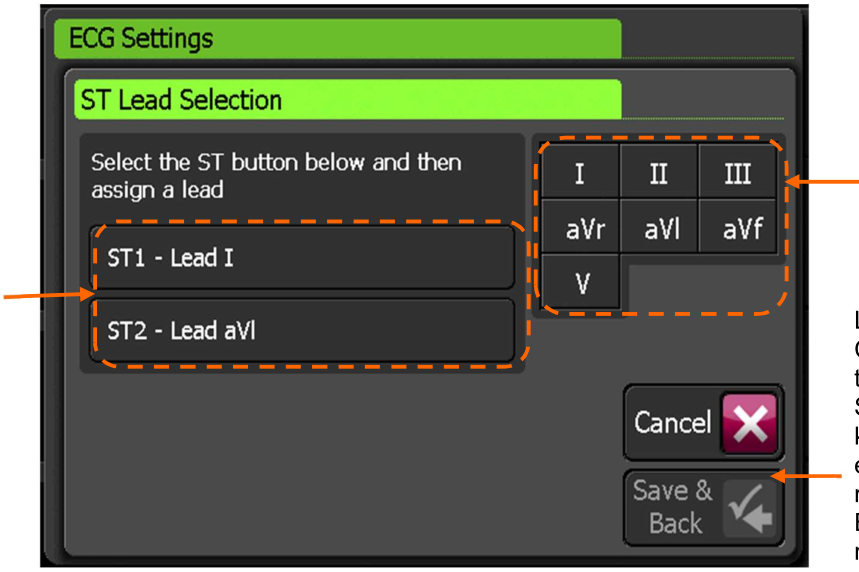


Note: When using a 3-lead ECG, the heart rate is derived from the waveform displayed:

</td></tr>
</table>

Menu option	Menu screen
Arrhythmia analysis and **ST/QT analysis** (on/off) – enable or disable these measurements. **ST lead selection** - when ST monitoring is active, change the ST leads that are displayed, see “ST lead selection”. **ST1 limits (mm)** – open the editor to change the ST alarm limits. **ST2 limits (mm)** – open the editor to change the ST alarm limits. **QT upper limit (ms)** – open the editor to change the QT alarm upper limit. **Cardiac wave speed (mm/s)** – select the desired speed of the waveform.	
HR/PR beat volume – volume of the audible tone that the Tempus emits every time a heartbeat is detected. Use the up and down arrows to adjust the volume from zero (no tone) to 100%. The tone is based on the **HR/PR source** setting (see above): when set to **ECG**, QRS beats trigger a single tone at 1 kHz. When set to **Pulse**, pulse beats trigger pulse tones – see “6.3.6 Pulse oximeter settings”. **Pacemaker indication** – turning this on will show a vertical dotted line on the ECG waveform when the ECG is connected to a patient wearing a pacemaker. **Monitoring filter** – use the arrows to select a filtering option: **Monitor**, **EMS**, **Diagnostic** or **Filtered diagnostic**. **12 lead filter** – use the arrows to select a filtering option: **Diagnostic** or **Filtered diagnostic**. **ECG power filter** – this is set to ON by default, switch it off if you wish the ECG to be presented without the Power noise being filtered out.	

ST lead selection screen



<u>"ST Lead Selection" screen</u>

Note	The ECG waveforms displayed on the home screen will change to match the selected ST leads.
Note	In order to reduce noise and artefact the Monitoring filter should be set to **EMS** and the Mains filter should be switched on. When attached to a mains power supply noise may be observed on the ECG if these filters are switched off. Consequently, they should be left switched on when operating under mains power or mains power should be disconnected if the filters need to be switched off.
Note	Users are reminded to have the mains filter frequency set to the correct value (50 Hz or 60 Hz) in order for it to be effective. This should be checked before use.

Mains frequency setting

Hz means Hertz, or cycles per second. In North America, mains electricity supplies operate at 60 Hz; most of the rest of the world uses 50 Hz. In aircraft the filter should normally be set to 50 Hz. In remote land and maritime applications, the local voltage could be either 50 Hz or 60 Hz. ECG systems can pick up interference from mains electricity supplies. This interference appears on the screen as regular interference patterns.EMS Heart Rate Mode

EMS Heart Rate Mode

If the organizational **EMS Heart Rate Mode** is selected, in periods where the rate maybe impacted by ECG signal artefact, the rate display may automatically switch between ECG derived rate and pulse oximeter derived rate. In the absence of an available pulse oximeter rate the heart rate may revert to displaying "**- - -**". This feature may temporarily override **HR/PR source**.

CAUTION ⚠	RDT recommends that this mode is used with **Arrhythmia analysis** set **on**. There is no impact on ECG arrhythmia detection or arrhythmia alarming.
CAUTION ⚠	If there is excessive noise when in EMS mode, heart rate will not be displayed.

6.1.10 Detecting arrhythmias during ECG monitoring

During ECG monitoring, the Tempus Pro analyzes the 2 ECG waveforms on the home screen to detect arrhythmias. All results are displayed as text on the results screen.

WARNING ⚠	**ECG monitoring arrhythmia analysis is a tool to be used in addition to patient assessment. Care should be taken to assess the patient at all times.**
WARNING ⚠	**The Automated ECG Arrhythmia Analysis software is designed to be used by qualified medical personnel.**
WARNING ⚠	**The clinician and/or medically qualified personnel are responsible for determining the clinical significance of each detected arrhythmia event or alarm.**
WARNING ⚠	**The arrhythmia detection feature on the Tempus Pro is not a diagnosis tool and is an additional tool to support the clinician in their own diagnosis only. The classification performance of the arrhythmia feature has been tested using the FDA recognized consensus standard. The analysis algorithm has also been tested and approved in other products. The monitor cannot be relied upon to detect all arrhythmias. Do not rely solely on the displayed arrhythmia data and alarms to assess the patient's condition. Always observe the patient closely and monitor all of their vital signs carefully.**
WARNING ⚠	**The arrhythmia analysis program is intended to detect ventricular arrhythmias. It is not designed to detect atrial or supraventricular arrhythmias. The program may incorrectly identify the presence or absence of an arrhythmia. A physician must analyze the arrhythmia information with other clinical findings.**
WARNING ⚠	**The system can filter out some interference but corrupted or excessively noisy signals may cause artefacts. If the ECG waveform is too noisy, Noise is displayed on screen. If no arrhythmias are detected and there is no noise in the signal, Arrhythmia Analysis On is displayed to ensure the operator is aware arrhythmia analysis is running.**
WARNING ⚠	**Heart parameters can be affected by the use of drugs. Patient drug use history should be reviewed when reviewing detected arrhythmia.**
WARNING ⚠	**For patients with pacemakers, ensure pacemaker indication is turned on.**
WARNING ⚠	**Arrhythmia analysis is sourced from the ECG signal (including detection of extreme bradycardia, extreme tachycardia), this is independent of the heart rate display source selection (ECG or pulse).**

CAUTION ⚠	Only start arrhythmia learning during periods of predominantly normal rhythm, and when the ECG signal is relatively noise-free. If arrhythmia learning takes place during ventricular rhythm, the ectopics may be incorrectly learned as the normal QRS complex. This may result in missed detection of subsequent arrhythmias.
CAUTION ⚠	If arrhythmia analysis is not restarted when a template changes: • Inaccurate arrhythmia alarms can be triggered. • Inaccurate heart rates can be recorded.

To optimize arrhythmia detection, select leads with the largest amplitude and the least amount of noise to be analyzed:

1. Turn on arrhythmia analysis in the ECG menu. Arrhythmia detection starts up and performs a learning period.
2. To change the waveforms being analyzed, press **Lead waveform selection** on the ECG menu and select the lead required.
3. If a patient's ECG template changes dramatically, turn arrhythmia analysis off and on.

Arrhythmia detection performs a relearning period based on the new template.

An arrhythmia detection relearning period can also be triggered when you:

- Change patients
- Change **Waveform / Lead selection** or **Heart rate lead** options

The following arrhythmia events are detected and recorded:

Arrhythmia	Screen text	Description	High priority patient alarm triggered	Event / waveform capture
Extreme bradycardia	Extreme Brady	The average heart rate is 5 bpm below the low heart rate set on the Tempus Pro (with a maximum of 60 bpm).	Yes	Yes
Extreme tachycardia	Extreme Tachy	The atrial and ventricular rates are equal. The heart rate is 5 bpm above the high heart rate limit set on the Tempus Pro (with a minimum of 100 bpm)	Yes	Yes
Ventricular tachycardia	Vent Tachy	Ventricular HR is greater or equal to the Tachycardia rate threshold and the number of PVCs is greater than 6.	Yes	Yes
Asystole	Asystole	No QRS complex for 5 consecutive seconds (in absence of ventricular fibrillation or chaotic signals)	Yes	Yes
Ventricular fibrillation	Vent Fib	Ventricular fibrillation occurs and persists for more than 6 seconds.	Yes	Yes

A medium priority alarm is triggered when:

- Arrhythmia analysis is turned **off**, and either the low heart rate or high heart rate alarm limits are breached.
- Arrhythmia analysis is turned **on**, and either the low heart rate or high heart rate alarm limits are breached

A high priority “Extreme Brady” or “Extreme Tachy” alarm is triggered when the heart rate continues past already breached alarm limits.

A 20 second waveform snapshot is captured for all arrhythmia events that trigger an alarm – see “8.1 Event capture”.

6.1.11 Measuring ST elevation and QT interval (optional)

WARNING	**ST measurements should always be verified by the clinician and/ or medically qualified personnel.**
WARNING	**Some clinical conditions may make it difficult to achieve reliable ST monitoring. For best results, consider the following: the ability of the patient to cooperate and be relaxed. Patients who are restless can produce noisy physiological signals. Noisy signals which can result in inaccurately high or low data measurements.**
WARNING	**QT measurements should always be verified by the clinician and/ or medically qualified personnel.**
WARNING	**The ST and QT measurements have not been qualified for use on pediatric or neonate patients and are therefore switched off when an age <18 is entered on the Tempus Pro.**
WARNING	**The QT algorithm has been tested for accuracy. The significance of the QT segment however needs to be determined by a clinician. This feature is to aid the clinician in monitoring the change in QT only.**
WARNING	**The ST algorithm has been tested for accuracy of the ST segment data. The significance of the ST segment changes however needs to be determined by a clinician. This feature is to aid the clinician in monitoring the change in ST and not diagnosing STEMI or Ischemic events.**
WARNING	**For very high heart rates (>250 bpm) no ST value will be reported.**
WARNING	**Some arrhythmia conditions may make it difficult to reliably produce ST measurements.**
WARNING	**With ECG waveforms with T-waves smaller than 0.1 mV, the accuracy of the QT measurement may be reduced, therefore select leads with large T-waves.**

Note	When using a 3-lead ECG cable only 1 ECG waveform is produced. Therefore, only 1 ST measurement value can be produced by the Tempus Pro. For optimum monitoring of ST, a 4- or more lead cable should be used so both ST measurements can be reported.

The Tempus Pro presents the user ST elevation (+) and depression (-) measurements as well as QT duration measurements for the electrocardiogram (ECG) monitoring waveform and displays it in numeric format on the results screen.

The Tempus makes ST and QT measurements on the 2 waveforms which are viewed on the home screen. These waveforms can be changed using the Lead Waveform selection option on the ECG menu. These measurements can be performed on any of the lead types available from the ECG cable being used. An ST measurement for each of the 2 waveforms is reported along with a single global QT measurement value.

The ST measurement is taken between the ECG isoelectric line (PR segment) and a point in the ST segment. The point taken for measurement is 80 milliseconds after the J point if the heart rate does not exceed 115 bpm and 60 milliseconds after the J point if the heart rate exceeds 115 bpm. The ST measurement value is updated every 10s. If the measurement is not available, the display will show “- - - “.

The ST value measured is a voltage; by convention the value is displayed as the equivalent measurement height for a standard ECG gain of 10mm/mV. In this convention 0.1mV will be displayed as 1.0mm. The ST measurement height gain is fixed at 10mm/mV and does not depend on the ECG gain setting.

A medium priority alarm will be triggered if the ST alarm limit is exceeded for more than 1 minute and the value will flash to indicate it is in alarm.

The QT interval is a measurement of the time between the start of the Q wave and the end of the T wave in the heart's electrical cycle. The Tempus Pro shows this measurement on the results screen. The QT value is reported in milliseconds (ms).

A medium priority alarm will be triggered if the QT alarm limit is exceeded for more than 1 minute and the value will flash to indicate it is in alarm.

6.1.12 Monitoring respiration rate with ECG cable

The Tempus Pro allows the user to monitor the patient's respiration through the ECG cable by measuring the change in the patient's impedance as their chest wall moves (impedance pneumography). This section will explain how to use this feature. The device defaults to impedance respiration being off. By turning Impedance Respiration on, it can be measured through one ECG lead on 3-, 4-, 5- or 12-lead cables, unless the Capnometer is in operation.

WARNING	**Do not operate the Tempus with any other monitor with respiration measurements on the same patient. The two devices could affect the respiration accuracy of each other.**
WARNING	**Do not use the Tempus as an apnoea monitor.**

Respiration rate from ECG leads

By default, the Tempus Pro will be set to monitor respiration rate through the ECG leads when it is supplied.

WARNING	**If capnography is used, the Tempus will always obtain its respiration rate measurement from this parameter. If capnography is not available, then respiration will be derived from the ECG leads using an impedance-based method. In this case, users are reminded that the reading represents a cyclic change in measured impedance which is interpreted to be the change of impedance as the chest moves out and in. Users should note that this is therefore an indirect reading of respiration.**
WARNING	**Impedance pneumography detects respiratory effort through changes in chest volume. However, No Breath episodes with continued respiratory effort may go undetected. Always monitor and set alarms for SpO_2 when using impedance pneumography to monitor respiratory function.**
WARNING	**With any monitor that detects respiratory effort through impedance pneumography, artefact due to patient motion, apnoea mattress shaking, or electrocautery use may cause apnoea episodes to go undetected. Always monitor and set alarms for SpO_2 when using impedance pneumography to monitor respiratory function.**
WARNING	**When using impedance pneumography, don't use the Tempus with another respiration monitor on the same patient, because the respiration measurement signals may interfere with one another.**
WARNING	**Impedance pneumography is *not* recommended for use on paced patients, because pacemaker pulses may be falsely counted as breaths.**

WARNING	**Impedance pneumography is not recommended for use with high frequency ventilation.**
WARNING	**Since impedance pneumography uses the same leads as the ECG channel, the Tempus unit determines which signals are cardiovascular artefact and which signals are the result of respiratory effort. If the breath rate is within five Percent of the heart rate, the monitor may ignore breaths and trigger a respiration alarm.**
WARNING	Impedance pneumography is not recommended for use on patients during active transport due to the impact on reliable reading which is susceptible to motion and vibrations.

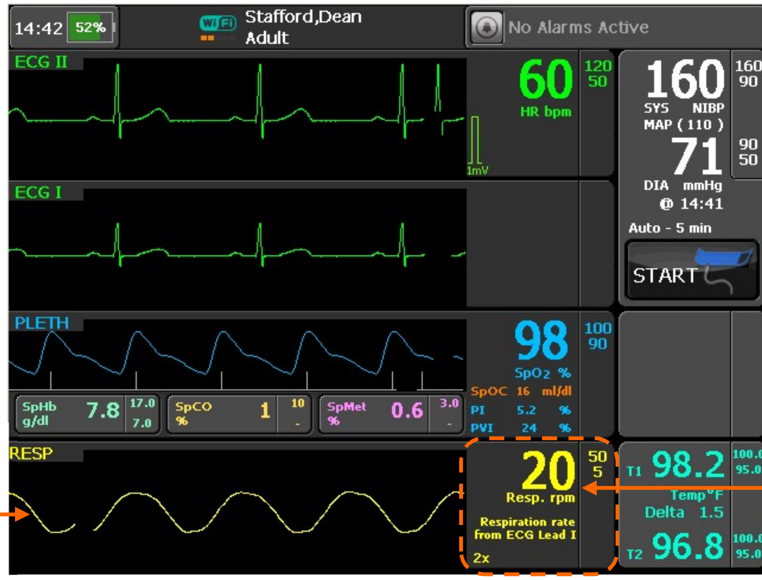
CAUTION	Users should note that in the case of supine patients, breathing may involve relatively greater movement of the abdomen (and consequently less expansion of the chest) during breathing. In this case chest impedance changes may not be significant to produce reliable readings.
CAUTION	If impedance respiration is being used, the ECG electrodes should be placed on the torso and not the limbs.
CAUTION	Impedance respiration measurements are highly subject to patient movement. The impedance changes across the chest during breathing can be easily masked by patient movement or muscle noise during movement.
CAUTION	In motion environments care should be taken to observe best practise on electrode site preparation, electrode and cable securement, securement of the Tempus and securement of the patient. RDT recommends using capnography to measure respiration in motion environments.

Impedance-based respiration monitoring is performed through the Lead II wires (RA-LL) by default. This can be changed to use Lead I (RA-LA). For the most reliable respiration measurement, users should select the lead which has the largest R-wave. Your choice depends on the ECG cable type connected and the lead chosen for ECG waveform monitoring:

ECG cable type connected	Lead selected for ECG waveform monitoring	Leads available for respiration monitoring
3-lead	Lead I	Lead I only
	Lead II	Lead II only
	Lead III	Lead I or Lead II
4-, 5- or 12-lead	Any	Lead I or Lead II (independent of ECG waveform)

Note	If capnography is in use, the waveform will automatically change to “CAPNO” and will display the end tidal CO_2. The results will change to additionally display the $ETCO_2$ reading.

Once the ECG cable is attached, respiration measurements will be started. The device will produce a reading once an average of 3 readings has been obtained. Consequently, with very low respiration rates (e.g. 5 per minute), a reading may not be obtained for 30 seconds or more. The product will as a consequence also have a slower level of responsiveness to marginal changes in respiration rates with patients with such low breathing rates.



Impedance respiration settings

Pressing anywhere on the Respiration area brings up the “Respiration Settings” menu.

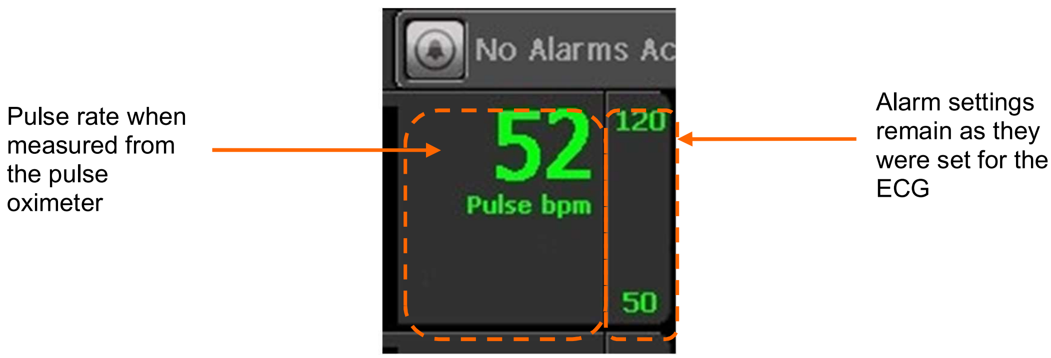
Note	If capnography is in use, the waveform shown will be labelled “CAPNO” and will display the end tidal CO_2. In this state pressing on the waveform will bring up the Capnometer Settings Menu.

Menu option	Menu screen
Respiration limits (rpm) – set the upper and lower alarm limits for respiration. **Respiration wave speed (mm/s)** – set wave speed to **3.1**, **6.25** or **12.5**. This setting is independent of other waveforms. **Respiration lead** – select which ECG lead will measure respiration rate. **Respiration wave gain** – use the up and down arrows to adjust wave height (gain). **Respiration monitoring** – turn impedance respiration off or on. Use this, for example, where patient movement is too great to allow stable impedance measurements to be taken.	

6.1.13 Monitoring heart/pulse rate with no leads

If the ECG is not attached to the patient, then the heart rate will be taken from the pulse oximeter if this is attached to the patient. Otherwise no heart rate will be shown.

Note	If the ECG is not connected but the pulse oximeter is, the heart rate will be show in the same place, in green but the signal will be derived from the pulse oximeter. In this event the reading will be labelled as shown below. In this condition the systole (heart symbol) is not shown and the systole tone will be replaced by the pulse ox tone. High and low heart rate alarms will operate as before.



Pulse rate measured from the pulse oximeter

If the ECG is inoperable then the ECG will either show a “Leads off” error for the specific Lead that is unavailable or will show a technical error as described in section “7.4 Technical alarms”.

6.2 Non-invasive blood pressure (NIBP)

WARNING	**This device should not be used when oscillometric pulses may be altered by other devices or techniques such as External Counter pulsation (ECP) or Intra-Aortic Balloon Pump Counter pulsation.**
WARNING	**DO NOT use the Blood pressure monitor for any purpose other than specified in this manual.**
WARNING	**DO NOT attach the cuff to a limb being used for IV infusions as the cuff inflation can block the infusion, potentially causing harm to the patient.**

WARNING	**You must use the right size of blood pressure cuff to suit the patient. The cuff's bladder length should be at least 80% of the limb's circumference which the cuff width should be equal to 40% of the limb's circumference. Selecting a cuff that is too small will result in measurements higher than the patient's actual blood pressure. Conversely using too large a cuff will result in too low a reading.** **The appropriate cuff sizes are:** • Disposable neonate cuff size 1 (3-6 cm), • Disposable neonate cuff size 2 (4-8 cm), • Disposable neonate cuff size 3 (6-11 cm), • Disposable neonate cuff size 4 (7-13 cm), • Disposable neonate cuff size 5 (8-15 cm), • Disposable infant cuff (8-13 cm), • ORANGE infant cuff (8-13 cm), • GREEN child (12-19 cm), • GREEN child LONG (12-19 cm), • ROYAL BLUE small adult (17-25 cm), • ROYAL BLUE small adult LONG (17-25 cm), • NAVY BLUE normal adult (23-33 cm), • NAVY BLUE normal adult LONG (23-33 cm), • BURGUNDY large adult (31-40 cm), • BURGUNDY large adult LONG (31-40 cm), • BROWN adult thigh (38-50 cm).

WARNING	**Always set the Tempus Pro to the correct patient age to ensure the initial inflation pressure is set correctly. The Tempus uses the definitions of patient age provided in IEC80601-2-30 as follows:** • **Neonate – Children 28 days old or less if born at term (based on 37-week gestation), otherwise up to 44 gestational weeks.** • **Pediatric (Child) – Children of 29 days up to and including 12 years old.** • **Adult – Individuals 13 years old or older.** **Failure to set the correct age will result in the Tempus using an inappropriate cuff inflation pressure which could cause serious harm to a neonate or infant. For pressures, see "14.1.5 Non-invasive blood pressure".** **Setting the patient age to 2 years or less will mean that inflation pressures, alarm defaults and alarm ranges will be set to those for a neonate (per IEC80601-2-30).** **If you do not set the patient age, the Tempus will use the default patient age group (see the *Tempus Pro Maintenance Manual*). Ensure that the patient age/type is always entered for non-adult patients – see "9.3.5 Entering patient details".**

WARNING	**The Tempus Pro non-invasive blood pressure monitor is not for use with pregnant, including pre-eclamptic, patients.**
WARNING	**The Tempus Pro may not operate effectively on patients who are experiencing convulsions or tremors, or who are moving or are in motion (transport) environments.**
WARNING	**Prolonged or repetitive use of the blood pressure cuff may harm skin integrity and circulatory status. Observe the limb concerned to check that circulation is not impaired.**
WARNING	**Do not attempt to take NIBP readings from patients undergoing cardiopulmonary bypass.**
WARNING	**If you suspect a non-invasive blood pressure measurement is invalid, repeat the measurement. If you remain unsure about the validity of the reading, then take another measurement using another piece of equipment or a different method.**
WARNING	**Do not apply the cuff over a wound, as this can cause further injury.**
WARNING	**Do not apply and pressurize the cuff on any limb where intravascular access or therapy such as an IV line, saline lock or an arterio-venous (A-V) shunt, is present as it may cause temporary interference with blood flow and result in injury to the patient.**
WARNING	**Do not apply and pressurize the cuff on the arm on the side of a mastectomy or lymph node clearance.**
WARNING	**Application of continuous cuff pressure due to kinking of the blood pressure hose can interfere with blood flow and result in harmful injury to the patient.**

CAUTION	Accuracy of any blood pressure measurement may be affected by the position of the subject, his or her physical condition and use outside of the operating instructions detailed in this manual. Interpretation of blood pressure measurements should be made only by a physician or trained medical staff.
CAUTION	Hoses of a certain material and/or durometer may cause the module to perform in an improper fashion. Only use hoses provided by RDT.
CAUTION	Incorrectly sized cuffs may cause measurement inaccuracy or errors.
CAUTION	If the blood pressure cuff is on the same limb as a pulse oximeter probe, the oxygen saturation results will be altered when the cuff occludes the brachial artery.
CAUTION	To obtain accurate blood pressure readings, the cuff must be the correct size, and also be correctly fitted to the patient. Incorrect size or incorrect fitting may result in incorrect readings.
CAUTION	When a cuff is going to be positioned on a patient for an extended length of time, be sure to occasionally check the limb for proper circulation.
CAUTION	Allergic exanthema (symptomatic eruption) in the area of the cuff may result, including the formation of urticaria (allergic reaction including raised edematous patches of skin or mucous membranes and intense itching) caused by the fabric material of the cuff.
CAUTION	Petechia (a minute reddish or purplish spot containing blood that appears in the skin) formation or Rumple-Leede phenomenon (multiple Petechia) on the forearm, following the application of the cuff, may lead to Idiopathic thrombocytopenia (spontaneous persistent decrease in the number of platelets associated with hemorrhagic conditions), or phlebitis (inflammation of a vein) may be observed.

CAUTION	Blood pressure readings can be affected by patient movement, speech, incorrect sizing and placement of the cuff and environmental factors (vibration, motion, noise etc.). Care should be taken to ensure that motion and other artefacts are removed or reduced when taking blood pressure readings.
CAUTION	Do not place the cuff on the same limb as the pulse oximeter sensor as the cuff may prevent the pulse oximeter sensor from reading reliably during NIBP measurement cycles.
CAUTION	Ensure the cuff is wrapped firmly around the arm; a loosely applied cuff can result in artificially high readings.

Note	Compression or restriction of the blood pressure hose or cuff, or induced movement or vibration or very low pulse volumes may prevent the monitor from taking a reading or may influence the accuracy of the reading obtained.
Note	The Tempus Pro non-invasive blood pressure monitor is designed to work with the cuffs and hoses supplied. Use of other cuffs and hoses may compromise performance and accuracy.
Note	The Tempus Pro non-invasive blood pressure monitor does not provide any specific burns protection parts for non-invasive blood pressure (in accordance with IEC80601-2-30 cl 201.7.9.2.101).

Note	
	Blood pressure results for patients with very weak pulses may be inaccurate or unavailable. The performance of the blood pressure monitor can be affected by extremes of temperature, humidity and altitude.

6.2.1 Getting started

The Tempus is intended to be used to take non-invasive blood pressure readings from neonates, children, small adults, adults and large adults. Blood pressure measurements can be affected by the position, physiologic condition, or activity level of the patient, who should be seated, standing or lying down depending on the usage model for the device or procedure being performed on the patient. If the patient is seated, they should have legs uncrossed, feet flat on the floor and back and arms supported. If the patient is lying (supine) they should not have their legs crossed.

The patient should be comfortable, relaxed as much as possible and not talking during the NIBP measurement. Before taking the first measurement, a period of at least 5 minutes should elapse to give time for blood pressure stabilization. When the pressure in the cuff increases, the patient should avoid excess movement during measurements. Let the cuffed arm hang or lay loosely, slightly away from the body. Avoid flexing the muscles or moving the hand and fingers of the cuffed arm. Avoid having the cuff or the hose next to vibrating surfaces.

The patient should be advised to relax as much as possible and not talk during the measurement procedure.

The operator should be positioned close to the device and the patient.

To take a non-invasive blood pressure reading, remove the cuff from the rear of the device. If a cuff is not attached to the hose, or the hose is not attached to the Tempus then make the necessary connections (twist and lock for the cuff and push/latch to attach the hose).

Ensure the cuff is wrapped firmly around the limb with the cuff placed centrally between the two joints of the limb i.e. 2-5 cm above the elbow joint or 5-10 cm above the knee. Ensure the end of the cuff wraps around within the two range lines – if it does not then use a different cuff. Ensure the cuff's artery marker is placed over the position of the artery e.g. between the bicep and the tricep for the brachial artery on the arm.

Ensure the hose is not kinked, crushed or damaged.

Remember to place the middle of the cuff at the same level as the right atrium of the heart. Placing the cuff substantially higher or lower than the heart will result in incorrect readings.

Remember that there may be differences in results between arms and legs, left and right side and can be affected by the patient's position (sitting, standing, lying down), exercise or their physiologic condition.

Once the cuff is attached, press the start button shown within the Blood Pressure area. The reading can be stopped at any time by pressing the button again (it will be labelled STOP).

If an unexpected reading is obtained, wait for a brief period and then take the reading again having checked the hose for leaks and kinks, the correct placement of the cuff and the patient's orientation.

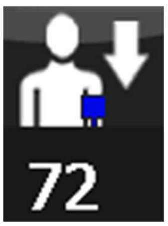
Remember that the pressure reading can be affected by common arrhythmias such as atrial or ventricular premature beats or atrial fibrillation, arteriosclerosis, poor perfusion, diabetes, age, pregnancy, pre-eclampsia, renal diseases, patient motion, trembling, shivering etc.

6.2.2 Taking readings

WARNING	**Cuff pressure displayed during the inflation/deflation cycle is not an indication of blood pressure. Blood pressure readings are displayed after the reading cycle has completed.**

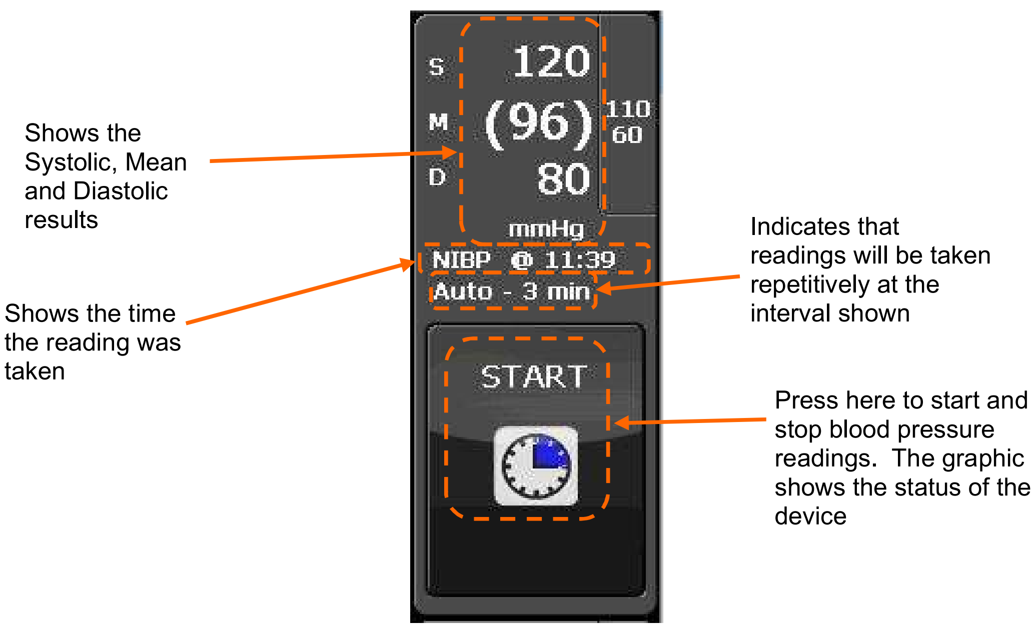
While the cuff is inflating and deflating, the icon in the start button will change to reflect the status of the reading i.e.:

- 

Cuff is inflating;
- 

Cuff is deflating.

The cuff pressure is an indicator of where the NIBP is in its cycle. Cuff pressure is not displayed when the start button is small, e.g. when three IP channels are active.

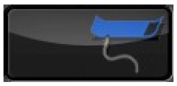


Once the measurement has been taken, the icon will change to indicate that the device is on timer as shown below:

-  Cuff is on timer;
-  Cuff is on timer;

-  Cuff is on timer;
-  Reading is about to start.

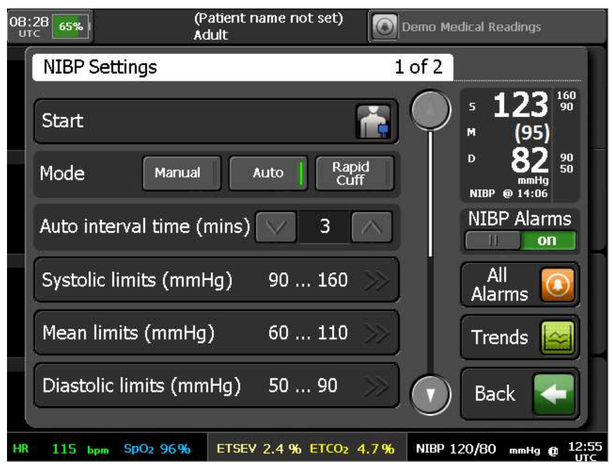
If the device is not set to take readings automatically then it will show the following:

-  Blood pressure monitor is idle.

6.2.3 NIBP settings

Pressing anywhere on the non-invasive blood pressure area brings up the “NIBP Settings” menu:

Note	The *Tempus Pro Maintenance Manual* describes how to set the Tempus to automatically clear the NIBP reading after 5 minutes if a new reading has not been taken.

Menu option	Menu screen
Start – start or stop NIBP readings. **Mode** – select **Manual**, **Auto** or **Rapid Cuff**. **Auto interval time (mins)** – when in **Auto** mode, set the time between NIBP readings. **Systolic limits**, **Mean limits** and **Diastolic limits (mmHg)** – set the lower and upper alarm limits.	
Initial inflation (mmHg) – change the initial inflation pressure. The default value is set by Tempus depending on patient age group. **Reading format** – change the format of displayed NIBP readings.	

NIBP mode

The NIBP can be set to read in three different modes:

- **Auto** – allows the user to set the Tempus to take readings automatically on a cycle (3 mins default, can be set to 2 mins, 3 mins, 5 mins, 10 mins, 15 mins, 30 mins and 60 mins).
- **Manual** – allows the user to take a single reading only every time the NIBP button is pressed.
- **Rapid Cuff** – where the Tempus will take as many readings as it can in 10 minutes and will then reset to Auto. If an NIBP technical alarm occurs while in Rapid Cuff mode, the alarm will be reported and the mode will reset to Auto.

CAUTION	Users should note that Rapid Cuff will only allow a short period between measurements for arterial recovery. Repeated use of the RapidCuff mode could have physiological effects including effects on the readings obtained.

Initial inflation

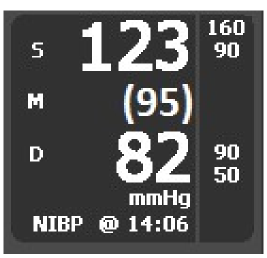
The user can adjust the initial NIBP inflation pressure. The default is set by the Tempus depending on patient age group – see "9.3.5 Entering patient details".

NIBP adjusts its inflation pressure dynamically between readings so this setting only applies to the first reading and when it is manually adjusted.

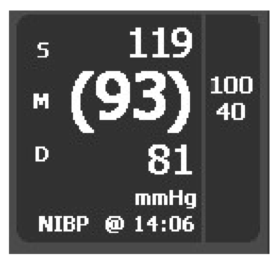
Patient Type	Default initial inflation	Range
Adult	160 mmHg	120 to 280 mmHg
Pediatric (29 days to 12 years)	140 mmHg	80 to 280 mmHg
Neonate (<29 days)	90 mmHg	60 to 140 mmHg

NIBP reading format

The **Reading format** control allows you to choose between two formats for displaying NIBP readings:



NIBP reading format S/D (M)



NIBP reading format (M) S/D

- S = systolic
- D = diastolic
- M = mean

Note	Regardless of the reading format chosen, the Tempus Pro will continue to monitor the systolic, mean and diastolic pressures and will still output alarms when thresholds are exceeded.

6.3 Pulse oximetry

WARNING	**Do not use this device in the presence of high EMI/RFI radiation. High EMI/RFI radiation may cause induced current to the SpO_2 sensor resulting in patient injury.**
WARNING	**Verify the compatibility of the monitor, probe and cable before use, otherwise patient injury can result.**
WARNING	**This device may give inaccurate readings in the presence of strong electromagnetic sources, such as electrosurgery equipment.**
WARNING	**This device may give inaccurate readings in the presence of computed tomography (CT) equipment.**
WARNING	**Prolonged use or the patient's condition may require changing the sensor site periodically. Change sensor site and check skin integrity, circulatory status, and correct alignment at least every 4 hours. Prolonged use may cause blisters, skin deterioration, and discomfort.**
WARNING	**Incorrectly applied sensors may give inaccurate readings.**
WARNING	**SpO_2 measurements may be inaccurate in the presence of high ambient light. Shield the sensor area (with a towel, for example) if necessary.**
WARNING	**Any condition that restricts blood flow, such as use of a blood pressure cuff or extremes in systemic vascular resistance, may cause an inability to determine accurate pulse rate and SpO_2 readings.**
WARNING	**Remove fingernail polish or false fingernails before applying SpO_2 sensors. Fingernail polish or false fingernails may cause inaccurate SpO_2 readings.**
WARNING	**Significant levels of dysfunctional hemoglobins, such as carboxyhemoglobin or methemoglobin, will affect the accuracy of the SpO_2 measurement.**
WARNING	**Tissue damage may result from overexposure to sensor light during photodynamic therapy with agents such as verteporphin, porfimer sodium and metatetrahydroxyphenylchlorin (mTHPC). Change the sensor site at least every hour and observe for signs of tissue damage. More frequent sensor site changes or inspections may be indicated depending upon the photodynamic agent used, agent dose, skin condition, total exposure time or other factors. Use multiple sensor sites.**
WARNING	**Optical cross-talk can occur when two or more sensors are placed in close proximity. It can be eliminated by covering each site with opaque material. Optical cross-talk may adversely affect the accuracy of the SpO_2 readings.**
WARNING	**Obstructions or dirt on the sensor's red light or detector may cause a sensor failure or inaccurate readings. Make sure there are no obstructions and the sensor is clean.**

WARNING	**Under certain clinical conditions, pulse oximeters may not display SpO_2 and/or pulse rate values. Under these conditions, pulse oximeters may also display erroneous values. These conditions include, but are not limited to: patient motion, low perfusion, cardiac arrhythmias, high or low pulse rates or a combination of the above conditions. Failure of the clinician to recognize the effects of these conditions on pulse oximeter readings may result in patient injury.**

WARNING	**Do not use the pulse oximeter for any other purpose than specified in this manual. It should NOT be used as an apnoea monitor.**
WARNING	**Pulse rate measurement is based on the optical detection of a peripheral flow pulse and therefore may not detect certain arrhythmias. The pulse oximeter should not be used as a replacement or substitute for ECG based arrhythmia analysis.**
WARNING	**A pulse oximeter should be considered an early warning device. As a trend towards patient deoxygenation is indicated, blood samples should be analyzed by a laboratory co-oximeter to completely understand the patient's condition.**
WARNING	**Tissue damage can be caused by incorrect application or use of the sensor, for example by wrapping the sensor too tightly. Inspect the sensor site as directed in the sensor *Directions for Use* to ensure skin integrity and correct positioning and adhesion of the sensor.**
WARNING	**Interfering Substances: Carboxyhemoglobin may erroneously increase readings. The level of increase is approximately equal to the amount of carboxyhemoglobin present.**
WARNING	**Reposition the oximeter probe at least once every hour to allow the patient's skin to respire. This time should be reduced in ambient temperatures over 41 °C. No other special actions are required for use in such ambient temperatures although users are advised to check the sensor site frequently in high temperatures to avoid skin damage.**
WARNING	**The SpO_2 sensor should snugly fit the finger without straining it and if not alternative fingers should be tried. The probe is sized for patients who weigh 20 kg /44 lbs or more. The probe can be used either on patient's fingers, thumbs or largest toe.**
WARNING	**The plethysmogram may become unstable on patients who are experiencing convulsions or tremors, who are moving or are in motion (transport) environments.**
WARNING	**Always read and follow the instructions on the packaging of SpO_2 sensors regarding their storage, use and disposal. In particular take into account any information regarding toxicity.**
WARNING	**For measurements of high or low SpHb readings, blood samples should be analyzed by laboratory instruments to completely understand the patient's condition.**
WARNING	**A pulse oximeter cannot measure elevated levels of COHb or MetHb. Increases in either COHb or MetHb will affect the accuracy of the SpO2 measurement.** **For increased COHb: COHb levels above normal tend to increase the level of SpO2. The level of increase is approximately equal to the amount of COHb that is present.** **For increased MetHb: the SpO2 may be decreased by levels of MetHb of up to approximately 10% to 15%. At higher levels of MetHb, the SpO2 may tend to read in the low to mid 80s. When elevated levels of MetHb are suspected, laboratory analysis (CO-Oximetry) of a blood sample should be performed.**

WARNING	**Hemoglobin synthesis disorders may cause erroneous SpHb readings.**
WARNING	**Elevated levels of Total Bilirubin may lead to inaccurate SpO2, SpMet, SpCO, SpHb, and SpOC measurements.**
WARNING	**Motion artefact may lead to inaccurate SpMet, SpCO, SpHb, and SpOC measurements.**
WARNING	**Severe anaemia may cause erroneous SpO2 readings.**
WARNING	**Very low arterial Oxygen Saturation (SpO2) levels may cause inaccurate SpCO and SpMet measurements.**
WARNING	**With very low perfusion at the monitored site, the readings may read lower than core arterial oxygen saturation.**
WARNING	**Do not use tape to secure the sensor to the site; this can restrict blood flow and cause inaccurate readings.**
WARNING	**Use of additional tape can cause skin damage or damage the sensor.**
WARNING	**If the sensor is wrapped to tightly or supplemental tape is used, venous congestion/pulsations may occur causing erroneous readings.**
WARNING	**Venous congestion may cause under reading of actual arterial oxygen saturation. Therefore, assure proper venous outflow from monitored site. Sensor should not be below heart level (e.g. sensor on hand of a patient in a bed with arm dangling to the floor).**
WARNING	**Venous pulsations may cause erroneous low readings (e.g. tricuspid valve regurgitation).**
WARNING	**Loss of pulse signal can occur when:** • The sensor is too tight. • The patient has hypotension, severe vasoconstriction, severe anaemia, or hypothermia. • There is arterial occlusion proximal to the sensor. • The patient is in cardiac arrest or is in shock.
WARNING	**The pulsations from intra-aortic balloon support can be additive to the pulse rate on the oximeter pulse rate display.**
WARNING	**Verify patient's pulse rate against the ECG heart rate.**

WARNING	**Misapplied sensors or sensors that become partially dislodged may cause either over or under reading of actual arterial oxygen saturation.**
WARNING	**Avoid placing the sensor on any extremity with an arterial catheter or blood pressure cuff.**
WARNING	**Before use, carefully read the sensor's *Directions for Use*.**
WARNING	**To avoid cross contamination only use Masimo single use sensors on the same patient.**
WARNING	**Unless otherwise specified, do not sterilize sensors or patient cables by irradiation, steam, autoclave or ethylene oxide.**
WARNING	**See the cleaning instructions in the directions for use for the Masimo re-useable sensors.**

CAUTION	Unplug the sensor from the monitor before cleaning or disinfecting to prevent damaging sensor or monitor, and to prevent user safety hazards.

CAUTION	Users are reminded that because pulse oximeter measurements are statistically distributed, only about two-thirds of measurements can be expected to fall within ±Arms of the value measured by a co-oximeter. Information on the clinical desaturation trials performed to validate the oximeter is available on request.
CAUTION	Use only Masimo oximetry sensors for SpO_2 measurements. Other oxygen transducers (sensors) may cause improper results.
CAUTION	Do not use the Pulse CO-Oximeter or oximetry sensors during magnetic resonance imaging (MRI) scanning. Induced current could potentially cause burns. The Pulse CO-Oximeter may affect the MRI image, and the MRI unit may affect the accuracy of the oximetry measurements.
CAUTION	Exercise caution when applying a sensor to a site with compromised skin integrity. Applying tape or pressure to such a site may reduce circulation and/or cause further skin deterioration.
CAUTION	Circulation distal to the sensor site should be checked routinely.

Note	SpO_2 averaging is the number of pulse beats over which the SpO_2 value is averaged; pulse averaging is the number of seconds over which the pulse value is averaged.
Note	DESAT trails were performed in the normal sensitivity mode.

Note	The plethysmogram is not normalized.
Note	The pulse oximeter probe is for use with intact skin only. The probe used is not sterile and contains no latex. Patient contact materials have undergone extensive biocompatibility testing; further information is available on request.
Note	Any condition that restricts blood flow, such as use of a blood pressure cuff (other than the Tempus Pro cuff used in accordance with the instructions herein) may cause an inability to determine accurate pulse and SpO_2 readings.
Note	SpO_2 measurements may be adversely affected in the presence of high ambient light levels (e.g. strong sunlight). This may be more noticeable with disposable probes. If necessary, shield the sensor area (e.g. with a towel) or a Masimo® light shield.
Note	Remove fingernail polish or false fingernails before applying SpO_2 sensors. Fingernail polish or false fingernails may cause inaccurate SpO_2 readings.
Note	Performance and safety test data are available on request from RDT, see “RDT contact details”.
Note	The graphical displays of pulse rate, SpO_2 and pulse strength are not proportional to the pulse volume. The amplitude of the waveform is adjusted on an on-going basis to provide the largest size waveform possible. Do not attempt to normalize the waveform to any scale.
Note	The SpO2$_2$ sensor should be on the opposite arm to the blood pressure cuff. The arm of the patient must be kept still and either be horizontal to the shoulder (if the patient is lying down) or below the shoulder (if the patient is sitting upright). If the finger selected does not give good results, this could be due to poor perfusion of blood. Ensure that the finger is inserted all way into the clip, or try taking a reading on another finger.
Note	If the accuracy of any measurement does not seem reasonable, first check the patient’s vital signs by alternate means and the check the MS board pulse oximeter for proper functioning.
Note	NO IMPLIED LICENSE - Possession or purchase of this device does not convey any express or implied license to use the device with unauthorized sensors or cables that would, alone or in combination with this device, fall within the scope of one or more of the patents relating to this device.
Note	Inaccurate measurements may be caused by: • Incorrect sensor application or use • Significant levels of dysfunctional hemoglobins. (e.g., carboxyhemoglobin or methemoglobin) • Intravascular dyes such as indocyanine green or methylene blue. • Exposure to excessive illumination, such as surgical lamps (especially ones with a xenon light source), bilirubin lamps, fluorescent lights, infrared heating lamps, or direct sunlight (exposure to excessive illumination can be corrected by covering the sensor with a dark or opaque material) • Excessive patient movement. • Venous pulsations. • Placement of a sensor on an extremity with a blood pressure cuff, arterial catheter, or intravascular line.

Note	Loss of pulse signal can occur in any of the following situation: • The sensor is too tight. • There is excessive illumination from light sources such as a surgical lamp, a bilirubin lamp, or sunlight. • A blood pressure cuff is inflated on the same extremity as the one with a SPO_2 sensor attached. • The patient has hypotension, severe vasoconstriction, severe anaemia, or hypothermia. • There is arterial occlusion proximal to the sensor. • The patient is in cardiac arrest or is in shock.
Note	High levels of COHb may occur with a seemingly normal SpO2. When elevated levels of COHb are suspected, laboratory analysis (CO-Oximetry) of a blood sample should be performed.

The following warnings, cautions and notes are reproduced verbatim from Masimo document R-CSD-1117 (revision P):

GENERAL

The pulse co-oximeter is to be operated by, or under the supervision of, qualified personnel only. The manual, accessories, directions for use, all precautionary information, and specifications should be read before use.

WARNING

- As with all medical equipment, carefully route patient cabling to reduce the possibility of patient entanglement or strangulation.
- Do not place the pulse co-oximeter or accessories in any position that might cause it to fall on the patient.
- Do not start or operate the pulse co-oximeter unless the setup was verified to be correct.
- Do not use the pulse co-oximeter during magnetic resonance imaging (MRI) or in an MRI environment.
- Do not use the pulse co-oximeter if it appears or is suspected to be damaged.
- Explosion hazard: Do not use the pulse co-oximeter in the presence of flammable anesthetics or other flammable substance in combination with air, oxygen-enriched environments, or nitrous oxide.
- To ensure safety, avoid stacking multiple devices or placing anything on the device during operation.
- To protect against injury, follow the directions below:
    - Avoid placing the device on surfaces with visible liquid spills.
    - Do not soak or immerse the device in liquids.
    - Do not attempt to sterilize the device.
    - Use cleaning solutions only as instructed in this operator's manual.
    - Do not attempt to clean the device while monitoring a patient.
- To protect from electric shock, always remove the sensor and completely disconnect the pulse co-oximeter before bathing the patient.
- If any measurement seems questionable, first check the patient’s vital signs by alternate means and then check the pulse co-oximeter for proper functioning
- Inaccurate SpCO and SpMet readings can be caused by:
    - Improper sensor application
    - Intravascular dyes such as indocyanine green or methylene blue
    - Abnormal hemoglobin levels
    - Low arterial perfusion
    - Low arterial oxygen saturation levels including altitude induced hypoxemia
    - Elevated total bilirubin levels
    - Motion artifact

WARNING

- Inaccurate SpHb and SpOC readings may be caused by:
    - Improper sensor application
    - Intravascular dyes such as indocyanine green or methylene blue
    - Externally applied coloring and texture such as nail polish, acrylic nails, glitter, etc.
    - Elevated PaO2 levels
    - Elevated levels of bilirubin
    - Low arterial perfusion
    - Motion artifact
    - Low arterial oxygen saturation levels
    - Elevated carboxyhemoglobin levels
    - Elevated methemoglobin levels
    - Hemoglobinopathies and synthesis disorders such as thalassemias, Hb s, Hb c, sickle cell, etc.
    - Vasospastic disease such as Raynaud's
    - Elevated altitude
    - Peripheral vascular disease
    - Liver disease
    - EMI radiation interference
- Inaccurate SpO2 readings may be caused by:
    - Improper sensor application and placement
    - Elevated levels of COHb or MetHb: High levels of COHb or MetHb may occur with a seemingly normal SpO2. When elevated levels of COHb or MetHb are suspected, laboratory analysis (CO-Oximetry) of a blood sample should be performed.
    - Elevated levels of bilirubin
    - Elevated levels of dyshemoglobin
    - Vasospastic disease, such as Raynaud’s, and peripheral vascular disease
    - Hemoglobinopathies and synthesis disorders such as thalassemias, Hb s, Hb c, sickle cell, etc.
    - Hypocapnic or hypercapnic conditions
    - Severe anemia
    - Very low arterial perfusion
    - Extreme motion artifact
    - Abnormal venous pulsation or venous constriction
    - Severe vasoconstriction or hypothermia
    - Arterial catheters and intra-aortic balloon
    - Intravascular dyes, such as indocyanine green or methylene blue
    - Externally applied coloring and texture, such as nail polish, acrylic nails, glitter, etc.
    - Birthmark(s), tattoos, skin discolorations, moisture on skin, deformed or abnormal fingers. etc.
    - Skin color disorders

WARNING	• Interfering Substances: Dyes or any substance containing dyes that change usual blood pigmentation may cause erroneous readings. • The pulse co-oximeter should not be used as the sole basis for diagnosis or therapy decisions. It must be used in conjunction with clinical signs and symptoms. • The pulse co-oximeter is not an apnoea monitor. • The pulse co-oximeter may be used during defibrillation, but this may affect the accuracy or availability of the parameters and measurements. • The pulse co-oximeter may be used during electrocautery, but this may affect the accuracy or availability of the parameters and measurements. • The pulse co-oximeter should not be used for arrhythmia analysis. • SpCO readings may not be provided if there are low arterial saturation levels or elevated methemoglobin levels. • SpO2, SpCO, SpMet, and SpHb are empirically calibrated in healthy adult volunteers with normal levels of carboxyhemoglobin (COHb) and methemoglobin (MetHb). • Do not adjust, repair, open, disassemble, or modify the pulse co-oximeter or accessories. Injury to personnel or equipment damage could occur. Return the pulse co-oximeter for servicing if necessary.

CAUTION	• Do not place the pulse co-oximeter where the controls can be changed by the patient. • Electrical shock and flammability hazard: Before cleaning, always turn off the device and disconnect from any power source. • When patients are undergoing photodynamic therapy they may be sensitive to light sources. Pulse oximetry may be used only under careful clinical supervision for short time periods to minimize interference with photodynamic therapy. • Do not place the pulse co-oximeter on electrical equipment that may affect the device, preventing it from working properly. • If SpO2 values indicate hypoxaemia, a laboratory blood sample should be taken to confirm the patient's condition. • If the Low Perfusion message is frequently displayed, find a better perfused monitoring site. In the interim, assess the patient and, if indicated, verify oxygenation status through other means. • Change the application site or replace the sensor and/or patient cable when a "Replace sensor" and/or "Replace patient cable", or a persistent poor signal quality message (such as "Low SIQ") is displayed on the host monitor. These messages may indicate that patient monitoring time is exhausted on the patient cable or sensor. • If using pulse oximetry during full body irradiation, keep the sensor out of the radiation field. If the sensor is exposed to the radiation, the reading might be inaccurate or the device might read zero for the duration of the active irradiation period. • The device must be configured to match your local power line frequency to allow for the cancelation of noise introduced by fluorescent lights and other sources. • To ensure that alarm limits are appropriate for the patient being monitored, check the limits each time the pulse co-oximeter is used.

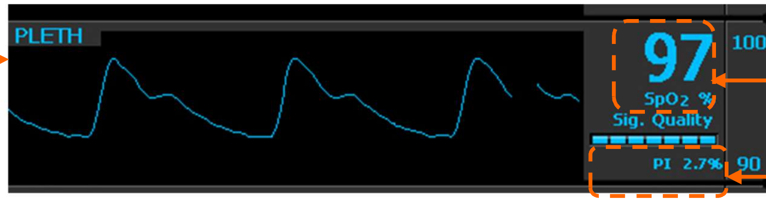
CAUTION	• Variation in hemoglobin measurements may be profound and may be affected by sampling technique as well as the patient's physiological conditions. Any results exhibiting inconsistency with the patient's clinical status should be repeated and/or supplemented with additional test data. Blood samples should be analyzed by laboratory devices prior to clinical decision making to completely understand the patient's condition. • Do not submerge the pulse co-oximeter in any cleaning solution or attempt to sterilize by autoclave, irradiation, steam, gas, ethylene oxide or any other method. This will seriously damage the pulse co-oximeter. • Electrical Shock Hazard: Carry out periodic tests to verify that leakage currents of patient-applied circuits and the system are within acceptable limits as specified by the applicable safety standards. The summation of leakage currents must be checked and in compliance with IEC 60601-1 and UL60601-1. The system leakage current must be checked when connecting external equipment to the system. When an event such as a component drop of approximately 1 meter or greater or a spillage of blood or other liquids occurs, retest before further use. Injury to personnel could occur. • Disposal of product - Comply with local laws in the disposal of the device and/or its accessories. • To minimize radio interference, other electrical equipment that emits radio frequency transmissions should not be in close proximity to the pulse co-oximeter. • Replace the cable or sensor when a replace sensor or when a low SIQ message is consistently displayed while monitoring consecutive patients after completing troubleshooting steps listed in this manual.

Note	• A functional tester cannot be used to assess the accuracy of the pulse co-oximeter. • High-intensity extreme lights (such as pulsating strobe lights) directed on the sensor, may not allow the pulse co-oximeter to obtain vital sign readings. • When using the Maximum Sensitivity setting, performance of the "Sensor Off" detection may be compromised. If the device is in this setting and the sensor becomes dislodged from the patient, the potential for false readings may occur due to environmental "noise" such as light, vibration, and excessive air movement. • Do not loop the patient cabling into a tight coil or wrap around the device, as this can damage the patient cabling. • Additional information specific to the Masimo sensors compatible with the pulse oximeter, including information about parameter/measurement performance during motion and low perfusion, may be found in the sensor's directions for use (DFU). • Cables and sensors are provided with X-Cal™ technology to minimize the risk of inaccurate readings and unanticipated loss of patient monitoring. Refer to the Cable or Sensor DFU for the specified duration of the patient monitoring time.

6.3.1 Getting started

To use the pulse oximeter, remove the soft finger probe from the back of the device. Attach it to a well-perfused finger of the patient taking care to ensure it is not the same arm which has a blood pressure cuff attached. The readings will begin a few seconds after the probe is attached to the patient.

The plethysmogram is shown continuously



The SpO_2 reading is provided shown here

The signal strength and perfusion index of the site are shown here

Pulse oximetry monitoring

The SpO_2 section gives the oxygen saturation of the blood and shows the plethysmogram, signal strength (in bar graph form) and the perfusion index.

The Signal Quality bar graph shows how well the pulse sensor is detecting the pulse. The amplitude of the indication indicates the quality of detection. If the indication on the Signal Strength meter is low, or becomes low, then the finger sensor should be repositioned. Similarly, the Perfusion Index gives a numerical indication of the level of arterial pulsatile blood at the sensor site.

The Pulse Rate displayed on the Tempus Pulse Oximeter may differ slightly from the heart rate displayed on ECG monitors due to differences in averaging times. There may also be a discrepancy between cardiac electrical activity and peripheral arterial pulsation. Significant differences may indicate a problem with the signal quality due to physiological changes in the patient or one of the instruments or application of the sensor or patient cable. The pulsations from intra-aortic balloon support can cause the pulse rate displayed on the Tempus to be significantly different than the ECG heart rate.

The perfusion index (PI) indicator provides a relative numeric indication of the pulse strength at the monitoring site. It is a calculated percentage between the pulsatile signal and non-pulsatile signal of arterial blood moving through the site. PI may be used to find the best perfused site and to monitor physiological changes in the patient. It displays an operating range of 0.02 % to 20.00 %. A percentage greater than 1.00 % is desired. Extreme changes in the display number are due to motion artefact and changes in physiology and blood flow.

It has been suggested that at extremely low perfusion levels, pulse oximeters can measure peripheral saturation, which may differ from central arterial saturation. This “localized hypoxemia” may result from the metabolic demands of other tissues extracting oxygen proximal to the monitoring site under conditions of sustained peripheral hypoperfusion. (This may occur even with a pulse rate that correlates with the ECG heart rate.)

CAUTION ⚠	If the low perfusion indication is frequently displayed, find a better-perfused monitoring site. In the interim assess the patient and if indicated verify oxygenation status through other means.

6.3.2 Pleth variability index

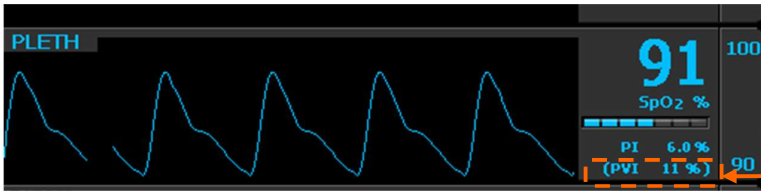
Pleth Variability Index or PVI® helps users to discern between patients who may respond to fluid from those who may not.

PVI has been shown to help clinician to predict fluid responsiveness in mechanically ventilated patients under general anesthesia, defined as a significant increase in cardiac output after fluid administration. PVI is an automatic measure of the dynamic change in Perfusion Index (PI) that occurs during the respiratory cycle.

$PVI = ((PI_{max}-PI_{min})/PI_{max}) \times 100$.

The greater the PVI, the more likely the patient will respond to fluid administration. Typically, a PVI of >14% prior to volume expansion is predictive that a mechanically ventilated patient will respond to fluid administration (81% sensitivity). A PVI prior to volume expansion is predictive that a mechanically ventilated patient will not respond to fluid administration (100% specificity).

PVI® is an option for Tempus Pro.



The PVI value is shown here

PVI value

6.3.3 Taking readings

Note	Depending on which of the features have been purchased, some of the features shown below may not be available on your Tempus Pro device.

1. Remove the patient cable from the RapidPak area of the Tempus.
2. Connect the SpO_2 cable to the Tempus Pro.
3. Attach the chosen sensor to the patient cable:

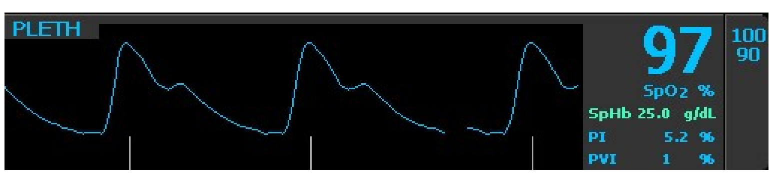
   SpO2 sensor - SpO2, PI and PVI

   SpCO sensor - SpCO, SpMet, Sp02, PI and PVI

   SpHb sensor - SpHb, SpMet, SpOC, Sp02, PI and PVI

Attach it to a well-perfused finger of the patient taking care to ensure it is not the same arm which has a blood pressure cuff attached. The measurement will begin as soon as the probe is attached to the patient but different values take different lengths of time to appear on the screen.

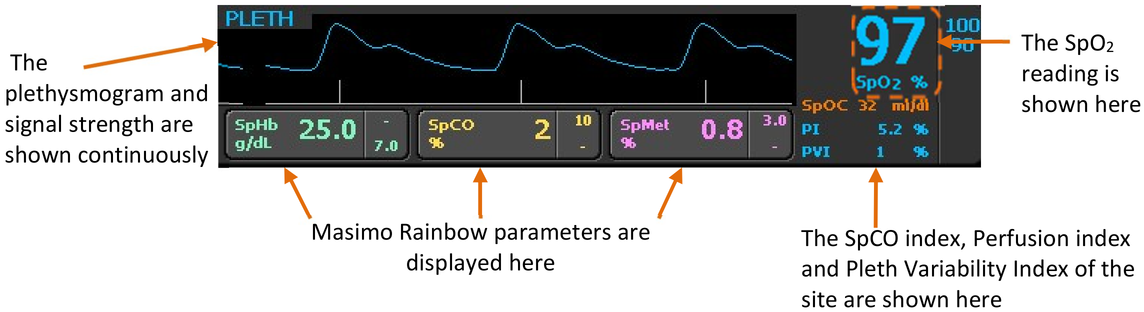
Standard waveform view



Large waveform view

Note	The vertical lines under the pleth waveform show the signal strength.

6.3.4 Features



<u>Pulse oximetry features</u>

Depending on which features have been purchased for your Masimo pulse oximetry and which sensor you are using will depend on which of the following features the Tempus Pro can monitor.

Masimo SET measurements supplied as standard:

- **SpO_2** - Intended for non-invasive measurement of arterial blood saturation and also provides pulse rate when ECG is not attached.
- **Pulse Rate (PR)**.
- **Perfusion Index (PI)** - Gives a numerical indication of the level of arterial pulsatile blood at the sensor site.

Masimo rainbow SET measurements purchased as optional upgrades:

- **SpCO** - Allows clinicians to noninvasively and immediately detect elevated levels of carbon monoxide.
- **SpMet** - Allows clinicians to noninvasively and immediately detect elevated levels of methemoglobin in the blood.
- **SpHb Index** - Gives continuous monitoring of the hemoglobin levels in blood.
- **SpOC** - Gives the patient’s oxygenation status by calculating the hemoglobin and oxygen saturation.

PVI purchased as an optional upgrade:

- **Pleth Variability Index (PVI)** - Helps clinicians noninvasively and continuously assess fluid status of patients.

The Signal Quality waveform (underneath the plethysmogram) shows how well the pulse sensor is detecting the pulse. The height of the peak indicates the strength of the signal, if the peaks are short the finger sensor should be repositioned.

The Pulse Rate displayed on the Tempus Pulse Oximeter may differ slightly from the heart rate displayed on ECG monitors due to differences in averaging times. There may also be a discrepancy between cardiac electrical activity and peripheral arterial pulsation. Significant differences may indicate a problem with the signal quality due to physiological changes in the patient or one of the instruments or application of the sensor or patient cable. The pulsations from intra-aortic balloon support can cause the pulse rate displayed on the Tempus to be significantly different than the ECG heart rate.

The device indicates perfusion. It has been suggested that at extremely low perfusion levels, pulse oximeters can measure peripheral saturation, which may differ from central arterial saturation. This “localized hypoxemia” may result from the metabolic demands of other tissues extracting oxygen proximal to the monitoring site under conditions of sustained peripheral hypoperfusion. (This may occur even with a pulse rate that correlates with the ECG heart rate).

CAUTION	If the low perfusion indication is frequently displayed, find a better-perfused monitoring site. In the interim assess the patient and if indicated verify oxygenation status through other means.

6.3.5 Pulse oximetry probes

The following pulse oximetry sensors may be used with the Tempus Pro: standard, SpCO or SpHb. Each sensor shows a different range of parameters. There is no single sensor that shows all parameters.

The standard adult SpO2 reusable sensor reads SpO2, PI and PVI

The standard adult SpO2 section

The SpCO sensor reads SpCO, SpMet, SpO2, PI and PVI

The pulse oximetry section with SpCO

The SpHb sensor reads SbHb, SpMet, SpOC, SpO2, PI and PVI

The pulse oximetry section with SpHb

Note	Depending on which sensor is used the Tempus will display only the available parameters.

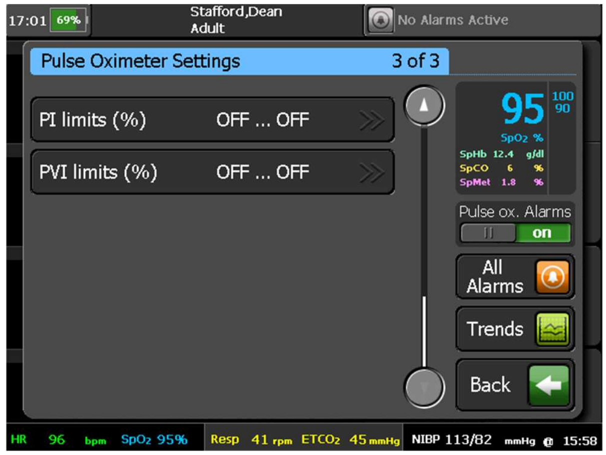
6.3.6 Pulse oximeter settings

Pressing anywhere in the plethysmogram area brings up the “Pulse Oximeter Settings” menu:

The SpCO, SpMet, SpHb and SpOC limits are only activated when their Masimo licenses have been installed and the appropriate probes are plugged in.

Note	PI and PVI alarms are only available in Tempus Pro devices with part numbers ending in "-R".

Menu option	Menu screen
SpO2, SpCO and SpMet limits (%) – set the upper and lower alarm limits for SpO2, SpCO and SpMet. **SpHb limits and settings** – open the SpHb Settings menu, see “6.3.7 SpHb settings”. **HR/PR source** – select **ECG** or **Pulse** oximeter as the heart/pulse rate source. **Large pleth waveform** – turn the large pleth waveform display on and off.	
Sensitivity mode – adjust sensitivity by pressing **Max**, **Norm** or **APOD**. **Averaging time (s)** – use the up and down arrows to adjust averaging time in seconds. **FastSAT** – turn FastSAT on or off. **Cardiac wave speed (mm/s)** – change wave speed. This setting is common to ECG, Sp02 and IP. **HR/PR beat volume** - volume of the audible tone that the Tempus emits every time a heartbeat is detected. **SpOC limits (ml/dl)** – set the upper and lower alarm limits for SpOC.	

Menu option	Menu screen
PI limits (%) – set the upper and lower alarm limits for Perfusion Index. **PVI limits (%)** – set the upper and lower alarm limits for Pleth Variability Index.	

Pulse tones

Pulse tones can be turned on from either the “Pulse Oximeter Settings” menu or the “ECG Settings” menu using the settings:

- HR/PR source = pulse
- HR/PR beat volume = 20% (or above)

When pulse tones are turned on Tempus Pro emits a beep for each pulse beat detected. The beep tone (frequency) depends on the value of pulse ox saturation (SpO2). Higher SpO2 values mean that each beat is emitted with a higher tone beep and vice-versa. So the tone drops as SpO2 (%) falls, with the lowest frequency tone at values of SpO2 = 80% or below.

Note	Alarm tones have priority over pulse tones. When pulse tones are enabled, user interface touch screen sounds are suppressed.
Note	Pulse tones are automatically enabled if the laryngoscope is used.

Averaging time

The pulse oximeter can be set to average its heart rate reading over different time periods. By default, this is 8 seconds. This can be adjusted to make the reading more or less responsive. The available settings are: (2-4 seconds, 4-6 seconds, 8 seconds, 10 seconds, 12 seconds, 14 seconds and 16 seconds).

Sensitivity mode

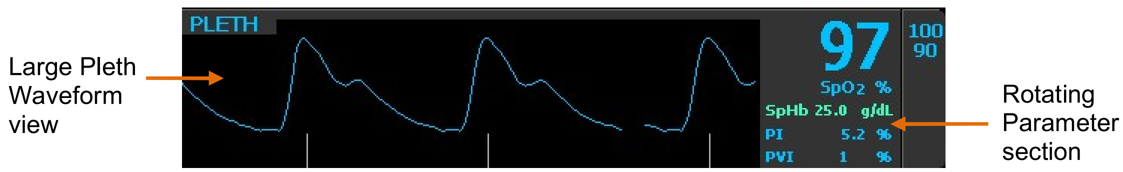
The Pulse Oximeter is equipped with 3 different sensitivity modes. Each mode allows the clinician to change the sensitivity settings of the device to meet the increased demands of the patient’s physiological condition or enable it to work during periods of low perfusion and/or motion. They are as follows:

- **Normal Sensitivity (Norm)** – This is the recommended mode for patients that are experiencing some compromise in blood flow or perfusion. It is advisable for care areas where patients are observed frequently, such as an intensive care unit (ICU).
- **Adaptive Probe Off Detection (APOD)** – This is the recommended start-up monitoring mode for most patients with acceptable perfusion or where a more robust sensor off detection is desired. It is the suggested mode for care areas where patients are not visually monitored continuously. This mode delivers enhanced protection against erroneous pulse rate and arterial oxygen saturation readings when a sensor becomes inadvertently detached from a patient.

- **Maximum Sensitivity (Max)** - This mode is recommended for patients with low perfusion or when the low perfusion or low signal quality message is displayed on the screen in APOD or normal sensitivity mode. This mode is not recommended for care areas where patients are not monitored visually, such as general wards. It is designed to interpret and display data at the measuring site when the signal may be weak due to decreased perfusion. When a sensor becomes detached from a patient, it will have compromised protection against erroneous pulse rate and arterial saturation readings. Also, after a power off and on cycle, the sensitivity will change from the MAX to the factory default or user configured default setting of APOD or NORM.

Large pleth waveform

Large pleth waveform changes the display to show a bigger waveform. This will remove the rainbow SET measurement parameters from the waveform section of the home screen and move them to the tile on the right. The parameters will then rotate every 5 seconds.



Large pleth waveform section of the home screen

FastSAT

FastSAT enables rapid tracking of arterial oxygen saturation changes. Arterial oxygen saturation data is averaged using pulse oximeter averaging algorithms to smooth the trend. When FastSAT is “On” the averaging algorithm evaluates all the saturation values providing an averaged saturation value that is a better representation of the patient’s current oxygenation status. With FastSat, the averaging time is dependent on the input signal. For the 2 and 4 second settings the averaging time may range from 2-4 and 4-6 seconds, respectively.

6.3.7 SpHb settings

Pressing the SpHb settings button (if activated) brings up the “SpHb Settings” screen:

Menu option	Menu screen
SpHb limits (g/dL) – set the upper and lower alarm limits for SpHb. **SpHb mode** – select **Arterial** or **Venous**. **SpHb averaging** – select **Short, Medium or Long**. **Sensor life remaining** – shows the remaining life of the pulse ox sensor.	

Sensor life remaining

Sensor life remaining (mins) shows the life of the sensor. A tech alarm will sound to warn when the remaining life is low. If the patient is still being monitored when the sensor life reaches zero, it will continue to monitor the patient but must be changed once monitoring has finished. A further tech alarm will sound once monitoring of patient has finished.

Note	If monitoring is complete, remember to disconnect the sensor from the patient cable as this may impact the life expectancy of the sensor.

6.4 Capnography

The capnography component of the Tempus unit is covered by one or more of the following US patents: 6,428,483; 6,997,880; 6,437,316; 7,488,229, 7,726,954; and their foreign equivalents. Additional patent applications pending.

NO IMPLIED LICENSE

Possession or purchase of this device does not convey any express or implied license to the device with unauthorized consumable CO_2 sampling consumable products, which would, alone, or in combination with this device, fall within the scope of one or more of the patents relating to this device and/or CO_2 sampling consumable products.

WARNING ⚠	**To ensure patient safety, do not place the monitor in any position that might cause it to fall on the patient.**
WARNING ⚠	**To ensure accurate performance and prevent device failure, do not expose the monitor to extreme moisture, such as rain.**
WARNING ⚠	**The monitor is a prescription device and is to be operated by qualified healthcare personnel.**
WARNING ⚠	**The Tempus Pro is not for apnoea detection. The Tempus Pro is not an apnoea monitor and does not provide apnoea alarming.**
WARNING ⚠	**If uncertain about the accuracy of any measurement, first check the patient's vital signs by alternative means, and then make sure the monitor is functioning correctly.**
WARNING ⚠	**Carefully route the FilterLine to reduce the possibility of patient entanglement or strangulation.**
WARNING ⚠	**Do not lift the monitor by the FilterLine, as it could disconnect from the monitor, causing the monitor to fall on the patient.**
WARNING ⚠	**The use of accessories, transducers, sensors and cables other than those specified may result in increased emission and/or decreased immunity of the equipment and/or system.**
WARNING ⚠	**CO_2 readings and respiratory rate can be affected by sensor application errors, certain ambient environmental conditions, and certain patient conditions.**
WARNING ⚠	**The FilterLine may ignite in the presence of oxygen if exposed to high heat.**
WARNING ⚠	**When using a sampling line for intubated patients with a closed suction system, do not place the airway adapter between the suction catheter and endotracheal tube. This is to ensure that the airway adapter does not interfere with the functioning of the suction catheter.**
WARNING ⚠	**Do not cut or remove any part of the sample line. Cutting the sample line could lead to erroneous readings.**

WARNING	**If too much moisture enters the sampling line (i.e., from ambient humidity or breathing of unusually humid air), the message *Clearing FilterLine* will appear in the message area. If the sampling line cannot be cleared, the message *FilterLine Blockage* will appear in the message area. Replace the sampling line once the FilterLine Blockage message appears.**
WARNING	**Loose or damaged connections may compromise ventilation or cause an inaccurate measurement of respiratory gases. Securely connect all components and check connections for leaks according to standard clinical procedures.**
WARNING	**Always ensure the integrity of the patient breathing circuit after insertion of the airway adapter by verifying a proper CO2 waveform (capnogram) on the monitor display.**
WARNING	**Do not lift the Tempus by the FilterLine as it could disconnect from the monitor causing the monitor to fall on the patient.**
WARNING	**Do not use the FilterLine H Set Infant/Neonatal during magnetic resonance imaging (MRI) scanning. Using the FilterLine H Set Infant/Neonatal during MRI scanning could harm the patient.**

CAUTION	During MRI scanning, the module must be placed outside the MRI suite. When the module is used outside the MRI suite, EtCO2 monitoring can be implemented using the FilterLine XL.
CAUTION	Use of a CO_2 sampling line with H in its name (indicating that it is for use in humidified environments) during MRI scanning may cause interference. The use of non H sampling lines is advised.
CAUTION	Microstream® EtCO2 sampling lines are designed for single patient use, and are not to be reprocessed. Do not attempt to clean, disinfect, sterilize or flush any part of the sampling line as this can cause damage to the monitor.
CAUTION	Dispose of sampling lines according to standard operating procedures or local regulations for the disposal of contaminated medical waste.
CAUTION	Before use, carefully read the Microstream $ETCO_2$ sampling line's *Directions for Use*.
CAUTION	In high-altitude environments, $ETCO_2$ values may be lower than values observed at sea level, as described by Dalton's law of partial pressures. When using the monitor in high-altitude environments, it is advisable to consider adjusting $ETCO_2$ alarm settings accordingly.
CAUTION	Use of monitoring during continuous nebulized medication delivery may result in damage to the Tempus Pro which is not covered by the warranty. Disconnect the Capnometer sample line from the Tempus Pro or switch off the Tempus Pro during medication delivery.
CAUTION	Do not use on patients that cannot tolerate the removal of 50 ml/min from their total minute ventilation.

Note	The following conditions can potentially cause a change in the flow rate: • Water, mucous or other patient contaminate has entered the sample tubing. • The sample tubing is crimped or pinched so that the sample flow rate has decreased. • Damage to the sample line. • The sample line has been cut, or split, causing the flow rate to increase.
Note	The Capnometer will require warming up if it is started from cold. The warm up time is specified in section 14.1.4. Once the Capnometer is warmed up, the capnogram will appear on the display.
Note	In order to avoid moisture build-up and sampling line occlusion during nebulization or suction for intubated patients, remove the sampling line luer connector from the Tempus.

The Tempus uses the Oridion® Microstream® FilterLine® and CapnoLine® systems to monitor CO_2.

Before using the Capnometer, select the appropriate cannula taking into account the patient age and the use of ventilation/intubation.

NSN	Oridion PN	Description	Application
6515-01-596-6547	015016	FilterLine H Set, Adult / Pediatric	Intubated sampling line and airway adapter for humid environments
6515-01-586-3075	015018	Smart CapnoLine Plus O_2	Non-Intubated Oral/ Nasal sampling with O_2 delivery for Adults
6515-01-596-7270	015021	FilterLine Set, Adult / Pediatric	Intubated sampling line and airway adapter for short term monitoring
6515-01-596-6391	015026	VitaLine H Set, Adult / Pediatric	Intubated sampling line and airway adapter for high ambient humidity
6515-01-596-7077	015027	Smart CapnoLine O_2 Pediatric	Non-Intubated Oral/ Nasal sampling with O_2 delivery for Pediatrics
6515-01-596-6387	015028	FilterLine H Set, Infant / Neonate	Intubated sampling line and airway adapter for humid environments, Infant/neonatal use

6.4.1 Getting started

To use the Capnometer, lift the door covering the Capnometer connector. The door will either be resting or will be pushed closed (in which case it will be latched). Lift the door by gently pressing on the latch end of the door and then levering the door up. The door is spring-loaded and will provide a gentle resistance to your finger. Letting go of the door will allow it to snap-shut. The door can be latched shut by applying gentle pressure to the latch end of the door to push it in – you will feel a click as the door latches.

CAUTION	The Capnometer door must be shut at all times when a cannula is not fitted. This is important to prevent dust, fluids or foreign objects from entering the Capnometer. Ensure the door is only kept open for the minimum time necessary to plug and unplug a cannula into the Tempus.
CAUTION	If the door snaps shut this will only provide functional protection against dust/fluids etc. to the Capnometer. For IP66 sealing the door must be pushed closed to engage the door's latch. If the Capnometer door is open the device has an IPX1 rating.
CAUTION	If the door is broken it must be replaced as soon as possible. The door and spring assembly can be removed and replaced with spare items by an appropriate technician without the unit being returned to RDT.

CAUTION ⚠	Allowing dust, fluids or foreign objects to enter the Capnometer can cause permanent damage to the device.

With the door open, attach a suitable patient airway adaptor (cannula) to the Capnometer socket of the Capnometer. An adult nasal cannula is typically supplied but a range of cannulas and other adaptors are available. Twist the cannula clockwise to insert and counter clockwise to remove. When inserting, ensure the cannula is fully inserted by twisting it until it is finger-tight.

CAUTION ⚠	Always ensure the cannula is fully inserted and fully tightened. Failing to tighten the cannula may result in the sample being diluted and an artificially low reading being produced.
CAUTION ⚠	Below the hole for the cannula inlet is a small hole for the Capnometer exhaust. Take care not to block this hole (an error will be caused if it becomes blocked) and ensure that dust, fluids or foreign objects do not enter it. Note that the Capnometer is not suitable for connection to a gas scavenging system.

Attach the cannula or airway adaptor to the patient following the instructions printed onto the pack of the adaptor. The Capnometer will start as soon as the probe is attached to the patient.

Applying a FilterLine set

The FilterLine set is intended for the CO_2 monitoring of intubated patients.

Before attaching the airway adapter to the breathing circuit, verify that the adapter is clean, dry and undamaged. Replace if necessary.

Place the airway adapter at the proximal end of the airway circuit between the elbow and the ventilator circuit wye. Do NOT place the airway adapter between the ET tube and the elbow as this may allow patient secretions to accumulate in the adapter.

If pooling does occur, the airway adapter and sampling line must be replaced.

Applying a Smart CapnoLine oral/nasal cannula

The oral/nasal cannulas are intended for monitoring CO_2 in non-intubated patients.

Oral/nasal sampling cannulas are especially valuable for patients who are prone to mouth breathing, since most (if not all) of the CO_2 is exhaled through the mouth. If a standard nasal CO2 sampling cannula is used on such patients, the $ETCO_2$ values and capnogram displayed will be substantially lower than the actual CO_2 levels present in the patient's expired breath.

Remove the cannula from the package. Verify that the cannula is clean, dry, and undamaged.

Replace if necessary.

6.4.2 Taking readings

Place the oral/nasal cannula onto the patient with two hands by first placing the prongs of the cannula into the patient's nostrils, then extend the lengths of the cannula around both sides of the patient's face routing the cannula tubing up to and then over both the patient's ears.

Proceed to loop the tubing around the back of the ears and bring the tubing under the ears and back towards where the patient's chin and throat meet.

Holding the tubes together with one hand under the chin, slide the collar around the cannula up so it secures both tubes of the cannula together firmly but comfortably under the patient's chin

CAUTION ⚠	Dispose of Microstream $ETCO_2$ consumables according to standard operating procedures or local regulations for the disposal of contaminated medical waste.

The capnography section of the home screen

Once the Capnometer is active it will run until the cannula is unplugged from the Tempus. Note that the wave speed of the capnogram is slower that the ECG and Pleth by default.

Once the Capnometer is started it will display the message “Warming up” for approximately 30 seconds. $ETCO_2$ readings will then display immediately and respiration readings will appear 2 breath cycles after the $EtCO_2$ readings appear. If the Capnometer is sufficiently warm, it can take as little as 7 seconds to return an $ETCO_2$ reading and 12 seconds to return a respiration rate reading (assuming a rate of >10 rpm).

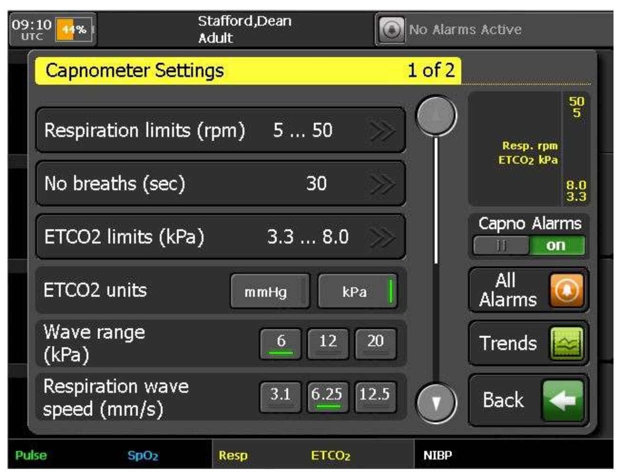
Once the Capnometer has started, check that the connections have been made properly by confirming the capnogram waveform appears correctly.

Note	If impedance respiration was in use, the waveform will automatically change to “CAPNO” and will display the end tidal CO_2 as soon as the cannula is connected. As soon as the cannula is disconnected (and the consequent error is cleared) the device will resume monitoring using impedance respiration.

After the Capnometer has started, it should draw a steady flow of breath gas down the cannula. If the cannula becomes blocked, the Capnometer will attempt to free the blockage by forcing low pressure air into the cannula. While this is occurring the waveform and readings will not be displayed and a message saying “Capnometer filter line purging” will be displayed in the waveform area. If the purging is successful then operation will resume automatically, if the blockage cannot be cleared (45 seconds approx.) then an error will be posted instructing to replace the cannula and the Capnometer will cease operation until the cannula is removed and replaced.

6.4.3 Capnometer settings

Pressing anywhere in the capnogram area brings up the “Capnometer Settings” menu:

Menu option	Menu screen
Respiration limits (rpm) – set the upper and lower alarm limits for respiration. **No breaths (sec)** – set the time that elapses after detection of no breaths until the alarm sounds. **ETCO2 limits** – set the upper and lower alarm limits for ETCO2. **ETCO2 units** – select **mmHg** or **kPa**. **Wave range** – adjust respiration wave height and amplitude. **Respiration wave speed (mm/s)** – set wave speed to **3.1**, **6.25** or **12.5**.	

Menu option	Menu screen
Operating mode – to switch off the pump (for suction or lavage), select **Pause 5 mins** or **Pause 10 mins**.	

Note	The ETCO2 units can be presented on the home screen in either mmHg or kPa but measurements in the Trends table and at i2i are only available in mmHg.

6.5 Contact temperature

WARNING	**The thermometer is only intended for use with YSI series 400 sensors. Disposable and reusable 400 series sensors may be used. Do not use series 700 sensors with the Tempus.**
WARNING	**Application and use of metal-jacketed temperature sensors that come into contact with conductive objects or clinical personnel during electrocautery may cause burns at the patient-probe/electrode contact points.**
WARNING	**Always read and follow the instructions on the packaging of temperature sensors regarding their storage, use and disposal. In particular take into account any information regarding toxicity.**

Note	The contact temperature probes must be attached to the parts of the body which their labelling shows they are indicated for.
Note	A single temperature channel (T1) is provided on the Tempus as standard. The second channel may be activated as an option using a software update.

6.5.1 Getting started

To use the thermometer, plug a YSI series 400 temperature sensor into the Tempus' ¼" socket marked "T1".

6.5.2 Taking readings

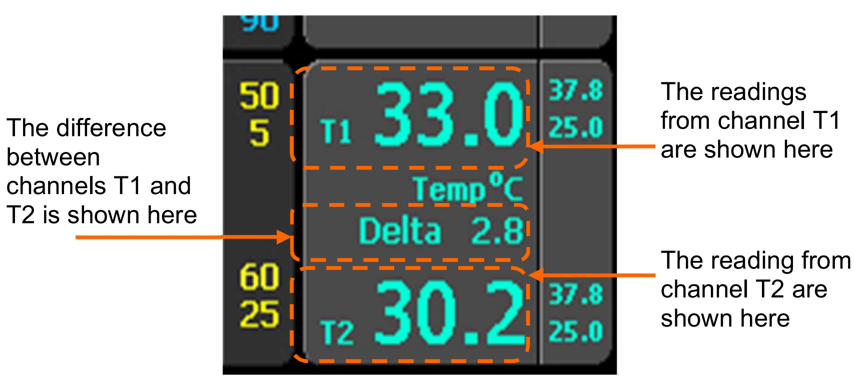
Attach the sensor to the patient following the instructions on the sensor's packaging.

The device will start to provide readings as soon as the sensor is connected (typically <10 seconds but up to 35 seconds according to IEC80601-2-56).

Users should monitor the temperature to observe when the reading settles (settling time will be dependent on environmental and patient factors).

Change the setting of the Tempus to °C or °F as required.

Note	Ensure the probe is suitably secured to the patient'



6.5.3 Contact temperature settings

Pressing anywhere on the contact temperature area brings up the “Temperature Settings” menu.

Note	The thermometer performs a self-check to 38.8 °C ±0.1 °C constantly to ensure that its readings are accurate and within specification.

If the Tempus Pro is configured to support only one temperature channel:

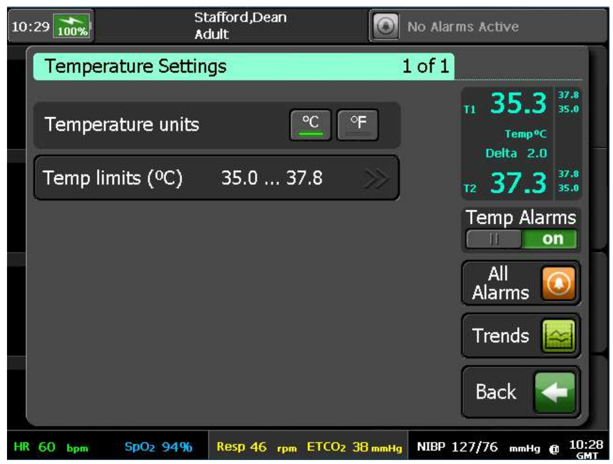
- The values in the Temperature Settings screen apply regardless of whether the temperature sensor is connected to socket T1 or T2.
- The maintenance user can still configure default alarm limits for two channels (T1 and T2), but the T2 default alarm limits are ignored and the initial temperature limits in the Temperature Settings screen are equal to the T1 default alarm limits.

The single channel “Temperature Settings” screen looks like this:

<table>
<tr><th>Menu option</th><th>Menu screen</th></tr>
<tr><td>

Temperature units – set the measurement units to °**C** or °**F**.

Temp limits – set the lower and upper temperature alarm limits.

</td><td>


</td></tr>
</table>

If the Tempus Pro is configured to support two channels of contact temperature:

- There are separate configured alarm limits for the T1 and T2 channels.
- There is a configured alarm limit for the difference between T1 and T2 (Delta).

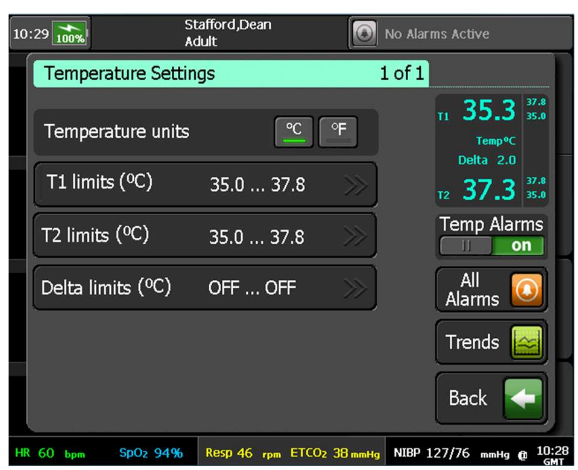
The two-channel “Temperature Settings” screen looks like this:

<table>
<tr><th>Menu option</th><th>Menu screen</th></tr>
<tr><td>

Temperature units – set the measurement units to °**C** or °**F**.

T1 limits and **T2 limits** – set the lower and upper temperature alarm limits for each channel.

Delta limits – set the delta temperature alarm limits, based on the difference between T1 and T2.

</td><td>


</td></tr>
</table>

6.6 Invasive pressure

WARNING ⚠	**DO NOT use the Invasive Pressure monitor for any purpose other than specified in this manual.**
WARNING ⚠	**If electrocautery is used, always avoid using any transducer with a conductive (metal) case connected to its ground shield. Using a conductive transducer that is connected to its cable shield risks high-frequency burns at the ECG electrodes if the transducer case becomes grounded to earth.**
WARNING ⚠	**Normal alarm functions will detect complete disconnections of invasive pressure transducers and faulty cables / transducers. Use only approved transducers and ensure that they are connected properly.**

WARNING ⚠	**Invasive Pressure is a tool to be used in addition to patient assessment. Care should be taken to assess the patient at all times.**
WARNING ⚠	**Do not touch metal parts of the transducer while it is in contact with the patient.**
WARNING ⚠	**Do not reuse any components that are labelled for single use only.**
WARNING ⚠	**Although complete disconnections of invasive pressure transducers will be detected by the normal alarm functions, partial disconnection will not be detected, nor will the use of some incompatible transducers. The user must exercise reasonable measures to ensure that approved transducers are used and that pressure transducers are connected properly.**
WARNING ⚠	**If electrocautery is used, always avoid using any transducer with a conductive (metal) case that is electrically connected to its cable shield. Using a conductive transducer case with such a shield connection risks high-frequency burns at the ECG electrodes if the transducer case becomes earth grounded.**
WARNING ⚠	**Always read and follow the instructions on the packaging of transducers regarding their storage, use and disposal. In particular take into account any information regarding toxicity.**
WARNING ⚠	**Remember to follow instructions provided with the transducers relating to use in defibrillation.**
WARNING ⚠	**Remember to differentiate the different transducers that are connected to the Tempus. The adaptor cables are labelled P1 & P2. The Tempus provides features to additionally label these channels with their site description on the display.**

CAUTION ⚠	Do not attempt to clean, disinfect, sterilize or in any way re-use transducers, catheters or other patient connected tubing which are labelled as single use only.
CAUTION ⚠	Dispose of all single use items according to standard operating procedures or local regulations for the disposal of contaminated medical waste.

Note	The invasive pressure meter will require warming up if it is started from cold. The warm up time is specified in section "14.1.6 Invasive pressure". Once the Tempus is turned on and the invasive pressure parameter functional, the pressure waveform will appear on the display.
Note	Check all cables and connections before use. Request that an appropriate service technician check the function of the device (including operation of all audible and visual alarms) on a regular basis.

The Tempus is compatible with a range of invasive pressure sensors – see "13.1.2 Invasive pressure accessories"

Always use the correct adaptor cable for the desired transducer to connect it to the Tempus.

The Tempus provides two channels of invasive pressure which can be extended to four channels via a USB IP Module. The Tempus Pro can then be used with one, two, three or four transducers.

6.6.1 Getting started

To measure invasive pressure, open the transducer packaging and inspect the transducer cable. If the cable shows signs of damage or wear, then replace it.

Attach the transducer to the patient following the instructions supplied with the device's packaging.

Attach the appropriate adaptor cable for the transducers you wish to use to the Tempus (white socket). Then connect the transducers to the cable, taking care to note and understand which transducer is attached to channel P1 and which to channel P2.

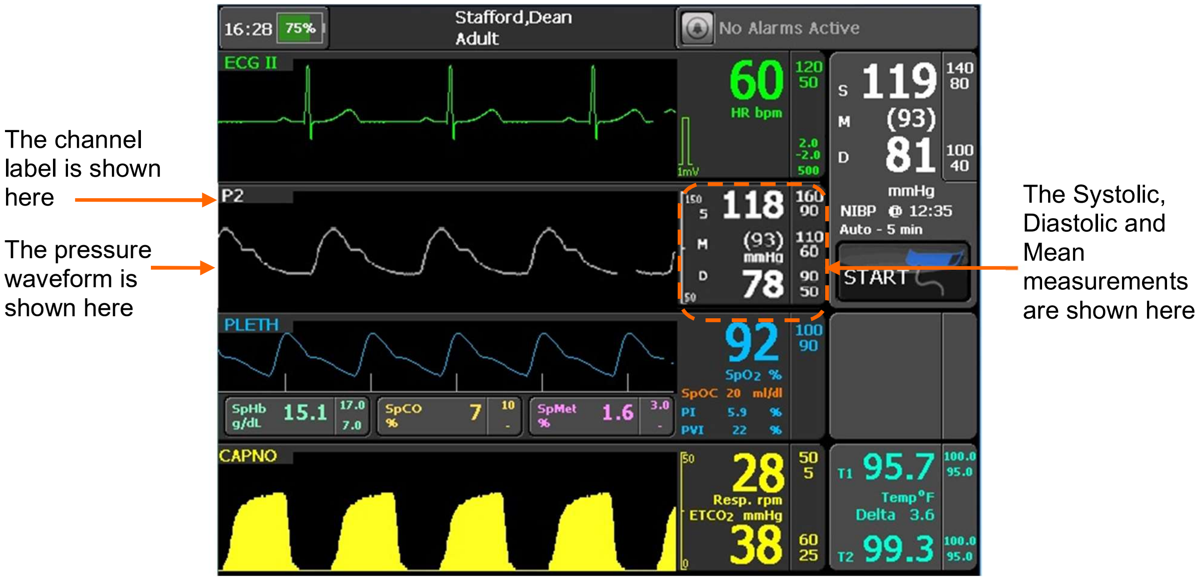
6.6.2 Taking readings

Then zero the transducers using the process described in section "6.6.5 Zeroing transducers".

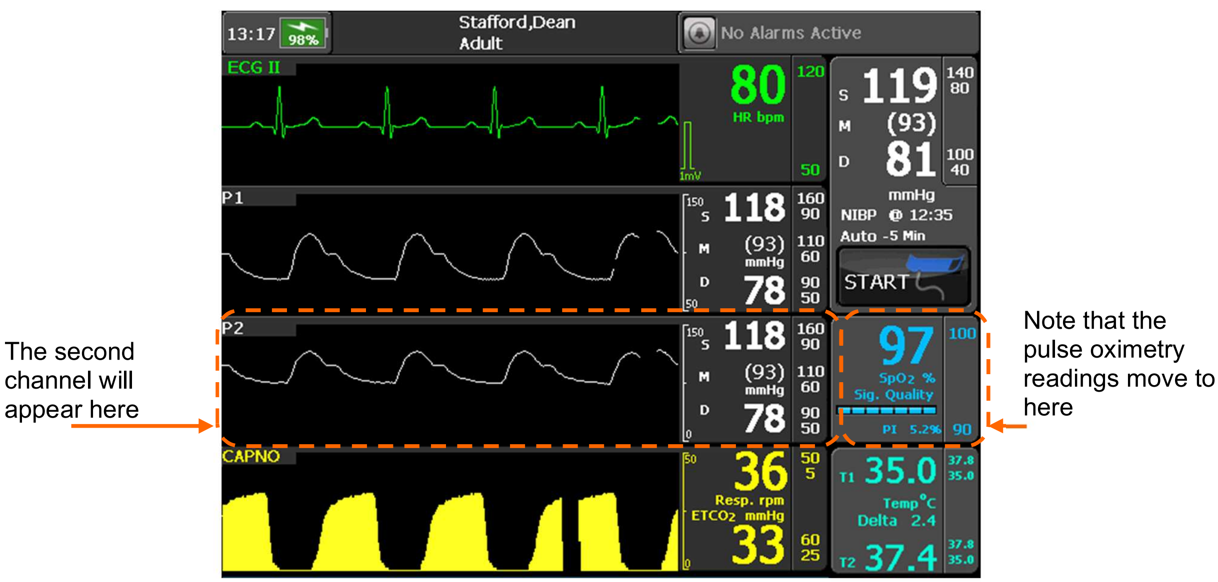
WARNING	**Always zero the transducer before using it.**

Then attach the transducer to the patient following the instructions on the transducer's packaging.

Note	The invasive pressure feature can be used to monitor a single channel at a time. In this case, either channel P1 or P2 can be used. See the example shown below where channel P2 is being used in isolation.



Once a second channel of invasive pressure is in use, it will appear as shown below.



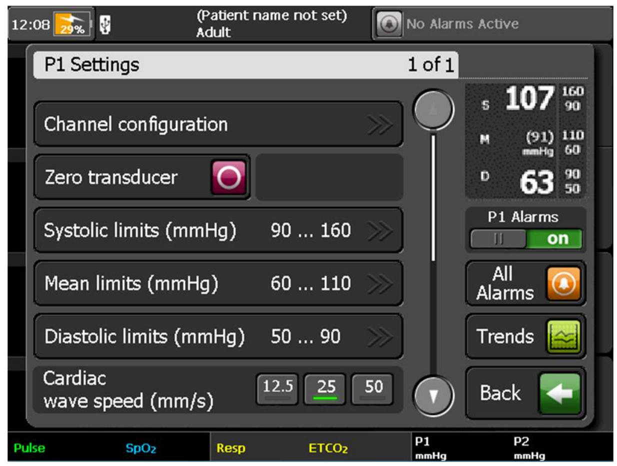
Monitoring two channels of invasive pressure

Note	If a second channel of invasive pressure is used, this channel will replace the pulse oximeter's results and plethysmogram in the results layout. The pulse oximeter results will move to the right into the previously unoccupied area. As soon as the either of the invasive pressure channels stop being used (when the transducer is disconnected from the adaptor cable and the alarm ceased), the pulse oximeter results and plethysmogram will return to their original location.

6.6.3 Invasive pressure settings

Pressing anywhere on the invasive pressure areas (P1 or P2) brings up the relevant settings menu, for example “P1 Settings”.

CAUTION ⚠	The settings of each invasive pressure channel can be set differently. Press on the area of the relevant channel to access its settings. Changes in settings made to one channel are never automatically made to any other channel.

<table>
<tr><th>Menu option</th><th>Menu screen</th></tr>
<tr><td>
The title bar shows the channel being configured (P1 in this example).
Channel configuration – label and reconfigure the channel, see “6.6.4 Configuring the transducer / channel”.
Zero transducer – zero the transducer, see “6.6.5 Zeroing transducers”.
Systolic limits, Mean limits and Diastolic limits – set the lower and upper alarm limits.
Cardiac wave speed (mm/s) – change wave speed. This setting is common to ECG, Sp02 and IP.
</td><td>


</td></tr>
</table>

Various conditions can affect the performance and accuracy of the invasive pressure measurement. These include:

- Clogging of the catheter.
- The placement of the catheter in the vasculature. Artefact such as catheter whip should be managed according to whatever clinical protocols are in place locally.
- The position of the transducer stopcock, catheter and flush port.
- The position of the transducer with respect to the patient’s hemostatic axis or the catheter tip.
- Patient movement.
- Saline line flushes which may temporarily interrupt accurate pressure measurement.
- Air bubbles in the catheter or in the transducer dome.

CAUTION ⚠	Flush the catheter regularly while taking invasive pressure measurements. Always view the waveform to ensure that pressure measurements are based on a physiological waveform.

6.6.4 Configuring the transducer / channel

Each channel can be configured according to its application. By default, both channels will appear in white, with results configured as “S, M and D” with a default scale (height) of the channel waveforms being 0-150 mmHg.

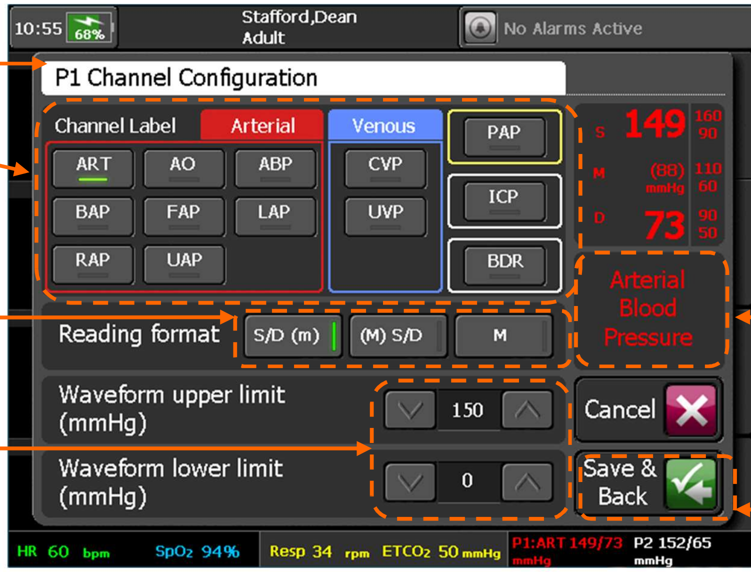
Accessing the “Channel Configuration” menu allows the channel labels to be individually named, the results format to be re-configured and the channel height to be adjusted.

The channel is labelled here

Press here to select the label and type of the channel

Press here to change the format of the measurement

Press to change the scale (height) of the channel's waveform – this can be changed independently of the default setting



Configuring an IP channel

The full description of the selected channel label will be shown here e.g. "Arterial Blood Pressure"

Press here to save the changes made – changes will not be made unless Save is pressed

CAUTION ⚠	The settings of each invasive pressure channel can be set differently. Changes in settings made to one channel are never automatically made to any other channel.

The menu allows the user to pick pre-set labels for the channel. These are grouped by color and application with RED being for arterial labels, BLUE being for venous labels, PALE YELLOW for pulmonary arterial pressure and WHITE being for other application labels.

Simply picking a label and then pressing Save will change the channel label and the color of the waveform and results. It will also change the waveform height and the Reading format and alarm limits. If a label is changed from one 'type' to another (e.g. from an arterial to a venous pressure), the reading format and waveform upper and lower limits will be set to defaults for the selected label. If a label is changed but stays within the same type (e.g. from ART to AO), any non-default reading format or waveform upper / lower limits will persist. The default settings are detailed below. For default alarm limits and alarm ranges, see "7.3 Patient alarms".

Note	Labelled channels will keep their labelling until either the patient is discharged, switched to previous patient or if the user manually changes the channel label.
Note	If an invasive pressure channel is labelled as Arterial and the critical alarm sounds pressing the alarm buttons will automatically set that channel to zero required.

Label	Application	Color	Reading format	Waveform height
ART – Arterial Blood Pressure	Arterial	Red	S/D (m)	0-150 mmHg
BAP – Brachial Artery Pressure	Arterial	Red	S/D (m)	0-150 mmHg
RAP – Radial Artery Pressure	Arterial	Red	S/D (m)	0-150 mmHg
AO – Aortic pressure	Arterial	Red	S/D (m)	0-150 mmHg
FAP – Femoral Artery Pressure	Arterial	Red	S/D (m)	0-150 mmHg
UAP – Umbilical Artery Pressure	Arterial	Red	S/D (m)	0-150 mmHg
ABP – Abdominal Aorta Pressure	Arterial	Red	S/D (m)	0-150 mmHg
LAP – Left Atrial Pressure	Arterial	Red	S/D (m)	0-150 mmHg

Label	Application	Color	Reading format	Waveform height
CVP – Central Venous Pressure	Venous	Blue	(M) S/D	0-30 mmHg
UVP – Umbilical Venous Pressure	Venous	Blue	(M) S/D	0-30 mmHg
PAP – Pulmonary Artery Pressure	Pulmonary artery	Pale yellow	S/D (m)	0-50 mmHg
ICP – Intra-Cranial Pressure	Intra-cranial	White	M	0-50 mmHg
BDR – Bladder Pressure	Bladder	White	M	0-30 mmHg

The reading format can be changed independently of the default settings. The reading can be formatted to read Systolic, Mean and Diastolic readings in the following orders:

S/D (m) – where Systolic and Diastolic are shown in larger font and the mean is shown in brackets in a smaller font.

(M) S/D – where Mean is shown first in brackets and the Systolic and Diastolic are then shown in a similar font.

M – where only the Mean is shown. When this setting is chosen (“M” only), Systolic and Diastolic alarming will be disabled for that channel.

The scale (height) of the channel waveform can be changed independently of the default settings. The scale can be formatted to read from -99 mmHg to 310 mmHg in pre-set levels.

Note	Users should review the default alarm and parameter settings and decide if these are compatible with the clinical protocols in place locally for attended and unattended monitoring of patients and make any procedural changes required in light of this review.

6.6.5 Zeroing transducers

WARNING	**Always zero the transducer before using it.**

Note	You can re-zero the transducer as often as required.
Note	If the transducer is disconnected from the cable, ensure a pause of at least 5 seconds is maintained before reconnecting the transducer (or before connecting a new transducer). Always zero a transducer after it has been it is connected to the Tempus (before use).

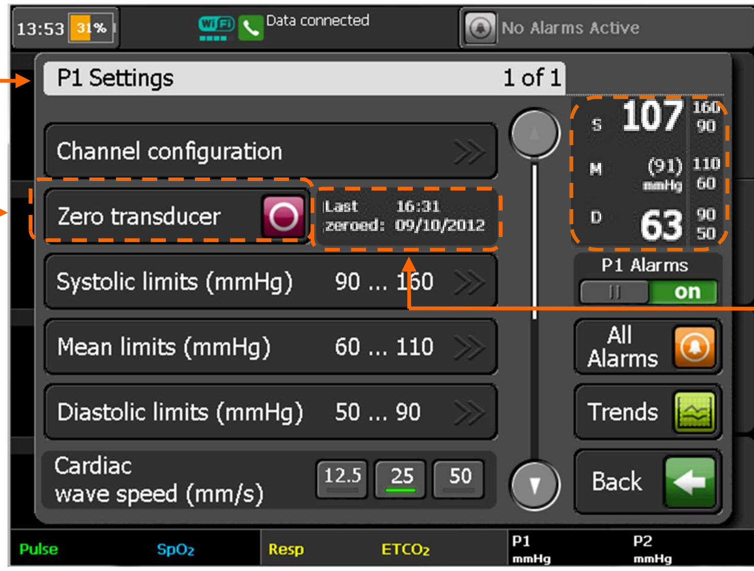
The invasive pressure transducers must be zeroed to ensure they provide accurate measurements. If a transducer is changed or moved, it must be zeroed again.

Follow the instructions on the transducer’s packaging while following the zeroing process detailed below.

The transducer should be placed at the same height as the patient’s left atrium before and during the zeroing process.

To zero, first close the transducer stopcock to the patient.

Then open the transducer’s venting stopcock to atmosphere. Allow the transducer a few seconds to settle.



Zeroing the transducer

CAUTION ⚠	Remember to zero the channel/transducer that you are using. Take care to differentiate between the two transducers and their settings.

If the process fails to measure a valid zero reading, check the transducer stop cock is open to atmospheric air and correctly connected to the Tempus. Then repeat the process. Replace the transducer if the error recurs.

6.6.6 Two channel USB invasive pressure module

2 Channel USB IP Module Connected to the Tempus Pro

2 Channel IP Module

When the 2 channel USB IP Module is plugged into the Tempus Pro USB socket and a patient cable and transducers are connected, channels 3 and 4 of will appear as shown below.

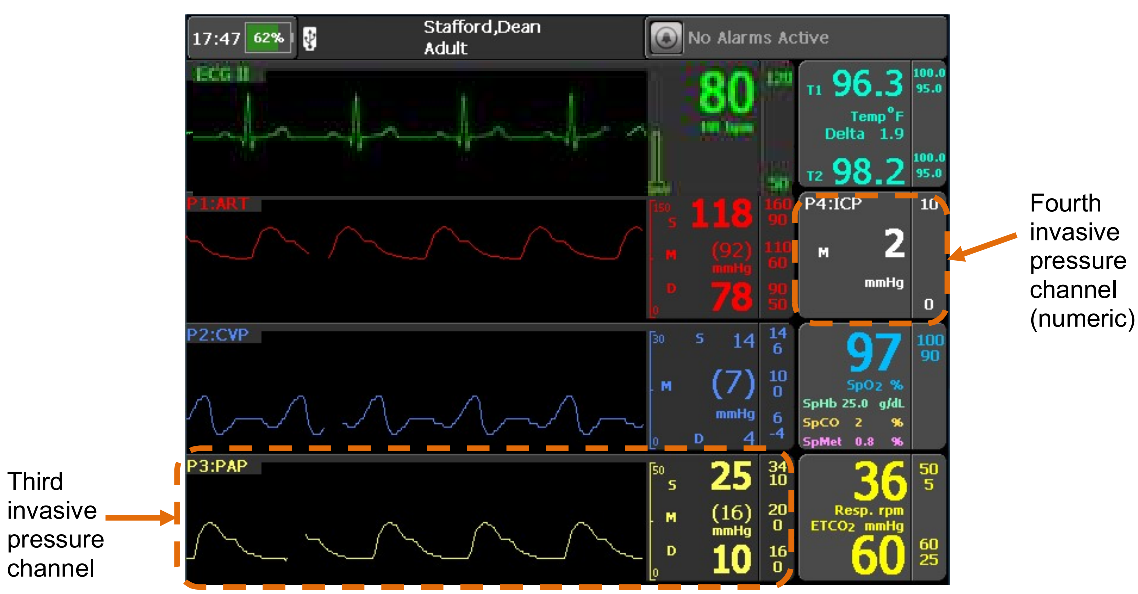
For part number details, see “12 Accessories list of the Tempus Pro”.

For specification details for the module, see “14 Specifications and standards”.

The USB module employs the same technology as the Tempus Pro’s internal Invasive Pressure measurement technology and therefore the same warnings, cautions and notes should be adhered to.

Four channel invasive pressure display

If four channels of invasive pressure are used, the third invasive pressure channel replaces the capnometer waveform and the fourth invasive pressure channel replaces the NIBP display:



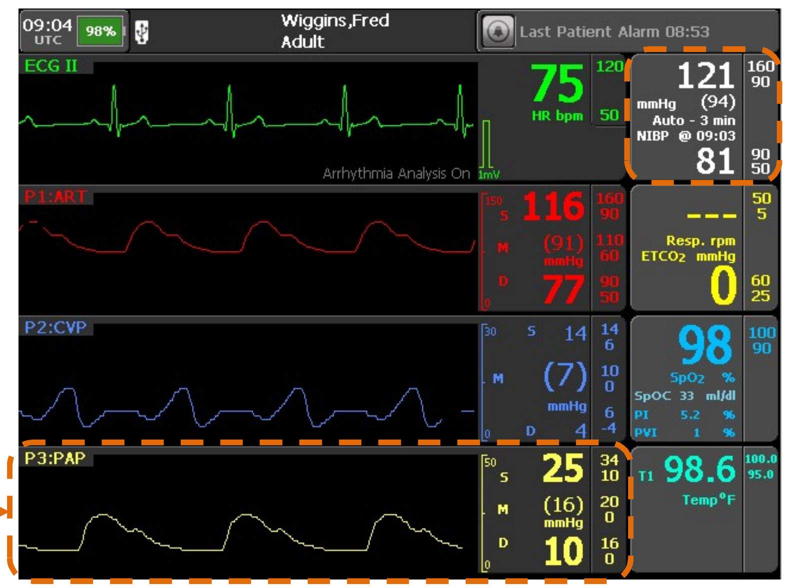
Home page showing four channels of invasive pressure

Note	If a fourth channel of invasive pressure is used, NIBP becomes unavailable and measurements are stopped.
Note	If you disable one of the four invasive pressure channels, the display reverts to three channel format (see below). If you disable two of the four channels, the capnometer waveform and NIBP results will return to their original location.

Note	The invasive pressure channels will always be displayed in numerical order on the results page, even if you plug in channel three and four before one and two. You can change which 3 of the 4 channels are displayed as waveforms by selecting if a waveform is required or not for that channel in the channel settings.

Three channel invasive pressure display

If three channels of invasive pressure are used, the third invasive pressure channel replaces the capnometer waveform and the NIBP is displayed in short form:



Home page showing three channels of invasive pressure

Note	If NIBP is active, the short form display does not include a Start button or a timer icon. To take a new NIBP reading without waiting for the auto-timer period, or to take a reading when in manual mode, press the NIBP display area and use the Start button at the top of the NIBP Settings menu.
Note	If you disable one of the three invasive pressure channels, the capnometer waveform and NIBP results will return to their original location.

7 Alarms

The Tempus produces both patient (physiological) alarms and technical alarms. Patient alarms are user configurable but technical alarms are not. Patient alarm settings are reset to factory defaults when a new patient is admitted or if you switch back to monitor a previous patient. If you turn the Tempus off and on and select to continue monitoring, the alarm settings will be retained (if the off period is less than 30 seconds) and the Tempus will revert to currently used settings.

Default alarm values will differ across different patient ages (adult, pediatric, and neonate) – see "7.3 Patient alarms".

Note that default alarms are configurable by the operating institution.

WARNING ⚠	**Do not silence the audible alarm if patient safety may be compromised.**
WARNING ⚠	**Always respond immediately to a system alarm since the patient may not be monitored during certain alarm conditions.**
WARNING ⚠	**Before each use, verify that the alarm limits are appropriate for the patient being monitored.**
WARNING ⚠	**Check the alarm suspend duration before temporarily suspending all the alarms. Alarm suspend feature pauses all audio and visual alarms.**
CAUTION ⚠	In high-altitude environments, EtCO2 values may be lower than values observed at sea level, as described by Dalton's law of partial pressures. When using the monitor in high altitude environments, it is advisable to consider adjusting EtCO2 alarm settings accordingly.

All alarms are "non-latching" i.e. the audible and visual alarm indicators will cease when the alarm condition ceases to exist. For example, if the heart rate goes from 80 to 105 and the alarm threshold has been set to 100, then both visual and audible alarms will be generated, when the heart rate drops below 100 the visual and audible alarms will cease automatically.

Alarms are audible in a 360° direction from the Tempus, however users should note that alarm volume will be loudest when the user is positioned facing the Tempus. Moving to the side or rear of the Tempus will produce quieter alarm volumes. Alarm volumes are specified at 1m from the front of the device.

The visual alarms are visible in a 360° direction around the Tempus. However, users should note that alarms will be most visible from the front of the device. While the alarms are visible from the sides (via light dispersed in the handle) and the rear, the visibility of the alarms is reduced in these positions.

The normal operating position of the operator should take into account the ability to see and hear the alarm signals taking into account local light and noise levels.

The alarm bar and the audible alarm will operate briefly soon after start up. Ensure both functions are operating before using the device.

WARNING	**If relying on the alarms, ensure that the display and alarm bar are viewable from the user's operating position. Ensure the handle and rear alarm light remain clean and unobstructed. Ensure the alarm speaker remains unobstructed and alarm volume is set to a level appropriate to the level of background noise expected.**
WARNING	**The alarm functions of the Tempus are intended to be used by the attendant user only. If the device is connected to a Response Center this is for the purpose of sharing vital signs data in real time, between two users for the purpose of obtaining additional clinical support. The system is not a distributed alarm system (e.g. nurse monitoring station system) in the terms of IEC60601-1-8. The i2i system at the Response Center is not equipped with alarm silencing or suspending controls.**
WARNING	**When connected to a PC running the i2i application at a Response Center, streamed data, such as the waveforms displayed on the Tempus Pro, will be transmitted and displayed automatically on that PC's display. Users are advised that streamed data are transmitted using the UDP protocol. The UDP protocol includes error checking but does not retransmit data. Therefore, any data that is dropped, lost or delayed during the streaming process will not be retransmitted. In the event that packets of waveform data are lost, they will appear as gaps in the waveform that is displayed on the i2i interface.**
WARNING	**Medical data (vital signs data, photos, ECG recordings, patient details, TCCC cards etc.) are transmitted using TCP/IP, this includes error checking and retransmission so missing or dropped packets are therefore retransmitted.**
WARNING	**Too many of the same or similar pieces of equipment in an area, with different alarm pre-sets, may make it difficult for the operator to know how the alarm system will operate.**
WARNING	**Users should check to ensure that any alarm setting is appropriate prior to use.**

CAUTION	Ensure alarm values are not changed to extreme levels simply to disable the alarms system. Ensure alarm levels are brought into a more sensitive state for severe or critical patients or for patients who will be left unattended.
CAUTION	The Tempus' alarms are intended to alert attendant users. They are not intended to alert users who remove themselves from the patient's vicinity. Users should consider the levels of background ambient noise and light if they will not be in close proximity to the patient. The alarms provided by the system do not replace user care.

Note	Users are reminded to ensure the alarm speaker is working every time they use the product. This can be verified by simply using the touchscreen – if the speaker is working then audible feedback will be given for each press of the screen.
Note	The Tempus will not report any alarm for the first 10 seconds after a sensor is attached. This is to prevent false positive alarms being generated due to patient/user/sensor artefact that can be commonly caused while a parameter's sensor is first fixed to the patient.

7.1 Audible alarm characteristics

The Tempus provides audible alarm indications through a speaker mounted on the front of the device.

WARNING	**If relying on the alarms, ensure that the display and alarm bar are viewable from the user's operating position. Ensure the handle and rear alarm light remain clean and unobstructed. Ensure the alarm speaker remains unobstructed and alarm volume is set to a level appropriate to the level of background noise expected.**
WARNING	If alarm volume is less than the ambient sound level, the operator may not recognize alarm conditions.
CAUTION	Do not attempt to clean beneath the grille with a sharp or rough implement; this can damage the water-proof sealing that is present beneath the grille and cause reduction or impairment of the ingress protection barrier.

Audible alarms are separated into either patient (physiological) alarms, "inop" (technical) alarms or alarm off/suspend reminder tones. These alarm signals are specified as follows:

<table>
<tr><th>Nominal Characteristics</th><th>Patient Alarm (Red)</th><th>Patient Alarm (Amber)</th><th>Technical Alarm</th><th>Reminder tone</th></tr>
<tr><td>Pulse sequence</td><td>10 x 175 ms pulses</td><td>3 x 125 ms pulses at 440 Hz (with 4 harmonics under 4 kHz)</td><td>2 x 250 ms pulses</td><td>1 x 500 ms pulse at 1 kHz</td></tr>
<tr><td>Repetition rate</td><td>Every 10 seconds</td><td>Every 5 seconds</td><td>Every 15 seconds</td><td>Every 3 minutes</td></tr>
<tr><td>Duration of gap between pulse sequences</td><td>50 ms silence between pulses 1 and 2, 2 and 3, 4 and 5, 6 and 7, 7 and 8, 9 and 10.
170 ms silence between pulses 3 and 4, 8 and 9.
625 ms silence between pulses 5 and 6</td><td colspan="3">250 ms silence between pulse sequences</td></tr>
<tr><td>Volume (dBA at 1 m)</td><td colspan="4">85 dBA (default) – the volume may be reduced using the All Alarms menu Alarm Volume settings: Min, Low, Mid and High. The maintenance user may have disabled one or more of these settings.</td></tr>
</table>

The alarm off/suspend reminder tone plays a beep every three minutes to remind the operator that alarms are off or suspended. "Off" only applies to parameters that are being monitored.

7.2 Visual alarm characteristics

When alarms occur, the Tempus Pro indicates their presence in the following ways:

Visual alarm manifestations

7.2.1 Alarm bar LEDs

The alarm bar LEDs illuminate in one of the following ways:

- Flashing yellow LED (for patient alarms);
- Solid yellow (for technical alarm);
- Flashing red (for specific high priority alarms).

CAUTION ⚠	The alarm bar can also be user-configured to show a solid green LED to indicate no active alarms (this feature is enabled and disabled in the Maintenance Menu, refer to the *Tempus Pro Maintenance Manual* for details).

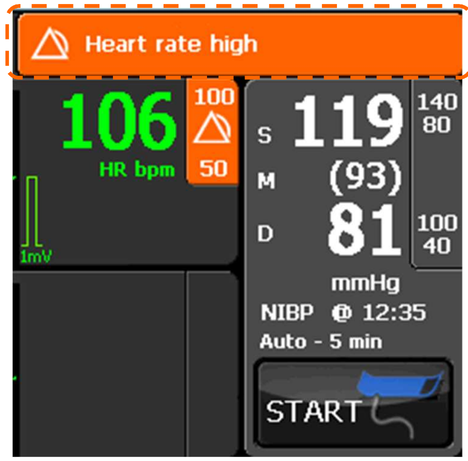
7.2.2 Alarm status area

For both patient and technical alarms, the Tempus shows the alarm symbol and a description of the alarm in the alarm status area at the top right of the display.

CAUTION ⚠	In strong daylight users should ensure that the alarm status area is visible.

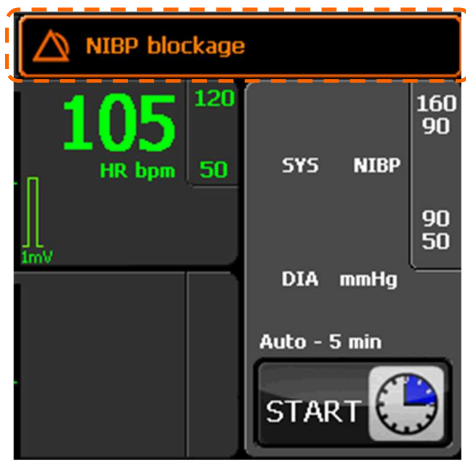
Note	If multiple alarms occur, they will be displayed in rotation in the alarm status area.

Press the alarm status area to view a list of all active alarms and to select technical alarms for clearing – see “7.7 Viewing and clearing alarms”.



Active patient alarms look like this

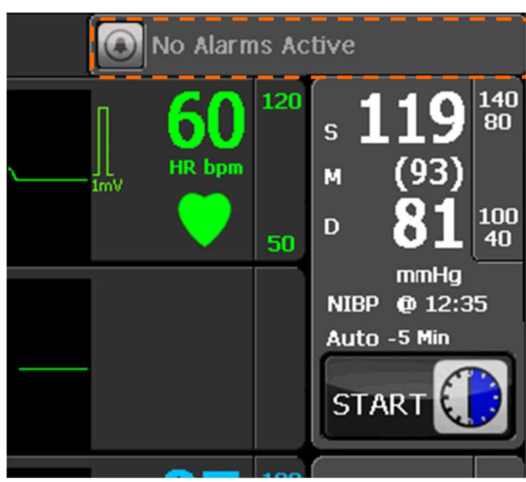
Alarm status area – patient alarm



Active technical alarms look like this

Alarm status area – technical alarm

When there is no alarm state, the alarm bar will display the time of the last patient alarm or “No Alarms Active” if there have not been any alarms.

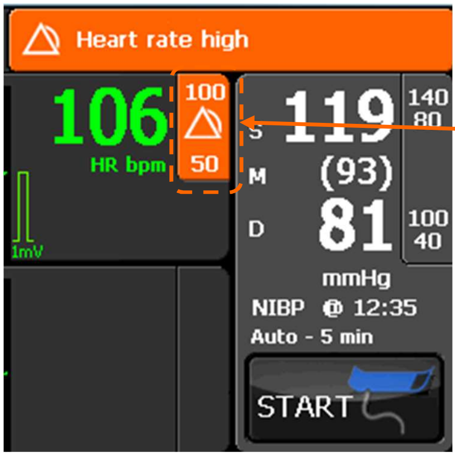


This shows that no alarms are active

Alarm status area with no alarms active

7.2.3 Alarm limits bar

For patient alarms only, on the Home screen, the Tempus highlights the alarm limits bar for the parameter that is alarming. It is highlighted in a solid (not flashing) yellow.



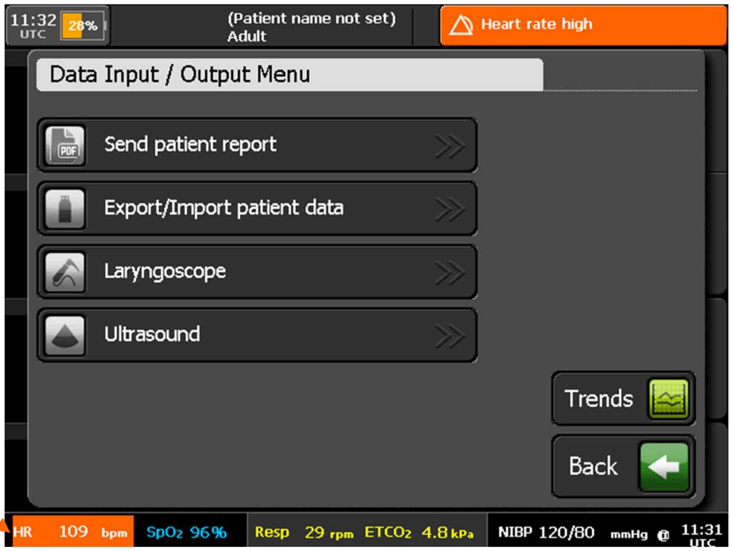
Alarm limits bar for the parameter that is alarming

Alarm limits bar

7.2.4 Bottom status bar

For patient alarms only, on screens that have a bottom status bar, the Tempus highlights the parameter that is alarming in flashing white text on an orange background.

The reading is shown in white text on an orange background to indicate that the value is outside limits



Reading outside Limits

7.3 Patient alarms

The table below lists all patient alarms:

Parameter	Measurement Range	Patient type	Default Alarm Low	Alarm Low Range	Default Alarm High	Alarm High Range
ECG Heart rate	30-300 bpm	Adult	50 bpm	30-238 bpm	120 bpm	32-239 bpm
	30-300 bpm	Pediatric	50 bpm	30-238 bpm	150 bpm	32-239 bpm
	30-300 bpm	Neonate	100 bpm	30-238 bpm	200 bpm	32-239 bpm
ST	-50 - +50 mm	Adult	-2.0 mm	-10 – +2 mm	+2.0 mm	-2 – +10.0 mm
	N/A	Pediatric	N/A	N/A	N/A	N/A
	N/A	Neonate	N/A	N/A	N/A	N/A
QT	1 – 2000 ms	Adult	N/A	N/A	500 ms	10 – 1990 ms
	N/A	Pediatric	N/A	N/A	N/A	N/A
	N/A	Neonate	N/A	N/A	N/A	N/A
PI	0-20%	Adult	OFF	Off, 0.1 to 18	OFF	0.2 to 19, Off
	0-20%	Pediatric	OFF	Off, 0.1 to 18	OFF	0.2 to 19, Off
	0-20%	Neonate	OFF	Off, 0.1 to 18	OFF	0.2 to 19, Off
PVI	0-100%	Adult	5%	Off, 1 to 98	40%	2 to 99, Off
	0-100%	Pediatric	5%	Off, 1 to 98	40%	2 to 99, Off
	0-100%	Neonate	5%	Off, 1 to 98	40%	2 to 99, Off
SpOC	0 to 35 ml/dl	Adult	10 ml/dl	Off, 1 to 33	25 ml/dl	2 to 34, Off
	0 to 35 ml/dl	Pediatric	10 ml/dl	Off, 1 to 33	25 ml/dl	2 to 34, Off
	0 to 35 ml/dl	Neonate	10 ml/dl	Off, 1 to 33	25 ml/dl	2 to 34, Off
SpO_2 pulse rate	25-239 bpm	Adult	50 bpm	30-238 bpm	120 bpm	32-239 bpm
	25-239 bpm	Pediatric	50 bpm	30-238 bpm	150 bpm	32-239 bpm
	25-239 bpm	Neonate	100 bpm	30-238 bpm	200 bpm	32-239 bpm
Impedance respiration	3-150 rpm	Adult	5 rpm	3-145 rpm	30 rpm	5-149 rpm
	3-150 rpm	Pediatric	5 rpm	3-145 rpm	30 rpm	5-149 rpm
	3-150 rpm	Neonate	12 rpm	3-145 rpm	80 rpm	5-149 rpm
SpO_2	0-100%	Adult	90%	50-98%	100%	52-100%
	0-100%	Pediatric	90%	50-98%	100%	52-100%
	0-100%	Neonate	85%	50-98%	95%	52-100%
SpCO	0-99%	Adult	OFF	1-97%	10%	2-98%
	0-99%	Pediatric	OFF	1-97%	10%	2-98%
	0-99%	Neonate	OFF	1-97%	10%	2-98%
SpHb	0-25 g/dl	Adult	7.0 g/dl	1.0-23.5 g/dl	17.0 g/dl	2.0-24.5 g/dl
	0-25 g/dl	Pediatric	7.0 g/dl	1.0-23.5 g/dl	17.0 g/dl	2.0-24.5 g/dl

Parameter	Measurement Range	Patient type	Default Alarm Low	Alarm Low Range	Default Alarm High	Alarm High Range
	0-25 g/dl	Neonate	7.0 g/dl	1.0-23.5 g/dl	17.0 g/dl	2.0-24.5 g/dl
SpMet	0.0-99.9%	Adult	OFF	0.1-99.0 %	3.0%	1.0-99.5 %
	0.0-99.9%	Pediatric	OFF	0.1-99.0 %	3.0%	1.0-99.5 %
	0.0-99.9%	Neonate	OFF	0.1-99.0 %	3.0%	1.0-99.5 %
Capnometer respiration	1-149 rpm	Adult	5 rpm	2-145 rpm	30 rpm	5-149 rpm
	1-149 rpm	Pediatric	5 rpm	2-145 rpm	30 rpm	5-149 rpm
	1-149 rpm	Neonate	12 rpm	2-145 rpm	80 rpm	5-149 rpm
Capnometer respiration – no breaths detected	-	Adult	30 seconds	10-60 seconds	-	-
	-	Pediatric	30 seconds	10-60 seconds	-	-
	-	Neonate	30 seconds	10-60 seconds	-	-
Capnometer $ETCO_2$	0-150 mmHg 0-20 kPa	Adult	25 mmHg 3.3 kPa	0-145 mmHg 0-19.3 kPa	60 mmHg 8 kPa	5-150 mmHg 0.7-20 kPa
	0-150 mmHg 0-20 kPa	Pediatric	25 mmHg 3.3 kPa	0-145 mmHg 0-19.3 kPa	60 mmHg 8 kPa	5-150 mmHg 0.7-20 kPa
	0-150 mmHg 0-20 kPa	Neonate	25 mmHg 3.3 kPa	0-145 mmHg 0-19.3 kPa	60 mmHg 8 kPa	5-150 mmHg 0.7-20 kPa
Non-Invasive Blood Pressure - Systolic	40-260 mmHg	Adult	90 mmHg	40-255 mmHg	160 mmHg	45-260 mmHg
	40-230 mmHg	Pediatric	70 mmHg	40-155 mmHg	120 mmHg	45-160 mmHg
	40-130 mmHg	Neonate	40 mmHg	40-125 mmHg	90 mmHg	45-130 mmHg
Non-Invasive Blood Pressure - Diastolic	20-200 mmHg	Adult	50 mmHg	20-195 mmHg	90 mmHg	25-200 mmHg
	20-160 mmHg	Pediatric	40 mmHg	20-155 mmHg	70 mmHg	25-160 mmHg
	20-100 mmHg	Neonate	20 mmHg	20-95 mmHg	60 mmHg	25-100 mmHg
Non-Invasive Blood Pressure - Mean	26-220 mmHg	Adult	60 mmHg	26-215 mmHg	110 mmHg	30-220 mmHg
	26-183 mmHg	Pediatric	50 mmHg	26-155 mmHg	90 mmHg	30-160 mmHg
	26-110 mmHg	Neonate	24 mmHg	26-155 mmHg	70 mmHg	25-160 mmHg
Contact temperature	20-45 °C / 68-113 °F	Adult	35°C / 95°F	20-44 °C / 68-111 °F	37.8 °C / 100 °F	21-45 °C / 69.8-113 °F
	20-45 °C / 68-113 °F	Pediatric	35°C / 95°F	20-44 °C / 68-111 °F	37.8 °C / 100 °F	21-45 °C / 69.8-113 °F
	20-45 °C / 68-113 °F	Neonate	35°C / 95°F	20-44 °C / 68-111 °F	37.8 °C / 100 °F	21-45 °C / 69.8-113 °F

Parameter	Measurement Range	Patient type	Default Alarm Low	Alarm Low Range	Default Alarm High	Alarm High Range
Invasive Pressure – Systolic Arterial	-99 – 310 mmHg	Adult	90 mmHg	-99 – 310 mmHg	160 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Pediatric	70 mmHg	-99 – 310 mmHg	120 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Neonate	40 mmHg	-99 – 310 mmHg	90 mmHg	-99 – 310 mmHg
Invasive Pressure – Diastolic Arterial	-99 – 310 mmHg	Adult	50 mmHg	-99 – 310 mmHg	90 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Pediatric	40 mmHg	-99 – 310 mmHg	70 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Neonate	20 mmHg	-99 – 310 mmHg	60 mmHg	-99 – 310 mmHg
Invasive Pressure – Mean Arterial	-99 – 310 mmHg	Adult	60 mmHg	-99 – 310 mmHg	110 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Pediatric	50 mmHg	-99 – 310 mmHg	90 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Neonate	24 mmHg	-99 – 310 mmHg	70 mmHg	-99 – 310 mmHg
Invasive Pressure – Systolic Venous	-99 – 310 mmHg	Adult	6 mmHg	-99 – 310 mmHg	14 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Pediatric	2 mmHg	-99 – 310 mmHg	10 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Neonate	2 mmHg	-99 – 310 mmHg	10 mmHg	-99 – 310 mmHg
Invasive Pressure – Diastolic Venous	-99 – 310 mmHg	Adult	-4 mmHg	-99 – 310 mmHg	6 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Pediatric	-4 mmHg	-99 – 310 mmHg	2 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Neonate	-4 mmHg	-99 – 310 mmHg	2 mmHg	-99 – 310 mmHg
Invasive Pressure – Mean Venous	-99 – 310 mmHg	Adult	0 mmHg	-99 – 310 mmHg	10 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Pediatric	0 mmHg	-99 – 310 mmHg	4 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Neonate	0 mmHg	-99 – 310 mmHg	4 mmHg	-99 – 310 mmHg
Invasive Pressure – Systolic PAP	-99 – 310 mmHg	Adult	10 mmHg	-99 – 310 mmHg	34 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Pediatric	24 mmHg	-99 – 310 mmHg	60 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Neonate	24 mmHg	-99 – 310 mmHg	60 mmHg	-99 – 310 mmHg
Invasive Pressure –	-99 – 310 mmHg	Adult	0 mmHg	-99 – 310 mmHg	16 mmHg	-99 – 310 mmHg

Parameter	Measurement Range	Patient type	Default Alarm Low	Alarm Low Range	Default Alarm High	Alarm High Range
Diastolic PAP	-99 – 310 mmHg	Pediatric	-4 mmHg	-99 – 310 mmHg	-4 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Neonate	-4 mmHg	-99 – 310 mmHg	-4 mmHg	-99 – 310 mmHg
Invasive Pressure – Mean PAP	-99 – 310 mmHg	Adult	0 mmHg	-99 – 310 mmHg	20 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Pediatric	12 mmHg	-99 – 310 mmHg	26 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Neonate	12 mmHg	-99 – 310 mmHg	26 mmHg	-99 – 310 mmHg
Invasive Pressure – Mean ICP	-99 – 310 mmHg	Adult	0 mmHg	-99 – 310 mmHg	10 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Pediatric	0 mmHg	-99 – 310 mmHg	4 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Neonate	0 mmHg	-99 – 310 mmHg	4 mmHg	-99 – 310 mmHg
Invasive Pressure – Mean BDR	-99 – 310 mmHg	Adult	-	-99 – 310 mmHg	10 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Pediatric	-	-99 – 310 mmHg	10 mmHg	-99 – 310 mmHg
	-99 – 310 mmHg	Neonate	-	-99 – 310 mmHg	10 mmHg	-99 – 310 mmHg

All patient alarms are medium priority alarm states as defined by IEC60601-1-8 with the exception of ECG arrhythmias and “catheter disconnected” for invasive pressure (mean arterial pressure **only**). In this event if the catheter pressure drops from a value of greater than 10 mmHg to less than 10 mmHg and remains there for a period longer than that specified in the table in section 7, then the red alarm bar LED will flash and a high priority alarm tone will be emitted.

7.4 Technical alarms

Technical alarms can relate to the state of the Tempus itself or its connection to the patient. Technical alarms are low priority alarm states as defined by IEC60601-1-8.

The majority of technical alarms are self-explanatory and should be addressed by following the on-screen instructions. Additional information is provided below for certain technical alarm conditions.

Technical alarms can be cleared by the operator – see “7.7 Viewing and clearing alarms”.

Condition	On-Screen Error Text	Comment
Bluetooth linking error	Attention - Bluetooth Unable to initialize Bluetooth. If the problem persists, use a wired headset.	Repeat the process taking care to follow on-screen and written instructions
Communications error – cable not plugged in	Attention - Data Cable The data cable is not plugged in correctly.	Repeat the process taking care to follow on-screen and written instructions
Communications notification	Attention - Data Connecting Initializing data encryption engine.	No action – initialization process lasts <15 seconds

Condition	On-Screen Error Text	Comment
Communications error – connection failed	Attention - Data Connection The data link was unable to connect at this time.	Repeat the process taking care to follow on-screen and written instructions
Data export option	Attention - Data Export There is patient record data from other devices/times. Do you want to over-write it?	Respond to instruction by pressing the appropriate on-screen button option
GPS error	Attention - GPS A GPS location has not yet been obtained; leave the GPS on and check this screen again in a few minutes.	Repeat the process taking care to follow on-screen and written instructions
Bluetooth headset linking error	Attention - Headset The headset has not linked to the Tempus. Repeat step 3 of the instructions.	Repeat the process taking care to follow on-screen and written instructions
Low battery	Attention - Low Battery Battery charge 25%, please connect the charger.	Temporary notice – follow onscreen instructions. Find another power source if available.
Low battery	Attention - Low Battery Battery charge 10%, please connect the charger. USB devices are disabled.	Temporary notice – follow onscreen instructions. Find another power source if available.
Low battery	Attention - Low Battery The battery is almost empty, USB devices are disabled.	Temporary notice – follow onscreen instructions. Find another power source if available.
Patient switching error	Attention – Switch To A Previous Patient To switch patient you must first stop monitoring the current patient.	Disconnect leads from the current patient before discharging/admitting a new patient
Voice connection not possible, a data connection must be made first	Attention - Voice Connection Headset voice connection will be started after connecting data.	Repeat the process taking care to follow on-screen and written instructions
Voice connection error	Attention - Voice Connection The voice link was unable to connect at this time.	Repeat the process taking care to follow on-screen and written instructions
WiFi connection error	Attention - WiFi WiFi was unable to connect at this time. If the problem persists restart the Tempus and attempt the connection again.	Repeat the process taking care to follow on-screen and written instructions
WiFi not activated	Warning - WiFi WiFi is not activated on this device	Refer to the maintenance manual.
GSM not activated	Warning – GSM (Cell-phone) GSM is not activated on this device	Refer to the maintenance manual.
Bluetooth pairing error	Bluetooth Pairing Error The required Bluetooth device not found	Repeat the process taking care to follow on-screen and written instructions

Condition	On-Screen Error Text	Comment
Bluetooth pairing error	Bluetooth Pairing Error Entered PIN is incorrect	Repeat the process taking care to follow on-screen and written instructions
Bluetooth pairing error	Warning - Bluetooth Bluetooth is not activated on this device.	Repeat the process taking care to follow on-screen and written instructions
System error	Internal Fault Detected A fault has been detected. Reason: exception. The Tempus will need to restart. If this problem persists please contact your supplier.	Log the exact details of the fault so they can be communicated to the supplier.
System error	Internal Fault Detected A fault has been detected. Reason: file open error. The Tempus will need to restart. If this problem persists please contact your supplier.	Log the exact details of the fault so they can be communicated to the supplier.
System error	Internal Fault Detected A fault has been detected. Reason: file read error. The Tempus will need to restart. If this problem persists please contact your supplier.	Log the exact details of the fault so they can be communicated to the supplier.
System error	Internal Fault Detected A fault has been detected. Reason: file write error. The Tempus will need to restart. If this problem persists please contact your supplier.	Log the exact details of the fault so they can be communicated to the supplier.
System error	Internal Fault Detected A fault has been detected. Reason:cOS exception. The Tempus will need to restart. If this problem persists please contact your supplier.	Log the exact details of the fault so they can be communicated to the supplier.
IP address error	Invalid IP Address Please check the IP Address is in the format x.x.x.x and multiple addresses (maximum of 3) are separated by ';'.	Correct the format of the IP address
Network Settings Error	Network Settings Error There are no network settings in this mode.	Enter network settings
Software update error	Options Key Check Failed One or more option key checks have failed. Some features may be disabled. If the problem persists contact the service representative.	A software upgrade may have occurred incorrectly. Log the exact details of the fault so they can be communicated to the supplier.
Low ambient temperature warning	Temperature Warning The device temperature is low, Tempus may shutdown.	Attempt to warm the device and then restart

Condition	On-Screen Error Text	Comment
USB connection error	USB - Connection The USB memory stick is not plugged into the Tempus USB socket, please plug in. If plugged in, check it's the USB and not the RJ45 socket.	Remove and re-insert the stick into the USB socket
USB error	USB - Connection Failed to connect with USB, please retry.	Remove and re-insert the stick into the USB socket
USB error	USB - Import The USB memory stick is empty; there is no handover data on it.	Use the correct stick or ensure the files are on the stick you have
USB error	USB - Import The data on the USB memory stick was exported for the current patient. You cannot re-import this data.	You are attempting to re-write the current patient record with itself, this cannot be done
Maintenance notification	Warning - Settings Settings were not saved at power down and have been reset to defaults.	Reset settings as required

iAssist and Technical alarms are low priority alarm states as defined by IEC60601-1-8.

7.5 Setting alarms

Alarm limits are set in either the individual parameter settings menu (accessible by pressing on that parameter area on the touchscreen) or through the All Alarms Menu (accessible from the main menu and parameter settings menus).

7.5.1 Default alarm settings

It should be noted that all patient alarms have a user configurable delay of 0-8 seconds (4 seconds default, 0 seconds minimum) to help prevent spurious alarms being generated – see "7.5 Setting alarms".

In addition to this user-configurable time, the Tempus will produce alarms after the durations specified below.

Alarm	Nominal Time to Delay (Excluding User Configurable Delay)
ECG heart rate high	2 seconds
ECG heart rate low	2 seconds
ECG heart rate out of range (low)	4 seconds (*)
Pulse rate (from Oximeter) high	3 seconds
Pulse rate (from Oximeter) low	3 seconds
Pulse rate (from Oximeter) out of range (low)	4 seconds
Pulse rate (from Oximeter) out of range (high)	4 seconds
SpO_2 high	3 seconds
SpO_2 low	3 seconds
SpHb low	2 seconds
SpHb high	2 seconds

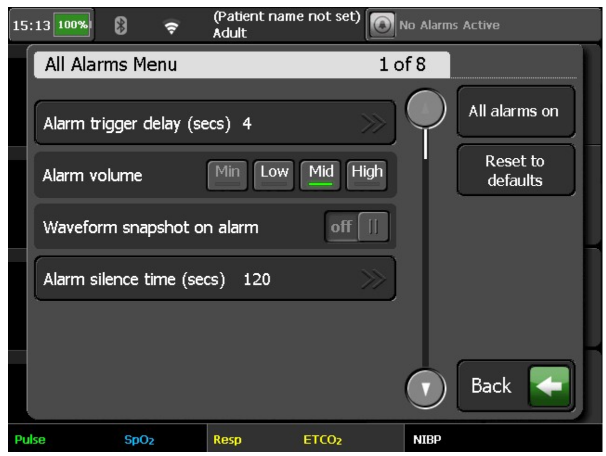
Alarm	Nominal Time to Delay (Excluding User Configurable Delay)
SpMet low	2 seconds
SpMet high	2 seconds
SpCO low	3 seconds
SpCO high	4 seconds
PI low	2 seconds
PI high	2 seconds
PVI low	2 seconds
PVI high	2 seconds
SpOC low	2 seconds
SpOC high	2 seconds
$ETCO_2$ high	4 seconds
$ETCO_2$ low	4 seconds
Respiration rate (from Capnometer) high	4 seconds
Respiration rate (from Capnometer) low	4 seconds
Respiration rate (from ECG) high	4 seconds
Respiration rate (from ECG) low	4 seconds
Capnometer no breaths detected	10 - 60 seconds (depending on delay setting)
Non-invasive blood pressure systolic high	0 seconds at end of measurement
Non-invasive blood pressure systolic low	0 seconds at end of measurement
Non-invasive blood pressure mean high	0 seconds at end of measurement
Non-invasive blood pressure mean low	0 seconds at end of measurement
Non-invasive blood pressure diastolic high	0 seconds at end of measurement
Non-invasive blood pressure diastolic low	0 seconds at end of measurement
Contact temperature low	4 seconds
Contact temperature high	4 seconds
Invasive pressure systolic high	4 seconds
Invasive pressure systolic low	4 seconds
Invasive pressure diastolic high	4 seconds
Invasive pressure diastolic low	4 seconds
Invasive pressure mean low	4 seconds
Invasive pressure mean high	4 seconds
ECG monitoring arrhythmia - Extreme Brady	6 seconds (*)
ECG monitoring arrhythmia - Extreme Tachy	7 seconds
ECG monitoring arrhythmia - Vent Tachy	5 seconds
ECG monitoring arrhythmia - Vent Fib	9 seconds
ECG monitoring arrhythmia - Asystole	6 seconds
QT Alarm	60 seconds

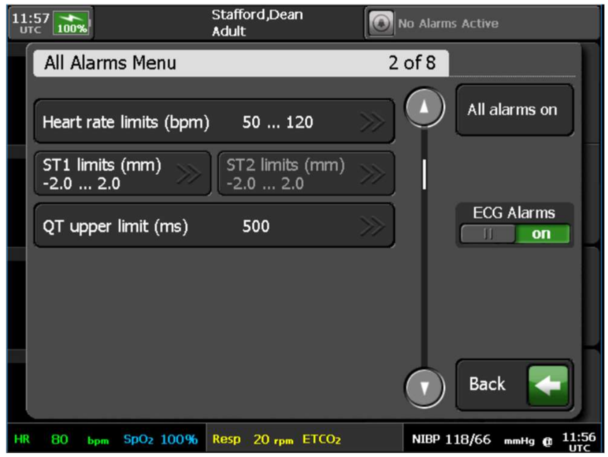
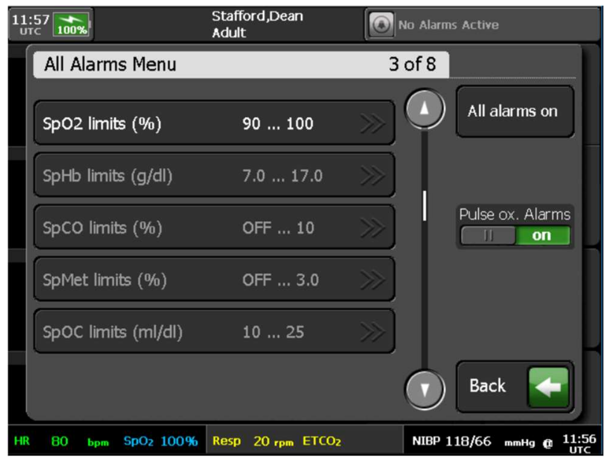
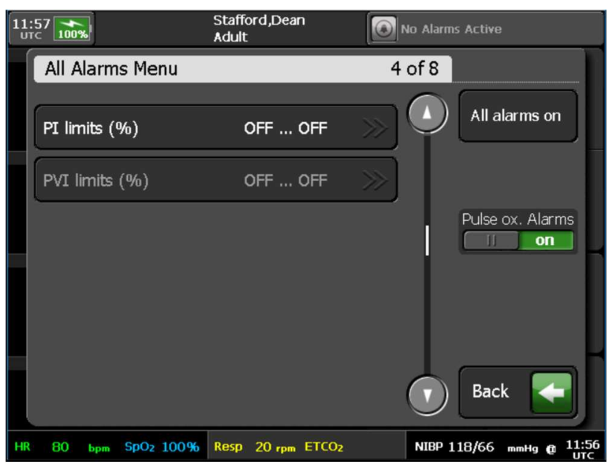
Alarm	Nominal Time to Delay (Excluding User Configurable Delay)
ST Alarm	60 seconds

(*) The user configurable delay is not applicable to this alarm

7.5.2 All alarms menu

CAUTION ⚠	The **Reset to defaults** button resets the following to their new patient default values: all alarm limits, Alarm trigger delay (Alarm suspend time), Alarm silence time, NIBP mode, NIBP adult inflation, NIBP pediatric inflation, NIBP neonate inflation, HR/PR source, Pacemaker indication, Arrhythmia analysis, ST/QT analysis, Monitoring filter, 12-lead filter and ECG power filter.

Menu option	Menu screen
Alarm trigger delay (secs) – set the delay between detection of an alarm condition and triggering of the audio and visual alarm. Range is from zero to the maximum allowed delay (set by the maintenance user, may be up to 8 seconds). **Alarm volume** – select **Low**, **Min**, **Mid** or **High** volume. The maintenance user may have disabled one or more of these settings. **Waveform snapshot on alarm** – store a snapshot of the waveform when an alarm is triggered (on or off). Overrides the default setting for the duration of the current patient incident only. **Alarm silence time** – set the time period in which audible alarms will be silenced when the Alarm Silence button is pressed. The available range is from 30 seconds up to the maximum value set in the Maintenance Menu. **All alarms on** – activate all alarms (present on all pages of this menu). **Reset to defaults** - reset all alarm limits and other parameters to their new patient default values.	 **Note:** You cannot set **Alarm trigger delay** to exceed your organization's maximum allowed delay (set by the maintenance user in the New Patient Defaults menu). If the maximum is zero, the **Alarm trigger delay** option is disabled here and alarms are triggered immediately on detection.

Menu option	Menu screen
Heart rate limits – set alarm limits. **ST1 limits** – set alarm limits. **ST2 limits** – set alarm limits. **QT upper limit** – set alarm limit. **ECG Alarms** – turn ECG alarms off or on.	
SpO2 limits – set alarm limits. **SpHb limits** – set alarm limits. **SpCO limits** – set alarm limits. **SpMet limits** – set alarm limits. **SpOC limits** – set alarm limits. **Pulse ox. Alarms** – turn pulse ox. alarms off or on.	
PI limits – set alarm limits. **PVI limits** – set alarm limits.	

<table>
<tr><th>Menu option</th><th>Menu screen</th></tr>
<tr><td>

ETCO2 limits – set alarm limits.

Capno no breaths – set alarm limit.

Respiration limits – set alarm limits.

T1 limits and **T2 limits** – set the lower and upper temperature alarm limits for each channel.

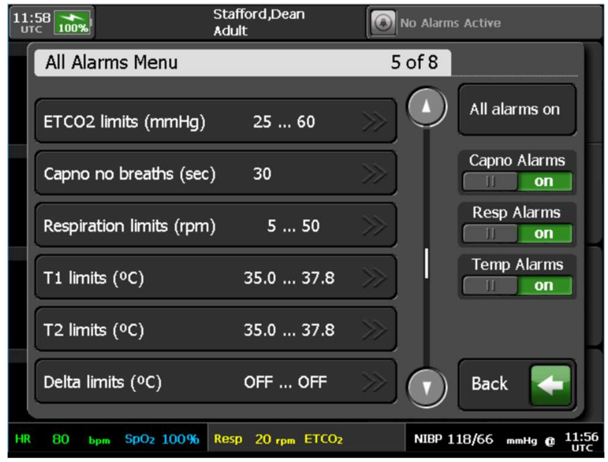
Delta limits – set the delta temperature alarm limits, based on the difference between T1 and T2.

Capno Alarms – turn capnometer alarms off or on.

Resp Alarms – turn respiration alarms off or on.

Temp Alarms – turn temperature alarms off or on.

</td><td>



Note: If there is only one contact temperature channel, the **Temp limits** control replaces **T1 limits, T2 limits** and **Delta limits**.

</td></tr>
<tr><td>

NIBP systolic limits – set alarm limits.

NIBP mean limits – set alarm limits.

NIBP diastolic limits – set alarm limits.

NIBP Alarms – turn NIBP alarms off or on.

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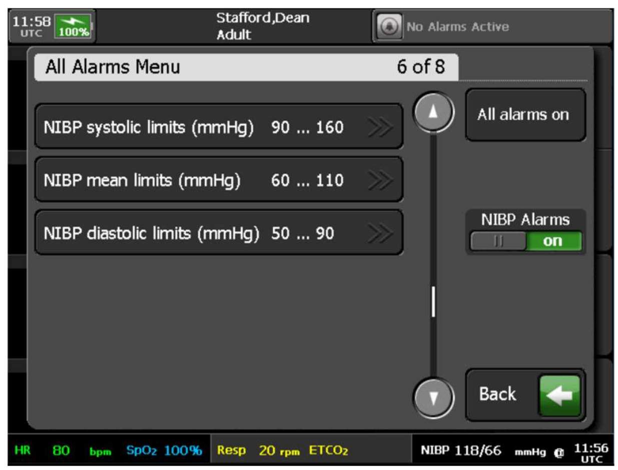


</td></tr>
<tr><td>

P1 systolic limits – set alarm limits.

P1 mean limits – set alarm limits.

P1 diastolic limits – set alarm limits.

P2 systolic limits – set alarm limits.

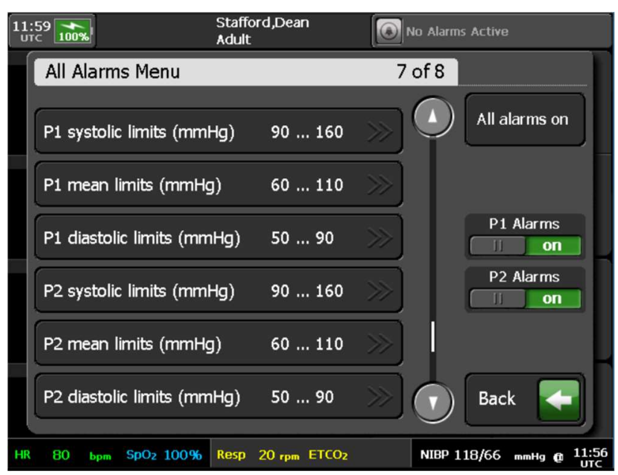
P2 mean limits – set alarm limits.

P2 diastolic limits – set alarm limits.

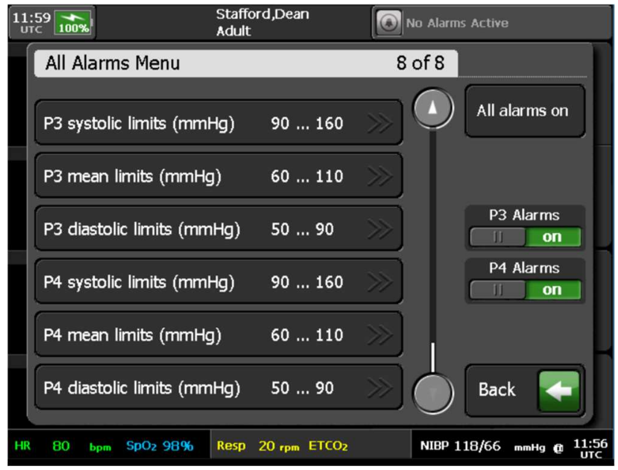
P1 Alarms – turn P1 alarms off or on.

P2 Alarms – turn P2 alarms off or on.

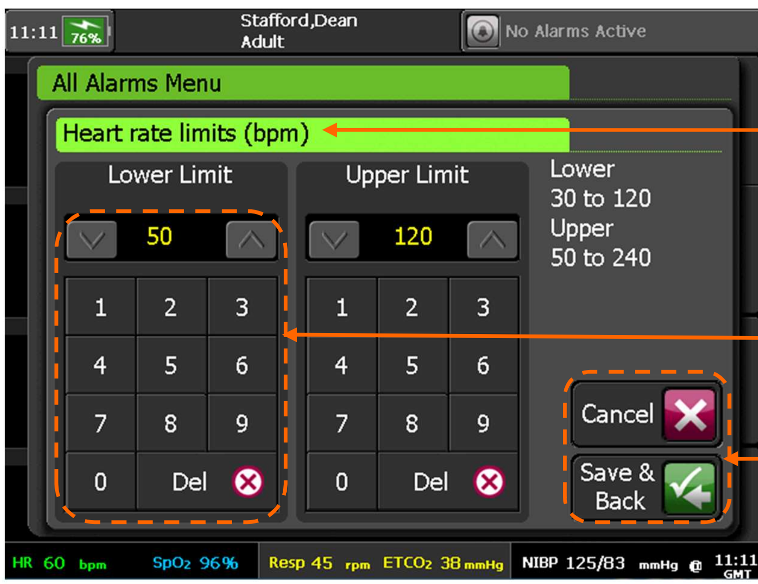
</td><td>



</td></tr>
</table>

Menu option	Menu screen
P3 systolic limits – set alarm limits. **P3 mean limits** – set alarm limits. **P3 diastolic limits** – set alarm limits. **P4 systolic limits** – set alarm limits. **P4 mean limits** – set alarm limits. **P4 diastolic limits** – set alarm limits. **P3 Alarms** – turn P3 alarms off or on. **P4 Alarms** – turn P4 alarms off or on.	

Simply press on the alarm that you wish to set and an editor will appear. This will allow you to edit both upper and lower limits of the alarm. The editor shows the range of the alarm settings. You change the alarm settings by simply typing in the desired value using the keypad or the up and down arrows. You can save the changes you have made or cancel them by pressing the relevant buttons at the bottom right.



The alarm being edited is shown in the title

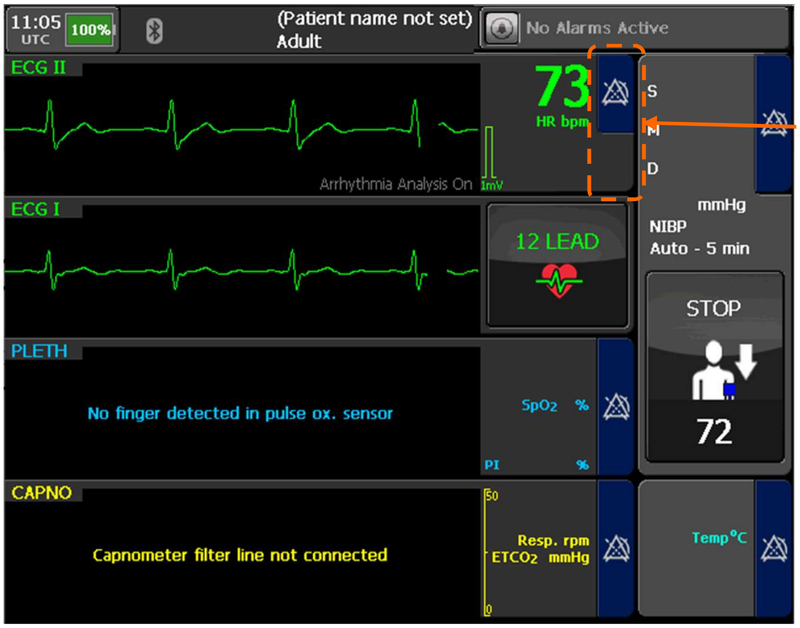
Change the upper or lower limits by typing the desired value into the keypad or use the arrows

Press here to cancel changes and go back or save changes and go back

Editing heart rate alarm limits

Note	The alarm limit shown is the number that the Tempus will display before the alarm is activated i.e. in the example shown above the tempus will display when the heart rate is less than 50 (49 or less) or greater than 120 (121 or more). The Tempus will not alarm at 50-120 inclusive. Therefore, it is possible to have the lower and upper limits set to the same value.

If any alarms are turned off, this will be shown on screen by the relevant parameter highlighted in blue.



The ECG Alarm bar shows the symbol for alarms inactive and is highlighted in blue

Alarms suspended

It should be noted that other system configuration changes e.g. ECG lead selection, QRS beat volume, gain or waveform speed changes, communications mode and communications settings, iAssist mode etc. are all settings which are stored by the Tempus. Therefore, any changes to the Tempus configuration **other than alarm settings** will remain set even after the Tempus has been switched off and on again (for periods less than 72 hours).

7.6 Silencing or suspending alarms

The audible alarm can be silenced by pressing the [Alarm Silence] button  on the front panel.

For most patient alarms, the [Alarm Silence] button mutes the audible alarm for 2 minutes (factory default), after which the alarm will sound again if the alarm state is still present. During this time the visual indications of the alarm will remain (the alarm LED will remain lit and the on-screen indications will remain).

Note	The maximum (and default) alarm silence time is configurable via the maintenance menu. For instructions, see the *Tempus Pro Maintenance Manual*. The factory default is 2 minutes with an available range from 1 to 5 minutes.
Note	The alarm silence time may be adjusted by the user from the All Alarms menu. The adjustment is limited in the range 30 seconds up to the value of alarm silence time set via the maintenance menu. The alarm silence time will reset to the preset maintenance value when a new patient is admitted, or when switching patients.
Note	During the alarm silence period, if a new alarm occurs (patient or technical) the alarm will sound as normal.

For certain patient alarms, pressing the Alarm Silence button ceases the alarm permanently until the condition is experienced again. This applies to the following alarms:

- Capnometer No Breaths Detected;
- NIBP patient alarms;
- Catheter Disconnected.

For technical alarms, the [Alarm Silence] button acknowledges and silences all active technical alarms and causes them to be displayed with this symbol:

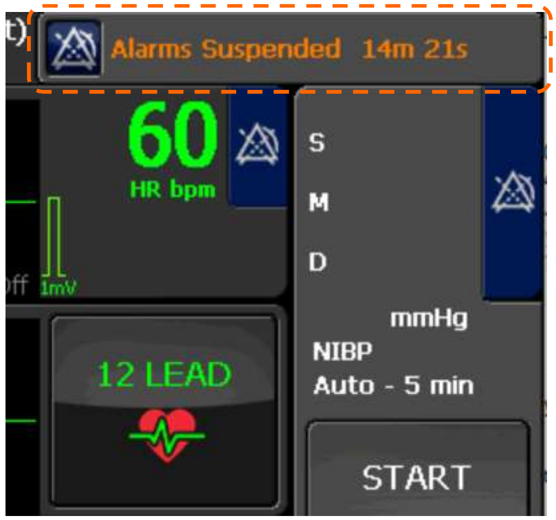
Alarm Suspend and Silence Buttons

Alarms can also be silenced before they occur using the [Alarm Suspend] button . This enables users to prevent alarms from occurring for two minutes due to intentional intervention with the patient e.g. removing or moving a patient sensor such as an ECG electrode or pulse oximeter probe.

The [Alarm Suspend] button is positioned above the [Alarm Silence] button and has a different function. Pressing it will disable all audible **and** visual alarm indications for **both** technical and patient alarms for two minutes. During this time the yellow LED above the button will remain lit and the alarm bars on the display will indicate that alarms are paused and show a countdown for when they will be resumed.

Note	If a new technical alarm condition occurs during an alarms suspended period, once that period has expired there may be a delay of up to 15 seconds before the technical alarm audio is sounded.

The alarm suspension state can be deactivated at any time during the countdown by pressing the [Alarm Suspend] button again.



The alarm bar shows that alarms have been suspended

Alarm bar showing alarms are suspended

When a patient alarm is silenced the visual indication will remain on screen until the condition has passed i.e. the patient's vital signs have returned to a level within the alarm thresholds.

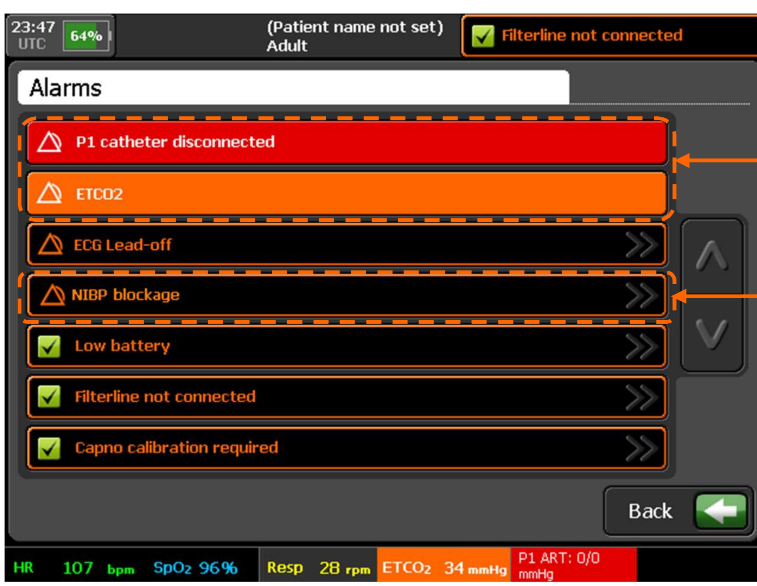
The non-invasive blood pressure is a periodic measurement. For this reason, an NIBP alarm is cleared by pressing the [Alarm Silence] or [Alarm Suspend] button. The alarm may then return when the next periodic reading is taken.

7.7 Viewing and clearing alarms

A list of active patient and technical alarms can be viewed on a screen, from which technical alarms can be selected to display more information and to clear them.

Note	Patient alarms cannot be cleared in this way.

To display the "Alarms" screen, press the touch screen in the alarm status area. The example screen shown below shows two patients and five technical alarms including three already silenced and acknowledged – see "7.6 Silencing or suspending alarms".



These are patient alarms

Press on any technical alarm to display its description

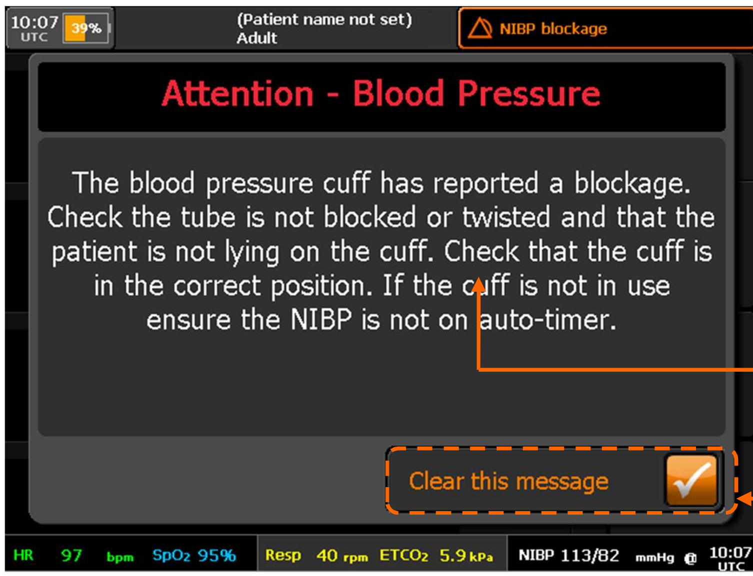
"Alarms" screen with multiple active alarms

Alarms are shown on the Alarm Status screen in the following priority order:

- Patient alarms are shown before technical alarms.
- Technical alarms are shown on a first in-first out basis.

Note	The following technical alarm conditions take priority over patient alarms: Battery Empty (lasts for 10 seconds or so before automatic shutdown occurs) and Ambient Temperature High or Low.

Press any technical alarm in the status screen to display a dialog which describes the nature of the condition alarm state (as shown below).



A title and detailed description will be shown in the center of the dialog box.

Press here to acknowledge the message and clear the alarm state

Clearing a technical alarm

To clear technical alarms, use one of the following methods:

- Either: press the "Clear this message" button on the Alarm Description Dialog screen;
- Or: correct the error that caused the alarm to be raised, for example refit a patient sensor that has been disconnected.

7.8 Tactical mode warning

If the Tempus is set to Tactical (Silent) Mode, then all audible alarms will be ceased (visual alarm indications will remain) – see "4.3.2 Tactical mode (optional)".

Setting Tactical Mode to **ON** will give the following warning:

Once Tactical Mode is set, this will be permanently displayed in the alarm status bar.

Tactical mode warning in status bar

Once set, Tactical Mode can be turned off by switching the physical switch back to its original position – see "4.3.2 Tactical mode (optional)".

WARNING	**Do not leave a unit in tactical mode without careful observation for alarm conditions on the screen.**

7.9 Testing alarms

RDT recommends that visual and audible alarms are tested before use.

The alarm bar will light and the audible alarm will sound on switch on (unless the tactical switch is enabled). If users do not perceive this test occurring, they should provoke a test to check that the visual and audible alarms are working before they begin to use the device. A simple way to do this would be to insert a finger into the pulse oximeter probe, wait for a reading to appear and then remove the finger. This will provoke a technical alarm which will test the alarm bar (will light solid yellow) and the alarm tone.

8 Summary Record of Care

Recording patient encounter data can be done with the Summary Record of Care (SRoC) feature which is accessed by pressing the Event button.

The Event button.

Event general, assessment, intervention, fluid and drug names and details are organizationally configurable. The default names and details can all be changed to match the organization's needs.

WARNING	**Event naming and detailed configuration must be reviewed and checked before the Tempus Pro is deployed in service. Read the details regarding event configuration in the *Tempus Pro Configuration Utility User Guide*.**

8.1 Event capture

The SRoC Events screen is displayed when the Event button is pressed.

8.1.1 General screen

The SRoC "General" screen allows you to quickly record a category marker or multiple events up to a maximum of 200 events. General events are a subset of events selected in the configuration from the full set of assessments, interventions, fluids and drugs. They are typically the most commonly needed events. All general events are also accessible from one of the other event tabs.

To record an event category for later editing, press one of the category markers on the left of the screen. The Tempus will record the category marker and automatically return to the monitoring screen.

Use the "General" screen to record events:

- To record an event with the current time but no other details, press and release the event button.
- To edit the time and details before recording the event, press and hold the event button. The Details Editor screen is displayed.

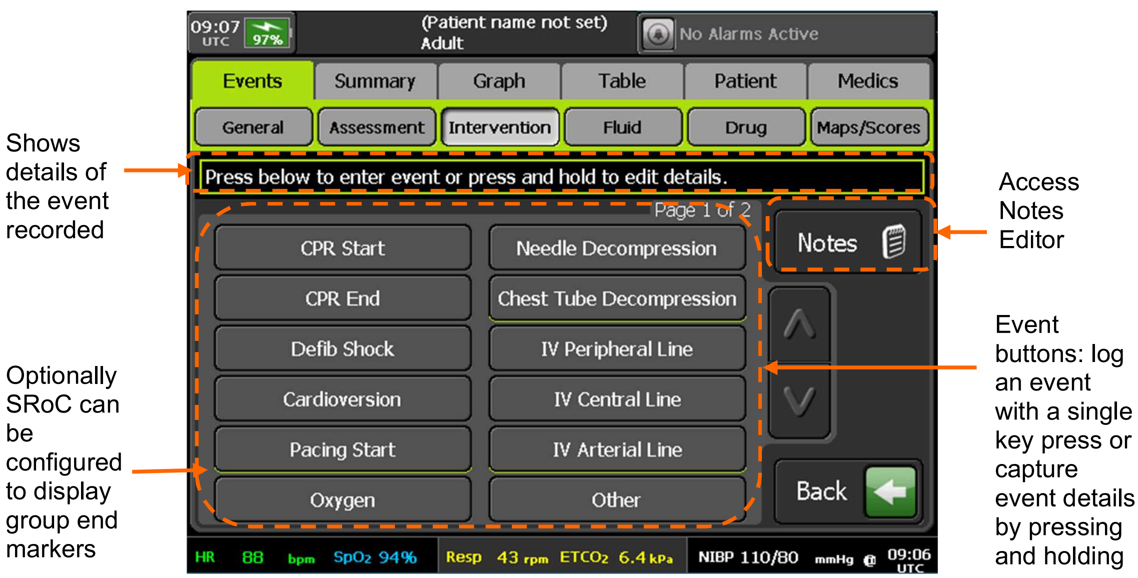
Note	If you subsequently change the time of a recorded event, the Tempus will clear the recorded vitals.

SRoC “General” screen

Note	The Tempus Pro can be configured to display group end markers in the SRoC “General” screen. For more information, see the *Tempus Pro Configuration Utility User Guide*.

8.1.2 Assessment, intervention, fluid and drug screens

The “Assessment”, “Intervention”, “Fluid” and “Drug” screens behave in a similar way to the “General” screen, allowing you to record multiple events of the corresponding type. Category markers are not available in these screens.



SRoC “Intervention” screen

Assessment events

Use the “General” or “Assessment” screens to record assessment events:

- To record an assessment event with the current time but no other details, press and release the event button.
- To edit the time and details before recording the assessment event, press and hold the event button. The assessment details editor screen is displayed, allowing you to edit the date and time, add an event note, or delete the event.

Assessment details editor

Intervention events

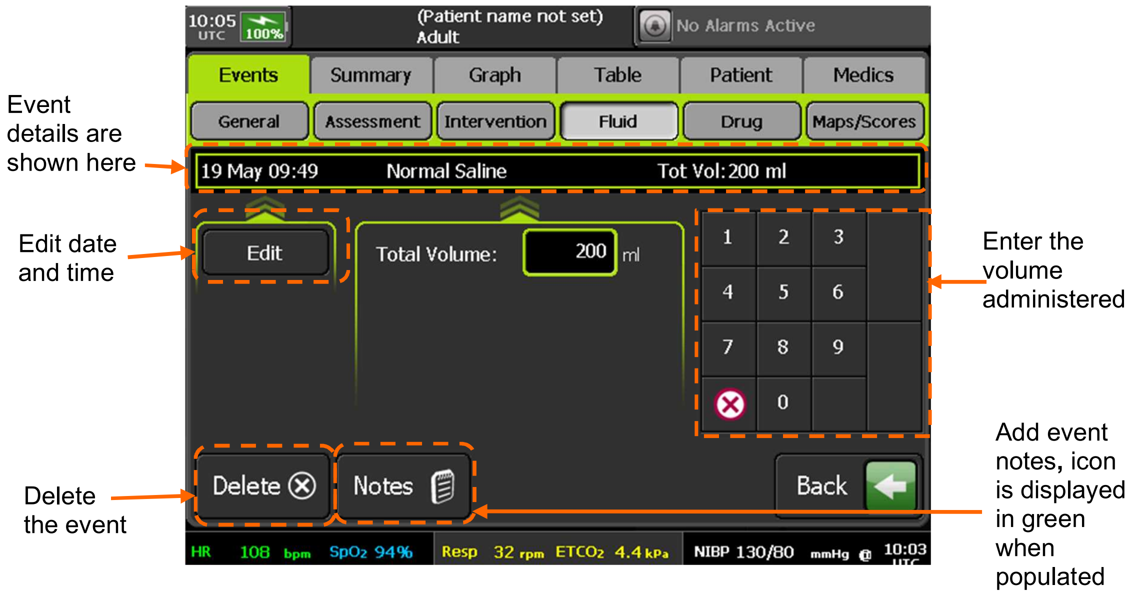
Use the “General” or “Intervention” screens to record intervention events:

- To record an intervention event with the current time but no other details, press and release the event button.
- To edit the time and details before recording the intervention event, press and hold the event button. The intervention details editor screen is displayed, allowing you to edit the date and time, add an event note, or delete the event. This screen contains the same controls as the assessment details editor (above).

Fluid events

Use the “General” or “Fluid” screens to record fluid events:

- To record a fluid event with the current time but no other details, press and release the event button.
- To edit the time and details before recording the fluid event, press and hold the event button. The fluid details editor screen is displayed, allowing you to edit the date and time, edit the volume, add an event note, or delete the event.

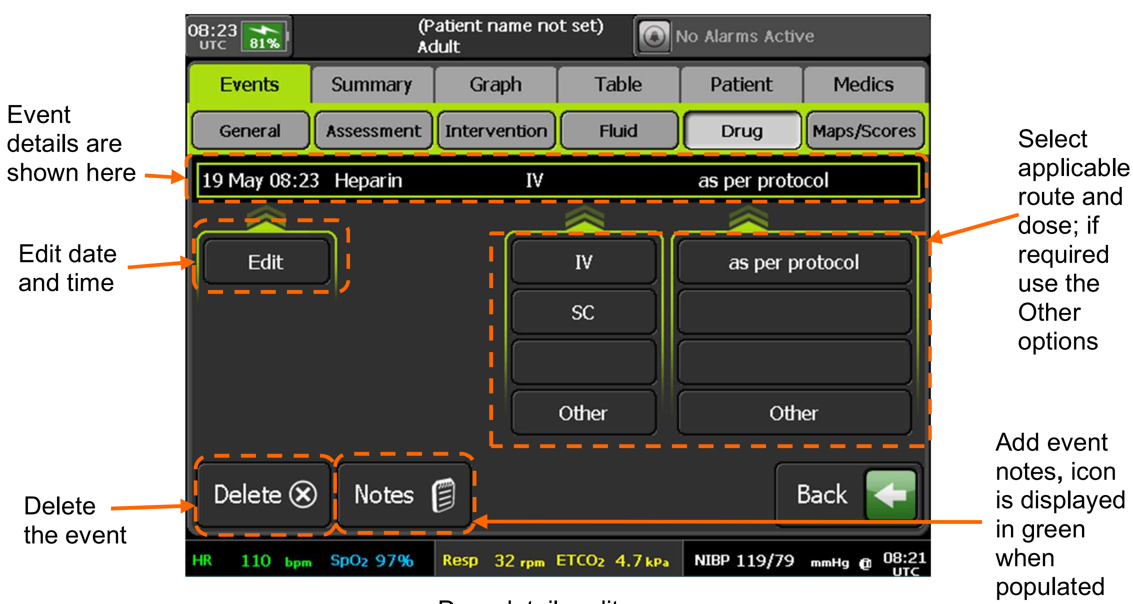


Fluid details editor

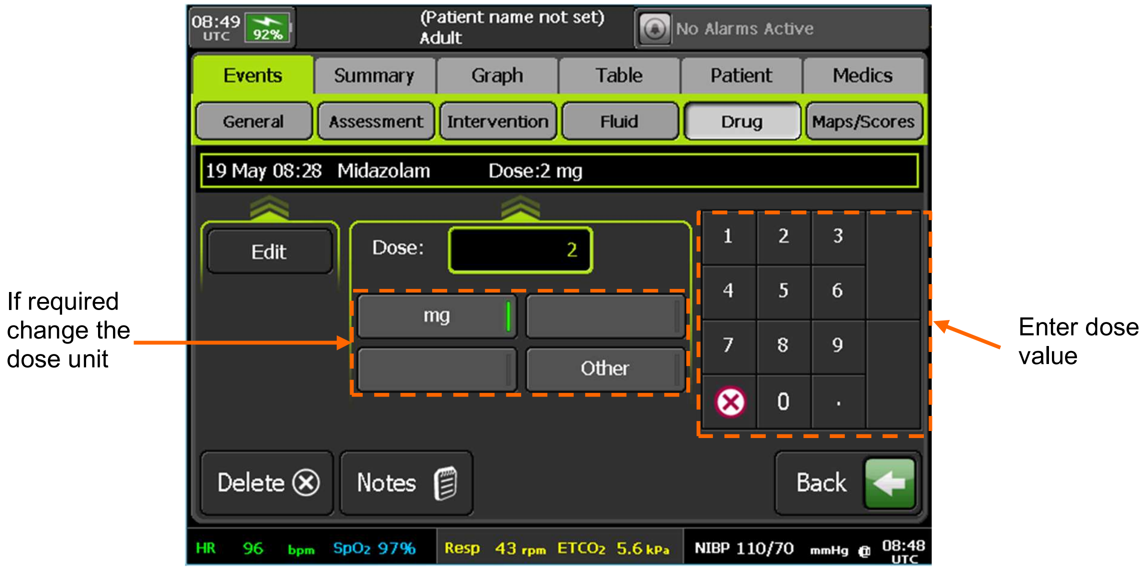
Drug events

Use the "General" or "Drug" screens to record drug events:

- To record a drug event with the current time but no other details, press and release the event button.
- To edit the time and details before recording the drug event, press and hold the event button. The drug details editor screen is displayed, allowing you to edit the date and time, edit the route, edit the dose, add an event note, or delete the event.



Drug details editor

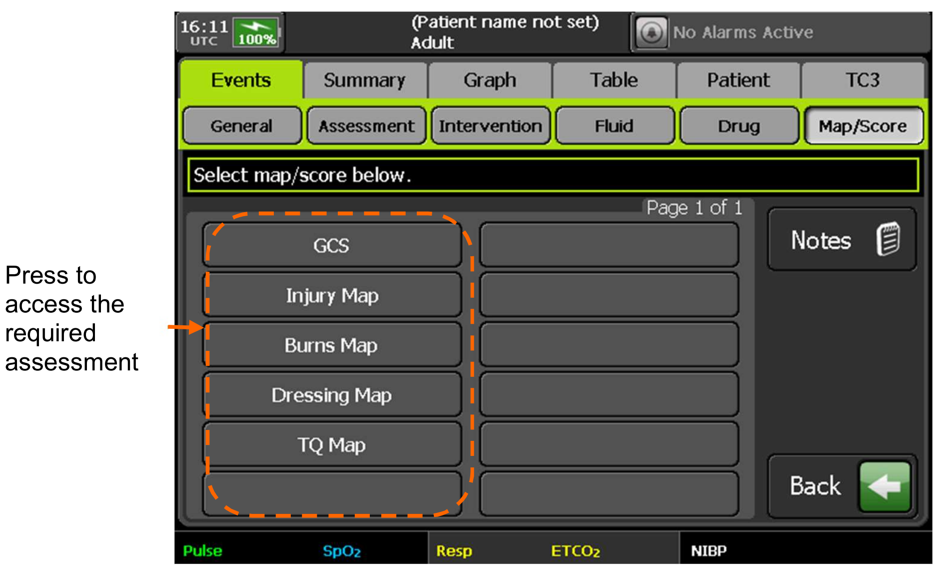


Drug dose editor

8.1.3 Map/score screen

The “Map/Score” screen provides access to the following editors:

- Glasgow Coma Scale (GCS)
- Injury Map
- Burns Map
- Dressing Map
- TQ Map

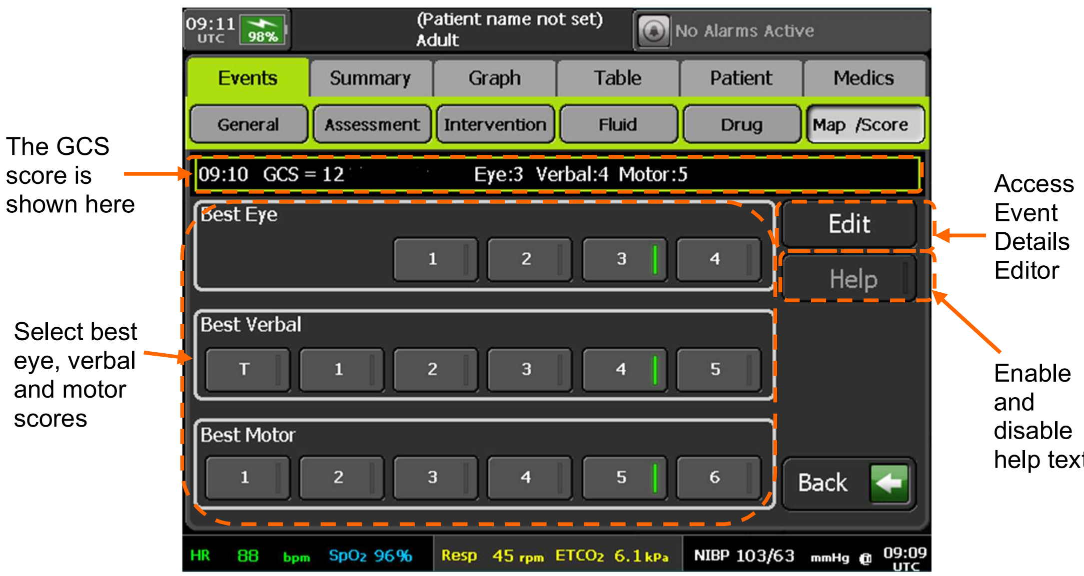


SRoC map/score screen

GCS screen

The Glasgow Coma Scale (GCS) screen allows you to enter scores for Best Eye, Best Verbal and Best Motor.   GCS is calculated by summation of the three scores. A ‘T’ verbal score indicates that the patient is intubated and provides a verbal score of 1.

CAUTION ⚠	If you exit the GCS screen without entering all three scores, then no GCS data is recorded.

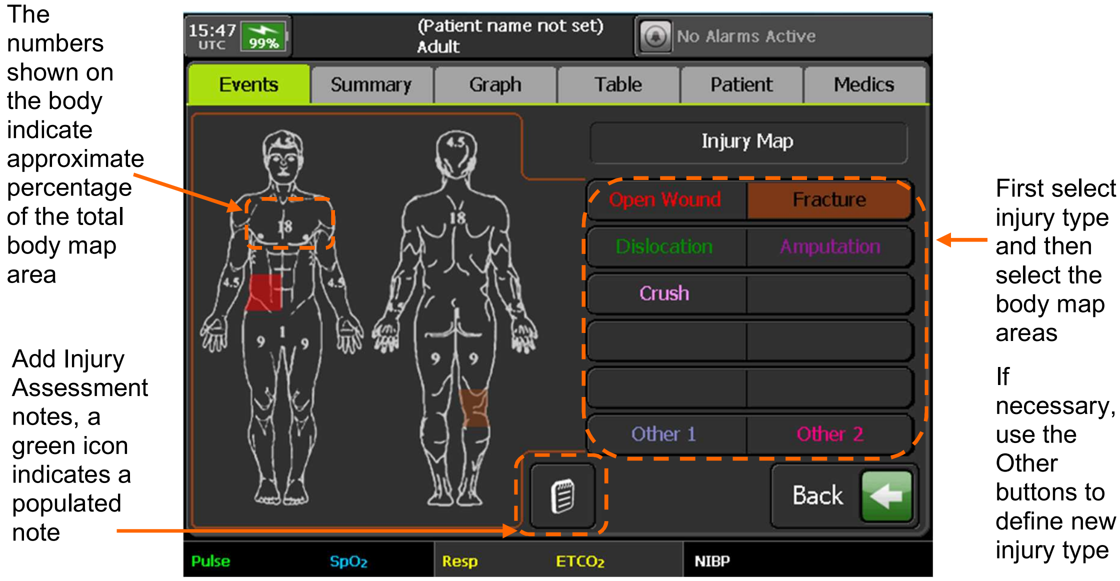


<u>GCS assessment screen</u>

To display help text, press the Help button. When you select a score with the Help button enabled, the Tempus will automatically move to the next entry screen.

Injuries, burns and dressings maps

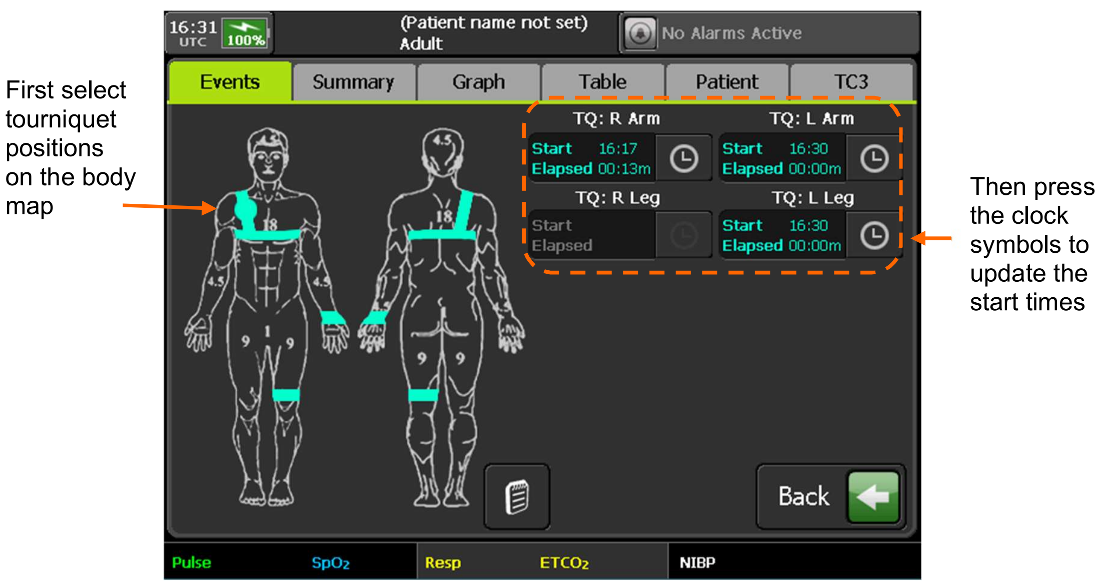
The Injury, Burns and Dressing Map screens allow you to record the corresponding encounter details on a body map which contains a graphical representation of the front and rear of the body. The Burns Map also allows you to record the total burns area as part of the burns assessment.



Injury map screen

TQ map

The TQ map screen allows you to record limb tourniquet positions and start times.



TQ map screen

8.1.4 Automatic event capture

SRoC is automatically augmented with patient encounter data when:

- An arrhythmia alarm is triggered - Tempus Pro automatically logs an event and records a waveform snapshot from 10 seconds before until 10 seconds after the event.
- A patient alarm is triggered - Tempus Pro automatically logs an event (if configured in the All Alarms menu). You can also store a snapshot of the waveform at the same time, see the All Alarms menu.

- A waveform is manually captured. To manually capture a snapshot of a waveform at any time, press and release the Camera & Waveform Snapshot button.
- A 12 Lead ECG is recorded.
- A camera or laryngoscope image is recorded. For laryngoscope details refer to *Tempus Pro Laryngoscope Supplement Guide*.
- An ultrasound image is recorded. Images can be taken during general and FAST exams. Refer to *Tempus Pro Ultrasound Supplement Guide*.

If the Tempus is fitted with an internal printer and automatic printing is enabled, it will print waveforms as soon as they are captured.

8.2 Events summary list

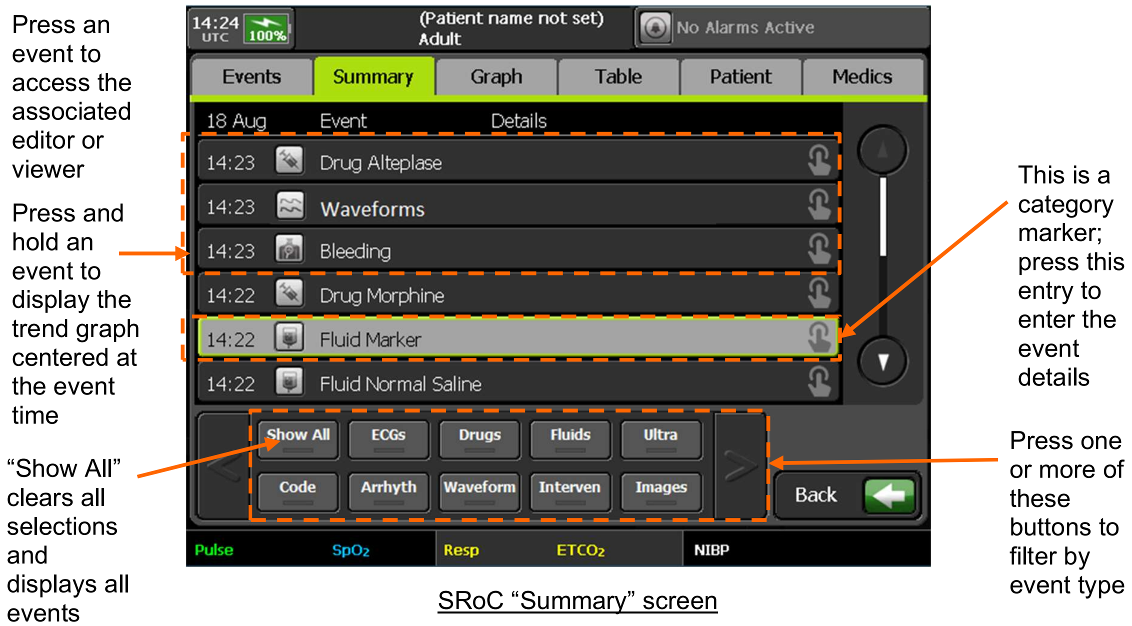
The SRoC Summary screen displays a list of all the events entered by the user together with any events automatically captured. Each event is shown with the associated event icon as detailed in the table below.

Event Type	Filter button	Icon
All	Show All	All
Arrhythmia Waveform	Arrhyth or Waveform	
12 Lead ECG	ECGs	
Camera Image	Images	
Drug	Drugs	
Fluid	Fluids	
Note	Notes	
Intervention	Interven	
Laryngoscope Image	Images	
Assessment	Assess	
Waveform (Manual)	Waveform	
Medics	Medics	
Alarm Waveform	Waveform	
Injury / Burns Dressing / TQ	Maps	
GCS	Scores	
Ultrasound Image	Ultra	
Category marker: assessment, intervention, fluid or drug	Markers	Icon matches marker type

Filters allow you to select which events are shown. Pressing an event will launch the associated editor /

viewer, automatically recorded events can only be viewed. Category event markers can be updated by pressing on the associated row. Pressing and holding an event that is less than 72 hours old will act to launch the trend graph centered at the event time, the graph will be set to show events and set to 45 minutes zoom.

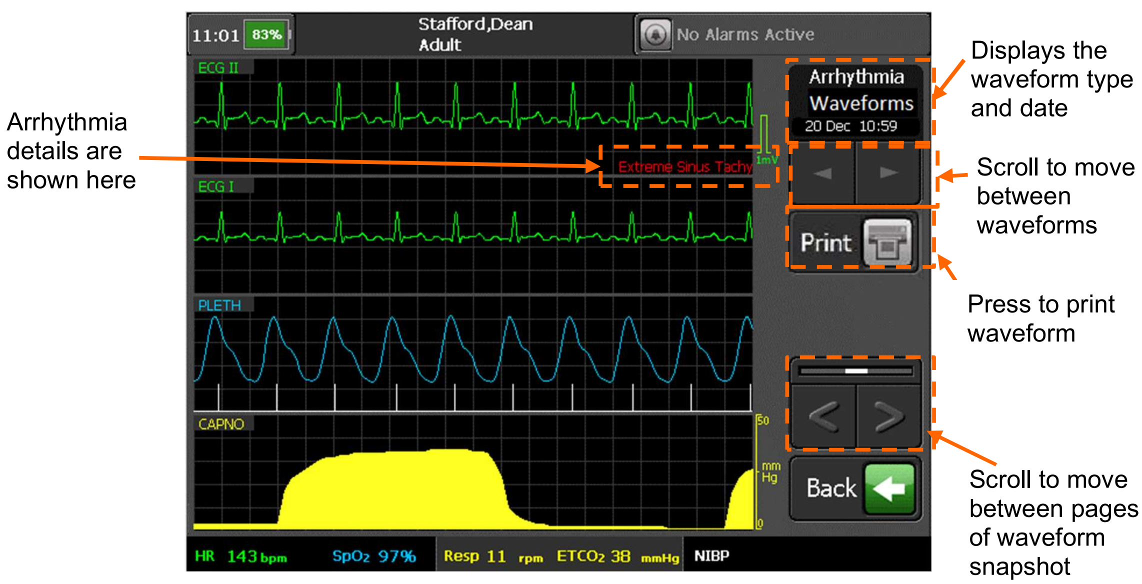
8.2.1 Summary screen



SRoC "Summary" screen

8.2.2 Waveform viewer

To view a waveform, press the waveform event in the SRoC Summary screen:



Waveform snapshot viewer

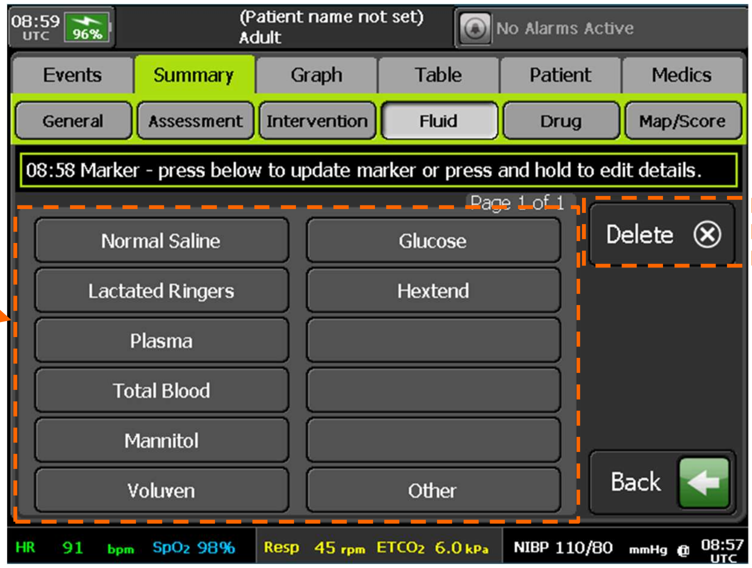
If the Tempus is fitted with an internal printer, you can print the waveform using the **Print** button.

If the Tempus Pro is fitted with an internal printer and automatic printing is enabled, it will print waveforms as soon as they are captured.

8.2.3 Updating a category marker

To convert a category marker to a fully recorded event, press the category marker entry in the SRoC Summary screen. The Tempus displays the entry screen for the relevant event category. Press an event button: this acts to update the marker. Any further events buttons pressed are recorded as new events.

To update marker press the required event. Press and hold required event to edit event details



Press to delete the marker

Updating category marker

8.3 Trends

Tempus automatically samples all continuous medical readings once per minute and stores these in the patient record. For readings taken less often such as Non-Invasive Blood Pressure all readings are stored in the patient record. The last 72 hours of trend data can be displayed on the Tempus as both graphical and tabular trend data. To access trend data, use one of the following methods:

- Press the [Event] membrane button then select the **Graph** or **Table** tab.
- Press the **Trends** button (displayed in all medical related menus as well as the main menu) then select the **Graph** or **Table** tab.
- Press and hold anywhere on the home screen. Depending on which waveform you press the graph will show you the results of that waveform.
- Press and hold on any event in the **Summary** tab. The graph will be centered at the time of the event you press.

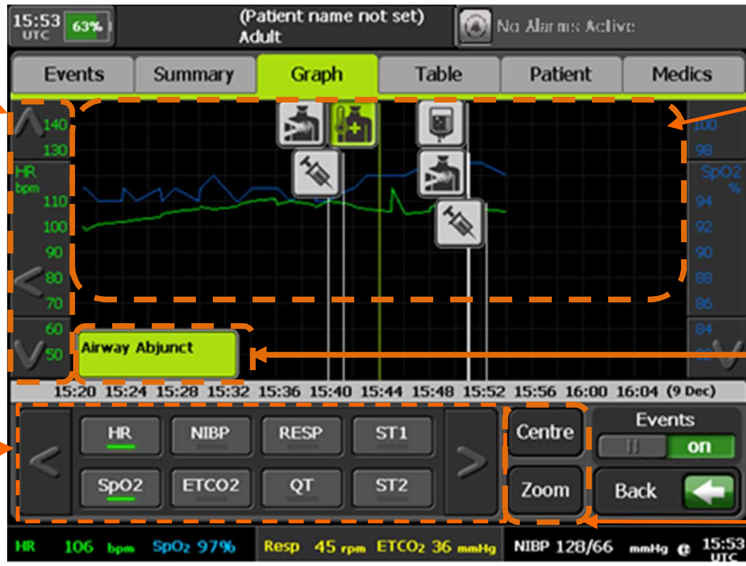
8.3.1 Trend graph

The trend graph allows you to select two vitals to be shown together, each vital having its own scale. The zoom button can be used to select a different time scale for the X axis. When enabled, icons for the following events are shown on the graph at the event time.

- Arrhythmia
- 12-Lead ECG
- Drugs
- Fluids
- Assessments
- Interventions

- GCS events, but not body maps
- Laryngoscope image

A maximum of 4 events can be displayed on the graph in a 4-minute interval and there can be a delay of up to 1 minute before events are displayed.



Trend graph

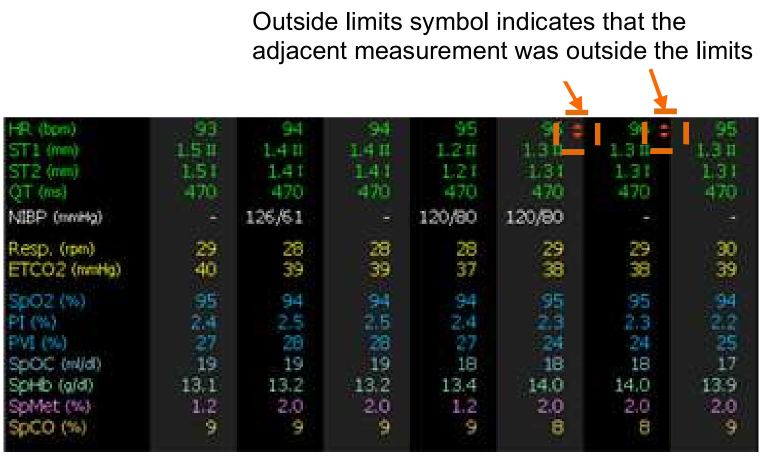
8.3.2 Trend table

The Trend Table allows you to scroll through all the 1-minute sample readings. When enabled the following events are also displayed:

- Waveform
- Arrhythmia
- Patient Alarm
- Camera Images
- Laryngoscope Images
- 12-Lead ECG
- Drugs
- Fluids
- Assessments
- Interventions
- Body Maps and GCS
- Ultrasound Images

Trend table

An outside limits symbol placed to the right of reading indicates that the value was outside the limits set when the measurement was taken. In the case of a reading containing multiple values such as non-invasive blood pressure the alarm symbol will not show which value was outside the limits.



Outside limits

The outside limits symbol will only be shown for patient alarms. It will not be shown to reflect any technical alarms such as low battery, finger out of SpO2 sensor etc. It will not be shown for invasive pressure “transducer removed” but will be shown for Capnometer “zero breaths detected”. The symbol shows only that the value at the time was outside the alarm limit that was set at the time – it will not change based on subsequent editing of the alarm threshold. It will not be shown for “out of range measurements”. It will not reflect if an audible alarm was present i.e. will not differentiate if an alarm is muted, alarms suspended etc. it will merely reflect that the measured parameter was outside of the alarm threshold set at the time.

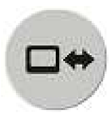
Note	The trend table shows a single reading where multiple readings may have occurred e.g. 60 individual readings for heart rate that are taken in a minute are represented as a single reading; this is the first reading recorded in that period. If a single alarm on a given parameter occurs within the time period set, then this will be shown in preference to the first reading taken within the period. If multiple alarms occur in the same period on the same parameter, then the trend table will show the first to occur.

9 Other features of the Tempus Pro

This section details the other features of the Tempus Pro.

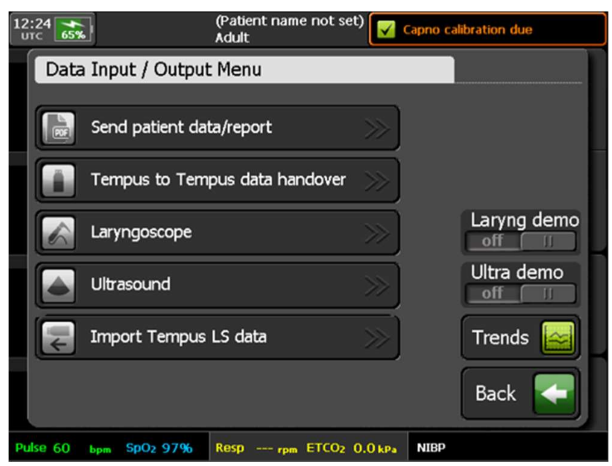
9.1 Data input and output

9.1.1 The Data Input and Output menu



The **Data Input/Output** button.

Pressing the **Data Input/Output** button launches a menu which allows you to do the following:

Menu option	Menu screen
Send patient data/report – send patient data or reports to external recipients, see "9.1.2 Send patient data/report". **Tempus to Tempus data handover** – export and import patient data for handover to or from another Tempus, see "9.1.3 Tempus to Tempus data handover". **Laryngoscope** – access laryngoscope functionality. For further details refer to *Tempus Pro Laryngoscope Supplement Guide*. **Ultrasound** – access ultrasound functionality, for further details refer to *Tempus Pro Ultrasound Supplement Guide*. **Import Tempus LS data** - for use with the Tempus LS system.	

9.1.2 Send patient data/report

The **Send patient data/report** option allows you to output the SRoC patient data or report to one of four types of recipient:

- USB memory device – patient report (PDF);
- External printer – patient report;
- Email address (if the feature is installed) – patient report (PDF);
- ePCR (electronic Patient Care Reporting system) (if the feature is installed) – patient data;
- Internal printer (if fitted) - summary report or 30 second waveform.

When sending the patient report to email, USB or external printer, you can select one of the following report formats:

- 12-lead Report
- Summary Report
- Detailed report

When sending to ePCR, all patient data is sent.

The contents of the report or data depend on the report destination, as shown in the following table:

Data	USB, External Printer and Email (detailed report)	ePCR (patient data)	Internal Printer (summary report)
SRoC events	All	All	Last 100 events
Trended vitals	Up to 72 hours	All	Last 200 minutes limited to HR, NIBP, Resp, ETCO2, SpO2, T1 and T2
12-lead ECG recordings	Latest (up to 10)	Latest (up to 10)	None
Camera / intubation images	Latest (up to 10)	Latest (up to 20)	None
Image annotations	Latest (up to 10)	None	None
Waveforms	Latest (up to 20)	Latest (up to 20)	None
Ultrasound images	Latest (up to 5) and FAST exam	Latest (up to 5)	None

Detailed reports are automatically formatted such that graphs and tables are not too long. Consequently, if the recorded incident is up to 40 minutes long, all vital signs graph and table data are shown sampled at one minute intervals. Longer incidents are reformatted as follows:

- Duration 40-200 minutes: sampled at 5 minute intervals.
- Duration over 200 minutes: sampled at 5 minute intervals (latest 200 minutes only).

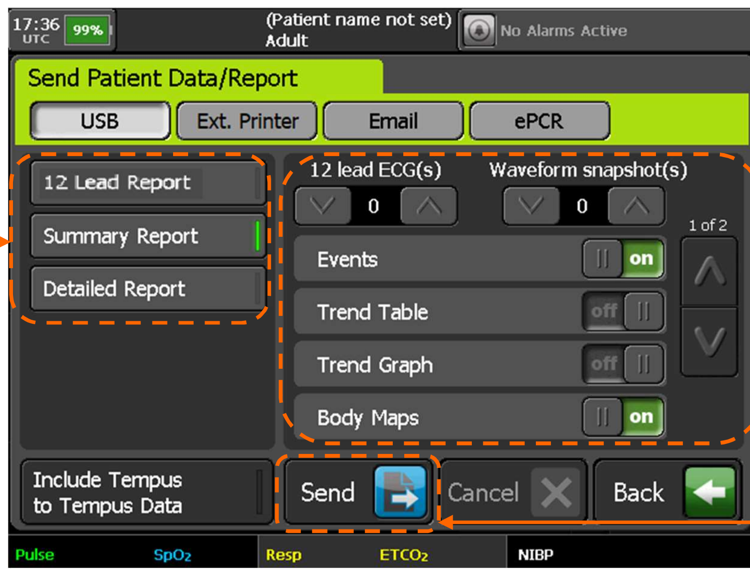
CAUTION ⚠	The reports created are medical records, the use and control of which may be subject to local regulations, e.g. HIPAA in the USA. It is the responsibility of the user to maintain compliance with these regulations. It is the responsibility of the individual or organization who created the record to maintain this information in a secure and confidential state where the integrity and security of the information is not in question.

CAUTION ⚠	Patient reports (PDFs) must be encrypted and password protected. The password must be preconfigured (Maintenance and Settings, Change Passwords, Patient Report Password) by following the procedure in the *Tempus Pro Maintenance Manual.* Configuration must be checked before deployment in service. The encryption type is AES128.

Note	RDT recommends only Adobe® Acrobat® PDF readers that are capable of reading Acrobat V5 or later files are used.
Note	RDT recommends users check that their selected PDF reader can open the created report before deploying the Tempus.

Send patient report to a USB memory device

To send the SRoC patient report (PDF) to a USB memory device, press **Send patient data/report** on the "Data Input/Output" menu, then press **USB** on the "Send Patient Data/Report" screen:



Sending patient report to a USB device

If handover data is required on the USB device, press **Include Tempus to Tempus Data**.

Press **Send**. The Tempus will return to the main monitoring screen. Progress information will be displayed at the top of the screen.

To check progress or cancel USB output, return to the Send Patient Data/Report screen. This screen will display a blue progress bar with status messages. If the Tempus has been configured to encrypt the SRoC patient report using a random password, then the password details will be displayed on the Send Patient Data/Report screen under the list of report formats.

Send patient report to an external printer

WARNING ⚠	**Printer connections should only be made when no connections are made to a patient. Do not connect to a patient and a printer at the same time as this could cause a leakage current hazard or could affect the clinical performance of the Tempus.**

The screen Send Patient Data/Report screen also has an option to send the SRoC patient report directly to a PCL3 compliant USB printer which operates in a similar way to sending to USB memory device. Ensure the printer is attached, switched on and provided with ink and paper before beginning. For printer instructions, see “9.9.1 External printer”.

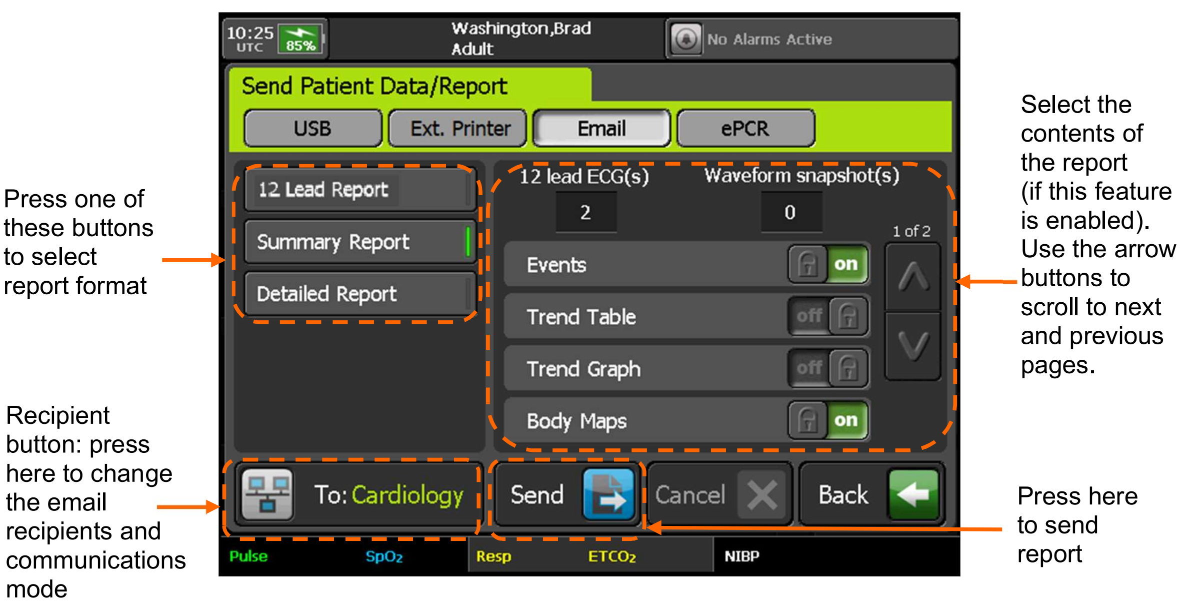
To send the SRoC patient report to an external printer, press **Send patient data/report** on the Data Input/Output menu, then press **Ext. Printer** on the Send Patient Data/Report screen.

Send patient report via email

CAUTION	Email services are generally very reliable but do not guarantee successful delivery. Email send times vary depending upon network conditions. You should always check that an emailed patient report has been received, e.g. by phoning the recipient.
CAUTION	The email service requires a valid email account on a server accessible from the internet. It is your responsibility to understand the local regulations regarding emailing patient data.
CAUTION	Tempus emails must be sent using secure communications.

Note	For email transmission, the maximum allowed patient report size is 8 megabytes. Email service provider limits may differ.
Note	The links below provide information about HIPAA compliant email: http://www.hipaahq.com/hipaa-compliant-email-explained/ http://www.hipaahq.com/hipaa-compliant-email-providers/.

To send the SRoC patient report (PDF) via email, press **Send patient data/report** on the “Data Input/Output” menu, then press **Email** on the “Send Patient Data/Report” screen:



Sending patient report via email

Press the recipient button to display the Email Recipients screen. Update email recipients and communications mode as required, then press **Back** to return to the Send Patient Data/Report screen.

Press the **Send** button. This will cause a PDF report of the selected format to be created and emailed. The Tempus will return to the main monitoring screen and display email status at the top of the screen.

To check progress or cancel emailing, return to the Send Patient Data/Report screen. This screen will display a blue progress bar with status messages. The top status area will display the PDF creation and network status details. When the email has been sent to the email server, the ‘report sent’ text will be displayed on the top status area for a further 30 seconds.

If the Tempus has been configured to use a random password, then a second email containing the random password will be sent to the recipients.

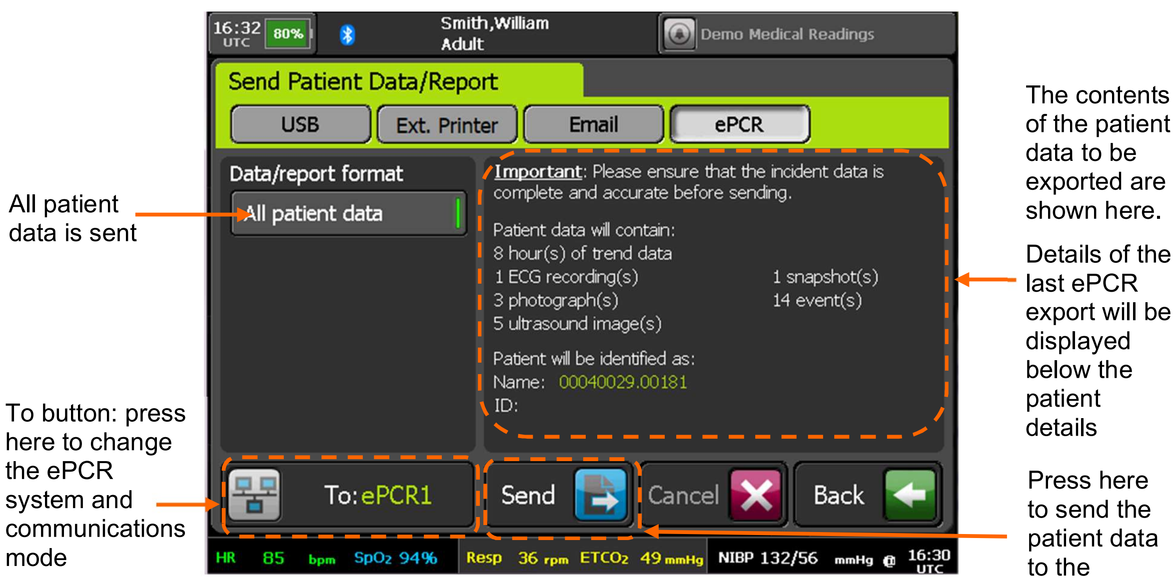
In the event of communication problems, the Tempus will attempt to resend for 20 minutes before aborting the send and displaying a warning message.

Send patient data to an electronic Patient Care Reporting System (ePCR)

CAUTION ⚠	Communication with the ePCR systems are generally reliable but do not guarantee successful delivery and storage of the patient data.
CAUTION ⚠	It is your responsibility to understand the local regulations regarding sending and storing patient data.

Note	Tempus patient data is sent using secure communications to one of the configured ePCR systems. The configuration must be checked before deployment in service.
Note	For ePCR data transmission, the maximum allowed patient data size is 8 megabytes on the Tempus Pro.
Note	The link below provides information about HIPAA compliant cloud storage providers: http://www.hipaahq.com/hipaa-compliant-cloud-storage-explained/.

To send the SRoC patient data files to an ePCR, press **Send patient data/report** on the “Data Input/Output” menu, then select the **ePCR** tab on the “Send Patient Data/Report” screen:



Sending patient data to an ePCR system

Press the **To** button to display the ePCR Selection screen. Select the ePCR system and communications mode as required, then press **Back** to return to the “Send Patient Data/Report” screen.

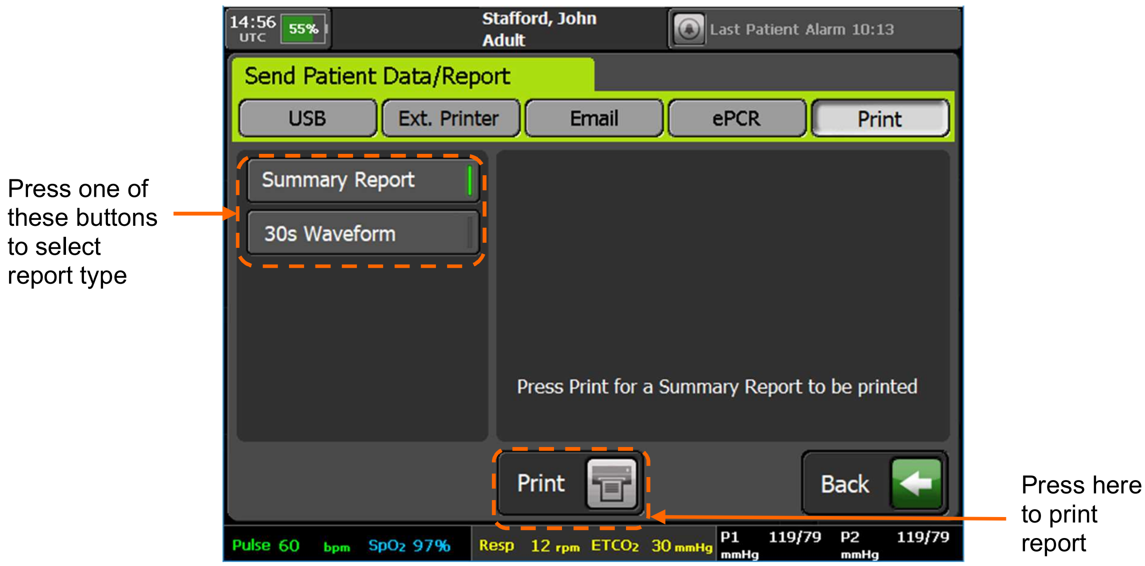
Press the **Send** button. This will cause patient data to be sent to the selected ePCR system.

To cancel sending ePCR data press the **Cancel** button.

In the event of communication problems, the Tempus will attempt to resend for 20 minutes before aborting the data export and displaying a warning message.

Send patient report to the internal printer (optional)

To send the summary report or 30 second waveform to the internal printer (if fitted), press **Send patient data/report** on the “Data Input/Output” menu, then press **Print** on the “Send Patient Data/Report” screen:



Sending patient report to the internal printer

Press a report type button: **Summary Report** or **30s Waveform**.

Press **Print**.

9.1.3 Tempus to Tempus data handover

The “Tempus to Tempus data handover” screen allows you to send or receive patient data using a USB memory device or Tempus to Tempus USB data cable (part number 01-2243). The output is a complete patient data package for handover from one Tempus to another. This allows Tempus data to move with the patient and for the next level care giver to see the history of injuries and treatment in the field.

CAUTION ⚠	Do not import patient records back onto the originating Tempus (directly or once-removed). Exporting / importing is for handing over data from one Tempus to another only.

Prerequisites – ensure that you have the following:

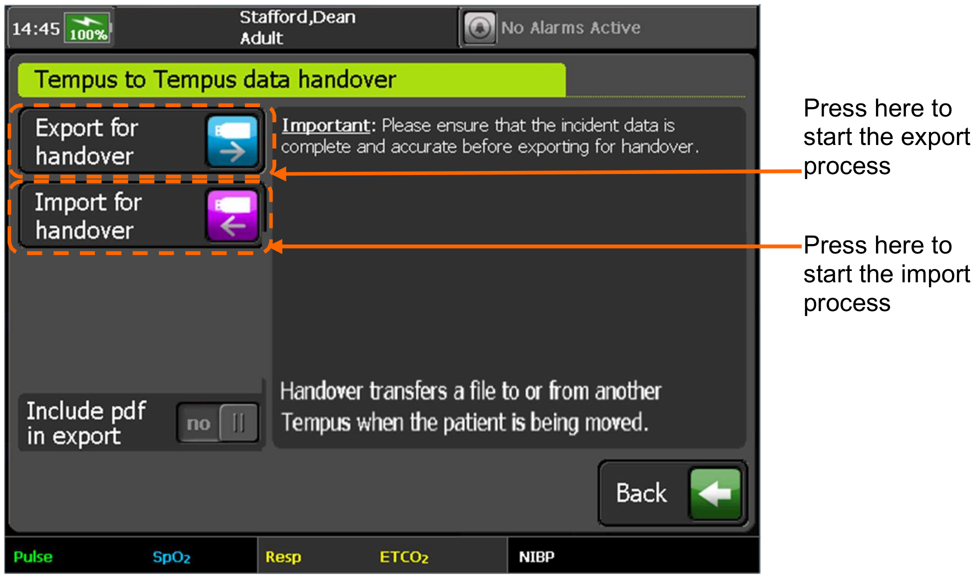
- Tempus Pro to send patient data;
- Tempus Pro to receive patient data;
- USB memory device **or** Tempus to Tempus USB data cable (part number 01-2243).

Handover via USB memory device

To export patient data:

1. Insert a USB memory device into the sending Tempus Pro. The Tempus automatically opens the “Data Input/Output Menu” screen.

2. Press **Tempus to Tempus data handover**. The Tempus opens the "Tempus to Tempus data handover" screen:



Tempus to Tempus data handover – USB memory device

3. If the PDF patient report is required by the recipient, set **Include pdf in export** to **yes**.
4. Press **Export for Handover**. If the USB device contains any existing Tempus patient data records, you will be asked if you want to keep or overwrite them.
5. The "Tempus to Tempus data handover" screen displays a blue progress bar with status messages. When the export progress bar displays "Finished", remove the USB memory device and hand it to the next level care giver.

To import patient data:

WARNING	**When importing, ensure that you select the correct patient record. Records can be identified by patient name (last and first), ID number, or incident start time. If you are not sure if the record you wish to select is the correct one, then select Admit as new patient. Mixing different patient records could lead to confusion and misdiagnosis.**

1. Insert the USB memory device into the receiving Tempus Pro. The Tempus automatically opens the "Data Input/Output Menu" screen.
2. Press **Tempus to Tempus data handover**. The Tempus opens the "Tempus to Tempus data handover" screen.
3. Press **Import for Handover**. The Tempus displays a list of all patient incidents found on the USB memory device. Select the patient incident record to be imported.
4. The Tempus displays the "Import Patient Record" screen. Press one of the following options:

   Merge – this merges the imported incident data into the current patient's data record.

   Admit as new patient – this discharges the current patient and imports the incident data as a new patient.
5. Press **Confirm**.

The "Tempus to Tempus data handover" screen displays a blue progress bar with status messages. When the import progress bar displays "Finished", remove the USB memory device.

Handover via USB data cable

Note	When the Tempus detects the USB data cable, it disables the **Include pdf in export** option.

To transfer the patient incident data from one Tempus to another:

1. Insert the Tempus to Tempus USB data cable (part number 01-2243) into the sending and receiving Tempus Pros.
2. Both Tempus devices automatically open the “Tempus to Tempus data handover” screen. Either press **Export for Handover** on the sending Tempus, or press **Import for Handover** on the receiving Tempus.
3. The “Tempus to Tempus data handover” screen displays a blue progress bar as it zips and sends the data.
4. The receiving Tempus displays the “Import Patient Record” screen. Press one of the following options:

   Merge – this merges the imported incident data into the current patient’s data record.

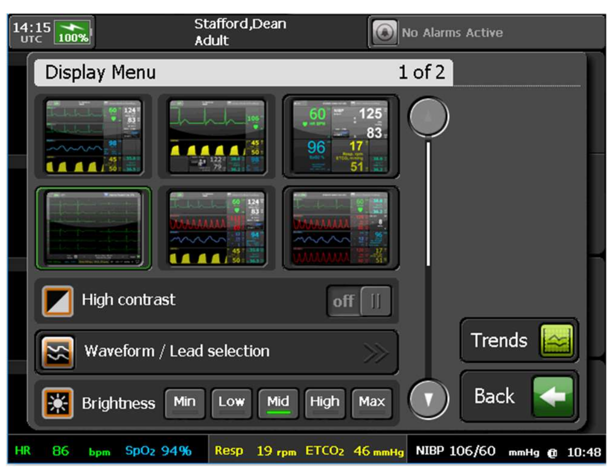
   Admit as new patient – this discharges the current patient and imports the incident data as a new patient.
5. Press **Confirm**.
6. The “Tempus to Tempus data handover” screen displays a blue progress bar with status messages. When the import progress bar displays “Finished”, remove the USB data cable.

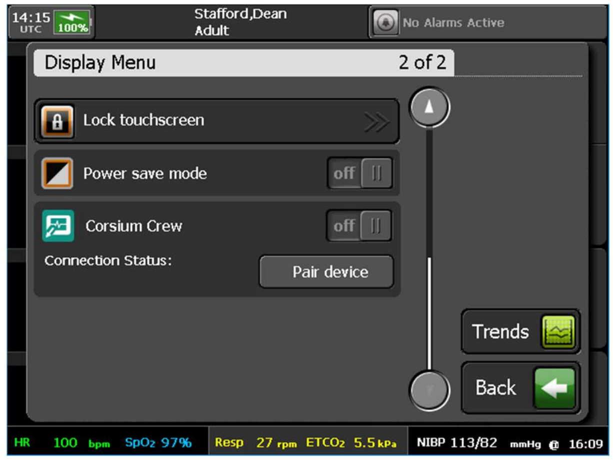
9.2 Display options

9.2.1 The Display menu

The [Display] button.

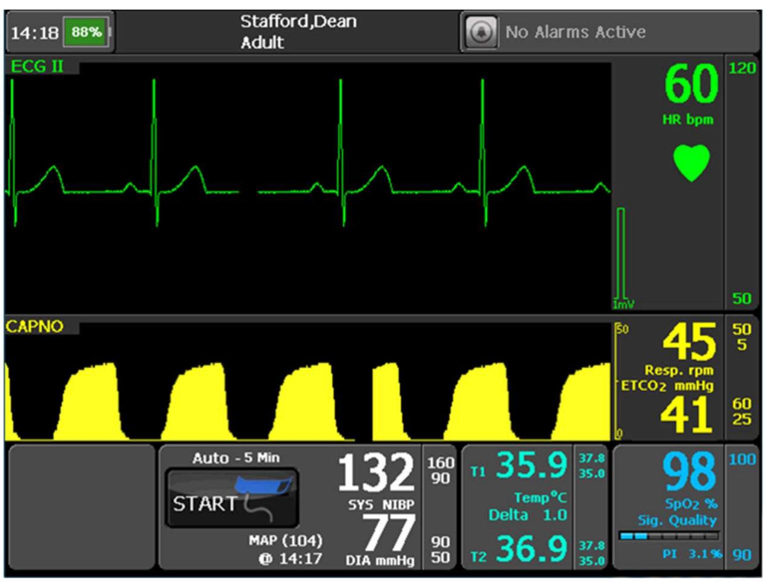
Pressing the [Display] button launches a menu which offers options on the display configuration.

Menu option	Menu screen
Select the home screen display view (from left to right): • Four channel view (the default). • Two channel view (large ECG). • Large numeric view. • Twelve lead diagnostic view. This is only active when the 12 lead ECG cable is attached to the device. See “9.2.2 Display modes”. **High Contrast** – turn high contrast display on or off, see “9.2.3 High contrast mode”. **Waveform / Lead Selection** – open the “Waveform Selection” menu, see “9.2.4 Waveform and lead selection”. **Brightness** – select the screen brightness level, see “9.2.5 Brightness control”.	

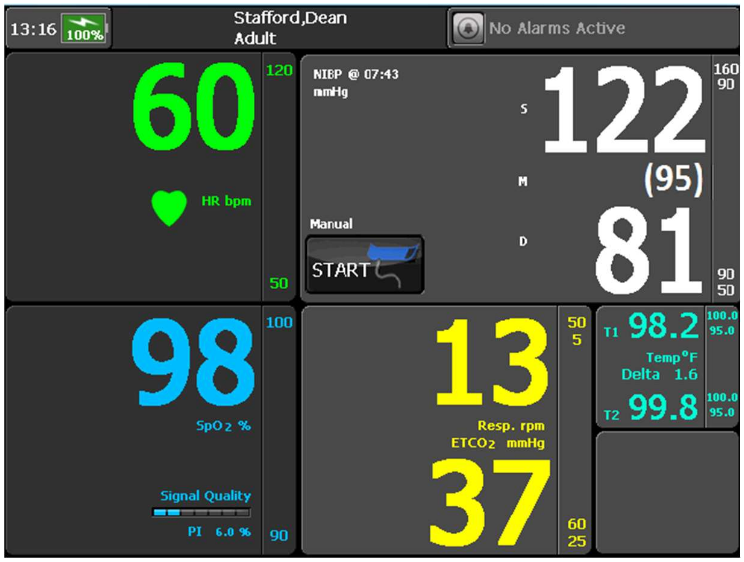
Menu option	Menu screen
Lock touchscreen – lock the touchscreen, see “9.2.6 Locking the touchscreen”. **Power save mode** – turn power saving on or off, see “9.2.7 Power save feature”. This setting does not persist after powering off the device. **Corsium Crew** - for use with the Corsium Crew extended display (optional). See the *Corsium Crew User Guide*.	

9.2.2 Display modes

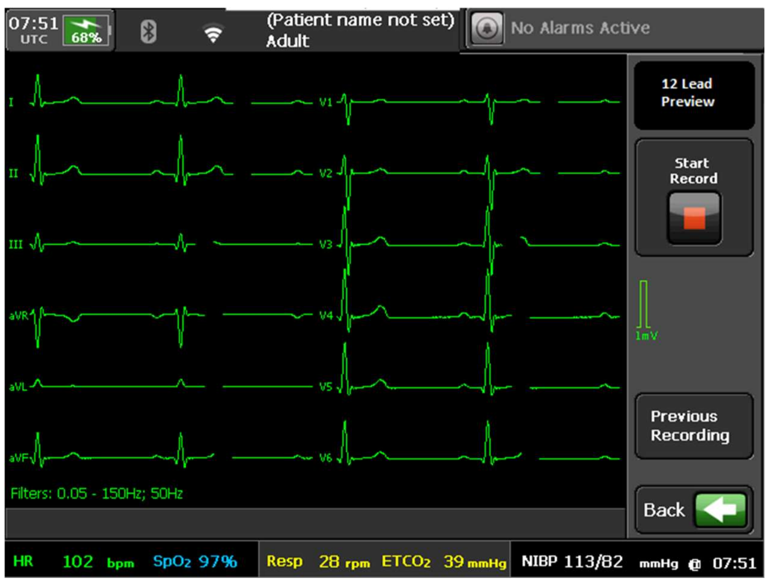
The “Display Menu” offers different ways to display the vital signs data on the home screen. Each home screen display option is shown as an icon. Pressing the icons changes the home screen display mode. In addition to the 4 channel display (described previously), the Tempus also offers the following home screen display options.



Large amplitude ECG waveform display



Large numeric display



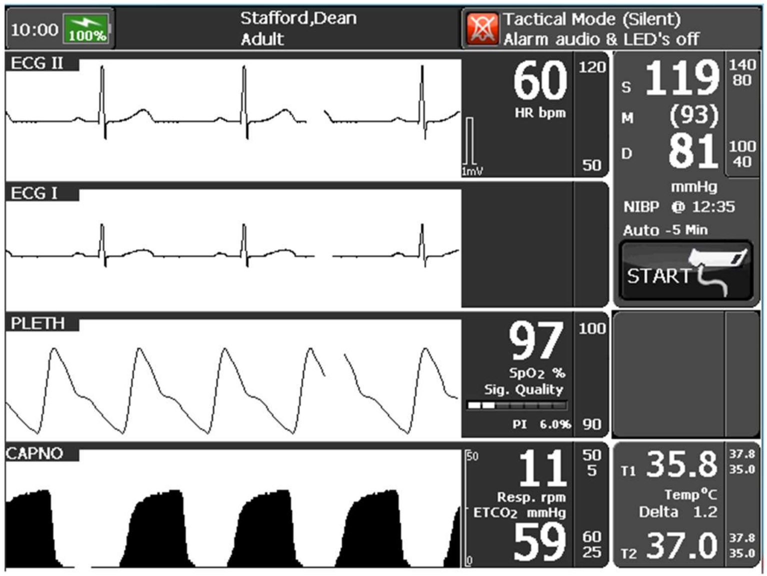
12-lead ECG preview

Note	The second waveform in the large ECG view defaults to display the impedance pneumography (ECG respiration) if the Capnometer is not plugged in.
Note	When switching between the 4 waveform and the large ECG views, the ECG gain and wave speed does not change except if the gain is set to 2 mm/mV on the 4 channel view in which case the ECG will be displayed as 4 mm/mV. The user can subsequently adjust the gain settings to optimize the available waveform height. For details on changing gain settings, see “6.1.8 ECG settings”.

9.2.3 High contrast mode

The first button below the different home screen views allows the user to enable a high contrast display mode on the results screen. Enabling this allows the Tempus to display vital signs data in a black on white display for high daylight conditions (shown below). The high contrast display mode can be enabled automatically by pressing and holding the button for 2 seconds. This allows the user to switch the high contrast display on by pressing a single button.

Note	The high contrast display is available for all Home screen display modes.



High contrast display activated

9.2.4 Waveform and lead selection

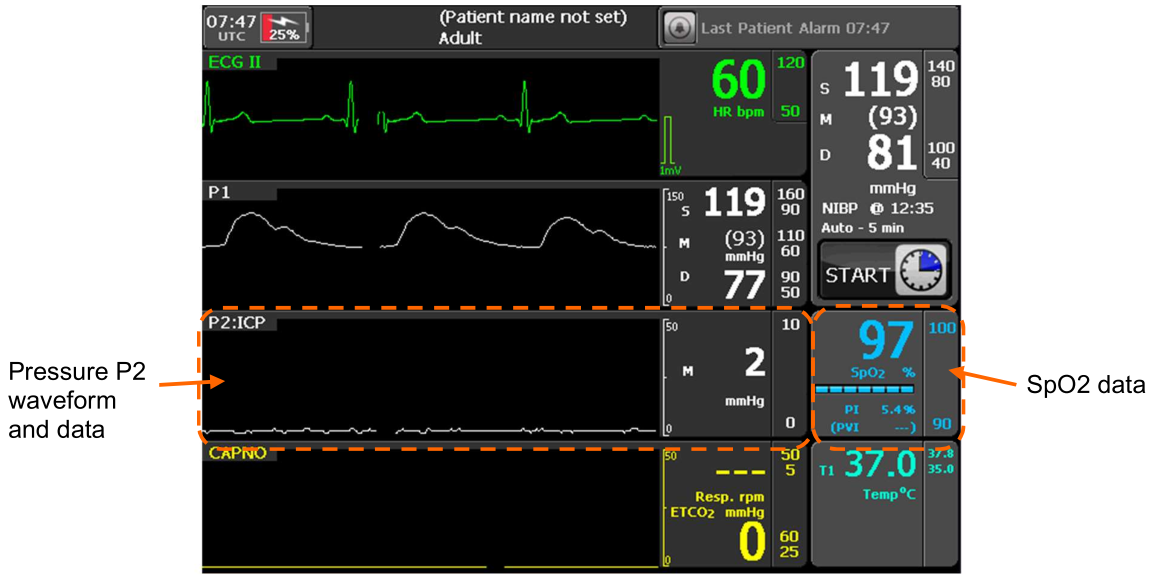
Use the "Waveform / Lead Selection" screen to customize the display of waveforms in the four-waveform home screen. To access this screen, either press the **Display** button and then press **Waveform / Lead Selection**, or from the "ECG Settings" menu press Waveform / **Lead selection**.

Selecting waveforms

Note	If Corsium Crew is enabled, the "Waveform / Lead Selection" screen will include **Crew App** waveforms. See the *Corsium Crew User Guide*.

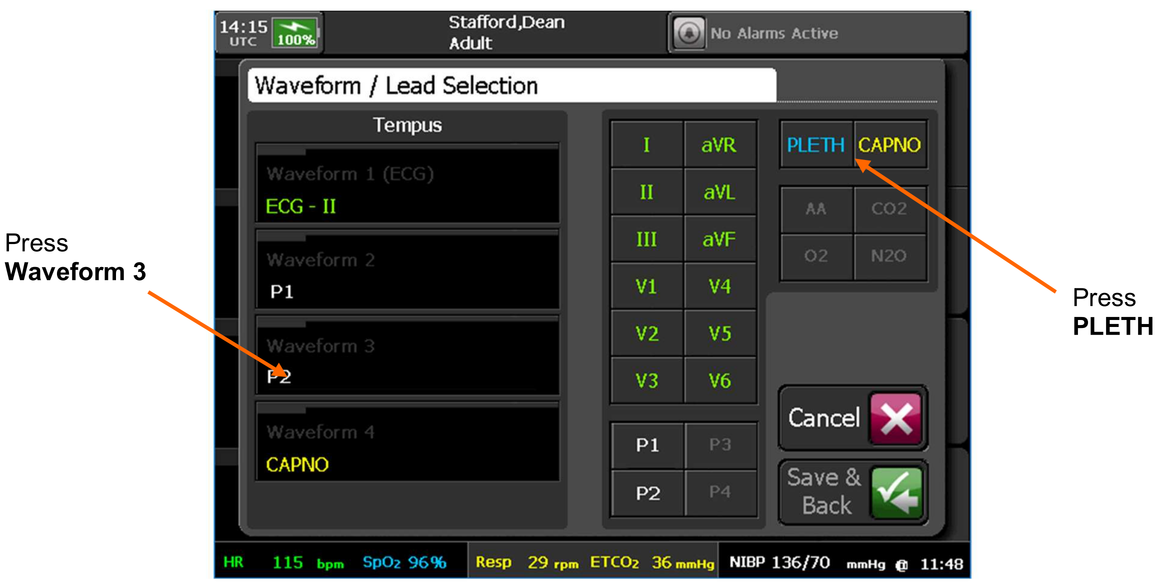
For example, you may want to move pressure P2 to a numeric area and display the plethysmogram waveform instead:

- Before this change is made, the home screen looks like this:



Standard home screen with 2 IP channels connected

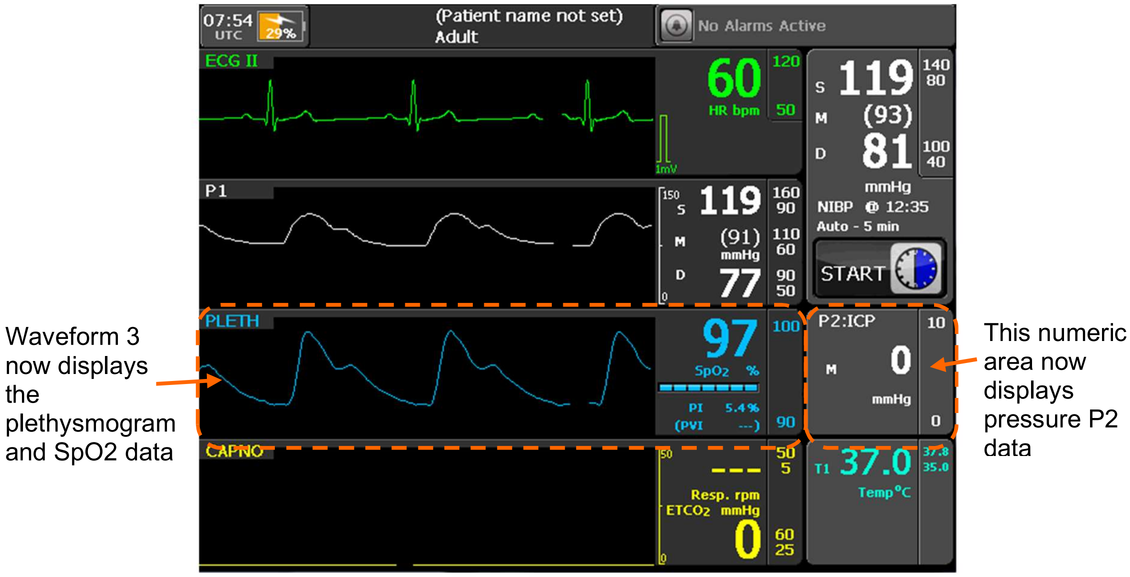
- From the Display Menu, press **Waveform/Lead Selection**. On the "Waveform/Lead Selection" screen, change the **Waveform 3** setting from **P2** to **PLETH**:



Setting Waveform 3 to PLETH

-

- After this change, the home screen looks like this:



Home screen with plethysmogram in Waveform 3

Note	You can swap two waveforms around by selecting both waveform regions. The waveform selection feature is not available when iAssist mode is activated

9.2.5 Brightness control

The "Display Menu" includes a screen brightness control. The screen's brightness has five available settings as follows:

- **Min** – 10%;
- **Low** – 30%;
- **Mid** (default) – 60%;
- **High** – 80%;
- **Max** – 100%.

Using a lower brightness setting will improve battery life.

WARNING	**The low display settings are intended for use by military users for scenarios where low emitted light is required or desired. Users are reminded that while these functions are enabled, the device will present a display that may be too dim to see in daylight conditions. Users should therefore ensure that they use this feature only when required and recognize that greater levels of patient care and supervision will be required.**
Note	Min, Low and Mid are compatible with NVGs (night vision goggles) – see "4.3.2 Tactical mode (optional)". The Max setting should only be used for strong sunlight applications.

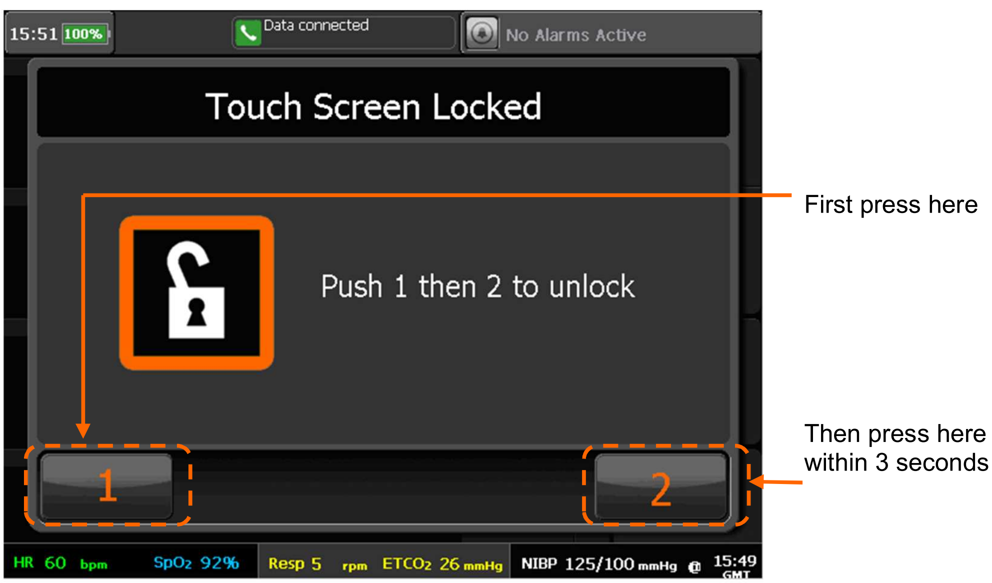
9.2.6 Locking the touchscreen

The Lock Touchscreen function allows the touchscreen to be intentionally disabled. If this feature is activated the Tempus will display the message shown below:

Touchscreen lock activated

This message stays on the screen for 2 seconds. After that point, if any button, control or the touchscreen are pressed the following message will be displayed. Some technical alarm conditions allow you access to the touch screen but once you return to the home screen the lock will still be active.

To disable the touchscreen lock, simply press the buttons marked “1” then “2” within 3 seconds of each other. If neither button is pressed, or the buttons are not pressed in the correct sequence, then the message will leave the screen after 5 seconds.



Touchscreen lock deactivation

9.2.7 Power save feature

When power save is activated the display turns to 'min' brightness after 3 minutes of no use (i.e. no user controls or alarms). If there is a further 2 minutes of no use the display turns off. The Tempus Pro continues monitoring etc. whilst the screen is off and the device's display will come back on at its original brightness as soon as an alarm is triggered, the touch screen is pressed or a keypad button is pressed.

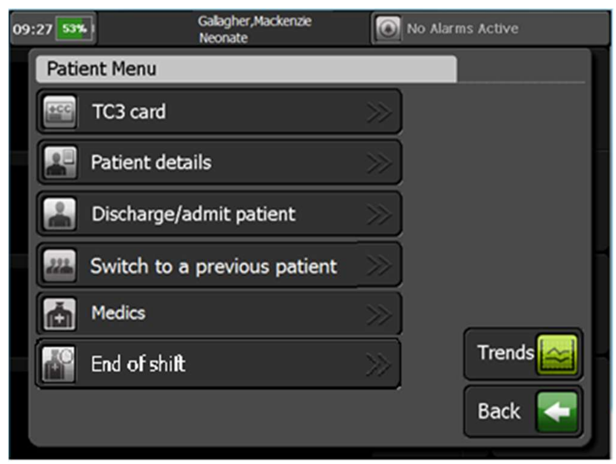
This control can be used to extend the battery life of the Tempus.

9.3 Patient information

9.3.1 The patient menu

The [Patient] button

Pressing the [Patient] button launches a menu which will offer options on the management of patient data. This menu allows you to:

Menu option	Menu screen
TC3 card – complete an electronic version of the military Tactical Combat Casualty Care (TCCC) card, see "9.3.2 Updating the TCCC card". **Patient details** – enter the patient's name, ID and allergies, see "9.3.5 Entering patient details". **Discharge/admit patient** – discharge the current patient and admit a new patient, see "9.3.3 Admitting a new patient". **Switch to a previous patient** – switch to a previously monitored patient, see "9.3.4 Switching to a previous patient". **Medics** - enter details of medics (first responders). **End of shift** – allows you to clear the patient list. This means that you will not have access to previous patient incidents.	 **Note:** The **TC3 card** option is only available on Tempus units sold to military users.

9.3.2 Updating the TCCC card

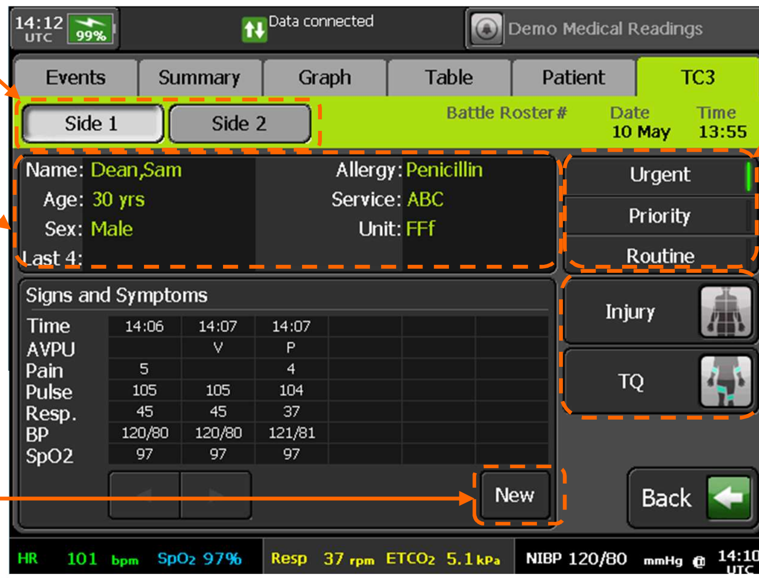
Pressing the **TC3 Card** button allows you to complete an electronic version of the military TCCC card. This feature is designed to replace the paper version used in the field. The screen is therefore laid out like a two-sided card and behaves the same way: you can insert, amend and delete TCCC data until the card is exported to a USB stick.

WARNING	**It is the user's responsibility to ensure the information recorded in the TCCC is accurate and complete. Omissions or inaccuracies could lead to mis-diagnosis resulting in death or serious injury.**

Press these buttons to switch between Side 1 and Side 2 of the card

Press this area to enter or edit the patient's name and details

Press the New button to bring up AVPU and Pain buttons to select from (time and vitals are recorded automatically)



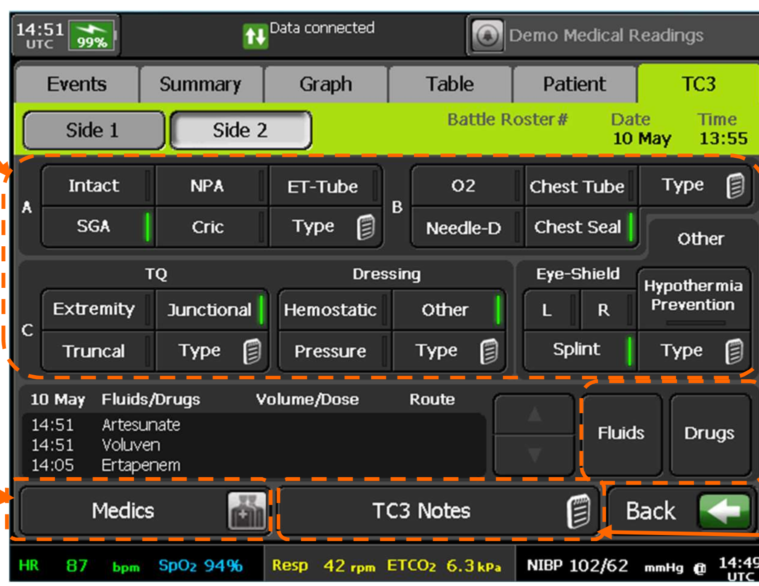
Press one of these buttons to record the urgency of the casualty

Press these buttons to record injuries and TQ locations

TCCC card side 1

Press this area to record ABC and other interventions

Press this button to enter medics details

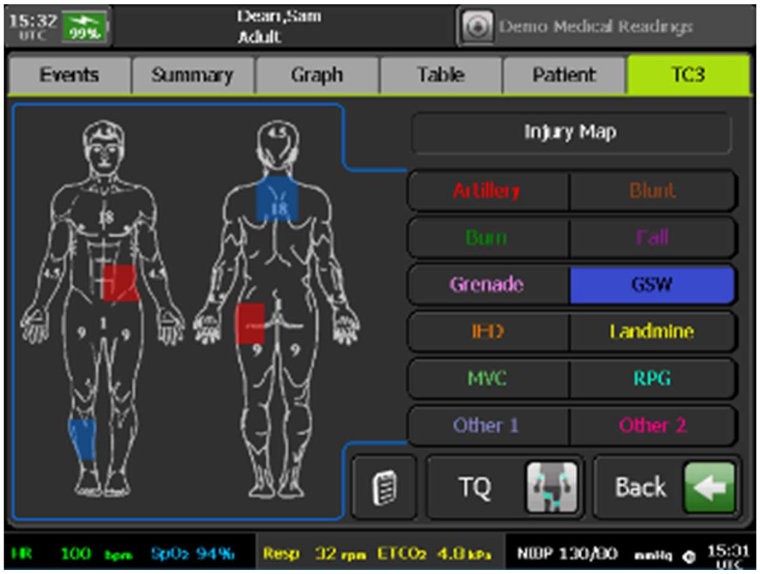


Press these buttons to enter fluids or drugs

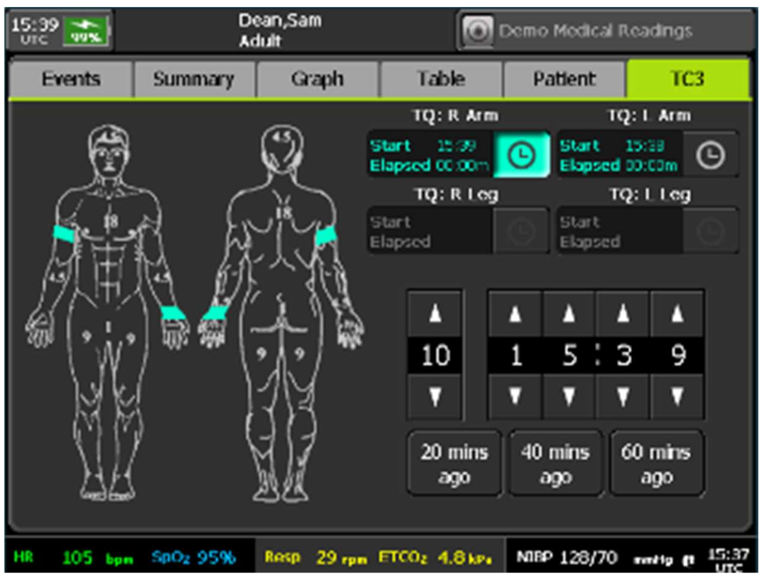
Press this button to enter notes

TCCC card side 2

The TCCC Injury and TQ editors provide body maps on which injuries and tourniquets may be recorded.



TCCC injury map

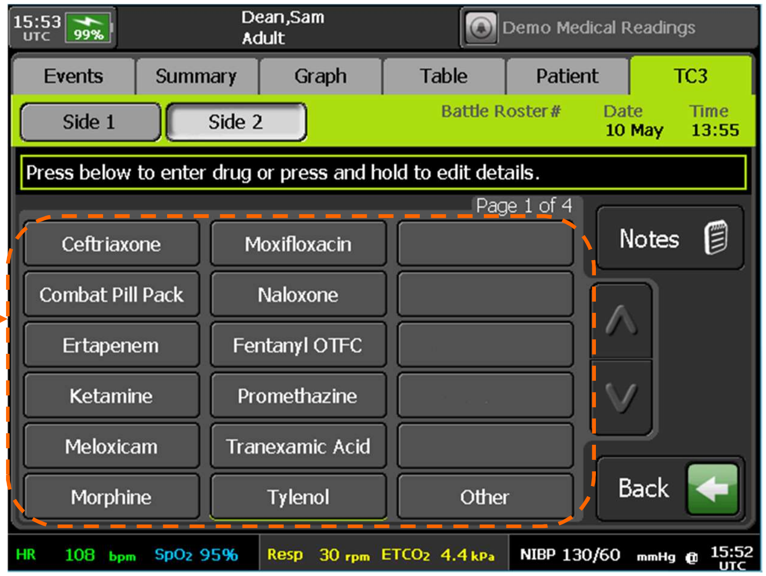


TCCC TQ map

The drugs editor can be configured to contain all the drugs provided by the military for standard medics. The dosages and routes can be edited, but by default should be those most commonly used.

Press and release a drug button to record a drug you have administered with the default time, route and dose.

Press and hold a drug button to edit the time, dose or route of a drug before it is recorded.

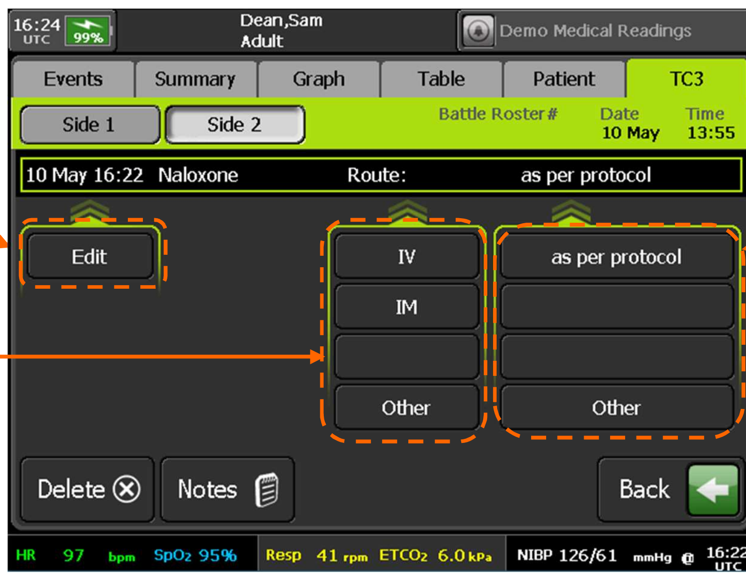


TCCC drugs editor

To edit the time, dose or route of administration of a drug before it is recorded, press and hold the drug button. This brings up an editor which allows any time (in the last 24 hours) to be selected and for the dose and route to be edited.

Press this button to enter the time of administration.

Press one of these buttons to enter the route of administration.



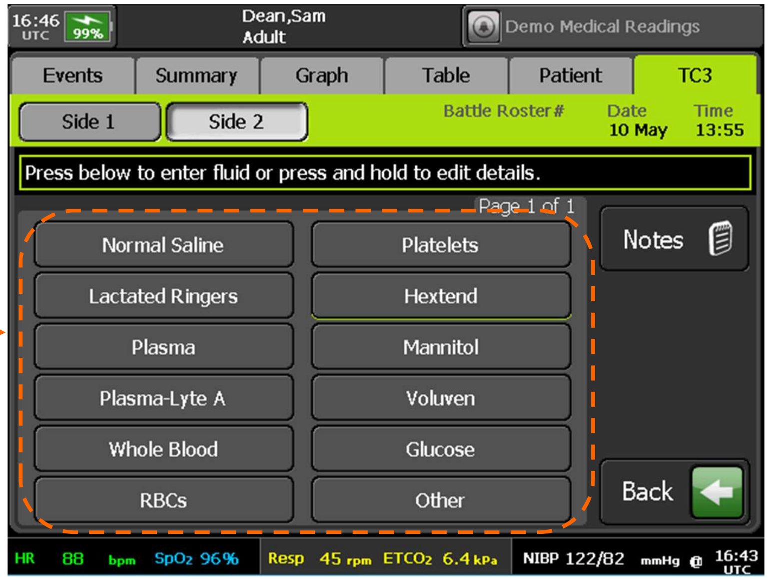
Press one of these buttons to enter the dose.

TCCC drugs time, does and route editor

The fluids editor can be configured to contain all the fluids provided by the military for standard medics. The volumes can be edited, but by default should be those most commonly used.

Press and release a fluid button to record a fluid you have administered with the default time and volume.

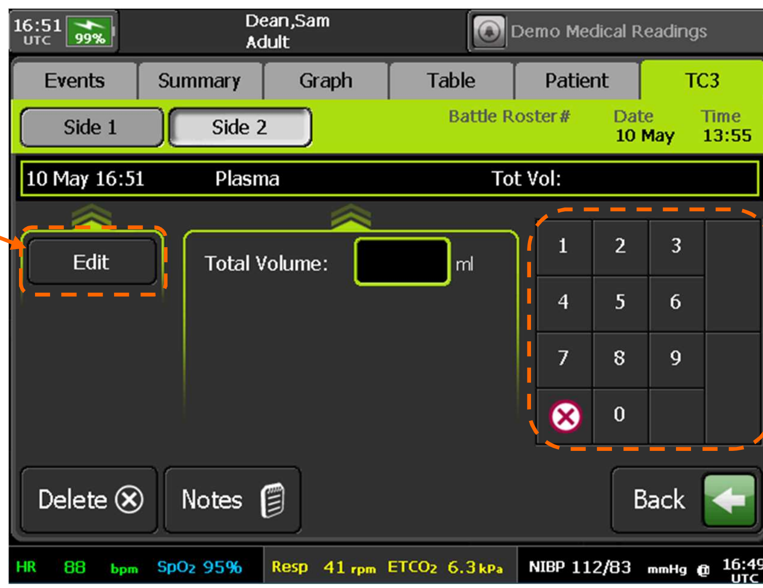
Press and hold a fluid button to edit the time or volume of a fluid before it is recorded.



TCCC fluids editor

To edit the time or volume of a fluid before it is recorded, press and hold the fluid button. This brings up an editor which allows any time (in the last 24 hours) to be selected and the volume to be edited.

Press this button to enter the time of administration.



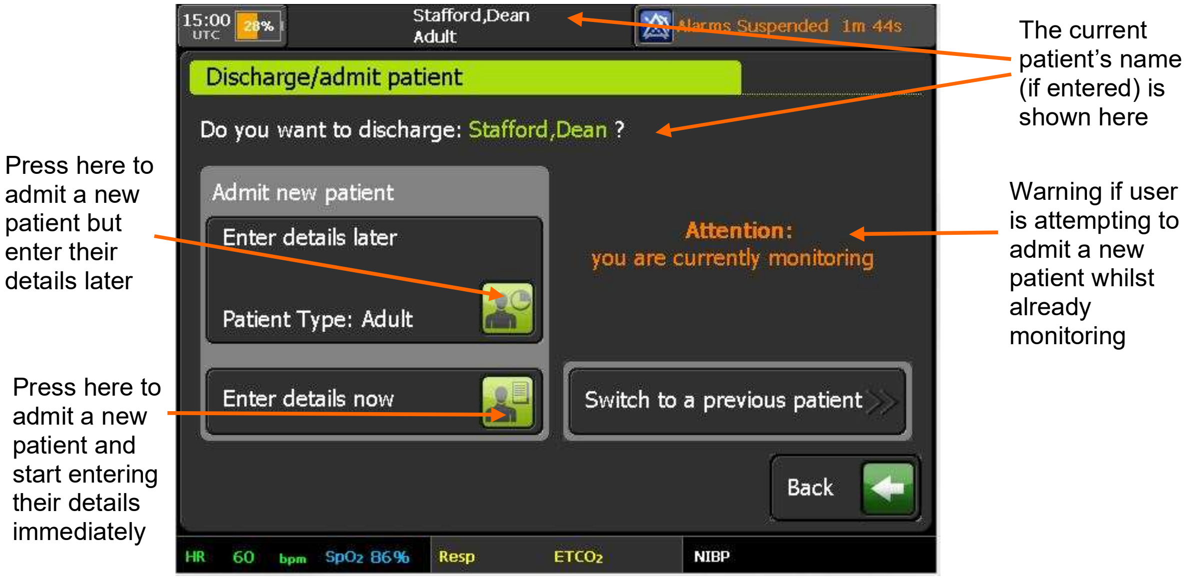
Use the keypad to enter the volume.

TCCC fluids time and volume editor

Note	If GPS is turned on and a fix is available, drug records will be geo-tagged with the location when the record is entered – see "9.5 GPS location".
Note	REMEMBER the TCCC card is a paper replacement system. Like a piece of paper, all information in the TCCC card can be edited or deleted up until the point the data is exported onto a USB stick – see "9.1.3 Tempus to Tempus data handover". Switching to a previous patient will allow the user to re-access the previous TCCC card and edit it – see "9.3.4 Switching to a previous patient". If the Tempus is connected to a Response Center they will not be able to see the TCCC card being edited but will be able to see a fixed update of the TCCC card on their system every 2 minutes. The Response Center user is not able to edit the TCCC card, only view it – see "9.6.5 Interacting with the Response Centre".

9.3.3 Admitting a new patient

If the **Discharge/admit patient** button is pressed, the Tempus will bring up a new dialog which asks you to confirm if the patient is new or not and gives the options to enter the new patient's name immediately or later. A warning is displayed if the current patient is still being monitored (see below).



Admitting a new patient

If a connection has not been established to a Response Center, all data for that patient can be transmitted later so long as a new patient is not admitted. For details on communications with a Response Center, see "10 Connecting to an alternate location".

Once a new patient is admitted, the recorded data cannot be transmitted to the Response Center.

WARNING	**Ensure that the Tempus Pro is disconnected from the old patient before confirming that a new patient is being monitored. Failing to do so will mean that real-time vital signs data from the old patient will be entered with the record for the new patient.**

CAUTION	If the Tempus is connected to a Response Center, discharging the old patient and starting to monitor a new one will create a new patient record at the Response Center also. Ensure the Response Center user understands who is being monitored.
CAUTION	Admitting a new patient will reset all alarm settings to the factory defaults – see "7 Alarms".

Note	Do not attempt to connect the Tempus Pro to other medical systems until such functionality is properly supported in software.

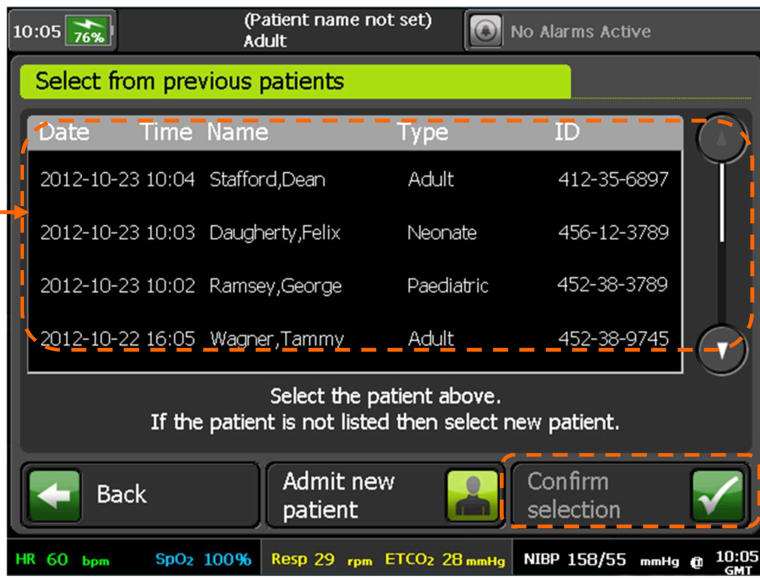
It should be noted that if the patient is removed from all real-time sensors then the Tempus will lose its real-time detection of the patient's presence. This scenario could be caused by users needing to move the Tempus from one patient (cease monitoring that patient) as they intend to start monitoring a new patient. In the instance that all real-time patient signals are lost for more than 5 minutes, the Tempus will anticipate that the user could be moving from one patient to a different one. It will therefore automatically give the user the option of switching to a new patient without having to press the Patient button (as described earlier in this section).

WARNING ⚠	**Ensure that you select the correct patient record. Records can be identified by patient name (last and first), ID number, or incident start time. If you are not sure if the record you wish to select is the correct one, then select new patient. Mixing different patient's records could lead to confusion and misdiagnosis.**

9.3.4 Switching to a previous patient

The Tempus allows you to switch between the current patient and a patient that has been monitored on the same Tempus device within the previous 72 hours. The Tempus will also show patients admitted more than 72 hours ago, but only up to a total of 20. This feature may be used, for example, in multi-casualty situations.

To switch between patients, simply press the **Switch to a previous patient** button and then select from any patient that is shown in the list. If no patients are shown, then there are no records to select from.



Selecting a previous patient

WARNING ⚠	**Ensure that you select the correct patient record. Records can be identified by patient name (last and first), ID number, or incident start time. If you are not sure if the record you wish to select is the correct one, then select new patient. Mixing different patient's records could lead to confusion and misdiagnosis.**
WARNING ⚠	**Ensure that the Tempus Pro is disconnected from the old patient before confirming that a new patient is being monitored. Failing to do so will mean that real-time vital signs data from the old patient will be entered with the record for the new patient.**

CAUTION ⚠	If the Tempus is connected to a Response Center, discharging the old patient and starting to monitor a new one will create a new patient record at the Response Center also. Ensure the Response Center user understands who is being monitored.
CAUTION ⚠	Admitting a new patient will reset all alarm settings to the factory defaults – see "7 Alarms".

9.3.5 Entering patient details

Pressing the **Patient Details** button allows the patient's name, ID and allergies to be entered.

Press the buttons to select what details you wish to enter

Predicted text options will appear here based on what is typed, press any option to select it

Press here to accept the details you have entered. You can go back and edit or delete any data later

Entering Patient Details

The patient's age (or age group) and sex can be entered at any time. If the age (in years) is entered by the user, the Tempus calculates the age group. If the patient's age is not known or is less than 1 year, the user can select the patient's age group. The Tempus uses the selected age group to update alarm limits.

WARNING	**Patient age group is an important factor for determining the correct initial inflation pressure (and maximum inflation pressure) for non-invasive blood pressure measurements in children of 12 years and under and also for neonates (28 days old or less) – see "6.2.3 NIBP settings".** **Failure to set the correct age group for these patients may result in them being subject to higher inflation pressures than is warranted. For new patients, the default settings are Adult/Male. Always check and enter the patient's age or select the correct age group.**

CAUTION	Patient age affects the settings used in some medical modules.

Note	Weight and height are included in the patient report but do not affect any monitoring settings in the Tempus Pro medical parameters.

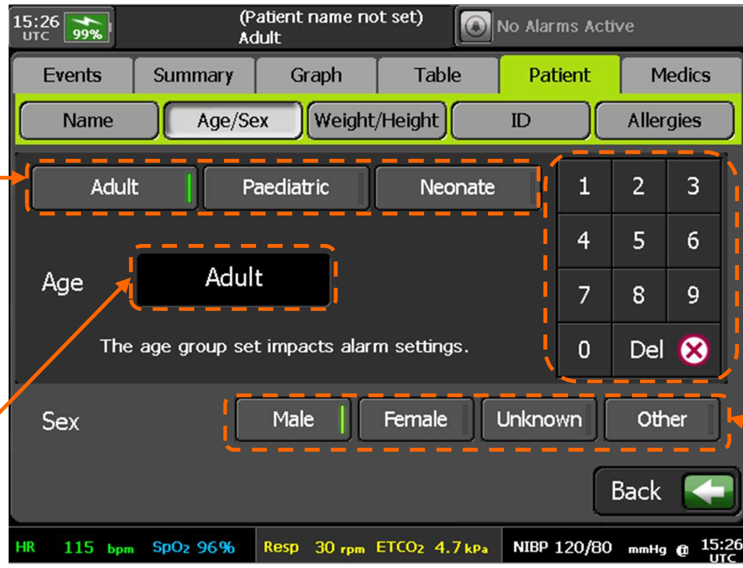
If patient's age is not known or is less than 1 year, press here to select the patient's age group:

Adult: > 12 years

Pediatric: 29 days to 12 years

Neonate: < 29 days

The entered age or selected age group is shown here



If patient's age is known, use this keypad to enter age in years (zero is interpreted as Pediatric)

Press here to record the patient's sex

Entering patient age & sex

9.4 Digital camera

the [Camera & Waveform Snapshot] button.

Holding the [Camera & Waveform Snapshot] button for more than 2 seconds launches a camera screen. The camera screen can also be launched from the main menu camera button. This shows the viewfinder of the camera and provides buttons to take a still picture, turn on the backlight or transmit video (i2i only).

It is possible to capture and send still digital pictures using the camera built into the device. If the Tempus is connected to a Response Center, then the image is sent immediately automatically. If it is not connected, the image is stored and can be uploaded later if a connection to the Response Center is made. Digital pictures are shown live on the Tempus Pro screen so that you can see what the camera is seeing. When you are happy with the displayed image, you capture the picture and can then send it to the Response Center. After pressing the "Take a photo" button the Tempus will display a 3 second countdown during which time the photo can be cancelled without being saved or transmitted.

A backlight is provided to help light the subject.

Real-time video can be transmitted if a data connection with sufficient bandwidth is connected to the Response Center (i2i ReachBak only).

Note	Do not look directly at the backlight or shine in eyes unnecessarily.
Note	The camera is intended to allow photos of the patient to be taken (as part of the care record) or to be transmitted to enable the user at the Response Center to see the patient (e.g. helping to decide the best care path and options will be aided by being able to see the patient). The Tempus Pro is NOT a teledermatology system or similar device.
Note	The transmission time for photographs is < 30 seconds in good conditions, depending on signal strength of the network.

Selecting camera and video options

Note	Moving video is intended to give the i2i Response Center the ability to see the patient moving or to see around the patient's environment. Users should remember that the resolution and quality of the received video stream will not be the same as they see on the screen of the Tempus Pro due to the effects of the video being compressed during transmission. This effect is lessened when the image being filmed is more stable or has less activity in it. Therefore, in order to ensure the received video is good quality, Users should try to move the camera slowly. If rapid movement of the camera is necessary then the received image is likely to have a temporarily lower level of resolution (will appear "blocky") while the camera is being moved around, this effect will reduce once the camera movement is reduced.
Note	The overall image quality and resolution of the camera is greater for still pictures than moving video. If the i2i Response Center require an image with a reasonable level of detail (such as a close-up image) then a still photo would probably be more suitable.
Note	Unlike still photos, moving video images are not stored on either the Tempus or at the Response Center. They are "real-time" images only.

Press the **Stop transmitting video** button to stop the transmission.

9.4.1 Annotation of digital pictures (i2i ReachBak only)

Images transmitted from the Tempus Pro can be altered using the software at the i2i Response Center. The altered image can then be sent back to the Tempus Pro to act as a support in the remote diagnostic procedure i.e. the physician can send pictures back that can be used to confirm exactly the issue being examined or discussed, thus avoiding the danger of misunderstanding verbal descriptions.

Images can be amended using the following tools:

- Addition of text
- Addition of circles
- Addition of lines and arrows
- Addition of free-form lines
- Selection of colors for added graphics

9.5 GPS location

The Tempus Pro has a built-in GPS receiver. You can access this feature from the Main Menu – see "5.2.1 The main menu".

The GPS may take up to 4 minutes to display a fix. If it is unable to obtain a fix (if the view of the sky is obstructed or if the unit is used indoors) then it will display an error. The most recent fix (with the time and date of the fix) is always displayed.

If the signal from only a limited number of GPS positioning satellites can be received by the Tempus (e.g. because of partial blocking of the sky by objects, buildings etc.) then the fix may be less accurate. In this case the fix will be labelled "Approximate fix" and the reading may be +/-2.5 km.

Fixing your position using GPS

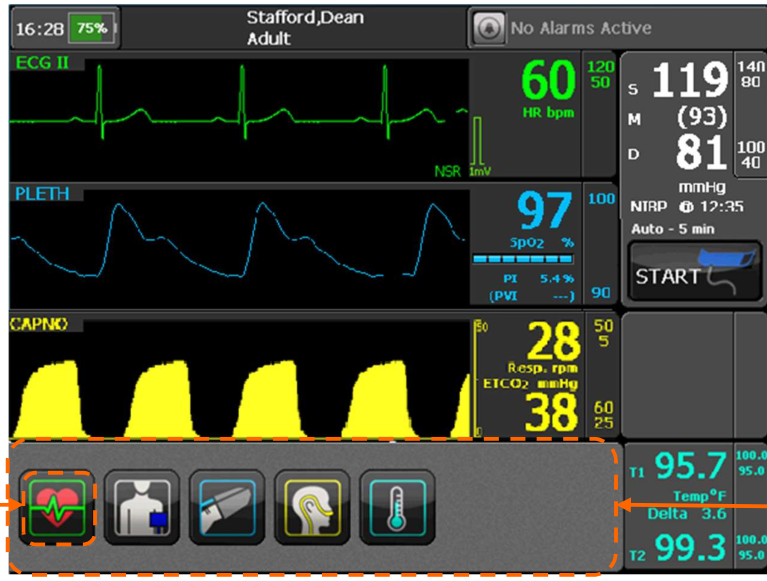
Note	The GPS operation will be limited if the Tempus Pro is not used outside with a clear view of the sky.
Note	The GPS is intended to provide the User and the Response Center with the patient's location. It should not be used as a guidance or navigation device.

9.6 iAssist processes

Note	The iAssist help processes on your Tempus Pro may differ from this example iAssist help process in the following sections. However, the process always follows the same key elements. Always ensure that you read the complete iAssist help process in order and do exactly what it requires.

The Tempus can provide additional on-screen instructions on how to use the device. This feature is called iAssist mode. The iAssist mode is intended to be set by more experienced users in order that they can pass the Tempus to less experienced colleagues knowing that their operation of the device will be supported by on-screen instructions. If **iAssist** is activated on the "Main Menu", the second ECG lead waveform will be removed and the plethysmogram and capnogram will be moved up to make room for a row of graphical buttons in the bottom-most waveform of the Tempus home screen – see "5.2.1 The main menu".

The first button is the ECG button



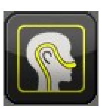
Each button is a graphical cue to allow the less experienced user to get on-screen instructions on how to deploy that particular parameter.

Home screen in iAssist mode

ECG – pressing this brings up instructions on how to use the ECG

Blood pressure – pressing this brings up instructions on how to take non-invasive blood pressure measurements

Pulse oximeter – pressing this brings up instructions on how to take blood oxygen measurements



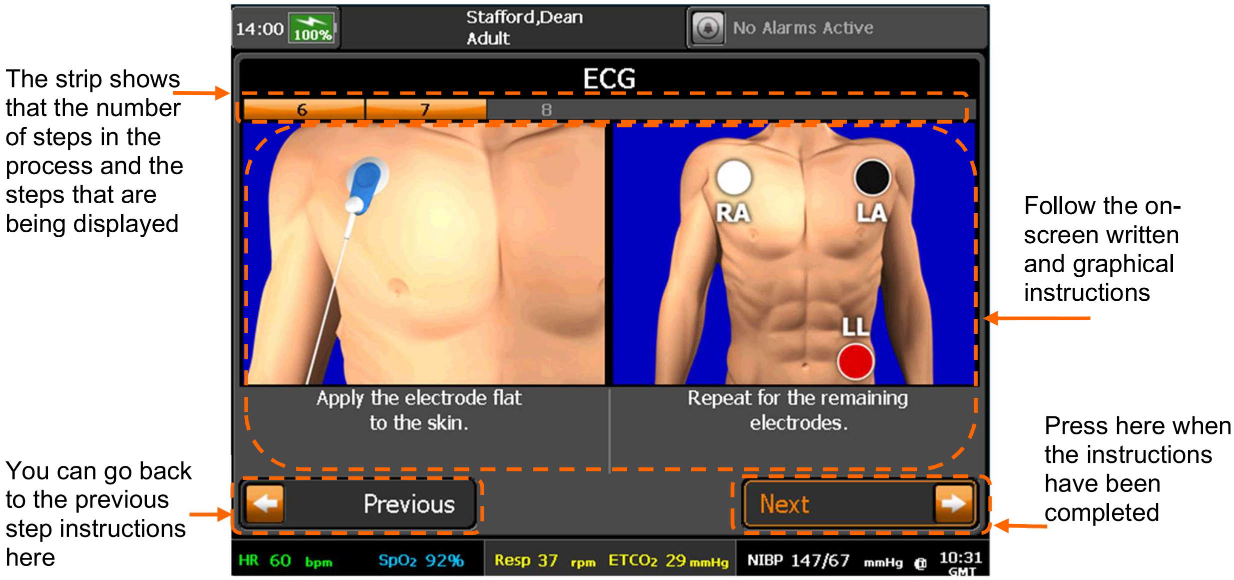
Capnometer – pressing this brings up instructions on how to take Capnometry measurements

Temperature – pressing this brings up instructions on how to take temperature measurements

When in iAssist Mode, the Tempus Pro provides the user with complete instructions on how to use it. Every step is detailed in pictures with accompanying text instructions. There are instruction processes for everything the user will need to do to use the parameter.

The example iAssist process shown below shows there are two distinct areas on the screen that give different types of information.

- Process Instructions - This area contains the graphical pictures and text instructions that show you how to use the device. This takes the user through each activity one or two steps at a time.
- Touch Screen Buttons - In this example there are two buttons at the bottom of the touchscreen. In all cases the user will press the button on the **bottom right** of the screen to progress onto the next step in the process.



Example of an iAssist process

When in iAssist Mode, the Tempus Pro breaks all processes down into small steps. These steps are shown on the screen in one or two at a time.

The user can see how many steps there are in any process by looking at the process ribbon near the top of the screen.

In the example shown above, the screen shows that the process has 8 steps and that the device is showing steps 6-7.

The user follows the instructions given on the screen, ensuring that they review <u>both</u> the image and the text. Once they have completed both steps they proceed onto the next steps by pressing the **Next** touchscreen button.

9.6.1 Monitoring ECG

To begin monitoring an ECG, press  button on the device.

The first step in the ECG help process will appear.

Follow the instructions provided on the iAssist help process to activate ECG.

9.6.2 Taking blood pressure measurements

To take blood pressure measurements, press the  button on the device. The first step in the Blood Pressure help process will appear.

Follow the instructions provided on the iAssist help process to activate blood pressure.

9.6.3 Taking pulse oximetry measurements

To take Pulse Oximetry measurements, press the  button on the device. The first step in the Pulse Oximeter help process will appear.

Follow the instructions provided on the iAssist help process to activate Pulse Oximeter.

9.6.4 Taking capnometry readings

To activate the Capnometer function, press  button on the device.

The first step in the Capnometer help process will appear.

Follow the instructions provided on the iAssist help process to activate Capnometer.

9.6.5 Taking temperature readings

To activate the thermometer, press  button on the device.

The first instructions will be displayed.

Follow the instructions provided on the iAssist help process to use the thermometer.

9.6.6 Using the camera

To activate the Camera, press  button on the device.

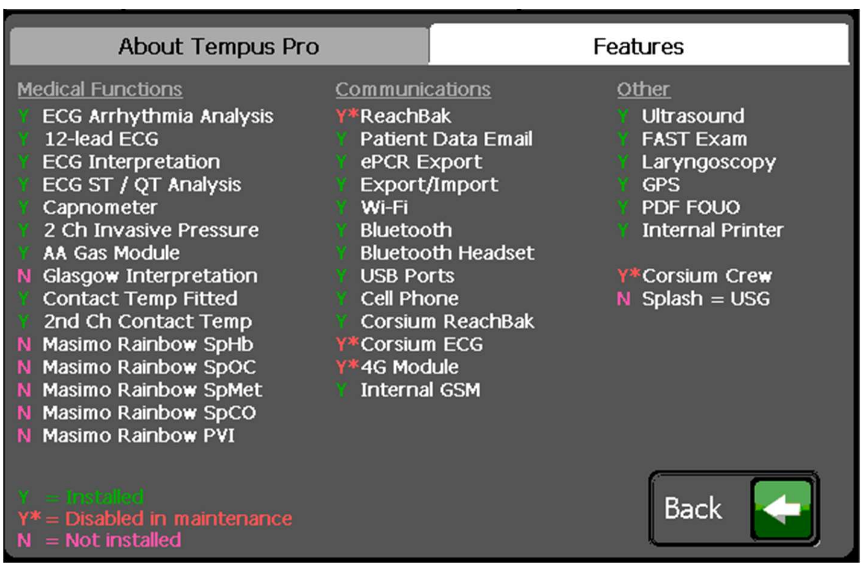
The first Camera help screen will appear.

A digital picture from the camera will appear on the Tempus Pro display in the position shown in the following picture.

Follow the instructions provided on the iAssist help process to take a photo.

9.7 View installed features

Use the “Main Menu” option **View Installed Features** and select the Features tab to view a list of features installed on Tempus Pro.



9.8 Printer maintenance

9.8.1 External printer configuration

You can use an external USB printer to print patient reports – see “9.1.2 Send patient data/report”.

To configure an external printer:

1. Select **Main Menu > Printer and Headset > External Printer Settings**.
2. Select the setting to change:
   Printer Type - PCL3 or PCL3-GUI.
   Paper - A4 or Letter (8.5 x 11“).
3. Press **Test Page** and check the resulting test print.
4. Press **Back**.

9.8.2 Internal printer test page (optional)

To print a test page on the internal printer (if fitted):

1. Select **Main Menu**  > **Printer and Headset** > **Internal Printer**
2. Press Test Page and check the resulting test print.
3. Press **Back**.

9.8.3 Changing the paper roll (internal printer only)

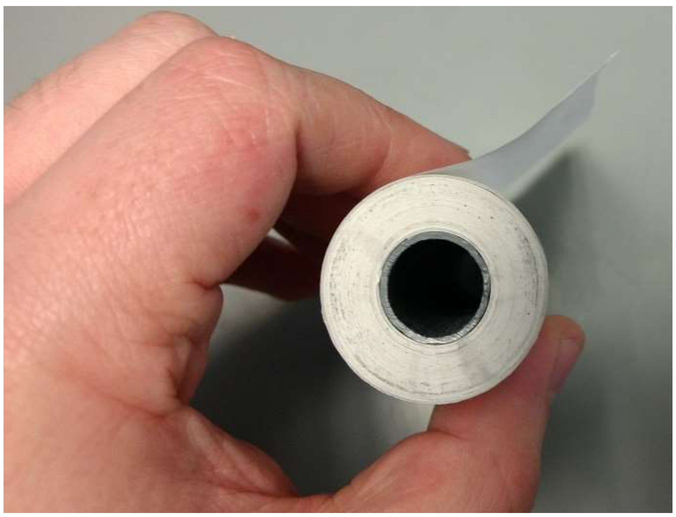
Note	To ensure ease of access to the printer roll, place the Tempus Pro on a flat surface, lying on its front. Do not force the lid to open at an angle greater than 90° from the back of the unit. Excessive force will detach the lid from the printer body: if this happens see "9.8.4 Fitting the printer lid".

To change the paper roll in the internal printer, proceed as follows:

1. Open printer lid by depressing the button in the printer housing. Open the lid to an angle not greater than 90° from the back of the unit. Do not force it to open any more than this:

2. Remove the old paper roll spindle:

3. Free the end of the new roll:



4. Place the roll into the housing, with the outside face of the paper against the printer mechanism:

5. Feed in one side of the roll, then the other:

6. Ensure the roll spindle is centered on the fixing springs, pull the roll to ensure it rotates freely:

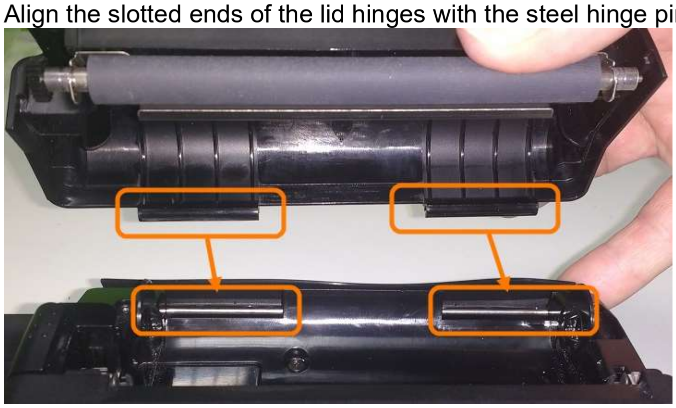
7. Close the lid and press it down on both corners, listening for a click. Tear off any extra paper:

8. Set the paper type (Plain or Grid). If required, print a test page.

9.8.4 Fitting the printer lid (internal printer only)

If the lid becomes detached from the printer housing, refit it as follows:

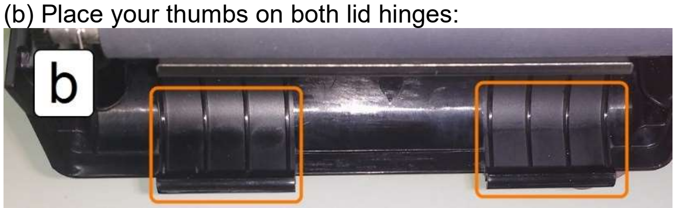
1. Align the slotted ends of the lid hinges with the steel hinge pins in the printer housing:



2. Lower the lid onto the printer housing, ensuring that the top rubber seal is not trapped. Do not press the lid into position yet:

3. (a) Hold the lid open at an angle of not more than 45° to the back of the unit.

(b) Place your thumbs on both lid hinges:



4. Push the slotted ends of the lid hinges down onto the steel hinge pins in the printer housing, ensuring they both click into place.

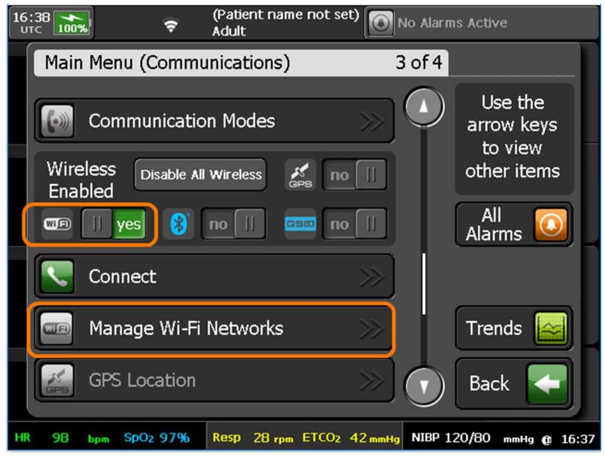
Important - ensure that the lid rotates freely on its hinges to open and close.

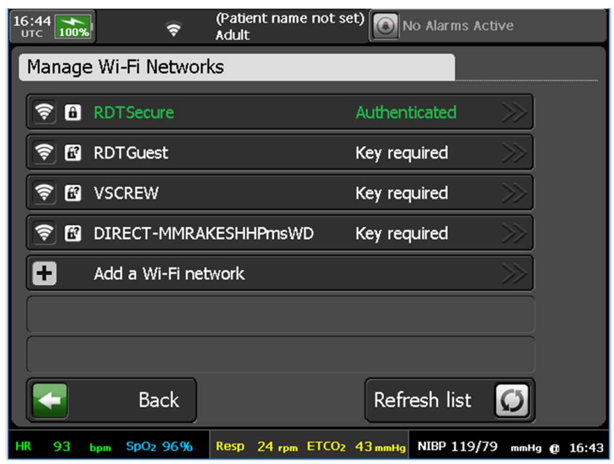
5. Close the lid and press it down on both corners, listening for a click:

9.9 Managing Wi-Fi networks

If Wi-Fi is the communication mode for your Tempus Pro, you can scan and connect to available Wi-Fi networks. Smart management of Wi-Fi enables Tempus Pro to automatically connect to any Wi-Fi network that it finds in range where the configuration details are already stored.

To manage your Wi-Fi networks, proceed as follows:

<table>
<tr><th>Step</th><th>Description</th><th>Screen</th></tr>
<tr><td>1.</td><td>
Press [Main Menu] .
In the Wireless Enabled control area, ensure that Wi-Fi is set to yes.
Press Manage Wi-Fi Networks.
</td><td>


</td></tr>
<tr><td>2.</td><td>
Tempus Pro displays the Manage Wi-Fi Networks menu. It may take up to 30 seconds to scan for and display the available networks. Each network is displayed with its connection status:
Connecting... - Tempus Pro is attempting to connect to this network.
Authenticated - Tempus Pro is now connected to this network. Press this button to view settings, update settings or forget the network (step 3a).
Saved - Tempus Pro knows the key for this network. Press this button to connect to the network, update settings or forget the network (step 3b).
Key required - Tempus Pro does not know the key for this network. Press this button to connect to the network (step 3c).
You can also connect to a hidden network (if you know its key and other settings) - press Add a Wi-Fi network (step 3d).
</td><td>


</td></tr>
</table>

<table>
<tr><th>Step</th><th>Description</th><th>Screen</th></tr>
<tr><td>3a.</td><td>

If you have selected the **Authenticated** network, choose an option:

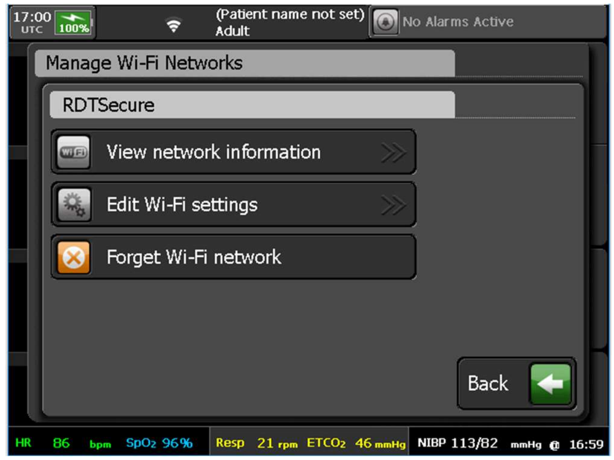
View network information - view IP address, IP subnet and IP gateway/DNS.

Edit Wi-Fi Settings - edit SSID, network key and hidden status.

Forget Wi-Fi Network - if you forget this network the Tempus will no longer be able to connect to it automatically.

Press **Back**.

</td><td>



</td></tr>
<tr><td>3b.</td><td>

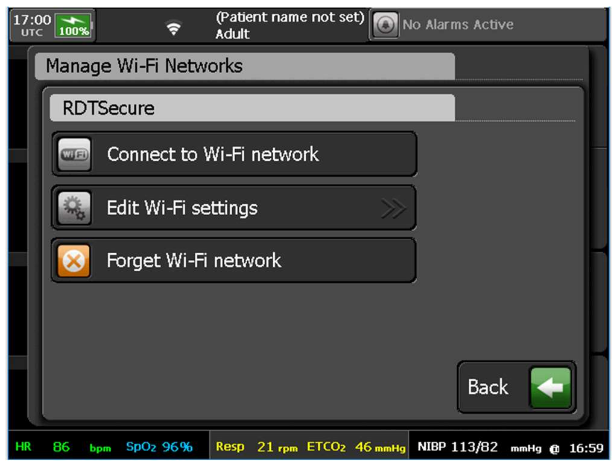
If you have selected a **Saved** network, choose an option:

Connect to Wi-Fi network - press this button to connect. Tempus Pro redisplays the Manage Wi-Fi Networks menu showing the status of this network as **Connecting....** If connection is successful, status changes to **Authenticated**.

Edit Wi-Fi Settings - edit SSID, network key and hidden status.

Forget Wi-Fi Network - if you forget this network the Tempus will no longer be able to connect to it automatically.

</td><td>



</td></tr>
<tr><td>3c.</td><td>

If you have selected a **Key required** network, choose this option:

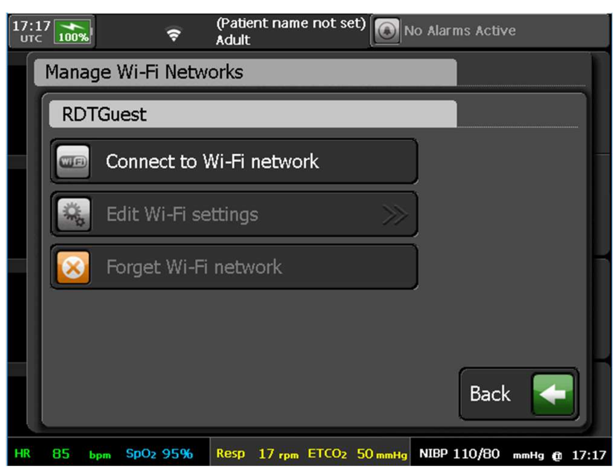
Connect to Wi-Fi network:

- Press this button.
- Enter the network key.
- Press **Connect**.

Tempus Pro redisplays the Manage Wi-Fi Networks menu showing the status of this network as **Connecting....** If connection is successful, status changes to **Authenticated**.

Note: If the "Wi-Fi error" message appears, check the connection assumptions in **Edit Wi-Fi settings** and try to connect again. If this does not help, press **Forget Wi-Fi network**, power cycle Tempus Pro and restart the connection to the required network.

</td><td>



</td></tr>
</table>

Step	Description	Screen
3d.	If you have selected **Add a Wi-Fi Network**: Enter the SSID, network key, hidden status and other details as requested. Press **Save**. Press **Back**.	

10 Connecting to an alternate location

This section describes how to connect the Tempus Pro to an alternate location to share data, typically a telemedicine Support Center or Response Center.

10.1 IntelliSpace Corsium (optional)

CAUTION	**IntelliSpace Corsium users:** If the Tempus LS software version is v1.3.4 or greater, automatic transfer of Tempus LS resuscitation data is active for devices set to connect to IntelliSpace Corsium. To ensure code review data is fully synchronized for update in IntelliSpace Corsium, keep your Tempus Pro powered up after using the Tempus LS with the TDL link on. Wait for the Tempus LS to synchronize with Tempus Pro at power off (as indicated on the Tempus LS shutdown screen) and leave Tempus Pro connected to IntelliSpace Corsium for a period to upload the data. This can also be done at any time within a few days of the resuscitation. When synchronization is complete, the sync icon disappears from the Corsium ReachBak menu: 
Note	**Tempus LS users:** Throughout these instructions, ‘Tempus LS’ refers to whichever defibrillator model is in use, either Tempus LS or Tempus LS-Manual. If the Tempus LS software version is v1.3.4 or greater, Tempus LS resuscitation data is automatically transferred to Tempus Pro over the TDL Link. You can also import data from the Tempus LS to Tempus Pro via a USB memory stick; use the **Import Tempus LS dat**a option in the Tempus Pro “Data Input and Output” menu.

10.1.1 Corsium ReachBak (optional)

The Corsium ReachBak feature allows Tempus Pro to send live monitoring data, SRoC events and 12-lead ECG review requests to the IntelliSpace Corsium support center. It also allows the support center to send ECG review results to the Tempus Pro, and for the operator to send review acknowledgements back to the support center.

Note	Tempus Pro and IntelliSpace Corsium may be in different time zones, for example local and UTC. If so, their displayed incident times will be different.

Corsium ReachBak connection status

The status bar displays information about the carrier network and the connection. For example when the Tempus Pro is connected via a Wi-Fi network and is streaming live data, the status bar looks like this:

The patient age group, patient name (if entered), device serial number and device name are displayed next to the IntelliSpace Corsium status icon.

If there is no carrier network connection:

- Check your Wi-Fi network - see "9.9 Managing Wi-Fi networks".
- Consider using a different connection mode, e.g. cell phone or Ethernet - press and then press **Connect in other ways**.

The connection status may be one of the following:

- Tempus Pro is not yet connected to IntelliSpace Corsium, or connection has been interrupted for more than two minutes.
- Tempus Pro is connected and streaming monitoring data to IntelliSpace Corsium.
- Connection has been interrupted for more than 30 seconds and less than 2 minutes.

Corsium ReachBak connection policies (optional)

The IntelliSpace Corsium administrator may set up the Tempus Pro connection policies. Depending upon which policies have been set, Tempus Pro connects to IntelliSpace Corsium when one of the following events occurs:

- A 12-lead ECG is recorded.
- The first trended vitals set is captured.
- Tempus Pro is connected to Tempus LS .

If you want to connect manually and begin streaming data before any of these events occur, press and then set **Connection to Corsium** to 'on'.

Corsium ReachBak menu

Press to display this menu:

Connection to Corsium - connect manually and begin streaming data to IntelliSpace Corsium (on/off). The sync icon indicates that there is pending data to be sent to IntelliSpace Corsium.

Support center - select a support center to receive data from this Tempus Pro.

ECG Summary - view a list of 12-lead ECG recordings made during this patient incident.

Connect in other ways - choose a different network carrier, e.g. cell phone or Ethernet.

(a) This panel shows the Tempus Pro device identity and device group.

(b) This panel shows the Corsium connection status. This is only displayed if **Show status in Corsium screen** is **on**.

Sending intervention data to IntelliSpace Corsium

Corsium ReachBak allows you to send completed intervention data from Tempus LS via Tempus Pro to IntelliSpace Corsium.

CAUTION ⚠	This procedure is only possible if Transmit Review Data in the Corsium Settings menu is enabled. If this feature is required, Contact your IntelliSpace Corsium system administrator or Tempus Pro maintenance personnel.

Proceed as follows:

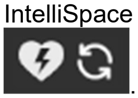
1. Export from Tempus LS:

- Plug a USB memory device into Tempus LS, select menu option **Menu > System > Intervention Management** and select one of the available options.
- Export the intervention data to USB.
- Remove the USB memory device from Tempus LS and plug it into Tempus Pro.

2. Import to Tempus Pro:

- Press **Data Input and Output**  and press **Import Tempus LS data**.
- Follow the instructions on the "Import Tempus LS Data" screens. On completion, Tempus Pro displays a message indicating the number of interventions imported.
- Remove the USB memory device from Tempus Pro.

3. Send from Tempus Pro to IntelliSpace Corsium:

- Press Connect  to display the Corsium ReachBak menu.
- If there are Tempus LS interventions or stored Tempus Pro waveforms waiting to be sent to IntelliSpace Corsium, the Connection to Corsium menu option will contain this icon: .
- This icon will disappear when all outstanding Tempus LS interventions and stored Tempus Pro waveforms have been successfully transfered to Corsium.
- Set **Connection to Corsium** to **on**. On successful connection, the intervention data will be automatically sent to IntelliSpace Corsium.

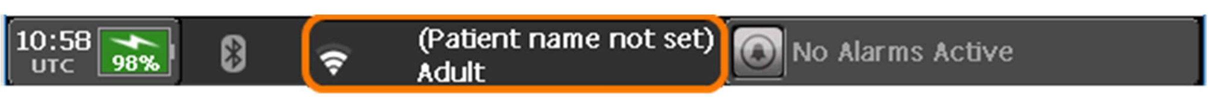
10.1.2 Corsium ECG (optional)

The Corsium ECG feature allows Tempus Pro to send 12-lead ECG review recordings to the IntelliSpace Corsium support center.

Note	The support center cannot send ECG review results back to the Tempus Pro; this is only possible if Corsium ReachBak is enabled in the Tempus Pro. Tempus Pro and IntelliSpace Corsium may be in different time zones, for example local and UTC. If so, their ECG recording times will be different.

Corsium ECG connection status

The status bar displays information about the carrier network and the IntelliSpace Corsium connection:



If there is no carrier network connection:

- Check your Wi-Fi network - see "9.9 Managing Wi-Fi networks".
- Consider using a different connection mode, e.g. cell phone or Ethernet - press and then press **Connect in other ways**.

Corsium ECG connection policy

When you send an ECG review request to IntelliSpace Corsium, the status indicator briefly changes to , then to and then it disappears when transmission is complete.

Corsium ECG menu

Press to display this menu:

Support center - select a support center to receive ECG data from this Tempus Pro.

ECG Summary - view a list of 12-lead ECG recordings made during this patient incident.

Connect in other ways - choose a different network carrier, e.g. cell phone or Ethernet.

(a) This panel shows the Tempus Pro device identity and device group.

(b) This panel shows the Corsium connection status. This is only displayed if **Show status in Corsium screen** is **on**.

10.1.3 ECG review with IntelliSpace Corsium (optional)

WARNING	**Do not delay patient care while waiting to correct a communications issue or while waiting for ECG review results.**

If the Tempus Pro has Corsium ReachBak capability, the ECG review and acknowledgement process is:

- Tempus Pro operator sends ECG recording to IntelliSpace Corsium support center.
- Support center user reviews ECG and sends results to Tempus Pro operator.
- Tempus Pro operator sends acknowledgement and optional estimated time of arrival back to support center.

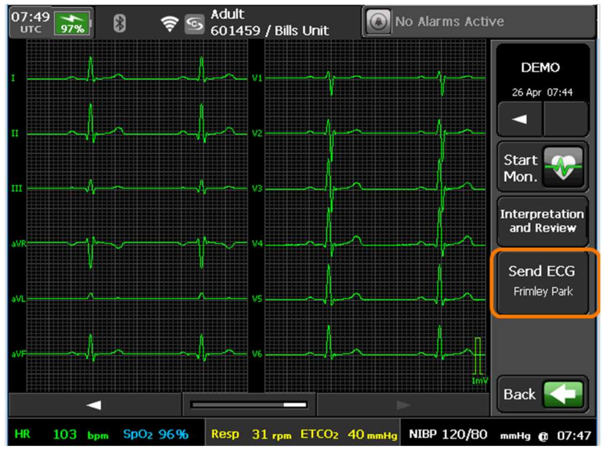
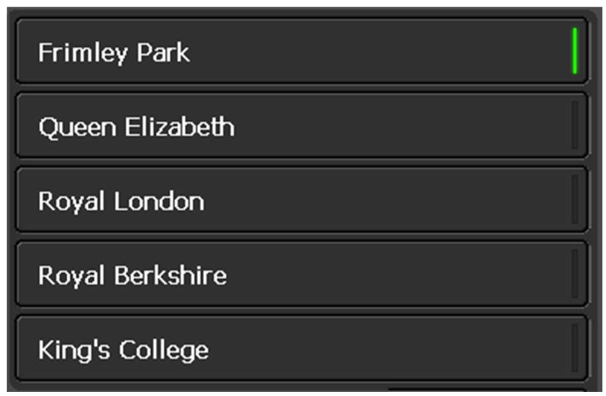
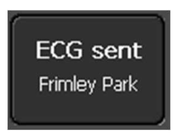
If the Tempus Pro is limited to Corsium ECG capability, the ECG review process is:

- Tempus Pro operator sends ECG recording to IntelliSpace Corsium support center.
- Support center user reviews ECG and records results.
- Results cannot be transmitted to Tempus Pro.

For connection instructions, see "10.1.1 Corsium ReachBak (optional)" or "10.1.2 Corsium ECG (optional)".

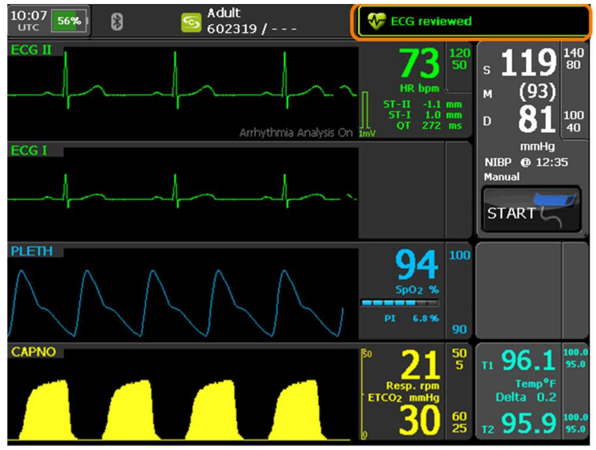
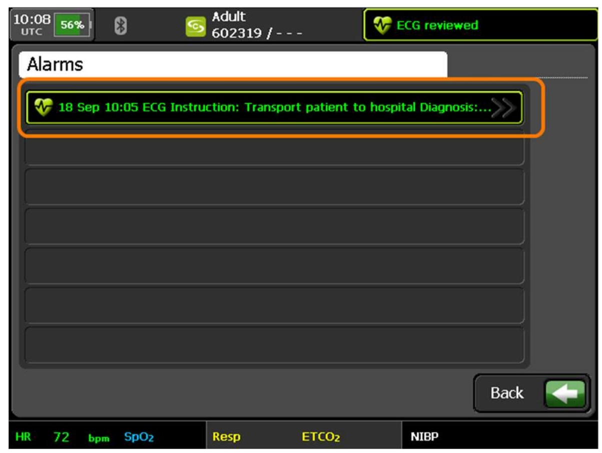
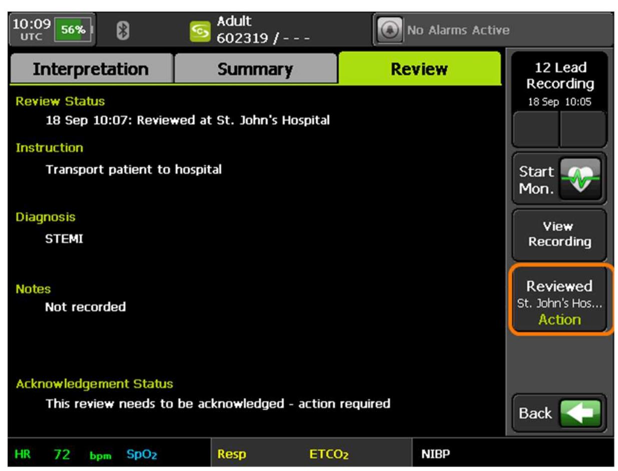
Sending an ECG report to IntelliSpace Corsium

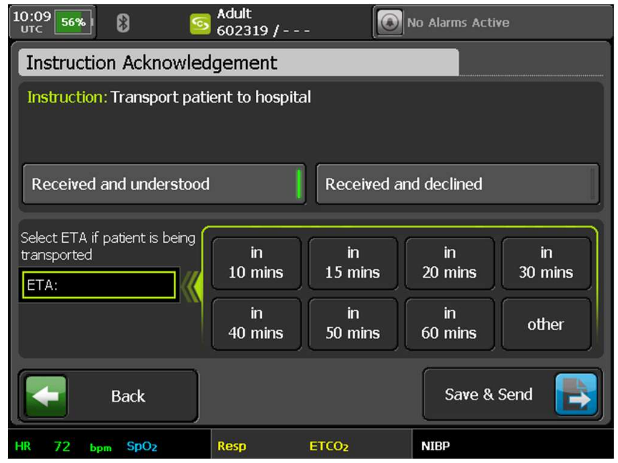
This feature is available with Corsium ReachBak and Corsium ECG.

Step	Description	Screen
1.	When you have taken a 10 second ECG recording, press **Send ECG**.	
2.	A list of available support centers is displayed with a green bar next to the default selection. If you want to send your ECG to a different support center, press one of the other buttons in the list. Press **Send**.	
3.	When transmission is complete, the send button text changes to **ECG sent**.	

Receiving and acknowledging the review result

This feature is available with Corsium ReachBak only.

Step	Description	Screen
1.	When your Tempus Pro receives ECG review results from the support center, the message "ECG reviewed" appears in the status bar. Press this message.	
2.	The active alarms are displayed. Press the ECG acknowledgement alarm.	
3.	The review results are displayed. If instructions are present, press **Action**.	

Step	Description	Screen
4.	If you have pressed **Action**, read the ECG reviewer's instructions and press either **Received and understood** or **Received and declined**. If you are transporting the patient to hospital, press an estimated time of arrival (ETA) button, for example **in 20 mins**. Press **Save & Send**.	

10.2 i2i ReachBak (optional)

In addition to being used as a vital signs monitor, the Tempus Pro can also be used to transmit the patient's vital signs, photos, moving video or position to an i2i Response Center (if the i2i ReachBak feature is enabled). The user of the Tempus can also speak to the Response Center with the Tempus.

To do this you will need to:

- Ensure the device is set to use an appropriate Communications Mode. The Tempus can connect to the same Response Center through a number of different, pre-set configuration Modes e.g. using Ethernet, WiFi or GSM (cellular data, if enabled) over a specific communications device. Normally the Tempus should be pre-configured to the correct mode but it may be necessary to change this from time to time – see "9.6.4 Communications modes".
- Make the data connection from the Tempus Pro to the Response Center.
- Fit the Headset comfortably in your ear. The operation of the headset is described in "10.3 Using the wireless headset".

The Tempus Pro can be left running with the data link connected but with the voice link disconnected i.e. if the Response Center physician wishes to continue monitoring the patient for a long duration but without keeping the voice link open with Tempus Pro User. In this case the voice link can be reconnected at any time by pressing the Connect button.

Note	Users are reminded that a valid cell phone network SIM card is required to be fitted to the Tempus in order to support cell phone connections.
Note	The characteristics of the communications network in use must be sufficient to enable the Tempus to communicate. Details of the Tempus' communications specifications are provided in "14.4 Communications". Ensure that the network is sufficient to enable the device's communications to operate as described in this manual.
Note	Should the communications network not be sufficient to enable reliable communications, the Tempus will attempt to re-establish communications for a limited number of attempts and will then show an error message. This situation is not hazardous but will prevent communications from occurring. In this situation the connection to the network should be checked, the network operation should be verified and if necessary/possible the network re-started before attempting communications with the Tempus again.

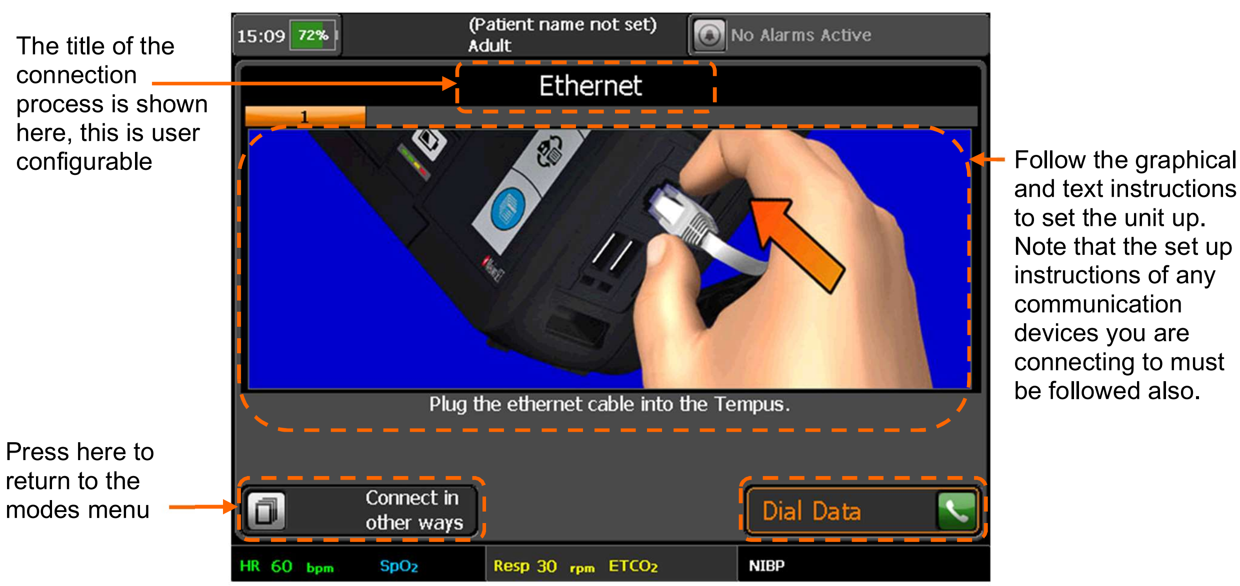
If the network fails during transmission the user can expect to see intermittent data transfer (complete failure or intermittent performance of data, voice or video). Again this is not hazardous in nature but users should wait to see if the network performance improves and then consider taking the steps detailed above to fix the problem.

CAUTION	In accordance with IEC60601-1 clause 14.13, responsible organizations should be conscious of the potential risks to patients e.g. risk of electric shock through the use of un-isolated electrical devices, the potential increase in summed leakage currents through the connection of devices etc. that could occur from connecting medical devices to communications networks. In addition, the impact of the use of the Tempus should be considered (e.g. bandwidth allocation). When making changes to the network (including adding or removing, updating or upgrading components), those responsible should take into account the bandwidth and other technical requirements (such as addressing and maintaining the appropriate ports in an open state) of the Tempus and ensure that these requirements are maintained. For details see "14.4 Communications".

10.2.1 Making the data connection

Note	The iAssist help processes on your Tempus Pro may differ from this example iAssist help process in the following sections. However, the process always follows the same key elements. Always ensure that you read the complete iAssist help process in order and do exactly what it requires'

Press the  button on the touch screen to get instructions on how to setup and make the data connection. Note that these instructions will appear by default every time you turn the unit on.



Example of the data connection process

Once the data connection process has been completed, press the **Dial Data** button to start the data connection process.

Note	These instructions are provided in iAssist format – see "9.7 iAssist processes". In the event that an alarm occurs, the Tempus will immediately revert back to the Home screen so that the alarm condition can be seen.
Note	The instructions shown above are examples, the on-screen instructions may differ depending on what Connection Mode the Tempus Pro is in. Users are reminded to read and follow the instructions on screen paying attention to both the written instructions and the graphics which show how to perform different elements of the task.

If the Response Center cannot be contacted, this could be due to errors in the way that the connection has been attempted. Help will be given in the form of an iAssist help process, follow the instructions given and wait for a few minutes before trying again. For Troubleshooting information, see "11.5 Troubleshooting".

If the Tempus cannot connect to the Response Center first time, it will automatically make a number of redial attempts (typically 3). The system will indicate the redial process by displaying a number over the Data Link status indicator.

10.2.2 Fitting the headset and making the voice connection

After starting the data connection, the Tempus will bring up instructions to make the voice connection. If a voice connection is not required press the Home button. The voice connection can be made using either the pre-attached Bluetooth® headset or the wired headset (accessory).

It is important to have attached the headset before dialling as the voice connection to the Response Center can be made quickly.

Remember that often the voice connection will be made over a satellite link so you may experience background noise or drop-outs. RDT recommends that you adopt a process of only one person speaking on the line at a time and then handing over to the other speaker by saying "over" or similar.

The voice connection can be setup only if the data connection is established first. The voice connection can be disconnected (hung up) while the data connection remains established but a voice connection cannot be made without a data connection. The voice connection is essentially a VoIP call. This is routed directly to a laptop/PC. Additional third-party hardware can be purchased to route the VoIP call to a landline – if this is required RDT should be contacted for technical support.

The voice connection is typically the means for the Tempus user to alert the personnel receiving the data that there is data to be received. Other means (such as a radio) can be used to notify personnel of an incoming data connection.

Voice connection process using a wired headset adaptor cable

Voice connection process using the wireless headset

To use the Bluetooth® headset, follow the on-screen instructions. When the headset is turned on, the Tempus will attempt to find it. During this time (20 seconds) a countdown will be displayed. Once the Tempus has located the headset it will confirm this on screen before resuming with the voice connection instructions.

Once the voice connection process has been completed, press the Dial Voice button to start the voice connection process.

Note	When using VoIP, the voice connection may take 10-15 seconds to setup after the "Dial Voice" button has been pressed.

10.2.3 Connection status indicators

The connection status indicators show whether the Tempus Pro is connected to the Response Center. There are separate indicators for the voice link and the data link.

The following symbols indicate the state of the links:

Data connection started (icon will flash)

Data connection to Data center (gateway) completed (icon solid), data not picked up by Response Center

Data connection to Response Center completed (icon solid and capital "C" on "Connected").

Voice connection in progress (icon flashing) or connected (icon solid)

The words 'connected' and 'disconnected' refer to whether there is a call in progress, NOT whether the Tempus Pro phone wires are plugged in.

Once the voice connection is in progress the User can hear its status through the headset e.g. ringing, connected etc.

WARNING	**When connected the patient name and age are not displayed.**

10.2.4 Communications modes

Tempus Pro can connect to the Response Center using a number of wired and wireless communications interfaces e.g. Ethernet or WiFi.

Note	The Tempus can only be pre-set to connect to a single Response Center. Configuring it to connect to new Response Center requires the support of RDT.

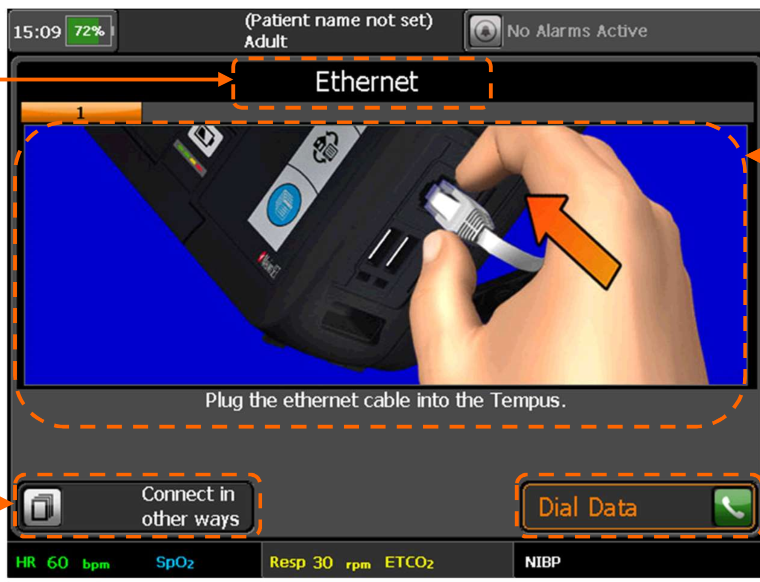
Connecting over wired interfaces such as Ethernet requires connecting the Tempus using a cable; connecting over wireless interfaces such as WiFi require no physical connection to be made.

To make it easy to switch between these types of connections, the Tempus Pro is pre-set to connect using different communication "Modes". Each Mode is supported by a full set of graphical connection iAssist help processes that provide the User with instructions specific to connecting using that technology.

The Tempus Pro shows what Mode it is in with a banner at the top of the Connection IAssist help process.

The Tempus Pro will stay in this Mode until it has been set to another Mode (even if it has been turned off and on again).

The title of the connection process is shown here, this is user configurable



Follow the graphical and text instructions to set the unit up. Note that the set up instructions of any communication devices you are connecting to must be followed also.

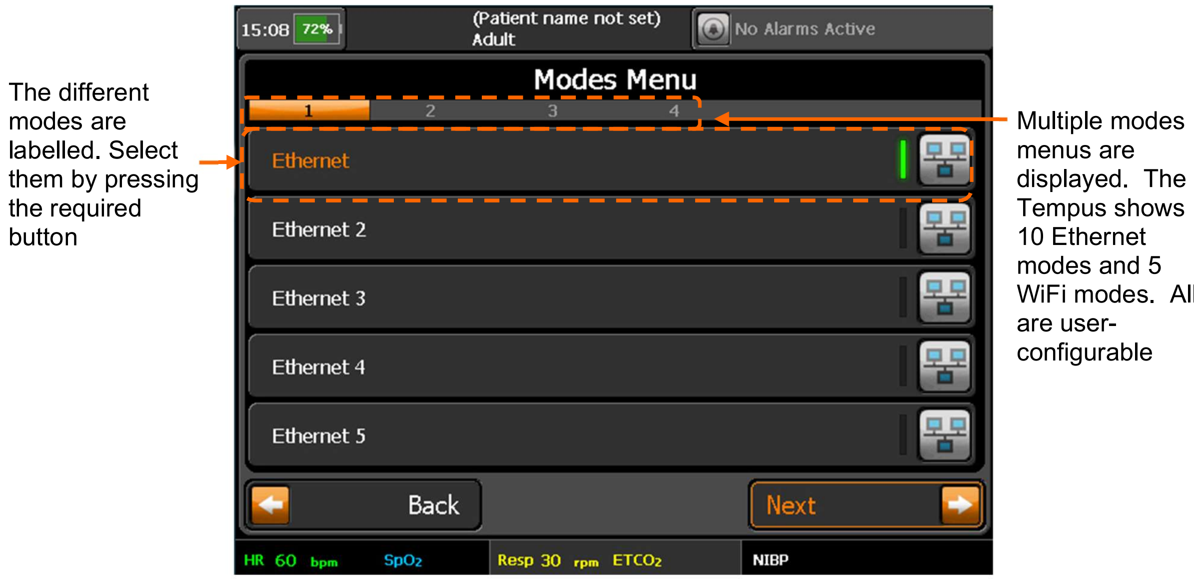
Press here to return to the modes menu

Press here to start the data connection process

Example of the data connection process

Changing modes

You can change the Communications Mode by pressing the **Connect in other ways** button. This will bring up the communications “Modes Menu”. This can also be reached through the **Communications Modes** button in the “Main Menu” – see “5.2.1 The main menu”.



Example Modes Menu

The mode that the Tempus is set to use will be indicated with a lit bar. To change mode, press the required mode. The instructions for that mode will be brought up on screen. The title of the new instructions will show what mode the Tempus is now set to use.

The modes that are available on each Tempus Pro are dependent on the requirements of each user. Refer to the “Modes Menu” on your Tempus Pro for specific details of each mode that is available.

Remember that each mode may have a different set of instructions for connecting, fault finding and repacking. Consequently, it is vital that you remember to read and follow what each iAssist help process says at all times.

It is also important to remember that if one mode cannot be used then another may be usable in its place.

Once the mode is selected the Tempus will display the instructions for connecting in that mode.

10.2.5 Interacting with the Response Center

The Response Center

Each Tempus Pro device is pre-configured to dial automatically to a specific Response Center. The center should be staffed 24 hours a day, 365 days a year and should always be able to receive your connection. If a connection cannot be established, you should wait a short time and attempt to connect again.

Note	It is the user’s responsibility to ensure their Response Center is properly staffed and equipped.

When interacting with the Response Center staff, you should also realize that they may be operating in a different time zone to the one where the incident is taking place. However, the clocks (and displayed times) of the both the Tempus Pro and the i2i software are synchronized.

Remote viewing and control

WARNING	**The alarm functions of the Tempus are intended to be used by the attendant user only. If the device is connected to a Response Center this is for the purpose of sharing vital signs data in real time, between two users for the purpose of obtaining additional clinical support. The system is not a distributed alarm system (e.g. nurse monitoring station system) in the terms of IEC60601-1-8. The i2i system at the Response Center is not equipped with alarm silencing or suspending controls.**

The Response Center operators will have exactly the same information on their screens as displayed on the Tempus Pro. Should the Tempus Pro display change e.g. if a new help screen is brought up, new data is displayed or an error message appears, the Response Center system will display exactly the same information a few seconds later. The only exceptions to this are when you are taking a digital picture or when you are monitoring or recording a 12 Lead ECG, in these situations the Response Center only see that you are in the process of recording the ECG or taking the photo but they don't see the ECG or photo until it has been downloaded.

WARNING	**Users are reminded that the i2i system at the Response Center is not equipped with alarm silencing or suspending controls. The i2i user is able to see the alarm condition through the on-screen visual alarm manifestations but they are not provided with audible alarms (it is not a distributed alarm system) and they are not equipped with means to silence or suspend the alarms. Managing the alarm state of the product is the responsibility of the Tempus User.**

Note	While the i2i will display the same information as the Tempus, user should note that the actual display size and aspect ratio of waveforms will vary slightly depending on the native resolution of the display that the i2i user has i.e. the i2i user's display may have differently sized and shaped pixels to the display of the Tempus and so very minor differences may be present between the two displays.
Note	If the waveform or video stop appearing on the i2i application stop the connection and reconnect the Tempus.

If the Response Center needs to operate the Tempus Pro remotely, they should make you aware that they are activating a function of the device before they do so. Ideally the Response Center will only take control of the Tempus Pro if the operator is having difficulty with an operation.

10.2.6 Recording data off-line and transmitting on-line

If a connection has not been established to a Response Center, all data for that patient can be transmitted later so long as a new patient is not admitted. Once a new patient is admitted, the recorded data cannot be transmitted to the Response Center. For details on discharging the patient, see "9.3.3 Admitting a new patient".

10.3 Using the wireless headset (optional)

The Tempus Pro uses the Presence or VMX200 Bluetooth headset provided by Sennheiser®.

CAUTION	Volume must be adjusted to reduce constant exposure to high volume, as this may affect hearing.

Note that the headset will be supplied in a charged state and its charge level is automatically maintained so long as it is regularly docked on the charging pin on the back of the device.

It should also be noted that the headset is a Sennheiser® wireless device. It is supplied “paired” with the Tempus Pro. You cannot use different wireless headsets with the Tempus Pro and RDT recommends that you do not attempt to use the headset supplied with the Tempus Pro with any other wireless devices (including other Tempus Pros or other communications devices such as mobile phones).

Each headset is “paired” to only the Tempus Pro to which it is attached on delivery. While attaching the headset to other Tempus Pro units will not cause any damage, users should avoid this practise as it may cause confusion and ultimately prevent voice calls from being made when needed.

If the headset is lost or is damaged, contact RDT for a replacement.

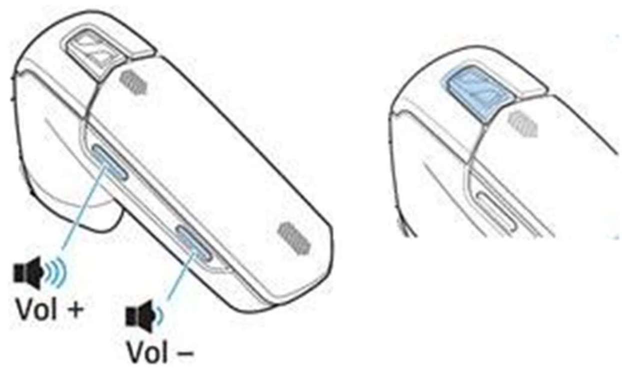
RDT recommends that the Sennheiser instructions (that are provided with the Tempus) are read in addition to the instructions below.

10.3.1 Introduction

The headset has 3 buttons:

- Volume up
- Volume down
- Multi-function button

Sennheiser Presence

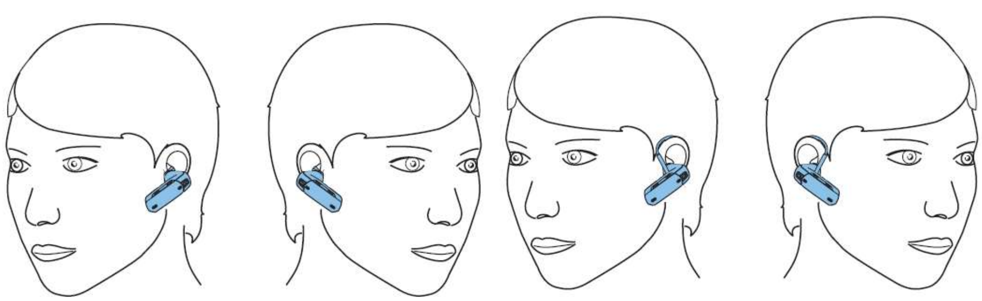


Sennheiser VMX200

Note	Users should avoid using the headset speaker volume controls. The speaker volume can be controlled through the Tempus and is preset to maximum.

10.3.2 Wearing the headset

To put the headset on, follow the instructions given on screen. Simply put the headset into your ear (either ear). It is small and light and should not require the use of an ear hook but this can be fitted if required (see Sennheiser instruction manual provided with the Tempus). You should ensure the speaker is well seated in your ear so it feels stable.



10.3.3 Controlling the wireless headset

To use the headset:

- Press the green dial button.
- Following the on-screen instructions, remove the headset from its dock.
- Following the on-screen instructions, press the Multi-function button once (for 1 second or less). The light on the headset will flash blue.

The Tempus will go through a process to link to the headset via Bluetooth®.

- Following the on-screen instructions put the headset into your ear.
- Following the on-screen instructions press the Connection button on the touchscreen.

Note	If you keep the talk button held down after the headset is on, you can put the headset into pairing mode. This is not desirable as it could potentially cause the headset to cease being paired with the Tempus Pro and thus prevent it from operating with the device. Pairing mode can be recognized by the indicator light slowly flashing red and then blue. If the headset is inadvertently put into pairing mode it should be placed back onto the docking pin on the Tempus to turn the headset off; the voice call should be disconnected and re-initiated.

You can check if your headset is on by pressing the talk button once. If the indicator light flashes blue, then this means the unit is on.

You do not need to switch the headset off; this is achieved by docking the headset back onto the Tempus Pro. If you do wish to turn the headset off, press and hold the Multi-function button for 3 seconds until the LED flashes blue three times.

Note	If you turn the headset off during a call, you will not be able to receive the call again with the headset i.e. if you turn the headset back on you will not hear the call again. If the headset is turned off during a call, disconnect the call using the Tempus Pro and then re-initiate the connection again following the on-screen instructions.
Note	RDT recommends that users do not use the Multi-function button for any other purpose than turning the headset on (as described above). If you use the Multi-function button, then other functions and features (as described in the Sennheiser manual) of the headset can be engaged – these may cause confusion.

To adjust the volume during a call, press the “vol+” button or “vol-“ button on the headset as shown above.

Note	If you press and hold the “vol-” button for 1 second you will mute the headset. To release muting, quickly press the “vol-” button on the headset. RDT does not recommend that you use the muting function as this could cause confusion during a call.

10.3.4 Changing headset settings

The performance of the headset can be changed if required. The headset settings are accessed using the **Headset Settings** button on the “Communications Settings” menu – see “5.2.2 Printer and Headset menu”.

Note	The headset settings are pre-set for the best performance for use in environments with a range of ambient noise levels. Users should change the settings if they are struggling to hear or be heard.

11 After using the Tempus Pro

11.1 Inspecting the Tempus Pro

After using the Tempus, always inspect the device, its accessories (including power supplies, mains cable, batteries and charger) for signs of damage or wear. While doing this ensure that you check the strain reliefs of cables and connectors to ensure they remain fit for use. Check that all connectors engage properly and that cable securements work acceptably. Also remember to check connector covers (if fitted) and particularly the Capnometer door, to ensure they close and latch acceptably.

Inform any signs of wear, damage or malfunction to your service department.

At least once per year completely inspect the whole device and in particular the mains power supply and battery charger, for signs of extreme damage. This inspection should be performed by appropriately trained and equipped personnel who are able to perform locally prescribed safety tests.

In addition, RDT recommends the device is tested by an appropriately trained and equipped Electro-Biomedical Engineer to confirm the medical and other functions of the device function within specification and within locally prescribed safety limits.

Note	Users should note that particular calibration curves exist for use of specific SpO_2 simulators with the Tempus' Masimo® pulse oximeter. Please contact RDT for details. Such functional testers should not be used to determine the accuracy of the device, only its correspondence to the particular calibration curve provided by the tester.

11.2 Cleaning the Tempus Pro

It is necessary to clean the Tempus Pro after use.

CAUTION	If you suspect that dirt or fluids have entered the battery compartment, see the *Tempus Pro Maintenance Manual*.
Note	If the Tempus is soiled, it is the user's responsibility to ensure it is withdrawn from use and properly decontaminated prior to re-deployment. In this instance users are recommended to replace the handle – for the appropriate part number, see "12 Accessories list of the Tempus Pro".

11.2.1 Cleaning the case of the Tempus Pro

To clean the Tempus, use a clean and dry cloth using an agent from the table below. Ensure that connector pins are covered during cleaning and remain dry. Wipe off any excess cleaning agent from the device with a different clean and dry cloth.

Approved Cleaning Agents:

- 70% IPA

- Warm water
- Liquid soap
- Wex-cide®
- Windex®

Never use any other cleaning agent than those described above. Use of more aggressive agents such as acetone and other solvents is not permitted.

The Tempus Pro instruments must be cleaned during the re-packing process.

In the event of contamination of the Tempus from human fluids or other waste matter the product should be thoroughly decontaminated and disinfected. In this event the handle should be removed and discarded and replaced with a new part sourced from RDT.

CAUTION ⚠	If the Tempus requires cleaning, users are advised to ensure that stiff or sharp objects are NOT used to remove detritus from underneath the louvered vents in the front and rear panels of the device. Doing so could compromise the IP66 sealing of the product. The front panel vent is located where shown in the image below. The rear panel vents are located behind the accessory pockets.

Position of vent

11.2.2 Cleaning the display

The screen may be cleaned using a proprietary screen cleaning wipe of the type used for other LCD screens. Under no circumstances should any abrasive substance be applied to the screen. It should be noted that the anti-reflective (AR) coating of the touchscreen may be impaired if unsuitable materials are used to clean the touchscreen. The table below lists different cleaning materials and their effect on the AR coating after a 30-minute exposure under ambient conditions:

Chemicals	Result
NaOH 3% aq. solution	Unchanged
Na4OH 3% aq. solution	Unchanged
HCl 3% aq. solution	Unchanged
Kitchen cleaner 5% aq. solution	Unchanged
Glass cleaner	Unchanged

CAUTION	Under no circumstances should any abrasive substance be applied to the screen.

11.2.3 Cleaning connectors

If connectors are wet, they should be rinsed with tap water and then dried. All solid material (e.g. dust or sand), should be removed with compressed air.

Cables or connectors with corroded contacts should be replaced.

11.2.4 Cleaning the NIBP cuffs and hose

The NIBP hose and reusable cuffs should be cleaned with a clean cloth after use with a disinfectant such as Clorox® (1:10 solution).

Ensure that any cleaning solution does not enter the hose or bladder. Ensure the cuff is completely dry before it is reused.

The cuffs may be machine washed at 40-50 °C (after plugging the air connector to prevent water entering the bladder).

11.2.5 Cleaning the SpO2, ECG and invasive pressure cables

Clean the sensor cables with a clean cloth after use with a disinfectant such as Clorox® (1:10 solution). Do not immerse any cables. Ensure the cables are completely dry before they are reused.

CAUTION	No cables or parts of the Tempus should be sterilized using any process (Steam, EO, Radiation etc.). Never immerse ECG cables, doing so could reduce the performance of defibrillation protection devices within the cable’

11.2.6 Cleaning the battery contacts

Remove the battery once a week. Inspect the electrical contacts of the Tempus Pro battery compartment and of the battery itself. If any battery contacts are dirty, clean them as described:

Approved cleaning agents:

- Isopropyl alcohol electronic cleaning solvent.

Cleaning:

1. Moisten a clean, dry cloth with the approved cleaning agent and wipe the battery contacts.
2. Wipe off any excess cleaning agent with a clean, dry cloth.
3. Refit the battery.

11.3 Disposal at end of life

The WEEE logo on the Tempus and its battery refers to the EU Directive on Waste Electrical and Electronic Equipment (WEEE). This Directive entered into force as European law on 13th February 2003; it resulted in a major change in the treatment of electrical equipment at end-of-life. The purpose of this Directive is, as a first priority, the prevention of WEEE, and in addition, to promote the reuse, recycling and other forms of recovery of such wastes so as to reduce the disposal of waste.

The symbol indicates that this product must not be disposed of or dumped with household waste. The owner of the equipment is liable to dispose of all electronic or electrical waste equipment by delivering to the specified collection point for recycling of such hazardous waste, collection and proper recovery of electronic and electrical waste equipment at the time of disposal will allow the producer to help conserve natural resources. Recycling of the electronic and electrical waste equipment will ensure safety of human health and the environment. For more information about electronic and electrical waste equipment disposal, recovery and collection points, please contact your local, waste disposal service or producer / distributor of this equipment.

Note	Tempus batteries must be disposed of in accordance with local regulations. Batteries should not be crushed or incinerated as they could present a risk of fire or explosion. Batteries should be handed to an appropriate organization that is competent in the disposal of such devices; this must be done in accordance with local regulations.

11.3.1 Disposal of single use devices

Any accessories that are designated as single-use devices must be discarded after use. No particular precautions are required when disposing of these items provided that they are not contaminated with bodily fluids. In case of such contamination, the items could present a bio-hazard and therefore should be disposed of in accordance with local regulations governing such matters.

12 Maintenance, servicing and troubleshooting

The Tempus Pro is designed to be as maintenance-free as possible. The only user-replaceable and user-serviceable parts in the Tempus Pro are those listed in this section of the manual.

More details on maintenance are given in the *Tempus Pro Maintenance Manual*, which is available from RDT, see "RDT contact details".

The *Tempus Pro Maintenance Manual* contains the necessary technical information, schematics, parts designation and process description for specified maintenance and testing to be performed.

WARNING	**Do not open the battery case or in any way disassemble, modify or repair the battery.**

CAUTION	All leads, cables and accessories should be checked before and after use to ensure that such parts are not subject to wear, fraying, tears, knots or other signs of damage. Damaged or worn parts should be replaced.
CAUTION	The ECG cables contain components to protect against the effect of a defibrillator. The ECG cables should be measured on an annual basis (or more frequently at the user's discretion if they are subject to a high number of defibrillation discharges) to ensure that they retain a resistance of 1 kΩ ±10% as per AAMI EC53. Cables should be replaced if this resistance is not maintained.

Note	If the Tempus Pro is no longer serviceable and is beyond repair, it may be scrapped. Scrapping the device and its accessories must be performed in compliance with applicable local regulations. It should be noted that special conditions may apply to the rechargeable battery if it is required to be scrapped. The battery should be discharged before scrapping and should not be crushed or incinerated.

12.1 Tempus Pro battery

12.1.1 The battery

The Tempus Pro contains a removable, rechargeable battery.

Example of the battery front

Example of the battery rear

In normal usage, the rechargeable battery provides power for at least 9 hours' continuous use when fully charged.

Note	Assessment of use is based on projections of reasonable device usage within a patient incident made by RDT.
Note	Do not attempt to fit batteries from previous RDT products (Tempus IC or Tempus IC Professional) into the Tempus Pro, they will not fit.

Every battery is provided with an integral battery life indicator which is also visible through the front panel of the case.

The battery life should be monitored periodically when the device is in storage and also before and after use.

Note	RDT recommends that the battery charge status should be checked at least every month and recharged if necessary. If the device is being used regularly then the battery should be checked, weekly or daily depending on the frequency of use. RDT also recommends that the battery be completely discharged and recharged once a year.
Note	The User should remember that battery capacity of older batteries will not be the same as new batteries.

By monitoring the remaining battery life, situations where the battery is too weak to power the Tempus Pro for the duration of an incident can be avoided. If the battery strength indicator shows less than 25% power remaining, you should change the battery if possible to ensure that there is adequate power for the next time it is needed.

Using the battery down to the point where it is completely empty will not cause any hazards or damage to the system.

The power of the battery can be extended by enabling a screen-saving feature on the Tempus. Enabling this feature will make the display switch off if 5 minutes pass with no operator intervention or alarms triggering. The Tempus will remain on and performing any monitoring function it has been set to do. If any patient or technical alarms occur, the Tempus display will come back on and the alarm functions sound as normal. While the display is off, if any operator controls are used then the display will come back in immediately.

Using the power save feature will enable the battery life to be extended significantly.

To enable this feature, go to the Display Menu and then turn Power Save to on – see “9.2.7 Power save feature”.

Checking the charge of the battery

The charge state of the battery can be obtained by pressing the button on the front .

The battery is provided with 4 charge state LEDs. Pressing the button will light one or more lights. Each light corresponds to 25% of the charge state of the battery in the order (from highest to lowest):

- Four green LEDs – 76-100%
- Three green LEDs – 51-75%
- Two green LEDs – 26-50%
- One green LED – 1-25%
- One green flashing LED – less than 10%
- No LEDs – no power due to deep discharge

Battery status is always displayed in the status bar – see “5.1.3 Battery status indicator”.

CAUTION	When the battery is low it should be recharged soon after use. Starting a Tempus on an empty battery or leaving batteries in a low or empty state for prolonged periods may put the battery into a deep-discharged state. In this state the battery LEDs will not light and the battery will require a prolonged period on charge in order to return it to normal operation.
CAUTION	If none of the LEDs light, then the battery may have been deep-discharged. It should be left on-charge on an external charger for 24 hours to bring the battery out of this state.

These will light cumulatively when the battery button is pressed i.e. only the one green light will light if the charge state is 1-25% after which the 2nd green light will light as well.

The Tempus Pro does not need to be turned on to check the battery.

If batteries do not retain their charge state, users should try charging them fully on an external charger. As batteries become old and over many charge/discharge cycles, users should expect to see a reduction in their capacity. Such batteries should be replaced at the user’s discretion.

12.1.2 Removing the battery from the Tempus Pro

WARNING	**Do not short-circuit the terminals of any battery. A short circuit can occur if the battery terminals come into contact with any metal or other electrically conductive object. The battery may be irreversibly damaged if it is short-circuited.**
WARNING	**Ensure the latches on both sides of the battery are fully engaged prior to using the Tempus - an incorrectly fitted battery could result in the Tempus losing power during use.**

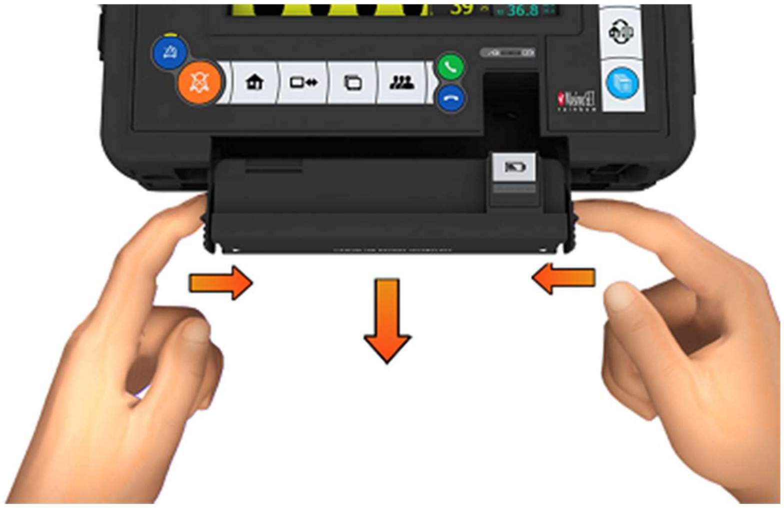
Note	Before removing the battery, you must switch off the Tempus Pro by pressing the power button. If necessary, you can force a hard shut-down by pressing and holding the power button for 10 seconds. Remember that the battery cannot be removed until the lamp on the front panel has gone out. The Tempus Pro should not be used without the battery fitted.

To replace the battery:

- First check the replacement battery has sufficient charge by checking its indicator.

Example of the battery indicator

- Next, ensure the Tempus is switched off. Then remove the battery by squeezing the two latches inwards and then pull the battery away.



Battery removal

- Slide the new battery all the way into the Tempus until it clicks into place on both sides.

Battery insertion

12.1.3 Charging the battery

The battery can be charged either when it is fitted to the Tempus Pro or when it is removed from the separate battery charger.

WARNING	**Do not attempt to charge the battery using any charger other than those supplied by RDT.**
WARNING	**Interruption of the mains supply for short periods e.g. 30 seconds, does not cause problems because the Tempus Pro automatically switches to battery power. If no battery is fitted (or the battery is completely flat) the Tempus Pro shuts down immediately. No patient data will be lost. If the device is restarted within 30 seconds, all patient settings remain unchanged. When mains power is resupplied, the Tempus Pro resumes as normal with default settings in place. You can switch back to monitoring the previous patient (and continue completing their record).**

CAUTION	When the battery is low it should be recharged soon after use. Starting a Tempus on an empty battery or leaving batteries in a low or empty state for prolonged periods may put the battery into a deep-discharged state. In this state the battery LEDs will not light and the battery will require a prolonged period on charge in order to return it to normal operation.

CAUTION	If none of the LEDs light, then the battery may have been deep-discharged. It should be left on-charge on an external charger for 24 hours to bring the battery out of this state.

Note	Users are reminded that the Tempus (when using its external power supply) should not be used to recharge batteries above 40°C – see "14.2 Physical characteristics and environmental specifications".

Charging the battery when attached to the Tempus Pro

When fitted to the Tempus Pro, the battery can be charged by connecting the power supply (part number 01-2049) to the connector on the right hand side of the Tempus.

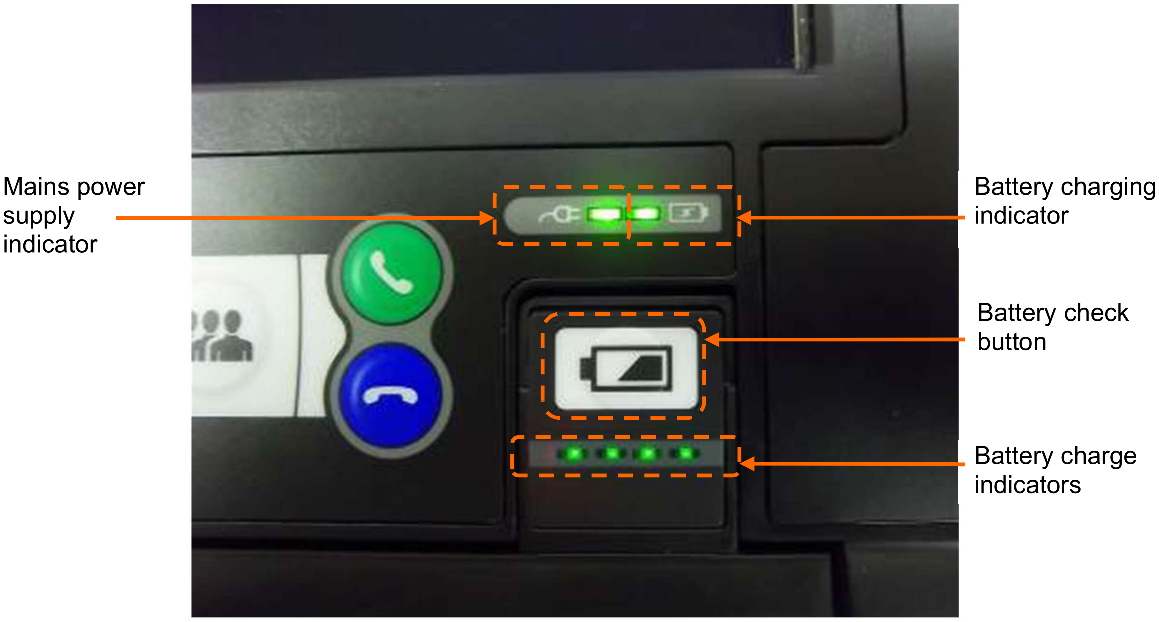
The PSU plug attached to the Tempus Pro connector

CAUTION	If the Tempus is powered using the mains power supply, ensure the mains power supply is accessible so the mains lead can be disconnected in the event that the device needs to be isolated from mains power. The means of isolation of mains power is to disconnect the mains lead from the power supply.
CAUTION	Where the integrity of the external protective conductor (earth) in the installation or its arrangement is in doubt, equipment shall be operated from its internal electrical power source.

Note	Interruption of the supply mains for short periods e.g. 30 seconds, will not cause a problem, as the Tempus will automatically switch to battery power. In the event that a battery is not fitted (or is completely flat) the device will shut down immediately. In this case no patient data will be lost. If the device is restarted within 30 seconds it will resume with all patient settings unchanged. If power is resupplied after that it will resume as normal with default settings in place. In this case the user can switch back to monitoring the previous patient (and continue completing their record) as described in sections 4.3.3 and 9.3.4.

When the power supply is attached to the Tempus Pro, the green power light on the Tempus Pro front panel will turn on.

If a battery is attached the green charge light will flash. The lights on the battery will light solidly up to the charge state of the battery at the time.



CAUTION ⚠	When the battery is low it should be recharged soon after use. Starting a Tempus on an empty battery or leaving batteries in a low or empty state for prolonged periods may put the battery into a deep-discharged state. In this state the battery LEDs will not light and the battery will require a prolonged period on charge in order to return it to normal operation.
CAUTION ⚠	If none of the LEDs light, then the battery may have been deep-discharged. It should be left on-charge attached to a battery charger (separate from the Tempus) for 24 hours to bring the battery out of this state.

Note	Charge times of the battery will vary depending on the how the Tempus Pro is being used. If the Tempus is switched off charging will be faster than if the Tempus is on and all features are being used. Charging a completely empty battery will take 4 hours when the Tempus Pro is switched off.

Using the battery charger

When the battery is separate from the Tempus Pro, the battery may be charged by connecting it to the battery charger (part number 01-1012). To attach the charger to the battery, the clip must be firmly pressed onto the connections of the battery. Note that the clip of the charger can only be connected to the battery in one way.

The battery connector attached to the battery

Clip the charger to the battery (the clip only attaches in one way).

Attach the charger to the main supply.

The LED on the charger will light orange (for approximately 0-85% charge), change to yellow during charging (at approximately 86-100% charge) and will turn green when finished. If the battery is only partially discharged, then the LED may start on yellow.

CAUTION	When the battery is low it should be recharged soon after use. Starting a Tempus on an empty battery or leaving batteries in a low or empty state for prolonged periods may put the battery into a deep-discharged state. In this state the battery LEDs will not light and the battery will require a prolonged period on charge in order to return it to normal operation.
CAUTION	If none of the LEDs light, then the battery may have been deep-discharged. It should be left on-charge attached to a battery charger (separate from the Tempus) for 24 hours to bring the battery out of this state.

Note	Battery charger is rated at 100-240 V 50-60 Hz 0.9 A. Recharging the battery with the external charger takes up to 5 hours for a fully discharged battery.

12.2 Wireless headset battery

The headset contains a rechargeable battery. The battery of the headset is not user-replaceable and does not require user intervention. In the unlikely event that the headset's battery becomes completely exhausted and can no longer hold charge, a replacement headset can be purchased.

12.2.1 Charging the headset

RDT do not supply a separate charger for the headset. The charger is built into the Tempus Pro.

You must follow the repacking instructions provided by the Tempus Pro on screen. These will instruct you to clean the headset after use and to replace it on its docking pin before shutting down.

If you do not replace the headset, then the Tempus will show an error advising that the headset should be refitted.

Placing the headset onto the docking pin enables the Tempus Pro to recharge it. The Tempus Pro recharges the headset for up to 9 hours (approx.) every time the headset is replaced. The charging cycle will continue regardless if the Tempus Pro is switched on or off. Charging is started as soon as the headset is fitted to the docking pin. The indicator light on the headset will light red for the duration of the charging process. Once the headset is charged its LED will flash blue every 5 seconds. The indicator lights will go off when charge cycle is complete (9 hours).

In addition, the Tempus Pro will top up the charge of the headset approximately every 97 days. This occurs when the Tempus is switched off and lasts for up to 9 hours (approximately).

CAUTION	Always maintain a level of at least 10% in the Tempus battery. Leaving the Tempus battery in a low charge state can risk it being depleted by the headset charging process.
CAUTION	Do not attempt to charge the headset using any other charging device. This will automatically suspend the warranty and could be dangerous.

12.2.2 General guidelines for safe use

Do not drop or try to alter the shape of your headset.

Do not expose the headset to liquid or moisture. Unlike the Tempus, the headset has no protection against ingress of solids or liquids.

Do not expose your headset to extreme temperatures. The temperature range of the headset is 10-40 °C.

Do not try to disassembly your headset. Service and Maintenance can only be performed by RDT.

Do not let children play with the headset since it contains small parts that could become detached and create a choking hazard.

WARNING	**Danger of explosion if battery is incorrectly replaced. Do not attempt to repair or replace the battery.**

CAUTION	If the battery is worn out, a new headset is required from RDT. Dispose of the headset in accordance with local regulations. Do not dispose as household waste.

The user manual for the Sennheiser headset is supplied on the same CD-ROM as this manual. Details from that manual have been reproduced in this section courtesy of Sennheiser®.

12.2.3 Disposal of batteries

Dispose of batteries in accordance with the applicable local regulations (these can vary from country to country).

Note	In most countries, it is illegal to dispose of used batteries in residual waste (landfill). Users are required to dispose of used batteries through the proper channels, for example through not-for-profit organizations mandated by local governments.

12.3 Daily checks

RDT recommends that the following daily checks are performed on the Tempus, its cables and accessories:

- Ensure they are clean (with no fluid spills) and free of visible damage. Clean and replace as required.
- Check for signs of damage, excessive wear (cuts in insulation, fraying, broken wires, dirty or bent connector pins). Replace if damaged.
- Ensure that the device is equipped with appropriate quantities of all disposable supplies (e.g. ECG monitoring electrodes etc.).
- Ensure that any accessories or consumables are within the shelf life printed on their packages.
- Ensure that the internal printer (if fitted) is not damaged and contains sufficient paper for at least one day's use.
- Verify that a charged battery is fully inserted into the Tempus.

- Ideally ensure that a fully charged spare battery is available.
- Ensure the Tempus powers on and that all controls (as described in section 5.1) operate as described.

12.4 Weekly checks

RDT recommends that the following weekly checks are performed on the Tempus:

CAUTION ⚠	To prevent battery failure, perform this check every week when the Tempus Pro is in use.

- Remove the battery from the Tempus Pro and inspect its contacts. If contacts are damaged, replace the battery. If contacts are dirty, clean them - see "1.1.6 Cleaning the battery contacts".

12.5 Troubleshooting

Occasionally, problems may occur with the Tempus Pro. Operator error, sensor problems or a failure within the Tempus Pro could cause these problems. In most instances, the Tempus Pro will display an error message on the screen. This section describes the possible error messages and what they mean.

All of the error messages take the form of a window which appears in the middle of the screen.

The window contains the following text:

- A title which identifies the sensor or system which is having trouble;
- A description of the problem;
- The effect that the error will have on the performance of the Tempus;
- Which button to press to clear the error message off the screen.

CAUTION ⚠	In the event that the Tempus Pro displays an error that is not described within this manual e.g. application errors, turn the Tempus Pro off and then on again. This should clear the error and allow normal operation to resume. Do not continue to use the device if such an error is displayed. If symptoms persist, please contact RDT.

13 Accessories list of the Tempus Pro

13.1 Parts list

The following user-replaceable accessories and consumables are available from RDT:

13.1.1 Temperature accessories

Product name & description	Part number
YSI-400 Series Reusable 401AC ContactTemp Probe Autoclavable	01-2153
YSI-400 Series Disposable 4492 12hr Contact Temp Probe	01-2078
Redy-Med TA008-A Contact Temperature Adaptor Cable	01-2316

13.1.2 Invasive pressure accessories

Product name & description	Part number
Part number 01-2113 connects to the Tempus Pro only via the universal cable (part number 01-2108). Part number 01-2108 accepts up to two adaptor cables for two IBP channels:	
Tempus Pro 2-Channel Invasive Pressure Universal Cable	01-2108
Transpac & Art-Line Transducer Adaptor Cable	01-2113
2 Channel Invasive Pressure Module	01-2017

The following invasive pressure transducers are approved for use with the Tempus Pro:

Transducer manufacturer	Transducer model	Adaptor cable
ICU Medical (Abbott)	Transpac IV or IT	RDT P/N: 01-2113 C&S part number 10369 (*)
Biometrix	Art-Line	RDT P/N: 01-2113 C&S part number 10369 (*)
Smiths-Medical (Medex)	TranStar	RDT P/N: 01-2113 C&S part number 10369 (*)
Smiths-Medical (Medex)	LogiCal	C&S part number 10366 (*)
Merit Medical (Argon Medical / BD)	Meritrans / DTXPlus	C&S part number 10367 (*)
Codan PVB (ITL Healthcare)	Xtrans (DPT 9000)	Not yet available
Codan PVB (ITL Healthcare)	DPT-6000	C&S part number 10371 (*)

Transducer manufacturer	Transducer model	Adaptor cable
Utah Medical	Deltran	C&S part number 10370 (*)
Biosensors International	Biotrans	C&S part number 10370 (*)
Biosensors International	Accutrans	C&S part number 10370 (*)
Edwards Lifesciences (Baxter)	TruWave	C&S part number 10368 (*)

(*) www.CablesandSensors.com

13.1.3 Non-invasive blood pressure accessories

Product name & description	Part number
Tempus Blood Pressure Cuff – adult cuff (23 – 33 cm)	01-1002
Tempus Blood Pressure Cuff – large adult cuff (31 – 40 cm)	01-1003
Tempus Blood Pressure Cuff – child cuff (12 – 19 cm)	01-1004
Tempus Blood Pressure Cuff – small adult cuff (17-25cm)	01-2119
Tempus Blood Pressure Cuff – infant (8 - 13 cm)	01-2021
Tempus Blood Pressure Cuff – adult thigh cuff (38 – 50 cm)	01-1032
Tempus Neonate Patient Hose Adaptor CPC/Luer	01-2067
Tempus Blood Pressure Cuff – neonate cuff (3 – 6 cm) Single Use (Box of 20)	01-2030
Tempus Blood Pressure Cuff – neonate cuff (4 – 8 cm) Single Use (Box of 20)	01-2031
Tempus Blood Pressure Cuff – neonate cuff (6 – 11 cm) Single Use (Box of 20)	01-2032
Tempus Blood Pressure Cuff – neonate cuff (7 – 13 cm) Single Use (Box of 20)	01-2033
Tempus Blood Pressure Cuff – neonate cuff (8 – 15 cm) Single Use (Box of 20)	01-2034
Tempus Blood Pressure Hose 4ft	01-1006
Tempus Pro NIBP Cuff Kit (adult, large adult, thigh and child cuffs)	01-2155
Tempus Pro Blood Pressure Hose 8ft	01-2074
NIBP Infant Disposable Cuff (Box of 20)	01-2230
NIBP Child Disposable Cuff (Box of 20)	01-2231
NIBP Small Adult Disposable Cuff (Box of 20)	01-2232
NIBP Adult Disposable Cuff (Box of 20)	01-2233
NIBP Large Adult Disposable Cuff (Box of 20)	01-2234
NIBP Thigh Disposable Cuff (Box of 20)	01-2235
Tempus NIBP One-Piece Cuff - Small Adult Plus (18-29cm)	01-2270
Tempus NIBP One-Piece Cuff - Adult Plus (28-40cm)	01-2271
Tempus NIBP One-Piece Cuff - Large Adult Plus (40-55cm)	01-2272

13.1.4 Masimo accessories

Product name & description	Part number
Masimo rainbow SET cables	
MasimoSET Rainbow Patient Cable 12 ft 25-Pin R/A RC12 RA	01-2138
MasimoSET Rainbow Patient Cable 4ft 25-Pin R/A RC4 RA	01-2088 or 989803182411

Product name & description	Part number
Masimo rainbow SET DCI	
Masimo Rainbow DCI Adt 3ft (SpO2, SpCO, SpMet) [1] - Clip	01-2086
Masimo Rainbow DCI SC-400 Adt 3ft (SpO2, SpHb, SpMET) [1] - Clip	01-2092
Masimo rainbow SET DCIP	
Masimo Rainbow DCIP SC-400 Ped 3ft (SpO2, SPCO, SpMet)[1] - Clip	01-2136
Masimo Rainbow DCIP Ped 3ft (SpO2, SpCO, SpMet) [1] - Clip	01-2137
Masimo rainbow SET R1	
Masimo Rainbow R1 25 Adt 1ft (SpO2, SpHb, SpMET) [10] - Adh	01-2128
Masimo Rainbow R1 20 Ped 1ft (SpO2, SpHb, SpMET) [10] - Adh	01-2129
Masimo Rainbow R1 20L Inf 1ft (SpO2, SpHb, SpMET) [10] - Adh	01-2130
Masimo Rainbow R1 25L Neo/Adt 1ft (SpO2, SpHb, SpMET) [10] - Adh	01-2131
Masimo rainbow SET R20	
Masimo Rainbow R20 Ped 1ft (SpO2, SpCO, SpMET) [10] - Adh	01-2133
Masimo Rainbow R20-L Inf 1ft (SpO2, SpCO, SpMET) [10] - Adh	01-2134
Masimo rainbow SET R25	
Masimo Rainbow R25 Adt 1ft (SpO2, SpCO, SpMET) [10] - Adh	01-2132
Masimo Rainbow R25-L Neo/Adt 1ft (SpO2, SpCO, SpMET) [10] - Adh	01-2135
Masimo SET M-LNCS	
MasimoSET M-LNCS DBI Adt SpO2 Reusable Sensor [1] - Boot	01-2089
MasimoSET® M-LNCS Adtx-3 Adult Sensor (20 box)	01-2090
MasimoSET® M-LNCS Neo-3 Neonatal/Adult Sensor (20 box)	01-2091
MasimoSET M-LNCS Pdtx-3 Ped 3ft [20] - Adh	01-2095
MasimoSET M-LNCS DCI Adt 3ft [1] - Clip	01-2123
MasimoSET M-LNCS DCIP Ped 3ft [1] - Clip	01-2124
MasimoSET M-LNCS INF-3 Inf 3ft [20] - Adh	01-2126
MasimoSET M-LNCS NEOPT-3 Neo 3ft [20] - Adh	01-2127
MasimoSET Rainbow Sample Pk (All M-LNCS-Adtx-3, pdtx-3, Neo-3)	01-2140
Masimo E1 (Adult Single Use Ear Sensor)	01-2205
Masimo RD:	
Masimo RD rainbow SET Patient Cable 5ft 25-Pin R/A	01-2268
Masimo RD SET DBI Adult Soft Reusable Sensor 3ft	01-2269
Masimo RD SET DCI Adult Reusable Sensor 3ft	01-2273
Masimo RD SET DCIP Pediatric Reusable Sensor 3ft	01-2274
Masimo RD SET Adt CS-2 Adult [20] - Adh 2ft	01-2275
Masimo RD SET Adt CS-3 Adult [20] - Adh 3ft	01-2276
Masimo RD SET Pdt CS-2 Pediatric [20] - Adh 2ft	01-2277
Masimo RD SET Pdt CS-3 Pediatric [20] - Adh 3ft	01-2278
Masimo RD SET In CS-3 Infant [20] - Adh 3ft	01-2279

Product name & description	Part number
Masimo RD SET Neo CS-3 Neonatal/Adult [20] - Adh 3ft	01-2280
Masimo RD SET NeoPt CS-3 Neonatal [20] - Adh 3ft	01-2281
Masimo RD rainbow SET-2 Adt SpO2 SpMet SpHb [20] - Adh 2ft	01-2282
Masimo RD rainbow SET-2 Pdt SpO2 SpMet SpHb [20] - Adh 2ft	01-2283
Masimo RD rainbow SET Patient Cable 12ft 25-Pin R/A	01-2284
Masimo RD rainbow SET Patient Cable 8ft 25-Pin R/A	01-2285
Masimo RD to M-LNC Adaptor Cable 1.5f	01-2286

13.1.5 Masimo rainbow SET field upgrades

Product name & description	Part number
Masimo Rbow Total Hemoglobin (SpHb+SpOC) Field License Kit	01-2328
Masimo Rbow Methemoglobin (SpMet) Field License Kit	01-2329
Masimo Rbow Carboxyhemoglobin (SpCO) Field License Kit	01-2330
Masimo Rbow Pleth Variability Index (PVI) Field License Kit	01-2331

13.1.6 Ultrasound accessories

Product name & description	Part number
Tempus Pro USB 3.5 MHz General Abdominal Ultrasound Probe (GP)	01-2008
Tempus Pro USB 7.5 MHz Vascular Ultrasound Probe (LP)	01-2042

13.1.7 Video laryngoscope accessories

Product name & description	Part number
Tempus Pro USB C-MAC S Imager Video Laryngoscope	01-2044
Video Laryngoscope Single Use D blade (pack of 10)	01-2063
Video Laryngoscope Single Use Size 3 Macintosh blade (pack of 10)	01-2010
Video Laryngoscope Single Use Size 4 Macintosh blade (pack of 10)	01-2011

13.1.8 Capnometer accessories

Product name & description	Part number
Smart CapnoLine Plus Adult/Intermediate 02 25UN (P/N 015018)	01-2143
Smart CapnoLine Pediatric 02 25units (P/N 015027)	01-2144
VitaLine H Set Adult/Pediatric 25units (P/N 015026)	01-2145
FilterLine H Set Adult/Pediatric 25units (P/N 015016)	01-2146
FilterLine Set Adult/Pediatric 25units (P/N 015021)	01-2147
FilterLine H Set Infant/Neonates 25units (P/N 015028)	01-2148
Filter Set ADU/PED 25UN	01-2150
Smart CapnoLine Plus Adult/Intermediate 02 25UN (P/N009822)	01-2151
Smart CapnoLine Pediatric 02 25units (P/N007269)	01-2247

Product name & description	Part number
VitaLine H Set Adult/Pediatric 25units (P/N010787)	01-2248
FilterLine H Set Adult/Pediatric 25units (P/NXS04624)	01-2249
FilterLine H Set Infant/Neonates 25 units (P/N006324)	01-2250

13.1.9 Electrocardiogram accessories

Product name & description	Part number
Tempus Pro 3-Lead ECG Cable (AAMI) 8 ft	01-2068
Tempus Pro 5-Lead ECG Cable (AAMI) 8 ft	01-2069
Tempus Pro 12 Lead ECG Cable (AAMI) 8 ft	01-2070
Tempus Pro 5-Lead ECG Cable (AAMI) 10ft.	01-2084
Tempus Pro 3-Lead Neonatal ECG Cable (AAMI) 6ft	01-2203
Tempus Pro 4-Lead ECG Modular Trunk Cable (AAMI) 8 ft	01-2177
Tempus Pro 6-Lead ECG Limb Leads for Modular Cable (AAMI)	01-2179
Tempus Pro 3-Lead ECG Cable (IEC) 8 ft	01-2071
Tempus Pro 5-Lead ECG Cable (IEC) 8 ft	01-2072
Tempus Pro 12 Lead ECG Cable (IEC) 8 ft	01-2073
Tempus Pro 3-Lead Neonatal ECG Cable (IEC) 6ft	01-2202
Tempus Pro 4-Lead ECG Modular Cable (IEC) 8 ft	01-2181
Tempus Pro 6-Lead ECG Limb Leads for Modular Cable (IEC)	01-2183
Ambu BlueSensor 'T' Adult Foam ECG Electrode - 10 Pack	01-2080
Ambu BlueSensor R 25units (P/N 1559024)	01-2152
Glasgow ECG algorithm	05-2075

13.1.10 Power and charging accessories

Product name & description	Part number
Tempus Lithium-ion Battery – one supplied with unit	01-2051
Tempus Mains Power Supply (PSU) – one supplied with unit	01-2049
When ordering a PSU, select the mains cable for the country of operation:	
• Switzerland	01-2058
• UK	01-2056
• Euro/Schuko	01-2057
• USA	01-2055
• Australia	01-2060
• Nigeria (as per UK)	01-2056
• Israel	01-2062
• South Africa	01-2054
• Thailand	01-2287
Tempus Battery Charger (including 2-core figure 8 mains cable)	01-1012
When ordering a battery charger, select the mains cable for the country of operation:	
• Switzerland	01-2059
• UK	01-2159
• Euro/Schuko	01-2160
• USA	01-2161
• Australia	01-2061
• Nigeria (as per UK)	01-2159
• Israel	01-2160
• South Africa	01-2162
Tempus DC Vehicle Power Supply	01-2053

13.1.11 Telemedicine accessories

Product name & description	Part number
Sennheiser Presence Bluetooth® Headset (Tempus Pro Kit)	01-2254
Tempus Wired Headset	01-1019
Tempus Ethernet Cable	01-2025
Tempus Pro Tactical Headset Adaptor Cable	01-2041
Inseego USB8 4G Dongle Kit (*)	01-2298
Inseego USB8 4G Dongle (*)	01-2302
Inseego USB8 4G Dongle Mounting Kit (*)	26-4000

(*) The 4G dongle and its accessories are compatible with printer models only. The 4G dongle must be installed by maintenance personnel only.

13.1.12 Miscellaneous accessories

Product name & description	Part number
Tempus Pro Litter Mount	01-2035
Tempus Pro Hard Transit Case	01-2400
Tempus Soft Transit Bag	01-2037
Tempus to Tempus USB Data Cable	01-2243
Tempus Pro SMART mount (order PSU and power cable separately)	01-2244
Tempus Pro Printer paper blank 100mm (1x roll)	01-2184
Tempus Pro Printer paper with grid 100mm (1x roll)	01-2186
Tempus Pro shoulder strap (with locking mech) and 20G fixings kit	01-2197
Tempus Pro shoulder strap only (with locking mech)	01-2200
Left Saddlebag for Tempus Pro	01-2241
Left Tactical Saddlebag for Tempus Pro	01-2259
Right Saddlebag for Tempus Pro	01-2238
Right Tactical Saddlebag for Tempus Pro	01-2260

13.1.13 Anesthetic gas monitoring accessories

Product name & description	Part number
ISA OR+ sidestream Analyzer	01-2167
Tempus to ISA OR+ adaptor cable	01-2168
Nomoline 2.0m (Box of 25) Sampling line with male luer lock connector, single-patient use, Adult/Pediatric/Infant. Requires T adaptor	01-2169
Nomoline Adapter 0.1m (Box of 25) Sampling line with female luer lock connector. Adult/Pediatric/Infant. Requires Nomo Extension and T adaptor	01-2170
Nomoline Airway Adapter Set 2.0m (Box of 20) Sampling line with straight airway adapter, single-patient use, Adult/Pediatric	01-2171
Nomo Extension 2.0m (Box of 25) Sampling line with male luer lock connector	01-2172
Nomo Extension 3.0m (Box of 25) Sampling line with male luer lock connector	01-2173
T-adapter (Box of 25) Airway adapter with female luer lock connector, Adult/Pediatric	01-2174
AA gas exhaust port kit (Masimo #4409)	01-2255
ISA OR+ Ferno Mount Bracket	01-2258

13.1.14 Corsium Crew accessories

Product name & description	Part number
Corsium Crew Cable (USB-A to USB-C)	01-2252

13.1.15 Manuals and software accessories

Product name & description	Part number
Tempus Pro User Manual Set (CD-ROM) – includes Configuration Utility software	43-2001
Tempus Pro Maintenance Manual CD-ROM)	43-2003

14 Configuring the Tempus Pro

For information on configuring communications settings and pairing Bluetooth® peripherals, see the *Tempus Pro Maintenance Manual*.

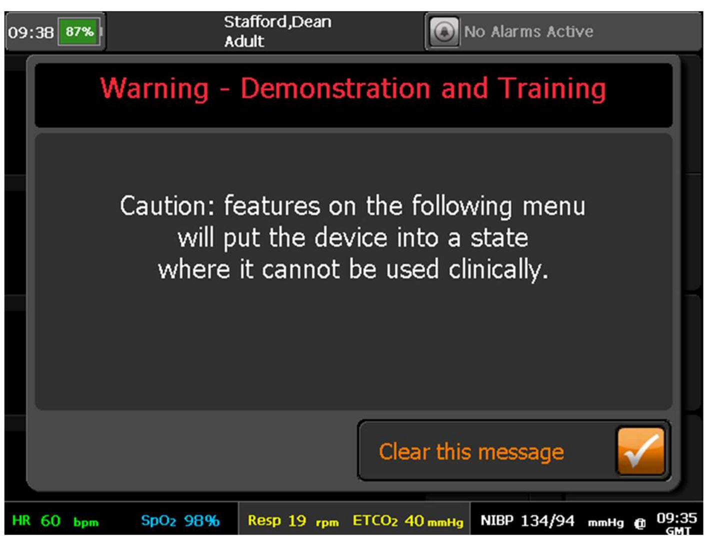
14.1 Demonstration and training

The Tempus can be set so that it can be used in a training environment. Once set for this purpose, the Tempus can simulate:

- All medical readings including ECG monitoring and recording - – in this case the readings are randomly generated by the Tempus regardless of whether a patient is connected or not;
- All other features such as the transmission of photos, videos, GPS positioning etc. – in this case the readings are randomly generated by the Tempus;
- Data and voice connections – where the device simulates the appearance of a connection even when none has been made.

WARNING	**Training and demonstration features are only for use in non-clinical settings i.e. training environments. Activation of these features in clinical settings could lead to confusion of the patient's real vital signs.**

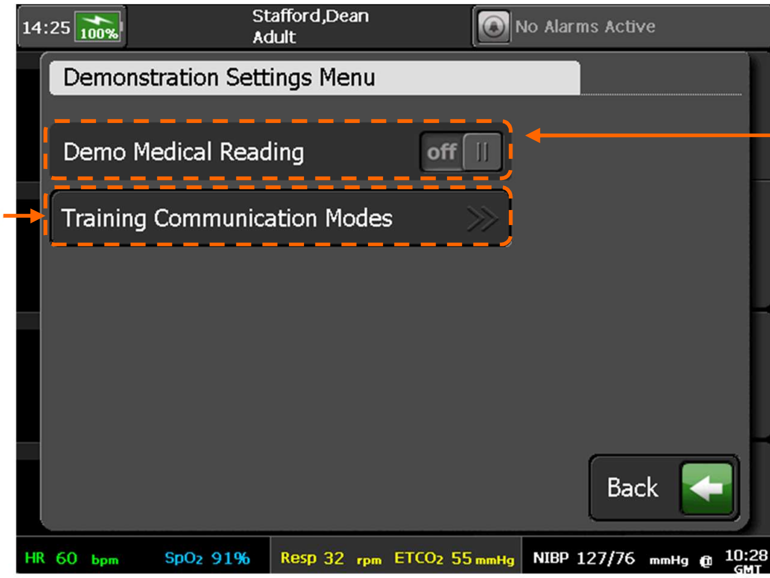
From the Main Menu, press Demonstration and Training button - the Tempus will display a caution.



Demonstration and Training Caution

Clearing this message brings up the Demonstration and Training menu.

Pressing here brings up the Training Modes Menu. This gives a range of communication modes for use in training, in which the Tempus simulates a full data and voice connection but does not require any real equipment to connect to



Pressing here turns ALL the medical and non-medical readings to demo mode – all readings will be simulated

Demonstration and Training Menu

If the Tempus is set to operate in Demonstration Mode, there will be a label “Demo” placed on the home screen and on the 12 Lead ECG recording screen (where the 50/60Hz mains filter buttons would normally be).

If the Tempus is put into a Training Communications Mode, this will be indicated by the word “Training” to the left of the iAssist connection process instruction.

In either case, the Tempus can be put back into normal operational mode by returning to the Demonstration and Training Menu and resetting Demo Mode to “Off”. The correct communications mode can be reset by accessing the Communications Mode Menu from the Main Menu.

WARNING	**If a Tempus is set into either Demonstration or Training mode, it will remain set in this way after the power has been cycled and after a new patient has been admitted.**

15 Specifications and standards

15.1 Specifications

Note that all figures quoted are based on room temperature, pressure and humidity unless otherwise stated.

15.1.1 ECG monitoring

Regulatory standard	Meets ANSI/AAMI EC13:2007 & IEC60601-2-27:2011
Cable	For use with RDT's proprietary 3-, 4-, 5- and 12-lead cables with 4mm snap disposable electrodes
Gain/Sensitivity	2, 4, 5, 10 (default), 1, 20, 30 & 50 mm/mV (scale 8:10)
Input Range	±5 mV
Input Impedance	>100 MΩ
DC Offset	±300 mVdc
Time bases	12.5, 25, 50 mm/s
Aspect ratio range	5:50 – 20:12.5 (mm/mV : mm/s) (scale 8:10)
Heart rate detection range	30 – 300 bpm
Accuracy	±3%
Acquisition sample rate	500 Hz
Monitoring bandwidth	0.67 Hz – 20 Hz : EMS 0.5 Hz – 35 Hz : Monitor 0.05 Hz – 150 Hz : Diagnostic 0.05 Hz – 35 Hz: Filtered diagnostic
Frequency response	0.05 to 175 Hz (-3 dB)
Defibrillator protection	Patient leads are isolated from system and operator, with 5kV protection
Electrosurgery protection	Protected
Common Mode Rejection	95 dB
Leads Off Indicators	Connection status for each lead is shown on acquisition screen
Permanent Filters	High Pass: 0.05 Hz 1st order Low Pass: 150 Hz 1st order Baseline Wander: Baseline reset by adaptive zeroing algorithm
Notch filter (Mains Noise Rejection)	50 Hz 4th order Butterworth,

	49.1Hz - 50.9 Hz, 60 Hz 4th order Butterworth, 59.1 Hz - 60.9 Hz
Low pass (Muscle Artifact Filter)	35 Hz 4th order
ST measurement range	-10 mm to 10 mm (-1 to 1 mV)
ST accuracy	±0.53 mm (±53 uV)
ST resolution	0.1 mm
ST measurement definition	J+80 ms (for heart rates < 115 bpm) J+60 ms (for heart rates > 115 bpm) ISO electric line taken from PQ segment
QT measurement range	10 ms - 1990 ms
QT accuracy	±25 ms mean 30 ms standard deviation
QT resolution	1 ms
Compliance	IEC60601-2-27:2011 AAMI EC57

15.1.2 12-lead diagnostic ECG viewing and recording

Note	These specifications relate to the performance of the product when it is configured to display and record the 12 lead diagnostic ECG view. The filters in this section apply only when viewing a 12-lead ECG recording

Regulatory standard	Meets ANSI/AAMI EC11:1991 inc R:2001 & IEC60601-2-25:2011
Cable	For use with RDT’s proprietary 12 lead (10 wire) cable with 4mm snap disposable electrodes
Gain/Sensitivity	2, 4, 5, 10 (default), 1, 20, 30 & 50 mm/mV
Input Range	±5 mV
Input Impedance	>100 MΩ
DC Offset	±300 mVdc
Acquisition sample rate	500 Hz
12-lead bandwidth (preview and recording)	0.05 Hz – 40 Hz : Filtered diagnostic 0.05 Hz – 175 Hz : Diagnostic **Note** ECG tested at 150 Hz per EC11 and with a 175 Hz 3dB bandwidth measurement. A conservative baseline stabilization filter is applied to the 12-lead ECG signals; the 12-lead recorded ECG and measurements have diagnostic quality suitable for ST segment analysis.

Frequency response	0.05 to 175 Hz (-3 dB)
Defibrillator protection	Patient leads are isolated from system and operator, with 5kV protection
Electrosurgery protection	Protected
Common Mode Rejection	95 dB
Leads Off Indicators	Connection status for each lead is shown on acquisition screen
Notch filter (Mains Noise Rejection)	50 Hz 4th order Butterworth, 49.1Hz - 50.9 Hz, 60 Hz 4th order Butterworth, 59.1 Hz - 60.9 Hz
Low pass (Muscle Artifact Filter)	40 Hz 4th order

15.1.3 Impedance pneumography (respiration)

Range:	30-150 rpm
Accuracy	±2 rpm or ±2% whichever is greater
Excitation current	<20 µA
Excitation frequency	56 kHz
Measurement Leads available	Lead I & Lead II

15.1.4 $ETCO_2$ sensor

The Capnometer is automatically compensated for local atmospheric pressure.

CO_2 Units	mmHg or kPa
$EtCO_2$ Range	0-150 mmHg
CO_2 Waveform Resolution	0.1 mmHg
$EtCO_2$, Resolution	1 mmHg
CO_2 Accuracy	0-38 mmHg: ± 2 mmHg 39-150 mmHg: ± (5% of reading + 0.08% for every 1 mmHg above 38 mmHg) **Note** For operation above 35°C, add an additional ± 1 mmHg or ± 2% (whichever is greater).
Respiration Rate Range	0-149 bpm
Respiration Rate Accuracy	0-70 bpm: ±1 bpm 71-120 bpm: ±2 bpm 121-149 bpm: ±3 bpm
Flow Rate	50 (42.5 ≤ flow ≤ 65) ml/min, flow measured by volume

Waveform Sampling	20 samples/s
Response Time (warmed up)	7 s (typical) for ETCO2, <13 seconds for respiration (assuming a breath rate of 15 rpm)
Calibration Interval	Annual performance verification may be performed if required and calibration then performed if the system requires. Calibration required after 4000 hours' usage of the Capnometer.

15.1.5 Non-invasive blood pressure

Range:	Adult Systolic 40 - 260 mmHg
	Pediatric Systolic 40-230 mmHg
	Neonate Systolic 40-130 mmHg
	Adult Diastolic 20 - 200 mmHg
	Pediatric Diastolic 20-160 mmHg
	Neonate Diastolic 20-100 mmHg
	Adult MAP 26 - 220 mmHg
	Pediatric MAP 26-183 mmHg
	Neonate MAP 26-110 mmHg
Accuracy:	± 3 mmHg
Resolution:	1 mmHg
Pulse rate range:	30-220 bpm ± 2% or ±3 bpm (whichever is greater)
Maximum inflation:	300 mmHg (150 mmHg neonate)
Initial inflation pressure:	160 mmHg adult, 140 mmHg pediatric, 90 mmHg neonate
Method:	Oscillometric, diastolic values correspond to Phase 5 Korotkoff sounds

15.1.6 Invasive pressure

Channels	up to 4 (2 as standard)
Sensitivity	5 µV/V/mmHg
Bridge	180 Ω minimum
Response	0-20 Hz (-3 dB)
Filters	50-60 Hz notch
Range:	-99 – 310 mmHg
Accuracy:	± 2% or ±2 digits including RDT's adaptor cable but excluding the transducer
Resolution:	1 mmHg
Warm up time	10 seconds

15.1.7 Masimo pulse oximetry

Technology	Masimo SET for use in motion and low-perfusion applications
Pulse Range	25-239 bpm
Pulse accuracy	Accuracy (all ages): no motion ≤3 digits, motion ≤5 digits
Pulse resolution	1 bpm
Pulse averaging	8 seconds (configurable)
SpO_2 Range	1-100%
SpO_2 accuracy	Accuracy (adults/child): no motion or low perfusion ±2 digits 70-100%, motion ±3 digits 70-100% Accuracy (neonate): motion, no motion and low perfusion ±3 digits 70-100%
SpO_2 resolution	1%
Data update rate	3 seconds
Type	Functional saturation (test methods available upon request)
Wavelength range	Red 660 nm, infra-red 905 nm Information about wavelength range can be useful to users, especially those performing photodynamic therapy.
Perfusion Index range	0.02-20%
Response	<1 second delay
Signal strength range	0-7 bars
Plethysmogram	1 – 100, auto-gain
Patient population	Reusable soft walled probe – for use in clinical and pre-hospital applications, on fingers and toes for children/adults over 8 years old (other probe types are available from Masimo®)

SpO_2 was determined by testing on healthy adult volunteers in the range 60%-100% SpO2 against a laboratory CO-Oximeter. SpO_2 accuracy was determined on 16 neonatal NICU patients ranging in age from 7 to 135 days old and weighting between 0.5 and 4.25 kg. Seventy-nine (79) data samples were collected over a range of 70 - 100% SaO2 with a resultant accuracy of 2.9% SpO_2.

The arterial oxygen saturation accuracy during no motion only applies to LNOP® Blue SpO_2 adhesive sensors.

Masimo SET technology with LNOP Adt sensors has been validated for motion accuracy in human blood studies on healthy adult male and female volunteers with light to dark skin pigmentation in induced hypoxia studies while performing rubbing and tapping motions, at 2 to 4 Hz at an amplitude of 1 to 2 cm and a non-repetitive motion between 1 to 4Hz at an amplitude of 2 to 3 cm in induced hypoxia studies in the range of 70-100% SpO2 against a laboratory CO-Oximeter and ECG monitor. This variation equals plus or minus one standard deviation which encompasses 68% of the population. The saturation accuracy of the neonatal sensors was validated on adult male and female volunteers with light to dark skin pigmentation and 1% was added to account for the properties of foetal hemoglobin.

The Masimo sensors have been validated for motion accuracy in human blood studies on healthy adult male and female volunteers with light to dark skin pigmentation in induced hypoxia studies while performing rubbing and tapping motions, at 2 to 4 Hz at an amplitude of 1 to 2 cm and a non-repetitive motion between 1 to 5 Hz at an amplitude of 2 to 3 cm in induced hypoxia studies in the range of 70-100% SpO_2 against a laboratory CO-Oximeter and ECG monitor. This variation equals plus or minus one standard deviation. Plus or minus one standard deviation encompasses 68% of the population.

Masimo SET technology has been validated for low perfusion accuracy in bench top testing against a Biotek Index 2 simulator and Masimo's simulator with signal strengths of greater than 0.02% and a % transmission of greater than 5% for saturations ranging from 70 to 100%. This variation equals plus or minus one standard deviation which encompasses 68% of the population.

Masimo sensors have been validated for pulse rate accuracy for the range of 25-240 bpm in bench top testing against a Biotek Index 2 simulator. This variation equals plus or minus one standard deviation which encompasses 68% of the population.

15.1.8 Masimo rainbow SET

Carboxyhemoglobin (SpCO)	
Measurement Range:	0 – 99%
Accuracy (Adults/Pediatrics/Infants):	1 – 40% ± 3%
Resolution:	1%
Alarm:	1-98%
Methemoglobin saturation (SpMet)	
Measurement Range:	0 – 99.9%
Accuracy (Adults/Pediatrics/Infants/Neonates):	1 – 15% ± 1%
Resolution:	0.1%
Alarm:	1-99.5%
Total hemoglobin concentration (SpHb)	
Measurement Range:	0 – 25 g/dL
Accuracy (Adults/Pediatrics):	8 – 17 g/dL ±1 g/dL (arterial or venous)
Resolution:	0.1 g/dL
Alarm:	1-24.5 g/dL
Total oxygen content (SpOC)	
Measurement Range:	0 – 35ml of O2/dL of blood

SpO2, SpCO, and SpMet accuracy was determined by testing on healthy adult volunteers in the range of 60% - 100% SpO2, O% - 40% SpCO, and 0% - 15% SpMet against a laboratory CO-Oximeter. SpO2 and SpMet accuracy was determined on 16 neonatal NICU patients ranging in age from 7-135 days old and weighing between 0.5-4.25 kg. Seventy-nine (79) data samples were collected over a range of 70-100% SaO2 and 0.5-2.5% MetHb with a resultant accuracy of 2.9% SpO2 and 0.9% SpMet.

The Masimo sensors have been validated for no motion accuracy in human blood studies on healthy adult male and female volunteers with light to dark skin pigmentation in induced hypoxia studies in the range of 70-100% SpO2 against a laboratory Co-Oximeter and ECG monitor. This variation equals plus or minus one standard deviation. Plus or minus one standard deviation encompasses 68% of the population.

The Masimo SET Technology has been validated for low perfusion accuracy in bench top testing against a Biotek Index 2 simulator and Masimo's simulator with signal strengths of greater than 0.2% and transmission of greater than 5% for saturations ranging from 70 to 100%. This variation equals plus or minus one standard deviation which encompasses 68% of the population.

The Masimo sensors have been validated for pulse rate accuracy for the range of 25-240 bpm in bench top testing against a Biotek Index 2 simulator. This variation equals plus or minus one standard deviation which encompasses 68% of the population.

SpHb accuracy has been validated on healthy adult male and female volunteers and on surgical patients with light to dark skin pigmentation in the range of 8-17 g/dl SpHb against a laboratory CO-oximeter. This variation equals plus or minus one standard deviation which encompasses 68% of the population. The SpHb accuracy has not been validated with motion or low perfusion.

The following substances may interfere with pulse CO-Oximetry measurements:

- Elevated levels of Methemoglobin (MetHb) may lead to inaccurate SpO2 and SpCO measurements
- Elevated levels of Carboxyhemoglobin (COHb) may lead to inaccurate SpO2 measurements
- Very low arterial Oxygen Saturation (SpO2) levels may cause inaccurate SpCO and SpMet measurements
- Severe anaemia may cause erroneous SpO2 readings
- Dyes, or any substance containing dyes, that change usual blood pigmentation may cause erroneous readings.
- Elevated levels of total bilirubin may lead to inaccurate SpO2, SpMet, SpCO and SpHb readings.

15.1.9 Contact temperature

Channels	2 (first standard, second is optional)
Compatibility:	YSI series 400
Measurement range:	20-45 °C / 68-113 °F
Accuracy:	±0.1 °C / ±0.2 °F
Resolution:	±0.1 °C / ±0.2 °F
Measurement time	<10 seconds

15.2 Physical characteristics and environmental specifications

Size and weight – non-printer model:	Standalone size and weight of Tempus Pro part number 00-1004-R with its battery and RapidPak clip but without external cables, sensors or accessories: 263 mm (10.3”) x 216 mm (8.5”) high x 100 mm (3.9”) (width x height x depth) 2.9 kg (6.4 Ib) nominal The non-printer model with a Bluetooth headset is slightly heavier (part number 00-1007-R).
Size and weight – printer model:	Standalone size and weight of Tempus Pro part number 00-1026-R with its battery and printer but without external cables, sensors or accessories: 263 mm (10.3 ”) x 216 mm (8.5 ”) x 102 mm (4 ”) (width x height x depth) 3.2 kg (7 Ib) nominal Printer models without invasive pressure are slightly lighter (part number 00-1024-R).
Altitude:	-200 to 5486 m (-656 to 18000 ft)
	(can be used at higher physical altitudes provided the local atmosphere is no higher than 5486 m, e.g. in a pressurized aircraft cabin)
	Note The power supply’s altitude rating must be adhered to. Do not use the power supply outside of its specifications.
Relative humidity:	15%-95% (non-condensing)
Operating temperature range:	0º C to 50º C all functions, 0º C to 40º C charging Short term operating low -20 °C for 60 mins after being stored at room temperature
Transport/Storage temperature range:	-37º C to +73.3º C
Transport/Storage pressure range:	-200 to 5486 m (-656 to 18000 ft) 104 kPa to 53 kPa
Start-up time in normal use	Time for switching on or starting until the equipment is ready for normal use: 42 seconds
Start-up time from storage	Time required for the equipment to warm from the minimum storage temperature between uses until the equipment is ready for intended use at 20°C: 180 minutes Time required for the equipment to cool from the maximum storage temperature between uses until the equipment is ready for its intended use at 20°C: 180 minutes
Mode of operation	Continuous

IP66 (dust and water ingress under pressure) – whole device. According to IEC60529, “IP” means ingress protection. The following two numerals refer to grades of ingress protection for solid objects (the first numeral) and water (the second numeral). The first numeral “6” means the device is “dust-proof” against the ingress of dust to a size <75 µm. The second numeral means the device is protected against the ingress of 100 l/m of water “jetting” at the device from 2.5 to 3 m.

15.2.1 Environmental performance and certification

Drop	1 m drop per IEC60601-1
Transit Drop	1.22 m (4 ft) MIL 810G, 26 drops all corners, faces and edges with and without accessories connected
Operational Drop	0.75 m (2.5 ft) IEC60068-2-32 Procedure 1, 6 drops to all faces with the device fully operational and all accessories connected
Impact	500 g steel ball from 1.3 m as per IEC60601 including the display

Temperature & altitude:

Category:	A1
Test Standard,	Temperature, RTCA/DO-160E Section 4, Para 4.5.1 to 4.5.4
Test Standard, Altitude,	RTCA/DO-160E Section 4, Para 4.6.1 and 4.6.2
Temperature:	Operating: 0 °C to +50 °C
	Storage: -37 °C to +73.3 °C
	Short Term High: +50 °C
Rapid Decompression:	2438 m to 5486 m (8,000 ft to 18000 ft) in 15 seconds

Temperature variation:

Test Standard:	RTCA/DO-160G Section 5 Cat C
Rate of Variation:	2 °C per minute.

Humidity:

Test Standard:	RTCA/DO-160E Section 6 Cat A
Transport/Storage:	15 to 95% RH Non-condensing (tested for 48 hours at 38-50 °C)
Operating:	15 to 95% RH Non-condensing (tested at the end of the storage cycle)

Operational shocks & crash safety:

Test Standard:	RTCA/DO-160E Section 7 Cat B
Operational Shock:	40 g per MIL-STD 810G 6 g per RTCA DO-160E (6g for 11ms saw-tooth wave, repeated 3 times in all axis).
Crash Safety:	20g in all directions (sustained).
Bump:	15 g per EN1789

Vibration:

Test Standard:	Test Standard: MIL-STD-810G rotary, fixed wing (jet and turboprop profile) Ground vehicle per EN1789

Explosion proofness:

Not tested. Not to be used in the presence of explosive gasses or vapours.

Water proofness:

Not tested to RTCA/DO-160G Section 10
Commercial qualification:

Fluids susceptibility:

Not applicable.

Sand & dust:

Commercial qualification:	IP66 (dust ingress under pressure) – whole device (excluding internal printer and optional 4G dongle, which are IP43)

Fungus resistance:

Not applicable.

Salt spray:

Not applicable.

Magnetic effect:

Not applicable.

Power input:

Not tested to RTCA/DO-160E Section 16.

Commercial qualification:	EN61000-3-2:2006 inc A1/A2:2009 Mains harmonics
	EN61000-3-3:2008 Mains flicker
	EN61000-4-11:2004 voltage dips and interruptions

Voltage spike:

Not tested to RTCA/DO-160G Section 17.

Commercial qualification:	IEC61000-4-4:2012 Fast transient bursts

Audio frequency conducted susceptibility – power inputs:

Not tested to RTCA/DO-160G Section 18.

Commercial qualification:	IEC61000-4-5:2006 Surges

Induced signal susceptibility:

Not tested to RTCA/DO-160G Section 19.

Commercial qualification:	IEC61000-4-6:2013 Conducted RF field

Radio frequency susceptibility (radiated & conducted):

Not tested to RTCA/DO-160G Section 20.

Commercial qualification:	IEC61000-4-3:2006 inc A2:2010 Radiated RF Interference

Emission of radio frequency energy:

Test Standard: RTCA/DO-160G Section 21 Cat M.

Lightning induced transient susceptibility:

Not applicable.

Lightning direct effects:

Not applicable.

Icing:

Not applicable.

ESD:

Not tested to RTCA/DO-160G Section 25.
Commercial qualification: IEC61000-4-2:2008 ESD

Fire and smoke hazards:

Main case material:	PC-ABS
Flame:	UL94V-0
Overmould material:	TPU (*)

(*) Contains 2-(2H-benzotriazol-2-yl)-4,6-ditertpentylphenol.

15.2.2 Use in high ambient temperatures

CAUTION ⚠	Since the Tempus is rated for use in high ambient temperatures, users are reminded to give consideration to the potential for outer surfaces of the Tempus to become hot in such ambient temperatures. Further to Table 23 of IEC60601-1:2005, plastic parts (such as the sides, touchscreen, handle etc.) which may be acceptable for long term physical contact at ambient temperatures up to 48°C need to be touched for shorter durations (e.g. less than 1 minute) when used in higher ambient temperatures. Similarly, users should note that while the rear heat sink may be touched permanently in ambient temperatures up to 36°C, in higher ambient temperatures contact with it should be reduced. Users should pay attention to this if instead of using the handle, they grip the device around its sides.
CAUTION ⚠	Do not use the Tempus Pro (when using its external power supply) to recharge batteries when ambient temperature exceeds 40°C.

Note	In ambient temperatures above 40 °C users should ensure that the device is used in an upright position and that there is air flow around the back of the unit.
Note	Users are reminded that the Tempus (when using its external power supply) should not be used to recharge batteries above 40°C – see "14.2 Physical characteristics and environmental specifications".
Note	Users are reminded that the specification of the power supply limits its use to 40°C maximum – see "14.3 Miscellaneous features and specifications".

15.3 Miscellaneous features and specifications

15.3.1 Invasive pressure USB module 01-2017 (optional)

Channels	2
Sensitivity	5 µV/V/mmHg
Bridge	180 Ω minimum
Response	0-20 Hz (-3 dB)
Filters	50/100/150 Hz & 50-60 Hz notch
Range:	-99 – 310 mmHg
Accuracy:	± 2% or ±2 digits including RDT's adaptor cable but excluding the transducer
Resolution:	1 mmHg
Warm up time	10 seconds
Ingress protection	IP56
Size	108 mm x 63 mm x 36 mm
Weight	116g
EMC	IEC 60601-1-2:2014 Class B
Altitude:	-200 to 5486 m (-656 to 18000 ft)
	(can be used at higher physical altitudes provided the local atmosphere is no higher than 5486 m, e.g. in a pressurized aircraft cabin)
Relative humidity:	15%-95% (non-condensing)
Operating temperature range:	0º C to 40º C due to transducer specifications
Transport/Storage temperature range:	-37º C to +73.3º C
Transport/Storage pressure range:	-200 to 5486 m (-656 to 18000 ft); 104 kPa to 53 kPa
Drop	1 m drop per IEC60601-1
Transit Drop	1.22 m (4 ft) MIL 810G, 26 drops all corners, faces and edges with and without accessories connected (when not connected to the Tempus Pro)
Impact	500 g steel ball from 1.3 m as per IEC60601
Emission of radio Frequency energy:	RTCA/DO-160G Section 21 Cat M.
Temperature & Altitude:	
Category:	A1
Test Standard, Temperature,	RTCA/DO-160E Section 4, Para 4.5.1 to 4.5.4
Test Standard, Altitude,	RTCA/DO-160E Section 4, Para 4.6.1 and 4.6.2
Temperature:	Operating: 0 °C to +40 °C Storage: -37 °C to +73.3 °C
Rapid Decompression:	2438 m to 15545 m (8,000 ft to 51,000 ft) in 1 second

Temperature Variation:	
Test Standard:	RTCA/DO-160E Section 5 Cat C
Rate of Variation:	2 °C per minute.
Humidity:	
Test Standard:	RTCA/DO-160E Section 6 Cat A
Transport/Storage:	15 to 95% RH Non-condensing (tested for 48 hours at 38-50 °C)
Operating:	15 to 95% RH Non-condensing (tested at the end of the storage cycle)
Operational Shocks & Crash Safety:	
Test Standard:	RTCA/DO-160E Section 7 Cat B
Operational Shock:	Para 7.2 (6g for 11ms saw-tooth wave, repeated 3 times in all axis).
Crash Safety:	20g in all directions (sustained).
Vibration:	
Test Standard:	MIL-810-F rotary, fixed wing (jet and turboprop profile)
Main case material:	PC-ABS
Flame:	UL94V-0

15.3.2 Rechargeable battery

Battery life	At least 10¾ hours with display brightness at 60% (default), ECG, SpO_2, $ETCO_2$, IP x 2, temp x 2 and NIBP every 15 minutes. At least 11½ hours with display brightness at 30%, ECG, SpO_2, $ETCO_2$, IP x 2, temp x 2 and NIBP every 15 minutes. Up to 14 hours with battery saving mode activated (display at default 60% brightness) – typically 12½ hours with the display active 50% of the time.
Nominal voltage	7.4 V
Charging voltage	8.4 V ±1%
Nominal capacity	10.2 Ah, 75.5 Wh
Weight	0.42 kg nominal
Dimensions	152 x 42 x 62 mm max
Shelf life	Approximately 75% remaining after 1 year (before the charge indicator light turns to Amber)
Operating temperature	-20°C to +60°C
Charge time	3 hours to 90% and approximately 4 hours to 100% Assumptions: • Battery fitted to Tempus Pro attached to mains power. • Subject to conditions of storage and use, times are approximate. • Tempus Pro is switched off while charging, charging takes longer when the device is active.

Note	Battery shelf life and run times are based on a new, fully charged battery stored and used at 20 °C. Run time is based on RDT's model of typical device usage in an incident.

15.3.3 Mains power supply

Note	Only the RDT mains power supply (part number 01-2049) can be used with the Tempus Pro.

Mains input voltage	100 – 240 V
Frequency	50/60 Hz & 400 Hz
Input current	~2 A (<0.5 A drawn when supplying the Tempus)
Output voltage	12 V dc
Output current	5 A
Relative humidity:	15%-95% (non-condensing)
Weight	0.42 kg nominal
Dimensions	133 x 60.7 x 41 mm (5.24” x 2.39” x 1.62”)
Operating temperature range:	0º C to 50º C
Transport/Storage temperature range:	-40º C to +85º C
Altitude:	0 – 4000 m (0 – 13,123 ft)
	Note The power supply's altitude rating of 4000m must be adhered to. Do not use the power supply outside of its specifications.

15.3.4 Battery charger

Note	Only the RDT Battery Charger (part number 01-1012) can be used with the Tempus Pro.

Mains input voltage	100-240 V
Frequency	50-60 Hz
Input current	0.9 A max (at 100 V approx.)
Output voltage	8.4 V dc
Output current	<2.73 A
Charge time (from empty)	5 hours
Weight	0.25 kg nominal
Dimensions	107 x 67 x 36.5 mm

15.3.5 Vehicle power supply

Battery input voltage	11V dc – 27 Vdc
Input connector type	Cigar plug
Output voltage	12 Vdc ±5%
Output current max	6 A
Rated Power	72 W
Efficiency	>85%
Operating temperature range:	0º C to 40º C
Transport/Storage temperature range:	-20º C to +85º C
Weight	0.21 kg nominal
Dimensions	105 x 40 x 26.5 mm
Operational shock	10g
Non-operational shock	60g

15.3.6 GPS

Antenna	Integral
Accuracy	±10 m (±2.5 km with <6 satellites – labelled as “Approximate Fix”)

15.3.7 Inseego 4G USB dongle (optional)

Note	The 4G dongle is available as an option on Tempus Pro printer models only.

Manufacturer	Inseego
Name	USB8
Model	MC800
Approvals	FCC (North America); ISED (Canada); PTCRB
Weight	68 g (2.4 oz)
Dimensions	91.1 x 37 x 11.4 mm (3.58” x 1.45” x 0.45”)
Chip set	QUALCOMM® SDX20
Interface type	Fold-away, rotating USB
SIM	4FF
Operating temperature	-10°C to +55°C (+14°F to +113°F)
Storage temperature	-20°C to +65°C (-4°F to +149°F)
Relativity humidity	5% to 90% over operating temperature
Drop	1 meter drop, no damage – fully operational
Vibration stability	5 Hz to 500 Hz, 0.1 octave/second

15.3.8 Other features

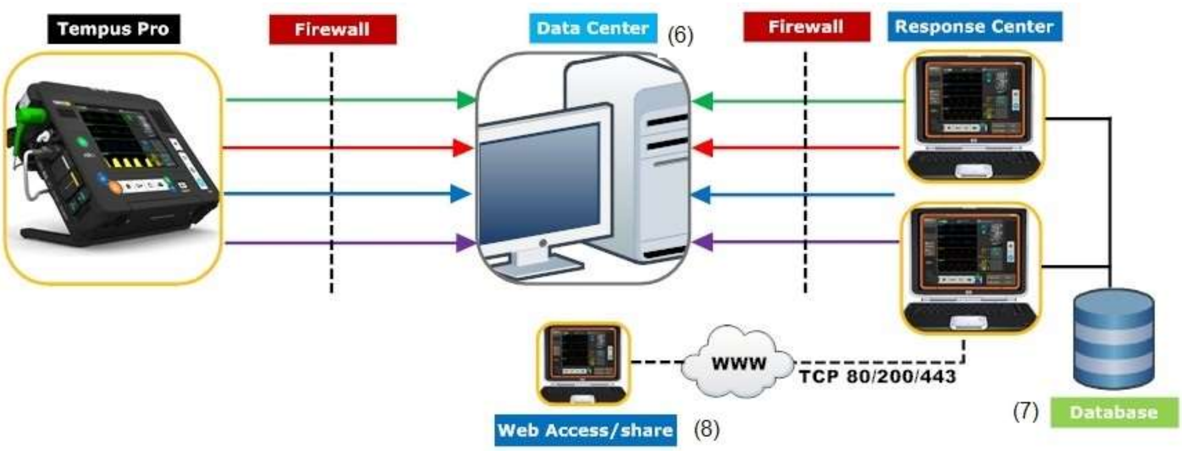
Display	Color 165 mm (6.5”) 640x480 pixels 130 klux daylight readable display
Printer (*)	High resolution 110 mm (4.3”) integrated thermal printer
Integral digital camera	Color 3.2M pixel camera Takes still pictures or video using the H264 algorithm (bandwidth dependent)

(*) Optional, additional feature

15.4 Communications

15.4.1 i2i ReachBak communications

When connecting to a Response Center via i2i ReachBak, the Tempus has the following communications requirements:



Tempus Connection – all data AES encrypted: Ethernet; WiFi – WEP, WPA, WPA-2; Cell – 3G (if enabled) , 4G (printer models only, if enabled).

Database – all PHI AES 256 encrypted.

Tempus to Data Center (Gateway)	Destination Port (FIPS 140-2 enabled)	Minimum data rate required			Response Center to Data Center (Gateway)	Destination Port (FIPS 140-2 enabled)	Minimum data rate required	
Medical Data (1)	TCP 2169	3 kbps	3 kbps		Medical Data (1)	TCP 52169	3 kbps	3 kbps
Waveforms (2)	UDP (SRTP) port 52508	6 kbps			Waveforms (2)	UDP (SRTP) port 52509	6 kbps	
Voice (VOIP) (3)	UDP (SRTP) 52505	12 kbps	12 kbps		Voice (VOIP) (3)	UDP (SRTP) 52511	12 kbps	12 kbps
Video (4)	UDP (SRTP) port 52506		40 kbps		Video (4)	UDP (SRTP) port 52507		40 kbps
Total (5)		21 kbps	55 kbps		Total (5)		21 kbps	55 kbps

Notes:

(1) Medical Data means heart rate, non-invasive blood pressure, SpO_2, $ETCO_2$, invasive pressure, respiration rate and temperature with still photos and 12 Lead ECG recordings transmitted on a store-and-forward basis and TCCC card data sent every 30 seconds.

(2) Waveforms means 1-2 channels of ECG, Plethysmogram and Capnogram and 1-2 channels of invasive pressure.

(3) VOIP - Speex codec .

(4) Video - H264 format CIF image transmitted at 5 frames per second - this provides a low-bandwidth video solution.

(5) Total rate is on a lossless data transfer; the bandwidth is likely to increase if data retry attempts are required due to packet loss on the network.

(6) The Data Center application is installed on the first or primary PC to use the i2i application. The Data Center application is simply installed (selected) during the i2i installation process. Ports will need to be opened on firewalls in and out of the DC and RC.

(7) A shared database is recommended between all RC installations. This requires a local wired network.

(8) Web access is optional and allows users with i2i installed on a PC (with internet access) to share the data elements of the real time display with other users via the web who don't have i2i. Web access is automatically terminated when the Tempus disconnects from the i2i.

i2i transmission rates

ECG data and digital pictures take an appreciable amount of time to send to the Response Center, approximate times are as follows:

- 12 lead ECG – < 30 seconds
- Digital photographs – < 30 seconds

These times are for guidance only and are based on a typical Ethernet network. Times may vary depending on network equipment, configuration and usage.

15.4.2 Ethernet specification

The Ethernet connection has the following specifications:

- IEEE 802.3 compliant
- RJ-45 connection
- DHCP or fixed IP, Mask, Gateway
- Network cable: CAT5 at least 50 m

15.4.3 WiFi specification

The WiFi technology used by the Tempus operates using IEEE 802.11b and 802.11g standard. It operates in the Industrial, Scientific and Medical (ISM) band between 2.412 GHz and 2.484 GHz.

WARNING	**Per IEC60601-1-2 cl 5.2.2.5 b), the Tempus may be interfered with by other systems even if they comply with CISPR emissions.**

The WiFi technology has the following features:

WiFi Specification	
The WiFi module has the following specifications:	
Range	~ 91 m (300 ft)
Data Rate:	CCK: 1, 2, 5.5, 11 Mb/s

WiFi Specification	
	OFDM 6, 9, 12, 18, 24, 36, 48, 54 MB/s
Security	WEP, TKIP, WPA and WPA2 AES/CCMP per IEEE 802.11 i
Baseband Modulation	802.11g: OFDM 802.11b: DSSS/CCK
Quality of Service	802.11 e EDCF

15.4.4 Integral 2G and 3G cell phone specification (if enabled)

The Cell Phone (GSM) technology used by the Tempus has the following specifications:

- Operating frequency range:

Parameter		Min	Max	Unit
Frequency Range Uplink (MS → BTS)	GSM 850	824	849	MHz
	E-GSM 900	880	915	MHz
	GSM 1800	1710	1785	MHz
	GSM 1900	1850	1920	MHz
	UMTS 850	824	849	MHz
	UMTS 900	880	915	MHz
	UMTS 1900	1850	1910	MHz
	UMTS 2100	1920	1980	MHz
Frequency Range Downlink (BTS → MS)	GSM 850	869	894	MHz
	E-GSM 900	925	960	MHz
	GSM 1800	1805	1880	MHz
	GSM 1900	1930	1990	MHz
	UMTS 850	869	894	MHz
	UMTS 900	925	960	MHz
	UMTS 1900	1930	1990	MHz
	UMTS 2100	2110	2170	MHz

- RF power:

Band	Max	Unit
GSM/GPRS 850/900	33	dBm
GSM/GPRS 1800/1900	30	dBm
EDGE 850/900	27	dBm
EDGE 1800/1900	26	dBm

UMTS 850/900/1900/2100	24	dBm

Note	This device contains GSM 900 MHz and GSM 1800MHz functions that are not operational in U.S. Territories.

15.4.5 USB dongle 4G cell phone specification (if enabled)

The 4G USB dongle (installed as an option on Tempus Pro printer models) has the following specifications:

Operating mode	Operating frequency range		Maximum transmit power (conducted) dBm
	Tx (MHz)	Rx (MHz)	
3G band I	1920 - 1980	2110 - 2170	24
3G band VIII	880 - 915	925 - 960	24
LTE band 1	1920 - 1980	2110 - 2170	24
LTE band 3	1710 - 1785	1805 - 1880	24
LTE band 7	2500 - 2570	2620 - 2690	24
LTE band 20	832 - 862	791 - 821	24
LTE band 28	703 - 748	758 - 803	24

15.4.6 Bluetooth® specification

WARNING	**Per IEC60601-1-2 cl 5.2.2.5 b), the Tempus may be interfered with by other systems even if they comply with CISPR emissions.**

The Bluetooth® module has the following specifications:

Indoor Range	~ 9 m (30 ft) (typical office environment)
Data Rate:	V2.0 up to 1 Mb/s EDR: 2.3 Mb/s
Baseband Modulation	V2.0: GFSK EDR: Pi/4 DQPSK, 8DPSK

15.4.7 Bluetooth® headset specification

The Tempus Pro uses the Sennheiser® Presence or VMX200 wireless headset. This is unmodified by RDT and is provided under FCC part 15C and an Industry Canada License 2099D. Users are reminded to refer to the Sennheiser user guide (attached to the CD-ROM provided with the Tempus) that provides instructions for use of the headset. These include environmental performance specifications which may differ from those of the Tempus.

Description	**General Specifications**
Bluetooth® type	V4.0 class 1 (Presence) V3.0 class 2 (VMX200)
Range	10 m max in an open field
Transmission frequency	2402 – 2480 MHz

Description	General Specifications
Bluetooth® Profiles	HSP, HFP, A2DP (Presence) HSP, HFP (VMX200)
Weight	13 g (Presence) 10 g (VMX200)
Size	51 x 19 x 23 mm (Presence) 55 x 26 x 58 mm (VMX200) (WxHxD)
Talk time	Up to 4 hours* (Presence) Up to 6 hours* (VMX200)
Stand by time	Up to 240 hours*
Operating temperature range	10 – 40 °C
Operating humidity range	20 – 85% non-condensing
Storage temperature range	-20 to +60 °C
Storage humidity range	10-95% non-condensing

*Battery shelf life and run times are based on a new, fully charged battery.

15.4.8 Communications security specification

The Tempus Pro is (optionally) provided with Mocana Federal Information Processing Standard (FIPS) 140-2 cryptographic module. This provides US government certified encryption on the voice, data and video transmission.

The FIPS 140-2 encryption provided on the Tempus Pro is Level 1 as specified by the FIPS 140-2 standard.

15.4.9 FCC and Industry Canada compliance

CAUTION ⚠	Do not disassemble the device. There are no user-serviceable parts inside. Refer servicing to the manufacturer. Changes or modifications not expressly approved by RDT could void the user's authority to operate the equipment.

This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

The user may find the following booklet helpful: *How to Identify and Resolve Radio-TV Interference Problems.* This booklet is available from U.S. Government Printing Office, Washington, D.C. 20402.

This equipment is also ETS 300 328, ETS 300 826, ETS 300 328-2, ETS EN301 489-1 and ETS EN301 489-17 compliant. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

FCC compliance

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation.

This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions:

- This device may not cause interference and
- This device must accept any interference, including interference that may cause undesired operation of the device.

IC complance

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Industry Canada Standard ICES-003. These limits are designed to provide reasonable protection against harmful interference in a residential installation.

This device contains license-exempt transmitter(s)/receiver(s) that comply with Innovation, Science and Economic Development Canada's license-exempt RSS(s). Operation is subject to the following two conditions:

1. This device may not cause interference.
2. This device must accept any interference, including interference that may cause undesired operation of the device.

Note	**4G USB dongle (printer models only):** For FCC and Industry Canada compliance information, refer to the user guide supplied with the Inseego USB8 4G Dongle.

Singapore compliance

Tempus Pro Singapore IMDA License Information:

Complies with IMDA standards DB100486 DA102408

DA102408 for wifi/WLAN/Bluetooth/NFC capability

DB100486 for 4G/GSM capability.

15.5 Tempus Pro device classification

The system is classified according to the requirements of EN60601-1:2006, the standard for Medical Electrical Equipment, Part 1, General Requirements for Safety, as:

- The Tempus Pro is internally (battery) powered. The external power supply is class I (earthed) as defined by the classification of the power supply specified and supplied by RDT. The battery charger is class II (double insulated).
- Applied parts type CF defibrillator proof. All patient coupled connections from the Tempus are designated as Patient Applied Parts per IEC60601-1.
- The Tempus Pro is rated IP66, protected against rainfall according to IEC60529. This means that the device is proof against the ingress of talcum powder into the body of the device (into its case) and also proof against the entry of water from a strong hose (akin to a garden hose) into the body of the device (into its case). All other parts are rated IPXX. The device is rated IP66 with all connectors either mated or unmated (except the Capnometer cannula which must be unconnected and its door clicked shut) and with dust covers fitted or not fitted and with the battery mated or unmated.
- No parts supplied sterile or suitable for/requiring sterilizing.

- Equipment not suitable for use in the presence of a flammable anesthetic mixture with air or with oxygen or nitrous oxide.
- Suitable for continuous use, for use with defibrillators and for use with electrosurgical systems.
- The expected service life of the device (excluding batteries) is 5 years. This can be extended through maintenance of the device.

15.5.1 Standards compliance

The Tempus Pro complies with the applicable parts of the following standards:

Standard	Title
IEC 60601-1 3rd edition	Medical electrical equipment -- Part 1-1: General requirements for safety - Collateral standard: Safety requirements for medical electrical systems
IEC 60601-1-2 4th edition	Medical electrical equipment -- Part 1-2: General requirements for safety - Collateral standard: Electromagnetic compatibility - Requirements and tests
IEC 60601-1-6	Medical electrical equipment -- Part 1-6: General requirements for safety - Collateral standard: Usability
EN 60601-1-8	Medical electrical equipment -- Part 1-8: General requirements, tests and guidance for alarm systems in medical electrical equipment and medical electrical systems
IEC 60601-2-49	Medical electrical equipment -- Part 2-49: Particular requirements for the safety of multifunction patient monitoring equipment
ISO 80601-2-56	Medical electrical equipment -- Part 2-56: Particular requirements for basic safety and essential performance of clinical thermometers for body temperature measurement
EN 60601-2-34	Medical electrical equipment -- Part 2-34: Particular requirements for the basic safety and essential performance of invasive blood pressure monitoring equipment
EN 60601-2-37	Medical electrical equipment - Part 2-37: Particular requirements for the basic safety and essential performance of ultrasonic medical diagnostic and monitoring equipment
IEC 60601-2-25	Medical electrical equipment - Part 2-25: Particular requirements for the basic safety and essential performance of electrocardiographs
IEC 60601-2-27	Medical electrical equipment - Part 2-27: Particular requirements for the basic safety and essential performance of electrocardiographic monitoring equipment
EN 80601-2-61	Medical electrical equipment -- Part 2-61: Particular requirements for basic safety and essential performance of pulse oximeter equipment
EN 80601-2-55	Medical electrical equipment -- Part 2-55: Particular requirements for the basic safety and essential performance of respiratory gas monitors
EN 62304	Medical device software -- Software life cycle processes
EN 62366	Medical devices -- Part 1: Application of usability engineering to medical devices

15.6 EMC information

The following tables provide information required to be provided under IEC60601-1-2.

15.6.1 Cable length of the sensors and the accessories

	RDT Part Number	Cable Length (typ.)	Tested Length
Ethernet cable	01-2025	2.1m	2.1m
SpO2 sensor	01-2014	2.4m	2.4m
3-lead ECG cable	Various	2.4m	2.4m
4-lead ECG cable	Various	2.4m	2.4m
5-lead ECG cable	Various	2.4m	2.4m
12-lead ECG cable	Various	2.4m	2.4m
IBP adaptor cable	Various	2.4m	2.4m
Wired headset	01-1019	1.2m	1.2m
Mains Power supply	01-2049	0.45m	0.45m
Mains lead	01-2055	2m	2m

WARNING ⚠	**The use of longer cable lengths may cause an increased emission or a reduced interference resistance. The use of other sensors or cables except the ones mentioned above is not allowed.**

CAUTION ⚠	For the purposes of compliance with IEC60601-1-2, all vital signs and communications functions were regarded as essential performance.

15.6.2 Manufacturer's Declarations

Electromagnetic Emissions (IEC EN 60601-1-2)

The Tempus Pro is intended for use in an electromagnetic environment as described below. The customer or user of the device should ensure that the device is used in such an environment.

Emission Measurements	Compliance	Electromagnetic Environment
HF emissions acc. to CISPR11	Group 2	The Tempus Pro must emit RF energy in order to perform its function. Nearby electronic devices may be affected. Note that the Tempus Pro can be configured for not to emit RF energy in which case it will be group 1 and will not be likely to cause any interference in nearby electronic equipment.
HF emissions acc. to CISPR11	Class B	The Tempus Pro is intended for use in all facilities including living quarters and such ones which are connected directly to a public power supply that supplies also buildings used for living purposes.

Emission Measurements	Compliance	Electromagnetic Environment
Emission of overtones acc. to IEC61000-3-2	Class B	
Emission of voltage fluctuation/flicker acc. to IEC61000-3-3	Complies	

Electromagnetic Emissions (IEC EN 60601-1-2)

The Tempus Pro is intended for use in an electromagnetic environment as described below. The customer or user of the device should ensure that the device is used in such an environment.

Interference Resistance Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment - Guidelines
Electrostatic discharge (ESD) acc. to IEC 61000-4-2	± 8 kV contact discharge ± 15 kV air discharge	± 8 kV contact discharge ± 15 kV air discharge	Floors should be of wood, concrete or ceramic tiles. If the floor is tiled with synthetic material the relative air humidity must have 30 % at least.
Fast transient electric disturbances / bursts acc. to IEC 61000-4-4	± 2 kV for power lines ± 1 kV for input and output lines	± 2 kV for power lines ± 1 kV for input and output lines	The quality of the supply voltage should conform to a typical business or clinic environment.
Surge voltage acc. to IEC 6100-4-5	± 1 kV normal mode voltage ± 2 kV common mode voltage	± 1 kV normal mode voltage ± 2 kV common mode voltage	Mains power should be that of a typical hospital or commercial environment.
Voltage drops, short interruptions and variations in supply voltage acc. to IEC 61000-4-11	< 5 % U_T (>95 % break of U_T) for 0.5 period 40 % U_T (60% break of U_T) for 5 periods 70 % U_T (30% break of U_T) for 25 periods < 5 % U_T (>95 % break of U_T) for 5 seconds	< 5 % U_T (>95 % break of U_T) for 0.5 period 40 % U_T (60% break of U_T) for 5 periods 70 % U_T (30% break of U_T) for 25 periods < 5 % U_T (>95 % break of U_T) for 5 seconds	Mains power should be that of a typical hospital or commercial environment. If the user of the Tempus Pro requires continued operation during power interruptions, then the battery may be used for periods up to 6 hours or a UPS may be used.
Magnetic field at the supply frequency (50/60 Hz) acc. to IEC 61000-4-8	30 A/m	100 A/m	Magnetic fields at the supply frequency should conform to the typical values as they occur in the business or clinic environment.
NOTE U_T is the AC mains voltage before the use of testing levels			

Electromagnetic Interference Resistance (IEC EN 60601-1-2)

The Tempus Pro is intended for use in an electromagnetic environment as described below. The customer or user of the device should ensure that the device is used in such an environment.

Interference Resistance Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment – Guidelines
			Portable and mobile RF communications equipment should be used no closer to the device including the cables than it is recommended by the equation for the frequency. **Recommended safety distance**
Conducted RF disturbances acc. to IEC61000-4-6	3 Vrms 150 kHz to 80 Mhz	3 Vrms	d = 1.2√P
	6 Vrms in ISM bands (*1)	6 Vrms	d = 1.2√P
Radiated RF disturbances acc. to IEC61000-4-3	3 V/m 80 MHz to 2.7 GHz	3 V/m for all parameters & 20 V/m for all parameters except invasive pressure per EN 1789	d = 1.2√P for 80 MHz to 800 MHz d = 2.3√P for 800 MHz to 2.7 GHz
	Frequency spots as listed below (*4)	Frequency spots as listed below (*4)	
			P is the nominal power of the transmitter in watt (W) according to the specifications of the transmitter manufacturer; d is the recommended safety distance in meters (m). The field strength of stationary transmitters should be lower than the Compliance level for all frequencies according to a testing on location (*2) (*3). Interference may occur in the vicinity of devices with the following symbol: 
NOTE 1: For 80 Hz and 800 MHz the higher frequency range applies. NOTE 2: These guidelines may not be applicable for all cases. The propagation of electromagnetic values is influenced by absorptions and reflections of buildings, objects and people.			

(*1) The ISM (industrial, scientific, and medical) bands between 150 kHz and 80 MHz are 6.765 MHz to 6.795 MHz; 13.553 MHz to 13.567 MHz; 26.957 MHz to 27.283 MHz; and 40.66 MHz to 40.70 MHz.

(*2) The field strength of stationary transmitters such as fixed parts of cellular phones and mobile radio sets, amateur radio stations, AM and FM radio and television cannot be determined exactly in theory. To detect the electromagnetic environment in regard to stationary transmitters a study of the location should be considered. If the measured field strength at the location where the device is being used exceeds the Compliance level above the device should be watched to verify the proper functions. If unusual features are watched additional actions might be necessary such as a modified orientation or another location of the device.

(*3) For the frequency range of 150 kHz to 80 MHz, the field strength should be lower than 10 V/m.

(*4) Immunity to proximity fields from RF wireless communications equipment:

Test frequency (MHz)	**Service**	**Modulation**	**Immunity test level** (V/m)
385	TETRA 400	Pulse modulation 18 Hz	27
450	GMRS 460, FR 460	FM ± 5 kHz deviation 1 kHz sine	28
710 745 780	LTE Band 13, 17	Pulse modulation 217 Hz	9
810 870 930	GSM 800/900, TETRA 800, iDEN 820, CDMA 850, LTE Band 5	Pulse modulation 18 Hz	28
1720 1845 1970	GSM 1800; CDMA 1900; GSM 1900; DECT; LTE Band 1, 3, 4, 25; UMTS	Pulse modulation 217 Hz	28
2450	Bluetooth, WLAN, 802.11 b/g/n, RFID 2450, LTE Band 7	Pulse modulation 217 Hz	28
5240 5500 5785	WLAN 802.11 a/n	Pulse modulation 217 Hz	9

15.6.3 Recommended safety distances

Recommended Safety Distances between portable and mobile RF Telecommunication Devices and the Tempus Pro (Tab. 206 according to IEC EN 60601-1-2).

The TEMPUS PRO is intended for use in an electromagnetic environment with controlled RF disturbances. The user of the device can help to avoid electromagnetic disturbances by keeping the minimum distance between portable and mobile telecommunication devices (transmitters) and the device - depending on the output power of the telecommunication devices as described below.

Safety Distance Depending on the Frequency in m			
Nominal power of the transmitter (W)	150 kHz to 80 MHz d = 1.2√P	80 MHz to 800 MHz d = 1.2√P	800 MHz to 2.7 GHz d = 2.3√P
0,01	0.12	0.12	0.23
0,1	0.38	0.38	0.73
1	1.2	1.2	2.3
10	3.8	3.8	7.3
100	12	12	23
For transmitters with a maximum nominal power not mentioned above: To detect the recommended safety distance use the equitation in the corresponding column. P is the maximum nominal power of the transmitter in watt (W) according to the specifications of the transmitter manufacturer.			
NOTE 1: For 80 Hz and 800 MHz the higher frequency range applies.			
NOTE 2: These guidelines may not be applicable for all cases. The propagation of electromagnetic values is influenced by absorptions and reflections of buildings, objects and people.			

15.7 Factory default settings

Setting	Range	Factory Default
Communications mode	Various – customer specific	Uppermost listed mode within the customer specific communications mode menu
Patient	Neonate, Child, Adult	Adult
NIBP mode	Manual, Auto, Rapid Cuff	Auto
NIBP automatic measurement interval	2, 3, 5, 10, 15, 30 and 60 minutes	Automatic - 3 minutes
QRS volume	0-100% in steps of 20	0%
ECG waveform 1	Lead I-III – 3-lead ECG cable Lead I-III, aVL, aVR & aVF – 4-lead ECG cable Lead I-III, aVL, aVR, aVF & V – 5-lead ECG cable Lead I-III, aVL, aVR, aVF, V1-6 – 12-lead ECG cable	Lead II
ECG waveform 2	Lead I-III, aVL, aVR & aVF – 4-lead ECG cable Lead I-III, aVL, aVR, aVF & V – 5-lead ECG cable Lead I-III, aVL, aVR, aVF, V1-6 – 12-lead ECG cable	Lead I
ECG wave speed	12.5, 25, 50 mm/s	25mm/s
ECG Gain	2, 4, 5, 10, 15, 20, 30 and 50 mm/mV	10mm/mV
ECG Monitoring filter	Monitor (0.5 Hz – 35 Hz) Filtered diagnostic (0.05 Hz – 35 Hz) Diagnostic (0.05 Hz – 150 Hz) EMS (0.67 Hz – 20 Hz)	Monitor (0.5 Hz – 35 Hz)
ECG Mains filter	ON-OFF	ON
Arrhythmia detection	ON-OFF	ON
Mains frequency	50-60 Hz	50Hz
Pulse Oximeter plethysmogram wave speed	12.5, 25, 50 mm/s	25mm/s
Capnograph wave speed	3.1, 6.25, 12.5 mm/s	6.25mm/s
Temperature readings	ºC - ºF	ºC
Invasive pressure	See "6.6.4 Configuring the transducer / channel".	P1 – White with arterial settings P2 – White with arterial settings P3 – White with arterial settings P4 – White with arterial settings

16 Linking Tempus LS to Tempus Pro

This chapter describes how to set up a link to allow data to be transferred from the Tempus LS defibrillator to the Tempus Pro patient monitor. This is known as a TDL link.

Note	Throughout these instructions, ‘Tempus LS’ refers to whichever defibrillator model is in use, either Tempus LS or Tempus LS-Manual. The Tempus LS or Tempus LS-Manual defibrillator may only be used if it is approved for commercial distribution in your territory.

16.1 TDL link

When the TDL link is on, Tempus LS automatically transmits event data to Tempus Pro, which records these events in its summary record of care. Tempus Pro can print these events manually (when requested by the user) or automatically (on units with an internal printer).

The TDL link may be wireless (via Bluetooth) or direct (via any standard USB-A to USB-B data cable).

Tempus LS must be authenticated by the maintenance user before it can be linked to Tempus Pro. For instructions, see the *Tempus Pro Maintenance Manual*.

WARNING	**Alarm limits must be selected carefully on Tempus Pro and Tempus LS when using them together on the same patient.**

Note	If the software versions of the Tempus LS and Tempus Pro are found to be incompatible on connection, the following dialog will be displayed by the Tempus Pro: "Attention - Tempus LS Connection Unable to connect to the Tempus LS This could be because the software versions on the Tempus Pro and Tempus LS are incompatible. Ensure the Tempus Pro and Tempus LS are both updated with the latest software." If this message appears, contact RDT to obtain the latest software versions. If the software versions of the Tempus Pro and Tempus LS are incompatible, the Pro and LS will pair via the TDL Link as normal, however, no event data will be automatically transmitted from the LS to the Pro via the TDL link. The event data will continue to be stored on the LS as normal and can be exported using the Export feature in the LS device.

16.2 Pairing Tempus Pro with Tempus LS

To allow the TDL link to operate wirelessly, the Tempus Pro and Tempus LS must be paired via Bluetooth. This pairing is normally performed once only, the first time the two devices are used together. Pairing is normally retained after this.

- If Tempus Pro and Tempus LS are both currently unpaired, you can pair them via either Bluetooth or a USB data cable.
- When connected via USB data cable, the devices silently pair via Bluetooth, meaning that when they are no longer connected via USB data cable they remain paired via Bluetooth.
- The *Tempus Pro Maintenance Manual* describes how to configure Tempus Pro to not persist Bluetooth pairing with Tempus LS after powering off. If **Persist Tempus LS pairing** is switched off, you will need to repeat Bluetooth pairing every time Tempus Pro is powered on.

Proceed as follows:

1. Put Tempus Pro and Tempus LS within 5 metres of each other (preferably in line of sight).
2. On the Tempus Pro, press and hold **Main Menu** for at least 2 seconds until the Bluetooth icon starts to blink (alternatively, select **Main Menu > TDL Settings > Connect to a Tempus LS**).
3. On the Tempus LS, press and hold the **Menu** button for at least 2 seconds until the Bluetooth icon starts to blink.
4. After 15 seconds (typical), the units pair and sound the link audio tones and flash their LEDs. The 'TDL' icon appears in the Tempus Pro status bar:

Event data created on the Tempus LS will now be pushed to the Tempus Pro and incorporated into the Summary Record of Care.

16.3 Using the TDL link

16.3.1 Connecting via Bluetooth

If Bluetooth pairing has been completed and if Tempus Pro is configured to persist pairing, the TDL link is active as soon as Tempus Pro and Tempus LS are powered on.

16.3.2 Connecting via USB cable

Tempus Pro and Tempus LS can also be connected via USB data cable (with or without Bluetooth).

To set up the TDL link via USB data cable, use either a direct or an indirect connection:

- Direct connection - plug a USB data cable into the USB ports of Tempus Pro and Tempus LS.
- Indirect connection - plug a USB data cable into the USB ports of the Smart Mounts of Tempus Pro and Tempus LS, then dock the two units in their Smart Mounts.

If Bluetooth is enabled on the Tempus Pro, it silently pairs with the Tempus LS when the devices are connected via USB data cable. After 15 seconds (typical), the units pair and sound the link audio tones and flash their LEDs. The 'TDL' icon appears in the Tempus Pro status bar:

16.3.3 Identifying the connected Tempus LS

On the Tempus Pro, select **Main Menu** **> TDL Settings > Identify connected Tempus LS**.

This triggers TDL link audio tones and LEDs on the connected Tempus LS.

16.4 Using the connected units on different patients

WARNING	**To avoid mixing up patients, always discharge and admit when moving between patients.**

If you need to use the connected Tempus Pro and Tempus LS on different patients:

1. On the Tempus Pro, select **Main Menu** **> TDL Settings**.
2. Set **TDL Link** to **off**.
3. When the confirmation dialog is displayed, press **Confirm**. Data from the current Tempus LS patient will not be written to the Tempus Pro summary record of care.
4. Press **Back**.

Note	When TDL Link is off, Tempus Pro and Tempus LS remain connected.

To revert to using the connected Tempus Pro and Tempus LS on the same patient:

1. On the Tempus Pro, select **Main Menu** **> TDL Settings**.
2. Set **TDL Link** to **on**.
3. When a dialog is displayed with the question "Are the Tempus Pro and Tempus LS being used on the same patient? ", press **Yes - same patient**. Data from the current Tempus LS patient will be written to the Tempus Pro summary record of care.
4. Press **Back**.

Certain events will cause Tempus Pro to display the question "Are the Tempus Pro and Tempus LS being used on the same patient? ". In response, select one of two options:

- **Yes - same patient** - data from the Tempus LS will be written to the Tempus Pro summary record of care.
- **No - different patient** - data from the Tempus LS will not be written to the Tempus Pro summary record of care.

16.5 Tempus LS event capture

If Tempus Pro is connected with a Tempus LS defibrillator and the TDL link is on, defibrillator events are recorded in the summary record of care (SRoC).

WARNING	The CPR sensor is not indicated for use on children < 25 kg (55 lbs) and/or children under 8 years.

CAUTION	Data from the Tempus LS will only be available if the connected Tempus Pro and Tempus LS are linked (that is, **TDL Link** is on) and sufficient time has been given for the data to transfer. When linked, a statement at the top of the SRoC screen will indicate if synchronization of data is in progress or incomplete. If the TDL link is not active, no Tempus LS data will be merged into the patient SRoC. The progress of data transfer does not affect the normal operation of the Tempus LS or Tempus Pro.

Note	The CPR start, stop and rate recorded in SRoC when the LifePoint sensor is not used is based on impedance measurements and therefore the rate is guidance only. For an accurate rate measurement use the LifePoint sensor. LifePoint is not supported in pediatric mode.

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The SRoC Summary tab displays the 'LS' icon against events captured from the Tempus LS, as shown in this example:

Tempus LS events captured in the SRoC include:

- Pads attached and detacted.
- Shock.
- Shock advised and no shock advised.
- Analysis requested.
- Pacer events.
- Mode switching.
- Waveform capture – this is triggered by pressing the **Snapshot / Print** button on Tempus LS.
- CPR events - the average speed of CPR compressions; if a LifePoint CPR device is used, the percentage of good compressions is also captured (this is only available for adult patients; quality may be displayed as 0% if the feedback device is used in pediatric mode on the Tempus LS).

Default settings on Tempus LS can be configured to automatically print waveforms for certain of these events.

Note	**Tempus Pro printer models:** The Tempus LS can be configured to print selected defibrillator events on the Tempus Pro internal printer. In Tempus LS (software v1.3 and above) when the ECG waveform is monitored in pacer mode, the waveform printed on the Tempus Pro with internal thermal printer will be the ECG waveform.
WARNING	In rare circumstances, the internal printer may not print as expected when linked to a Tempus LS. If an expected print does not start automatically then start the print manually from the Tempus Pro**.**

WARNING	In rare circumstances, the Tempus LS snapshot may contain a waveform of shorter duration than expected. If required, capture another snapshot or review the full incident in Corsium Code Review.

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17 End user license agreement

17.1 Tempus Pro EULA

This license covers RDT's Tempus Pro software.

Warning: The software contained herein is protected by copyright law and international treaties. Unauthorized reproduction, distribution or reverse engineering of this program, or any portion of it, may result in severe civil and criminal penalties, and will be prosecuted to the maximum extent possible under the law.

You acknowledge that you will read and adhere to the user manual and ensure that users receive proper training from RDT or an appropriately trained individual. You also acknowledge that you will maintain the software by installing new software updates supplied by RDT within 5 working days of receiving or being notified of them.

License:
You may transfer the program and license to another party if the party agrees to accept the terms and conditions of this agreement. You will not share the program with other parties and will keep the program and details of its functions confidential. You will ensure that access to the software and use of it will be restricted to properly trained and authorized personnel only. You will not try to copy or reverse engineer the software.

The Tempus Pro operates over third-party communications links, such as telephone lines, GSM or satellite links and the Internet. RDT does not accept liability for the failure of these links to reliably transmit information from RDT's products. Users are reminded that it is their responsibility to ensure that GSM network and other communications contracts are maintained and suitably setup and configured for the areas in which they need to be used.

In order to function correctly Tempus Pro needs to operate over a communications link such as satellite communications, GSM or a telephone line and other types of links. It is your responsibility to maintain these communications links. Such links may have security or other measures implemented on them such as firewalls. It is your responsibility to ensure that any such firewalls or other elements of the communication link are configured correctly to allow data from Tempus Pro to communicate over said link. RDT does not accept any responsibility for failure to transmit data or to transmit data reliably over such links if they have not been configured correctly. Support on configuring such links can be obtained from RDT upon request.

Neither the Tempus Pro or RDT are a "covered entity" under the Health Insurance Portability and Accountability Act of 1996 and the regulations promulgated thereunder ("HIPAA"). As a result, HIPAA does not apply to the transmission of health information by RDT or Tempus Pro to any third party. However, users of the Tempus Pro may be "covered entities" under the HIPAA act and so are reminded that they are responsible for acting accordingly. All users are reminded that it is their responsibility to maintain all patient records in a manner that is compliant with applicable local regulations.

The software gives users the ability to share and transmit medical data with third parties. Such activities are entirely the responsibility of the user.

RDT owns all proprietary rights to the Tempus Pro. RDT gives you a personal, revocable, non-assignable, and non-exclusive license to use the Tempus Pro.

Limited Warranty and Remedies:
In no event shall RDT or its distributors or agents, be liable for any damages resulting from loss of data, loss of revenue or for any incidental or consequential damages incurred arising out of or relating to the use of this

software product. Some jurisdictions do not allow the exclusion of implied warranties, so the above exclusion may not apply to you. This warranty gives you specific legal rights and you may have other rights that vary from region to region.

RDT’s Terms and Conditions apply.

RDT and its distributors make no representations with respect to the merchantability or fitness of the Tempus Pro software and the product is supplied "as is", without any warranty of any kind. Further, RDT reserves the right to revise its publications and program(s) without obligation to notify customer of such a revision.

You acknowledge that you have read this agreement, understood it, and agree to be bound by its terms and conditions.

Failure to enforce any provision will not constitute a waiver of that provision. If any provision is found unenforceable, it and any related provisions will be interpreted to best accomplish the unenforceable provision's essential purpose.

This agreement is governed by UK law. The exclusive venue for any dispute relating to this agreement is London UK. You and RDT consent to the personal jurisdiction of these courts. Nothing in this agreement limits either party's ability to seek equitable relief.

17.2 Mocana EULA

This license covers the Mocana FIPS 140-2 security technology.

CUSTOMER LICENSE AGREEMENT MINIMUM TERMS

1. No Warranty. RDT does not extend to its customers (on Mocana’s behalf) any warranty – express, implied or statutory – including without limitation the implied warranties of title, non-infringement, merchantability, fitness for a particular purpose, accuracy, and quiet enjoyment.

2. No Liability. RDT does not extend to its customers (on Mocana’s behalf) any liability, including liability for any incidental, special, consequential, indirect or direct damages of any kind (including damages for interruption of business, procurement of substitute goods, lost data, lost profits, or the like) regardless of the form of action, whether in contract, tort (including negligence), strict product liability, or any other legal or equitable theory.

3. No Reverse Engineering. Users are not permitted to in any way reverse engineer, decompile, or modify the Tempus Pro software, the i2i software or the Mocana software.

4. Export Control. The Tempus Pro (including the Mocana FIPS 140-2 encryption software) contains certain cryptographic functionality, the export and re-export of which is restricted under U.S. law.

Note	The Mocana FIPS security algorithm is provided as an option only and is not standard on all Tempus Pro devices.

17.3 MPEG4 EULA

NOTICE REGARDING VIDEO STANDARDS.

THIS PRODUCT IS LICENSED UNDER ONE OR MORE VIDEO PATENT PORTFOLIO LICENSES SUCH AS AND WITHOUT LIMITATION VC-1 AND MPEG4 PART2 VISUAL FOR THE PERSONAL AND NON-COMMERCIAL USE OF A CONSUMER TO:

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17.4 Masimo® EULA

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3. Assignment. Purchaser shall not assign or transfer this License, in whole or in part, by operation of law or otherwise, without RDT's prior written consent; any attempt without such consent, to assign any rights, duties or obligations arising hereunder shall be void.

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5. Use Restriction. As the Purchaser, you may physically transfer the products from one location to another provided that the software/firmware is not copied. You may not electronically transfer the software/firmware from the products to any other device. You may not disclose, publish, translate, release distribute copies of, modify, adapt, translate, reverse engineer, decompile, disassemble, or create derivative works based on the Masimo Product, the software/firmware, or the written materials without the prior written consent of RDT. Masimo Sensors that are designated for single use are licensed under Masimo patents for use on a single patient only, and are not sold. There is no license, implied or otherwise, that would allow use of single use Masimo Sensors beyond their intended single use. After use of single use Masimo Sensors, there is no further license granted by Masimo to use the sensors and they much be discarded.

6. Transfer Restrictions. The software/firmware is licensed to the Purchaser, and may not be transferred to anyone, except other end-users, without the prior written consent of RDT. In no event may you transfer, assign, rent, lease, sell, or otherwise dispose of the software/firmware or the products on a temporary basis.

7. Beneficiary. Masimo Corporation is a Beneficiary of this Agreement and has the right to enforce its provisions.

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17.8 US Government Devices – US Department of Defense (DoD) Notice and Consent Banner

The following statement is applicable to devices purchased by US government employees and which are used on SPR networks.

You are accessing a U.S. Government (USG) Information System (IS) that is provided for USG-authorized use only. By using this IS (which includes any device attached to this IS), you consent to the following conditions:

1. The USG routinely intercepts and monitors communications on this IS for purposes including, but not limited to, penetration testing, COMSEC monitoring, network operations and defense, personnel misconduct (PM), law enforcement (LE), and counterintelligence (CI) investigations.

2. At any time, the USG may inspect and seize data stored on this IS.

3. Communications using, or data stored on, this IS are not private, are subject to routine monitoring, interception, and search, and may be disclosed or used for any USG-authorized purpose.

4. This IS includes security measures (e.g., authentication and access controls) to protect USG interests--not for your personal benefit or privacy.

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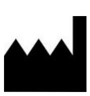
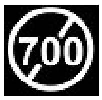
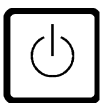
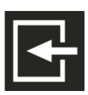
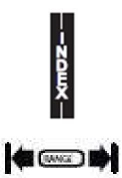
18 Symbols used on the Tempus Pro

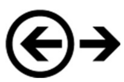
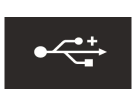
18.1 Symbols used

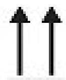
The following symbols are used either on the Tempus Pro or the accessories and packaging provided with it:

Symbol	Description
	Consult the instructions for use. These are provided in electronic format in the Philips Document Library.
	Consult the instructions for use (for an electronic IFU). These are provided in electronic format in the Philips Document Library.
EC REP	Name and address of the European Union Authorized Representative.
Rx Only	The device is for prescription use only
	Consult the instructions for use for important warning information
	Consult the instructions for use for important cautionary information
CE 0413	Signifies European technical conformity for a device/accessory, CE marked by notified body number 0413.
CE	Signifies European technical conformity for a device/accessory, For class I devices.
	Indicates the medical device manufacturer.

Symbol	Description
 YYYY-MM	Date of manufacture, where the year that the item was manufactured is represented by the year and then the month e.g. 2019-06 is June 2019.
LOT YYYY-MM	The manufacturer’s batch code. Where the year that the item was manufactured as a part of a larger batch is represented by the year and then the month e.g. 2019-06 is June 2019.
	Indicates the number of pieces in the package
	Keep away from sunlight
	Pause audible alarms temporarily
	Pause all alarm manifestations (visual and audible) temporarily
	Alarm off (either individual parameter or all parameters)
	General alarm symbol
	Type CF defibrillation-proof applied part complying with IEC 60601-1
CO_2	Capnometer inlet
TEMP	Contact temperature sockets
ECG	ECG Socket
P1/P2	Invasive pressure socket
NIBP by SunTech	NIBP socket

Symbol	Description
	Pulse oximeter socket
	Tactical Mode associated with low emitted light and low emitted sound
	Not for use with YSI 700 series sensors
	Battery Charge indicator – flashes green when the battery is on charge
	Battery power level
	System power on/off (push/push)
	Single use device only, discard item after use
	Capnometer Cannula inlet connection
	These symbols are used to check that the blood pressure cuff is the correct size, once wrapped around the arm the INDEX mark should be between the RANGE marks
	Center point of blood pressure cuff bladder - to be aligned with the patient's artery
 XX-YY cm	Diameter size for the blood pressure cuff
	Item does not contain any latex (blood pressure cuffs)

Symbol	Description
	Item does not contain any PVC (blood pressure cuffs)
IP66	IP66 (dust and water ingress under pressure) – whole device. According to IEC60529, “IP” means ingress protection. The following two numerals refer to grades of ingress protection for solid objects (the first numeral) and water (the second numeral). The first numeral “6” means the device is “dust-proof” against the ingress of dust to a size <75 µm. The second numeral “6” means the device is protected against the ingress of 100 l/m of water “jetting” at the device from 2.5 to 3 m.
 YYYY-MM	Use-by date, where the time that the unit must be used by is represented by the year and then the month e.g. 2019-06 is June 2019.
	EU Directive on Waste Electrical and Electronic Equipment (WEEE). This product must not be disposed of or dumped with household waste.
	Communications connections
	WiFi connection mode to Response Center
	Bluetooth® connection to medical modules
	Battery Connection – to indicate positive terminal polarity
	Global Positioning System (GPS)
	Global System for Mobile (GSM) communications
	Headset connector
	Ethernet Socket (RJ-45)
	USB 2.0 & 1.0 sockets
	Power Status (green indicates mains power is connected)

Symbol	Description
	Camera Backlight
	Device contains wireless transmitters
	DC connector
	Packing box to be stored upright
	Packing box contains breakable items – handle with care
	Recyclable packaging
	Packaging should not be discarded
	Packing box should not be exposed to rain/water/moisture
	Packing box storage humidity range
	Packing box storage temperature range
	Packing box storage pressure/altitude range

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